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Discrete Mathematics

MTH202

Virtual University of Pakistan

Knowledge beyond the boundaries

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Lecture No.23 Mathematical Induction

PRINCIPLE OF MATHEMATICAL INDUCTION:

Let P(n) be a propositional function defined for all positive integers n. P(n) is true for every positive integer n if

1.Basis Step:

The proposition P(1) is true.

2.Inductive Step:

If P(k) is true then P(k + 1) is true for all integers k ≥ 1.

i.e. $\forall$ k p(k) $\rightarrow$ P(k + 1)

EXAMPLE:

Use Mathematical Induction to prove that

$$1+2+3+\cdots+n=\frac{n(n+1)}{2} \quad \text{for all integers } n \geq 1$$

SOLUTION:

Let $P(n): 1+2+3+\cdots+n=\frac{n(n+1)}{2}$

1.Basis Step:

P(1) is true.

For n = 1, left hand side of P(1) is the sum of all the successive integers starting at 1 and ending at 1, so LHS = 1 and RHS is

$$R.H.S=\frac{1(1+1)}{2}=\frac{2}{2}=1$$

so the proposition is true for n = 1.

2. Inductive Step: Suppose P(k) is true for, some integers k ≥ 1.

(1) $$1+2+3+\cdots+k=\frac{k(k+1)}{2}$$

To prove P(k + 1) is true. That is,

(2) $$1+2+3+\cdots+(k+1)=\frac{(k+1)(k+2)}{2}$$

Consider L.H.S. of (2)

$$\begin{aligned}
1+2+3+\cdots+(k+1) &= 1+2+3+\cdots+k+(k+1) \\
&= \frac{k(k+1)}{2}+(k+1) \qquad \text{using (1)} \\
&= (k+1)\left[\frac{k}{2}+1\right]
\end{aligned}$$

$$= (k+1)\left[\frac{k+2}{2}\right]$$
$$= \frac{(k+1)(k+2)}{2} = \text{RHS of (2)}$$

Hence by principle of Mathematical Induction the given result true for all integers greater or equal to 1.

EXERCISE:

Use mathematical induction to prove that
$1+3+5+\ldots+(2n-1) = n^2$ for all integers $n \geq 1$.

SOLUTION:

Let P(n) be the equation $1+3+5+\ldots+(2n-1) = n^2$

1. **Basis Step**:

   P(1) is true
   For n = 1, L.H.S of P(1) = 1and
   R.H.S = 12 = 1
   Hence the equation is true for n = 1

2. **Inductive Step**:

   Suppose P(k) is true for some integer $k \geq 1$. That is,
   $1 + 3 + 5 + \ldots + (2k - 1) = k^2$ …………………(1)

To prove P(k+1) is true; i.e.,
$1 + 3 + 5 + \ldots + [2(k+1)-1] = (k+1)^2$ ………….……(2)

Consider L.H.S. of (2)

$$\begin{aligned} 1+3+5+\cdots+[2(k+1)-1] &= 1+3+5+\cdots+(2k+1) \\ &= 1+3+5+\cdots+(2k-1)+(2k+1) \\ &= k^2+(2k+1) \qquad \text{using (1)} \\ &= (k+1)^2 \\ &= \text{R.H.S. of (2)} \end{aligned}$$

Thus P(k+1) is also true. Hence by mathematical induction, the given equation is true for all integers $n \geq 1$.

EXERCISE:

Use mathematical induction to prove that
$1+2+2^2 + \ldots + 2^n = 2^{n+1} - 1$ for all integers $n \geq 0$

SOLUTION:

Let P(n): $1 + 2 + 2^2 + \ldots + 2^n = 2^{n+1} - 1$

1. **Basis Step:**

   P(0) is true.

For n = 0
L.H.S of P(0) = 1
R.H.S of P(0) = $2^{0+1} - 1 = 2 - 1 = 1$
Hence P(0) is true.

2. **Inductive Step:**

Suppose P(k) is true for some integer k ≥ 0; i.e.,

$1+2+2^2+\ldots+2^k = 2^{k+1} - 1$……………………(1)

To prove P(k+1) is true, i.e.,

$1+2+2^2+\ldots+2^{k+1} =$ 2k+1+1 – 1…………………(2)

Consider LHS of equation (2)

$1+2+2^2+\ldots+2^{k+1} = (1+2+2^2+\ldots+2^k) + 2^{k+1}$
$= (2^{k+1} - 1) + 2^{k+1}$
$= 2\cdot 2^{k+1} - 1$
$= 2^{k+1+1} - 1$ = R.H.S of (2)

Hence P(k+1) is true and consequently by mathematical induction the given propositional function is true for all integers n ≥ 0.

EXERCISE:

Prove by mathematical induction

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for all integers n ≥1.

SOLUTION:

Let P(n) denotes the given equation

1. **Basis step:**

P(1) is true
For n = 1
L.H.S of P(1) = 12 = 1

$$\text{R.H.S of P(1)} = \frac{1(1+1)(2(1)+1)}{6}$$

$$= \frac{(1)(2)(3)}{6} = \frac{6}{6} = 1$$

So L.H.S = R.H.S of P(1).Hence P(1) is true

2.Inductive Step:

Suppose P(k) is true for some integer k ≥1;

$$1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6} \qquad \ldots\ldots\ldots(1)$$

To prove P(k+1) is true; i.e.;

$$1^2 + 2^2 + 3^2 + \cdots + (k+1)^2 = \frac{(k+1)(k+1+1)(2(k+1)+1)}{6} \qquad \ldots(2)$$

Consider LHS of above equation (2)

$$
\begin{aligned}
1^2+2^2+3^2+\cdots+(k+1)^2 &= 1^2+2^2+3^2+\cdots+k^2+(k+1)^2\\
&= \frac{k(k+1)(2k+1)}{6}+(k+1)^2\\
&= (k+1)\left[\frac{k(2k+1)}{6}+(k+1)\right]\\
&= (k+1)\left[\frac{k(2k+1)+6(k+1)}{6}\right]\\
&= (k+1)\left[\frac{2k^2+k+6k+6}{6}\right]\\
&= \frac{(k+1)(2k^2+7k+6)}{6}\\
&= \frac{(k+1)(k+2)(2k+3)}{6}\\
&= \frac{(k+1)(k+1+1)(2(k+1)+1)}{6}
\end{aligned}
$$

EXERCISE:

Prove by mathematical induction

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1} \qquad \text{for all integers } n\geq 1$$

SOLUTION:

Let P(n) be the given equation.

1. Basis Step:

P(1) is true

For n = 1

L.H.S of P(1) = $\frac{1}{1\cdot 2}=\frac{1}{1\times 2}=\frac{1}{2}$

R.H.S of P(1) = $\frac{1}{1+1}=\frac{1}{2}$

Hence P(1) is true

2. Inductive Step:

Suppose P(k) is true, for some integer k≥1. That is

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{k(k+1)}=\frac{k}{k+1} \quad \ldots\ldots(1)$$

To prove P(k+1) is true. That is

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k+1)(k+1+1)}=\frac{k+1}{(k+1)+1} \quad \ldots\ldots..(2)$$

Now we will consider the L.H.S of the equation (2) and will try to get the R.H.S by using equation (1) and some simple computation.

Consider LHS of (2)

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{(k+1)(k+2)}$$
$$=\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}$$
$$=\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}$$
$$=\frac{k(k+2)+1}{(k+1)(k+2)}$$
$$=\frac{k^2+2k+1}{(k+1)(k+2)}$$
$$=\frac{(k+1)^2}{(k+1)(k+2)}$$
$$=\frac{k+1}{(k+2)}$$
$$=\text{RHS of (2)}$$

Hence P(k+1) is also true and so by Mathematical induction the given equation is true for all integers n ≥1.

EXERCISE:

Use mathematical induction to prove that

$$\sum_{i=1}^{n+1} i2^i = n\cdot 2^{n+2}+2, \qquad \text{for all integers } n\geq 0$$

SOLUTION:

1.Basis Step:

To prove the formula for n = 0, we need to show that

$$\sum_{i=1}^{0+1} i.2^i = 0\cdot 2^{0+2}+2$$

Now, L.H.S = $\sum_{i=1}^{1} i\cdot 2^i = (1)2^1 = 2$

R.H.S = $0\cdot 2^2 + 2 = 0 + 2 = 2$

Hence the formula is true for n = 0

2.Inductive Step:

Suppose for some integer n = k ≥0

$$\sum_{i=1}^{k+1} i\cdot 2^i = k\cdot 2^{k+2}+2 \qquad \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

We must show that $\sum_{i=1}^{k+2} i\cdot 2^i = (k+1)\cdot 2^{k+1+2}+2$ ………..(2)

Consider LHS of (2)

$$\sum_{i=1}^{k+2} i\cdot 2^{i} = \sum_{i=1}^{k+1} i\cdot 2^{i} + (k+2)\cdot 2^{k+2}$$
$$= (k\cdot 2^{k+2}+2)+(k+2)\cdot 2^{k+2}$$
$$= (k+k+2)2^{k+2}+2$$
$$= (2k+2)\cdot 2^{k+2}+2$$
$$= (k+1)2\cdot 2^{k+2}+2$$
$$= (k+1)\cdot 2^{k+1+2}+2$$
= RHS of equation (2)

Hence the inductive step is proved as well. Accordingly by mathematical induction the given formula is true for all integers n≥0.

EXERCISE:

Use mathematical induction to prove that

$$\left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\cdots\left(1-\frac{1}{n^2}\right)=\frac{n+1}{2n} \quad \text{for all integers } n \geq 2$$

SOLUTION:

1. **Basis Step**:

For n = 2

$$\text{L.H.S} = 1-\frac{1}{2^2} = 1-\frac{1}{4} = \frac{3}{4}$$
$$\text{R.H.S} = \frac{2+1}{2(2)} = \frac{3}{4}$$

Hence the given formula is true for n = 2

2. **Inductive Step**:

Suppose for some integer k ≥2

$$\left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\cdots\left(1-\frac{1}{k^2}\right)=\frac{k+1}{2k} \quad \ldots\ldots\ldots\ldots(1)$$

We must show that

$$\left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\cdots\left(1-\frac{1}{(k+1)^2}\right)=\frac{(k+1)+1}{2(k+1)} \quad \ldots\ldots\ldots\ldots(2)$$

Consider L.H.S of (2)

$$\left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\cdots\left(1-\frac{1}{(k+1)^2}\right)$$
$$=\left[\left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\cdots\left(1-\frac{1}{k^2}\right)\right]\left(1-\frac{1}{(k+1)^2}\right)$$
$$=\left(\frac{k+1}{2k}\right)\left(1-\frac{1}{(k+1)^2}\right)$$
$$=\left(\frac{k+1}{2k}\right)\left(\frac{(k+1)^2-1}{(k+1)^2}\right)$$
$$=\left(\frac{1}{2k}\right)\left(\frac{k^2+2k+1-1}{(k+1)}\right)$$
$$=\frac{k^2+2k}{2k(k+1)}=\frac{k(k+2)}{2k(k+1)}$$
$$=\frac{k+1+1}{2(k+1)}= \quad \text{RHS of (2)}$$

Hence by mathematical induction the given equation is true

EXERCISE:

Prove by mathematical induction

$$\sum_{i=1}^{n} i(i!)=(n+1)!-1 \qquad \text{for all integers } n\geq 1$$

SOLUTION:

1. **Basis step:**

For n = 1

L.H.S $=\sum_{i=1}^{n} i(i!)=(1)(1!)=1$

R.H.,S = (1+1)! - 1 = 2! - 1
= 2 -1 = 1

Hence

$$\sum_{i=1}^{1} i(i!)=(1+1)!-1$$

which proves the basis step.

2.Inductive Step:

Suppose for any integer k ≥1

$$\sum_{i=1}^{k} i(i!)=(k+1)!-1 \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

We need to prove that

$$\sum_{i=1}^{k+1} i(i!)=(k+1+1)!-1 \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$$

Consider LHS of (2)

$$\begin{aligned}\sum_{i=1}^{k+1} i(i!) &= \sum_{i=1}^{k} i(i!) + (k+1)(k+1)! && \text{Using (1)}\\ &= (k+1)! - 1 + (k+1)(k+1)!\\ &= (k+1)! + (k+1)(k+1)! - 1\\ &= [1+(k+1)](k+1)! - 1\\ &= (k+2)(k+1)! - 1\\ &= (k+2)! - 1\\ &= \text{RHS of (2)}\end{aligned}$$

Hence the inductive step is also true.
Accordingly, by mathematical induction, the given formula is true for all integers n ≥1.

<u>EXERCISE:</u>

Use mathematical induction to prove the generalization of the following DeMorgan's Law:

$$\overline{\bigcap_{j=1}^{n} A_j} = \bigcup_{j=1}^{n} \overline{A_j}$$

where $A_1, A_2, \ldots, A_n$ are subsets of a universal set U and n≥2.

<u>SOLUTION:</u>

Let P(n) be the given propositional function

Here $\overline{A}$ represents complement of A.

1.Basis Step:

P(2) is true.

$$\begin{aligned}\text{L.H.S of P(2)} = \overline{\bigcap_{j=1}^{2} A_j} &= \overline{A_1 \cap A_2} && \text{By DeMorgan's Law}\\ &= \overline{A_1} \cup \overline{A_2}\\ &= \bigcup_{i=1}^{2} \overline{A_j} = \text{RHS of } P(2)\end{aligned}$$

2.Inductive Step:

Assume that P(k) is true for some integer k ≥2; i.e.,

$$\overline{\bigcap_{j=1}^{k} A_j} = \bigcup_{j=1}^{k} \overline{A_j} \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

where $A_1, A_2, \ldots, A_k$ are subsets of the universal set U. If A_{k+1} is another set of U, then we need to show that

$$\overline{\bigcap_{j=1}^{k+1} A_j} = \bigcup_{j=1}^{k+1} \overline{A_j} \qquad \ldots\ldots\ldots\ldots\ldots\ldots..(2)$$

Consider L.H.S of (2)

$$\begin{aligned}
\overline{\bigcap_{j=1}^{k+1} A_j} &= \overline{\left(\bigcap_{j=1}^{k} A_j\right) \cap A_{k+1}} \\
&= \left(\overline{\bigcap_{j=1}^{k} A_j}\right) \cup \overline{A_{k+1}} && \text{By DeMorgan's Law} \\
&= \left(\bigcup_{j=1}^{k} \overline{A_j}\right) \cup \overline{A_{k+1}} \\
&= \bigcup_{j=1}^{k+1} \overline{A_j} \\
&= \text{R.H.S of (2)}
\end{aligned}$$

Hence by mathematical induction, the given generalization of DeMorgan's Law holds.

Lecture No.24 Mathematical Induction for Divisibility

MATHEMATICAL INDUCTION FOR
DIVISIBILITY PROBLEMS
INEQUALITY PROBLEMS

DIVISIBILITY:

Let n and d be integers and $d \neq 0$. Then n is divisible by d or d divides n written as $d \mid n$. iff $n = d \cdot k$ for some integer k.
Alternatively, we say that

n is a multiple of d
d is a divisor of n
d is a factor of n

Thus $d|n \Leftrightarrow \exists$ an integer k such that $n = d \cdot k$

EXERCISE:

Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer.

SOLUTION:

1. Basis Step:

For n = 1

$n^3 - n = 1^3 - 1 = 1 - 1 = 0$

which is clearly divisible by 3, since $0 = 0 \cdot 3$

Therefore, the given statement is true for n = 1.

2. Inductive Step:

Suppose that the statement is true for n = k, i.e., k^3-k is divisible by 3 for all $n \in Z+$
Then

$k^3-k = 3 \cdot q$……………………….(1)

for some $q \in Z$

We need to prove that $(k+1)^3 - (k+1)$ is divisible by 3.
Now

$$\begin{aligned}(k+1)^3 - (k+1) &= (k^3 + 3k^2 + 3k + 1) - (k + 1)\\ &= k^3 + 3k^2 + 2k\\ &= (k^3-k) + 3k^2 + 2k + k\\ &= (k^3 - k) + 3k^2 + 3k\\ &= 3 \cdot q + 3 \cdot (k^2 + k) \qquad \text{using(1)}\\ &= 3[q+k^2 + k]\end{aligned}$$

$\Rightarrow$ $(k+1)^3 - (k+1)$ is divisible by 3.
Hence by mathematical induction $n^3 - n$ is divisible by 3, whenever n is a positive integer.

EXAMPLE:

Use mathematical induction to prove that for all integers $n \geq 1$, $2^{2n}-1$ is divisible by 3.

SOLUTION:

Let P(n): 2^{2n} -1 is divisible by 3.

1. Basis Step:

P(1) is true

Now P(1): $2^{2(1)}$- 1 is divisible by 3.

Since $2^{2(1)}$- 1= 4 - 1 = 3

which is divisible by 3.

Hence P(1) is true.

2. Inductive Step:

Suppose that P(k) is true. That is 2^{2k}-1 is divisible by 3. Then, there exists an integer q such that

$2^{2k} - 1 = 3\cdot q$ …………………………(1)

To prove P(k+1) is true, that is $2^{2(k+1)}$- 1 is divisible by 3.

Now consider

$$
\begin{aligned}
2^{2(k+1)} - 1 &= 2^{2k+2} - 1 \\
&= 2^{2k}\, 2^2 - 1 \\
&= 2^{2k} 4 - 1 \\
&= 2^{2k}(3+1) - 1 \\
&= 2^{2k}\cdot 3+(2^{2k} - 1) \\
&= 2^{2k}\cdot 3+3\cdot q \qquad \text{[by using (1)]} \\
&=3(2^{2k} + q)
\end{aligned}
$$

$\Rightarrow 2^{2(k+1)} - 1$ is divisible by 3.

Accordingly, by mathematical induction. 2^{2n}- 1 is divisible by 3, for all integers $n \geq 1$.

EXERCISE:

Use mathematical induction to show that the product of any two consecutive positive integers is divisible by 2.

SOLUTION:

Let n and n + 1 be two consecutive integers. We need to prove that n(n+1) is divisible by 2.

1. Basis Step:

For n = 1

$n (n+1) = 1\cdot(1+1) = 1\cdot 2 = 2$

which is clearly divisible by 2.

2. Inductive Step:

Suppose the given statement is true for n = k. That is

k (k+1) is divisible by 2, for some $k \in Z+$

Then $k (k+1) = 2\cdot q$ ………………….(1) $q \in Z+$

We must show that

(k+1)(k+1+1) is divisible by 2.

Consider

$$
\begin{aligned}
(k+1)(k+1+1) &= (k+1)(k+2) \\
&= (k+1)k + (k+1)2 \\
&= 2q + 2 (k+1) \quad \text{using (1)} \\
&= 2(q+k+1)
\end{aligned}
$$

Hence (k+1) (k+1+1) is also divisible by 2.

Accordingly, by mathematical induction, the product of any two consecutive positive integers is divisible by 2

EXERCISE:

Prove by mathematical induction $n^3 - n$ is divisible by 6, for each integer $n \geq 2$.

SOLUTION:

1.Basis Step:

For $n = 2$

$n^3 - n = 2^3 - 2 = 8 - 2 = 6$

which is clearly divisible by 6, since $6 = 1\cdot 6$

Therefore, the given statement is true for $n = 2$.

2.Inductive Step:

Suppose that the statement is true for $n = k$, i.e., $k^3 - k$ is divisible by 6, for all integers $k \geq 2$.

Then

$$k^3 - k = 6\cdot q \ldots\ldots\ldots\ldots\ldots(1) \quad \text{for some } q \in Z.$$

We need to prove that

$(k+1)^3 - (k+1)$ is divisible by 6

Now $(k+1)^3 - (k+1) = (k^3 + 3k^3 + 3k + 1)-(k+1)$

$= k^3 + 3k^3 + 2k$

$= (k^3 - k) + (3k^3 + 2k + k)$

$= (k^3 - k) + 3k^3 + 3k$ Using (1)

$= 6\cdot q + 3k(k+1) \ldots\ldots\ldots\ldots\ldots\ldots..(2)$

Since k is an integer, so $k(k+1)$ being the product of two consecutive integers is an even number.

Let $k(k+1) = 2r$ $\quad r \in Z$

Now equation (2) can be rewritten as:

$(k+1)^3 - (k+1) = 6\cdot q + 3\cdot 2\, r$

$= 6q + 6r$

$= 6(q+r)$ $\quad q, r \in Z$

$\Rightarrow (k+1)^3 - (k+1)$ is divisible by 6.

Hence, by mathematical induction, $n^3 - n$ is divisible by 6, for each integer $n \geq 2$.

EXERCISE:

Prove by mathematical induction. For any integer $n \geq 1$, $x^n - y^n$ is divisible by $x - y$, where x and y are any two integers with $x \neq y$.

SOLUTION:

1. Basis Step:

For $n = 1$

$x^n - y^n = x^1 - y^1 = x - y$

which is clearly divisible by $x - y$. So, the statement is true for $n = 1$.

2. Inductive Step:

Suppose the statement is true for $n = k$, i.e.,

$x^k - y^k$ is divisible by $x - y \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$

We need to prove that $x^{k+1} - y^{k+1}$ is divisible by x - y

Now

$$x^{k+1} - y^{k+1} = x^k \cdot x - y^k \cdot y$$
$$= x^k \cdot x - x \cdot y^k + x \cdot y^k - y^k \cdot y \quad \text{(introducing } x.y^k\text{)}$$
$$= (x^k - y^k) \cdot x + y^k \cdot (x-y)$$

The first term on R.H.S=$(x^k - y^k)$ is divisible by x - y by inductive hypothesis (1).

The second term contains a factor (x-y) so is also divisible by x - y.

Thus $x^{k+1} - y^{k+1}$ is divisible by x - y. Hence, by mathematical induction $x^n - y^n$ is divisible by x - y for any integer $n \geq 1$.

PROVING AN INEQUALITY:

Use mathematical induction to prove that for all integers $n \geq 3$.

$$2n + 1 < 2^n$$

SOLUTION:

1. Basis Step:

For n = 3

L.H.S= 2(3) + 1 = 6 + 1 = 7

R.H.S = 2^3 = 8

Since 7 < 8, so the statement is true for n = 3.

2. Inductive Step:

Suppose the statement is true for n =k, i.e.,

$2k + 1 < 2^k$…………………(1) $k \geq 3$

We need to show that the statement is true for n = k+1,

i.e.;

$2(k+1) + 1 < 2^{k+1}$…………..(2)

Consider L.H.S of (2)

$$\begin{aligned}
&= 2(k+1) + 1 \\
&= 2k + 2 + 1 \\
&= (2k + 1) + 2 \\
&< 2^k + 2 && \text{using (1)} \\
&< 2^k + 2^k && (\text{since } 2 < 2^k \text{ for } k \geq 3) \\
&< 2 \cdot 2^k = 2^{k+1}
\end{aligned}$$

Thus $2(k+1)+1 < 2k+1$ (proved)

EXERCISE:

Show by mathematical induction

$$1 + n x \leq (1+x)^n$$

for all real numbers x > - 1 and integers $n \geq 2$

SOLUTION:

1. Basis Step:

For n = 2

L.H.S = 1 + (2) x= 1 + 2x

RHS $= (1 + x)^2 = 1 + 2x + x^2 > 1 + 2x \qquad (x^2 > 0)$

$\Rightarrow$ statement is true for n = 2.

2. Inductive Step:

Suppose the statement is true for n = k.

That is, for $k \geq 2$, $1 + k x \leq (1 + x)^k$……………….(1)

We want to show that the statement is also true for n = k + 1 i.e.,

$1 + (k + 1)x \leq (1 + x)^{k+1}$

Since $x > -1$, therefore $1 + x > 0$.

Multiplying both sides of (1) by (1+x) we get

$$
\begin{aligned}
(1+x)(1+x)^k &\geq (1 + x)(1 + kx) \\
&= 1 + kx + x + kx^2 \\
&= 1 + (k + 1)x + kx^2
\end{aligned}
$$

but

so $\begin{cases} x > -1, & \text{so } x^2 \geq 0 \\ \& k \geq 2, & \text{so } kx^2 \geq 0 \end{cases}$

$$(1+x)(1+x)^k \geq 1 + (k + 1)x$$

Thus $1 + (k+1)x \leq (1+x)^{k+1}$. Hence by mathematical induction, the inequality is true.

PROVING A PROPERTY OF A SEQUENCE:

Define a sequence $a_1, a_2, a_3, \ldots$ as follows:

$a_1 = 2$

$a_k = 5a_k-1$ for all integers $k \geq 2$ **……………**(1)

Use mathematical induction to show that the terms of the sequence satisfy the formula.

$a_n = 2\cdot5^{n-1}$ for all integers $n \geq 1$

SOLUTION:

1.Basis Step:

For n = 1, the formula gives

$a_1 = 2\cdot5^{1-1} = 2\cdot5^0 = 2\cdot1 = 2$

which confirms the definition of the sequence. Hence, the formula is true for n = 1.

2.Inductive Step:

Suppose, that the formula is true for n = k, i.e.,

$a_k = 2\cdot5^{k-1}$ for some integer $k \geq 1$

We show that the statement is also true for n = k + 1. i.e.,

$a_{k+1} = 2\cdot5^{k+1-1} = 2\cdot5^k$

Now

$$
\begin{aligned}
a_{k+1} &= 5\cdot a_{k+1-1} && \text{[by definition of } a_1, a_2, a_3 \ldots \text{ or by putting k+1 in (1)]} \\
&= 5\cdot a_k \\
&= 5\cdot(2\cdot5^{k-1}) && \text{by inductive hypothesis} \\
&= 2\cdot(5\cdot5^{k-1}) \\
&= 2\cdot5^{k+1-1} \\
&= 2\cdot5^k
\end{aligned}
$$

which was required.

EXERCISE:

A sequence $d_1, d_2, d_3, \ldots$ is defined by letting $d_1 = 2$ and $d_k = \dfrac{d_{k-1}}{k}$

for all integers k ≥ 2. Show that $d_n = \frac{2}{n!}$ for all integers n ≥ 1, using mathematical induction.

SOLUTION:

1.Basis Step:

For n = 1, the formula $d_n = \frac{2}{n!}$; n ≥1 gives

$$d_1 = \frac{2}{1!} = \frac{2}{1} = 2$$

which agrees with the definition of the sequence.

2.Inductive Step:

Suppose, the formula is true for n=k. i.e.,

$d_k = \frac{2}{k!}$ for some integer k ≥ 1…………….(1)

We must show that

$$d_{k+1} = \frac{2}{(k+1)!}$$

Now, by the definition of the sequence.

$$d_{k+1} = \frac{d_{(k+1)-1}}{(k+1)} = \frac{1}{(k+1)} d_k \qquad \text{using } d_k = \frac{d_{k-1}}{k}$$

$$= \frac{1}{(k+1)} \frac{2}{k!} \qquad \text{using (1)}$$

$$= \frac{2}{(k+1)!}$$

Hence the formula is also true for n = k + 1. Accordingly, the given formula defines all the terms of the sequence recursively.

EXERCISE:

Prove by mathematical induction that

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}$$

Whenever n is a positive integer greater than 1.

SOLUTION:

1. Basis Step: for n = 2

L.H.S$= 1 + \frac{1}{4} = \frac{5}{4} = 1.25$

R.H.S$= 2 - \frac{1}{2} = \frac{3}{2} = 1.5$

Clearly LHS < RHS
Hence the statement is true for n = 2.

2.Inductive Step:
Suppose that the statement is true for some integers k > 1, i.e.;

$$1+\frac{1}{4}+\frac{1}{9}+\cdots+\frac{1}{k^2}<2-\frac{1}{k} \qquad (1)$$

We need to show that the statement is true for n = k + 1. That is

$$1+\frac{1}{4}+\frac{1}{9}+\cdots+\frac{1}{(k+1)^2}<2-\frac{1}{k+1} \qquad (2)$$

Consider the LHS of (2)

$$1+\frac{1}{4}+\frac{1}{9}+\cdots+\frac{1}{(k+1)^2}=1+\frac{1}{4}+\frac{1}{9}+\cdots+\frac{1}{k^2}+\frac{1}{(k+1)^2}$$

$$<\left(2-\frac{1}{k}\right)+\frac{1}{\left(k+1\right)^2}$$

$$=2-\left(\frac{1}{k}-\frac{1}{(k+1)^2}\right)$$

We need to prove that

$$2-\left(\frac{1}{k}-\frac{1}{(k+1)^2}\right)\leq 2-\frac{1}{k+1}$$

or $$-\left(\frac{1}{k}-\frac{1}{(k+1)^2}\right)\leq -\frac{1}{k+1}$$

or $$\frac{1}{k}-\frac{1}{(k+1)^2}\geq \frac{1}{k+1}$$

or $$\frac{1}{k}-\frac{1}{k+1}\geq \frac{1}{(k+1)^2}$$

Now $$\frac{1}{k}-\frac{1}{k+1}=\frac{k+1-k}{k(k+1)}$$

$$=\frac{1}{k(k+1)}>\frac{1}{(k+1)^2}$$

Lecture No.25 Methods of proof

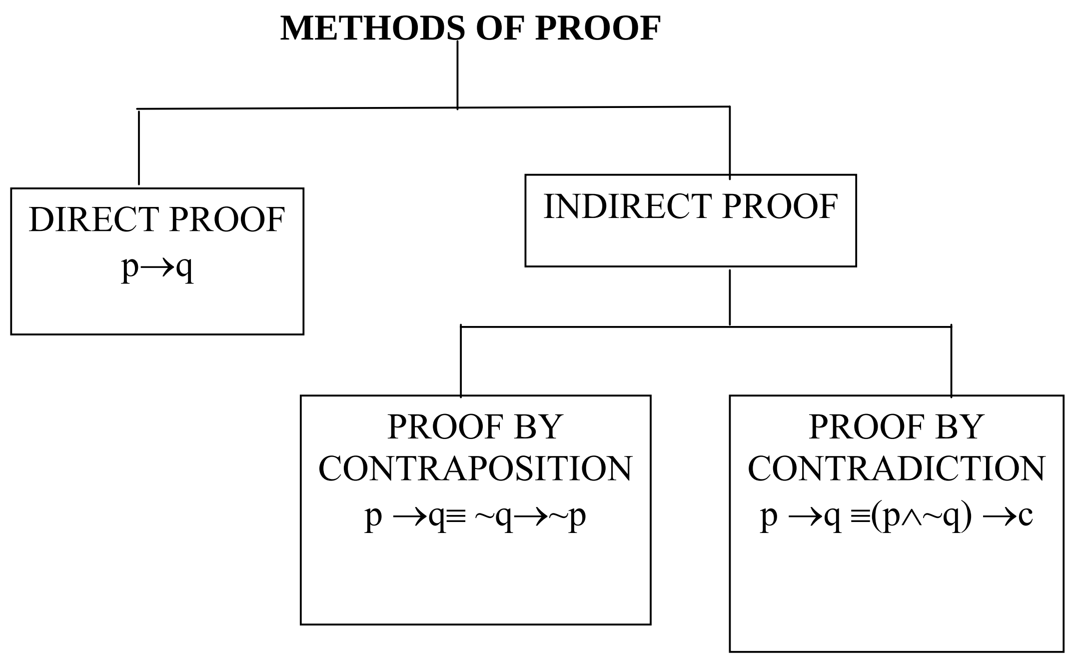
METHODS OF PROOF

-- DIRECT PROOF
-- DISPROOF BYCOUNTER EXAMPLE

<u>INTRODUCTION:</u>

To understand written mathematics, one must understand what makes up a correct mathematical argument, that is, a proof. This requires an under standing of the techniques used to build proofs. The methods we will study for building proofs are also used throughout computer science, such as the rules computers used to reason, the techniques used to verify that programs are correct, etc.

Many theorems in mathematics are implications, $p \rightarrow q$. The techniques of proving implications give rise to different methods of proofs.



<u>DIRECT PROOF:</u>

The implication $p \rightarrow q$ can be proved by showing that if p is true, the q must also be true. This shows that the combination p true and q false never occurs. A proof of this kind is called a direct proof.

p	q	$p\rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

SOME BASICS:

1. An integer n is even if, and only if, n = 2k for some integer k.
2. An integer n is odd if, and only if, n = 2k + 1 for some integer k.
3. An integer n is prime if, and only if, n > 1 and for all positive integers r and s, if n = r·s, then r = 1 or s = 1.
4. An integer n > 1 is composite if, and only if, n = r·s for some positive integers r and s with $r \neq 1$ and $s \neq 1$.
5. A real number r is rational if, and only if, $\frac{a}{b}$ for some integers a and b with $b \neq 0$.
6. If n and d are integers and $d \neq 0$, then d divides n, written d | n if, and only if, n = d.k for some integers k.
7. An integer n is called a perfect square if, and only if, $n = k^2$ for some integer k.

EXERCISE:

Prove that the sum of two odd integers is even.

SOLUTION:

Let m and n be two odd integers. Then by definition of odd numbers

$m = 2k + 1$ for some $k \in Z$

$n = 2l + 1$ for some $l \in Z$

Now $m + n = (2k + 1) + (2l + 1)$

$= 2k + 2l + 2$

$= 2 (k + l + 1)$

$= 2r$ where $r = (k + l + 1) \in Z$

Hence m + n is even.

EXERCISE:

Prove that if n is any even integer, then $(-1)^n = 1$

SOLUTION:

Suppose n is an even integer. Then n = 2k for some integer k.

Now

$(-1)^n = (-1)^{2k}$

$= [(-1)^2]^k$

$= (1)^k$

$= 1$ (proved)

EXERCISE:

Prove that the product of an even integer and an odd integer is even.

SOLUTION:

Suppose m is an even integer and n is an odd integer. Then

$m = 2k$ for some integer k

and $n = 2l + 1$ for some integer l

Now

$m \cdot n = 2k \cdot (2l + 1)$

$= 2 \cdot k (2l + 1)$

$= 2 \cdot r$ where $r = k(2l + 1)$ is an integer

Hence m·n is even. (Proved)

EXERCISE:

Prove that the square of an even integer is even.

SOLUTION:

Suppose n is an even integer. Then n = 2k

Now

$$\begin{aligned}\text{square of } n = n^2 &= (2\cdot k)^2\\ &= 4k^2\\ &= 2\cdot(2k^2)\\ &= 2\cdot p \quad \text{where } p = 2k^2 \in Z\end{aligned}$$

Hence, n^2 is even. (proved)

EXERCISE:

Prove that if n is an odd integer, then $n^3 + n$ is even.

SOLUTION:

Let n be an odd integer, then n = 2k + 1for some $k \in Z$

Now
$$\begin{aligned}n^3 + n &= n\,(n^2 + 1)\\ &= (2k + 1)\,((2k+1)^2 + 1)\\ &= (2k + 1)\,(4k^2 + 4k + 1 + 1)\\ &= (2k + 1)\,(4k^2 + 4k + 2)\\ &= (2k + 1)\,2.\,(2k^2 + 2k + 1)\\ &= 2\cdot(2k + 1)\,(2k^2 + 2k + 1) \qquad k \in Z\\ &= \text{an even integer}\end{aligned}$$

EXERCISE:

Prove that, if the sum of any two integers is even, then so is their difference.

SOLUTION:

Suppose m and n are integers so that m + n is even. Then by definition of even numbers

$$m + n = 2k \qquad \text{for some integer } k$$
$$\Rightarrow \quad m = 2k - n \;\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Now
$$\begin{aligned}m - n &= (2k - n) - n \qquad \text{using (1)}\\ &= 2k - 2n\\ &= 2\,(k - n) = 2r \qquad \text{where } r = k - n \text{ is an integer}\end{aligned}$$

Hence m - n is even.

EXERCISE:

Prove that the sum of any two rational numbers is rational.

SOLUTION:

Suppose r and s are rational numbers.

Then by definition of rational

$$r = \frac{a}{b} \quad \text{and} \quad s = \frac{c}{d}$$

for some integers a, b, c, d with $b \neq 0$ and $d \neq 0$

Now
$$r+s = \frac{a}{b}+\frac{c}{d}$$
$$= \frac{ad+bc}{bd}$$
$$= \frac{p}{q}$$
where p = ad + bc $\in$Z and q =bd $\in$Z
and q $\neq$0
Hence r + s is rational.

EXERCISE:

Given any two distinct rational numbers r and s with r < s. Prove that there is a rational number x such that r < x < s.

SOLUTION:

Given two distinct rational numbers r and s such that
r < s ………………….(1)
Adding r to both sides of (1), we get
r + r < r + s
2r < r + s
$\Rightarrow$ $$r < \frac{r+s}{2} \qquad \text{………………….(2)}$$
Next adding s to both sides of (1), we get
r + s < s + s
$\Rightarrow$ r + s < 2s
$\Rightarrow$ $$\frac{r+s}{2} < s \qquad \text{………………………(3)}$$
Combining (2) and (3), we may write
$$r < \frac{r+s}{2} < s \qquad \text{……………………..(4)}$$
Since the sum of two rationals is rational, therefore r + s is rational. Also the quotient of a rational by a non-zero rational, is rational, therefore $\frac{r+s}{2}$ is rational and by (4) it lies between r & s.
Hence, we have found a rational number
such that r < x < s. (proved)

EXERCISE:

Prove that for all integers a, b and c, if a|b and b|c then a|c.

PROOF:

Suppose a|b and b|c where a, b, c $\in$Z. Then by definition of divisibility
b=a·r and c=b·s for some integers r and s.
Now c = b·s
= (a·r)·s (substituting value of b)
= a·(r·s) (associative law)
= a·k where k = r·s $\in$ Z
$\Rightarrow$ a | c by definition of divisibility

EXERCISE:

Prove that for all integers a, b and c if a|b and a|c then a|(b+c)

PROOF:

Suppose a|b and a|c where a, b, c $\in Z$

By definition of divides

$b = a\cdot r$ and $c = a\cdot s$ for some $r, s \in Z$

Now

$b + c= a\cdot r + a\cdot s$ (substituting values)

$= a\cdot(r+s)$ (by distributive law)

$= a\cdot k$ where $k = (r + s) \in Z$

Hence $a|(b + c)$ by definition of divides.

EXERCISE:

Prove that the sum of any three consecutive integers is divisible by 3.

PROOF:

Let n, n + 1 and n + 2 be three consecutive integers.

Now

$n + (n + 1) + (n + 2)= 3n + 3$

$= 3(n + 1)$

$= 3\cdot k$ where $k=(n+1)\in Z$

Hence, the sum of three consecutive integers is divisible by 3.

EXERCISE:

Prove the statement:

There is an integer $n > 5$ such that $2^n - 1$ is prime

PROOF:

Here we are asked to show a single integer for which 2^n -1is prime. First of all we will check the integers from 1 and check whether the answer is prime or not by putting these values in 2^n-1.when we got the answer is prime then we will stop our process of checking the integers and we note that,

Let n = 7, then

$2^n - 1 = 2^7 - 1 = 128 - 1 = 127$

and we know that 127 is prime.

EXERCISE:

Prove the statement: There are real numbers a and b such that

$\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

PROOF:

Let $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

Squaring, we get $a + b = a + b + 2\sqrt{a}\sqrt{b}$

$\Rightarrow$ $0 = 2\sqrt{a}\sqrt{b}$ canceling a+b

$\Rightarrow$ $0 = \sqrt{ab}$

$\Rightarrow$ $0 = ab$ squaring

$\Rightarrow$ either a = 0 or b = 0
It means that if we want to find out the integers which satisfy the given condition then one of them must be zero.
Hence if we let a = 0 and b = 3 then

$$R.H.S = \sqrt{a+b} = \sqrt{0+3}$$
$$R.H.S = \sqrt{3}$$
$$Now\ L.H.S = \sqrt{a} + \sqrt{b}$$
By putting the values of a and b we get
$$= \sqrt{0} + \sqrt{3}$$
$$= \sqrt{3}$$

From above it quite clear that the given condition is satisfied if we take a=0 and b=3.

PROOF BY COUNTER EXAMPLE:
Disprove the statement by giving a counter example.
For all real numbers a and b, if a < b then $a^2 < b^2$.

SOLUTION:
Suppose a = -5 and b = -2
then clearly - 5 < - 2
But $a^2 = (-5)^2 = 25$ and $b^2 = (-2)^2 = 4$
But 25 > 4
This disproves the given statement.

EXERCISE:
Prove or give counter example to disprove the statement.
For all integers n, $n^2 - n + 11$ is a prime number.

SOLUTION:
The statement is not true
For n = 11
we have , $n^2 - n + 11= (11)^2 - 11 + 11$
$= (11)^2$
$= (11) (11)$
$=121$
which is obviously not a prime number.

EXERCISE:
Prove or disprove that the product of any two irrational numbers is an irrational number.

SOLUTION:
We know that $\sqrt{2}$ is an irrational number. Now $(\sqrt{2})(\sqrt{2}) = (\sqrt{2})^2 = 2 = \frac{2}{1}$

which is a rational number. Hence the statement is disproved.

EXERCISE:
Find a counter example to the proposition:
For every prime number n, n + 2 is prime.

SOLUTION:

Let the prime number n be 7, then

$n + 2 = 7 + 2 = 9$

which is not prime.

Lecture No.26 Proof by Contradiction

PROOF BY CONTRADICTION:

A proof by contradiction is based on the fact that either a statement is true or it is false but not both. Hence the supposition, that the statement to be proved is false, leads logically to a contradiction, impossibility or absurdity, then the supposition must be false. Accordingly, the given statement must be true.

This method of proof is also known as reductio ad absurdum because it relies on reducing a given assumption to an absurdity.

Many theorems in mathematics are conditional statements ($p\rightarrow q$). Now the negation of he implication $p\rightarrow q$ is

$$\begin{aligned} \sim(p\rightarrow q) &\equiv \sim(\sim p\vee q) \\ &\equiv \sim(\sim p) \wedge (\sim q) \qquad \text{DeMorgan's Law} \\ &\equiv p \wedge \sim q \end{aligned}$$

Clearly if the implication is true, then its negation must be false, i.e., leads to a contradiction.

Hence $\sim(p\rightarrow q) \equiv (p \wedge \sim q) \rightarrow c$, where c is a contradiction.

Thus to prove an implication $p \rightarrow q$ by contradiction method, we suppose that the condition p and the negation of the conclusion q, i.e., $(p \wedge \sim q)$ is true and ultimately arrive at a contradiction.

The method of proof by contradiction, may be summarized as follows:

1. Suppose the statement to be proved is false.
2. Show that this supposition leads logically to a contradiction.
3. Conclude that the statement to be proved is true.

THEOREM:

There is no greatest integer.

PROOF:

Suppose there is a greatest integer N. Then $n \leq N$ for every integer n.

Let $M = N + 1$

Now M is an integer since it is a sum of integers.

Also $M > N$ since $M = N + 1$

Thus M is an integer that is greater than the greatest integer, which is a contradiction.

Hence our supposition is not true and so there is no greatest integer.

EXERCISE:

Give a proof by contradiction for the statement:

"If n^2 is an even integer then n is an even integer."

PROOF:

Suppose n^2 is an even integer and n is not even, so that n is odd.

Hence $n = 2k + 1$ for some integer k.

Now $n^2 = (2k + 1)^2$

$= 4k^2 + 4k + 1$

$= 2\cdot(2k^2 + 2k) + 1$
$= 2r + 1$ where $r = (2k^2 + 2k) \in Z$

This shows that n^2 is odd, which is a contradiction to our supposition that n^2 is even. Hence the given statement is true.

EXERCISE:

Prove that if n is an integer and $n^3 + 5$ is odd, then n is even using contradiction method.

SOLUTION:

Suppose that $n^3 + 5$ is odd and n is not even (odd). Since n is odd and the product of two odd numbers is odd, it follows that n^2 is odd and $n^3 = n^2$. n is odd. Further, since the difference of two odd number is even, it follows that

$$5 = (n^3 + 5) - n^3$$

is even. But this is a contradiction. Therefore, the supposition that $n^3 + 5$ and n are both odd is wrong and so the given statement is true.

EXERCISE:

Prove by contradiction method, the statement: If n and m are odd integers, then n + m is an even integer.

SOLUTION:

Suppose n and m are odd and n + m is not even (odd i.e by taking contradiction).

Now $n = 2p + 1$ for some integer p
and $m = 2q + 1$ for some integer q

Hence $n + m = (2p + 1) + (2q + 1)$
$= 2p + 2q + 2 = 2\cdot (p + q + 1)$

which is even, contradicting the assumption that n + m is odd.

THEOREM:

The sum of any rational number and any irrational number is irrational.

PROOF:

We suppose that the negation of the statement is true. That is, we suppose that there is a rational number r and an irrational number s such that r + s is rational. By definition of ration

$$r = \frac{a}{b} \quad \ldots\ldots\ldots\ldots\ldots(1)$$

and

$$r + s = \frac{c}{d} \quad \ldots\ldots\ldots\ldots\ldots(2)$$

for some integers a, b, c and d with $b \neq 0$ and $d \neq 0$.
Using (1) in (2), we get

$$\frac{a}{b} + s = \frac{c}{d}$$

$$\Rightarrow \quad s = \frac{c}{d} - \frac{a}{b}$$

$$s = \frac{bc - ad}{bd} \qquad (bd \neq 0)$$

Now bc - ad and bd are both integers, since products and difference of integers are integers. Hence s is a quotient of two integers bc-ad and bd with bd ≠ 0. So by definition of rational, s is rational.

This contradicts the supposition that s is irrational. Hence the supposition is false and the theorem is true.

EXERCISE:

Prove that $\sqrt{2}$ is irrational.

PROOF:

Suppose $\sqrt{2}$
is rational. Then there are integers m and n with no common factors so that

$$\sqrt{2} = \frac{m}{n}$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2}$$

Or $m^2 = 2n^2$ ……………………….(1)

This implies that m^2 is even (by definition of even). It follows that m is even. Hence

$m = 2k$ for some integer k (2)

Substituting (2) in (1), we get

$$(2k)^2 = 2n^2$$
$$\Rightarrow \quad 4k^2 = 2n^2$$
$$\Rightarrow \quad n^2 = 2k^2$$

This implies that n^2 is even, and so n is even. But we also know that m is even. Hence both m and n have a common factor 2. But this contradicts the supposition that m and n have no common factors. Hence our supposition is false and so the theorem is true.

Substituting (2) in (1), we get

$$(2k)^2 = 2n^2$$
$$\Rightarrow \quad 4k^2 = 2n^2$$
$$\Rightarrow \quad n^2 = 2k^2$$

This implies that n^2 is even, and so n is even. But we also know that m is even. Hence both m and n have a common factor 2. But this contradicts the supposition that m and n have no common factors. Hence our supposition is false and so the theorem is true.

EXERCISE:

Prove by contradiction that $6-7\sqrt{2}$ is irrational.

PROOF:

Suppose $6-7\sqrt{2}$ is rational.
Then by definition of rational,

$$6-7\sqrt{2}=\frac{a}{b}$$

for some integers a and b with $b \neq 0$.
Now consider,

$$7\sqrt{2}=6-\frac{a}{b}$$

$$\Rightarrow \quad 7\sqrt{2}=\frac{6b-a}{b}$$

$$\Rightarrow \quad \sqrt{2}=\frac{6b-a}{7b}$$

Since a and b are integers, so are 6b-a and 7b and $7b \neq 0$;
hence $\sqrt{2}$ is a quotient of the two integers 6b-a and 7b with $7b \neq 0$.
Accordingly, $\sqrt{2}$ is rational (by definition of rational).
This contradicts the fact because $\sqrt{2}$ is irrational.
Hence our supposition is false and so $6-7\sqrt{2}$ is irrational.

EXERCISE:

Prove that for any integer a and any prime number p, if p|a, then $P \nmid (a + 1)$.

PROOF:

Suppose there exists an integer **a** and a prime number **p** such that p|a and p|(a+1).
Then by definition of divisibility there exist integer r and s so that

$$a = p\cdot r \quad \text{and} \quad a + 1 = p\cdot s$$

It follows that

$$\begin{aligned} 1 &= (a + 1) - a \\ &= p\cdot s - p\cdot r \\ &= p\cdot(s-r) \qquad \text{where } s-r \in Z \end{aligned}$$

This implies p | 1.
But the only integer divisors of 1 are 1 and -1 and since p is prime p>1. This is a contradiction.

Hence the supposition is false, and the given statement is true.

EXERCISE:

Prove that $\sqrt{2}+\sqrt{3}$ is irrational.

SOLUTION:

Suppose $\sqrt{2}+\sqrt{3}$ is rational. Then, by definition of rational, there exists integers a and b with b≠0 such that

$$\sqrt{2}+\sqrt{3}=\frac{a}{b}$$

Squaring both sides, we get

$$2+3+2\sqrt{2}\sqrt{3}=\frac{a^2}{b^2}$$

$$\Rightarrow \qquad 2\sqrt{2\times 3}=\frac{a^2}{b^2}-5$$

$$\Rightarrow \qquad 2\sqrt{6}=\frac{a^2-5b^2}{b^2}$$

$$\Rightarrow \qquad \sqrt{6}=\frac{a^2-5b^2}{2b^2}$$

Since a and b are integers, so are therefore $a^2 - 5b^2$ and $2b^2$ with $2b^2 \neq 0$. Hence $\sqrt{6}$ is the quotient of two integers $a^2 - 2b^2$ and $2b^2$ with $2^2 \neq 0$. Accordingly, $\sqrt{6}$ is rational. But this is a contradiction, since $\sqrt{6}$ is not rational. Hence our supposition is false and so $\sqrt{2}+\sqrt{3}$ is irrational.

REMARK:

The sum of two irrational numbers need not be irrational in general for

$$(6-7\sqrt{2})+(6+7\sqrt{2})=6+6=12$$

which is rational.

THEOREM:

The set of prime numbers is infinite.

PROOF:

Suppose the set of prime numbers is finite.
Then, all the prime numbers can be listed, say, in ascending order:

$p_1 = 2,\ p_2 = 3,\ p_3 = 5,\ p_4 = 7,\ \ldots,\ p_n$

Consider the integer

$N = p_1.p_2.p_3.\ \ldots\ .\ p_n + 1$

Then N > 1. Since any integer greater than 1 is divisible by some prime number p, therefore p | N.
Also since p is prime, p must equal one of the prime numbers
$p_1, p_2, p_3, \dots, p_n$.
Thus
$P \mid (p_1, p_2, p_3, \dots, p_n)$

But then

$$P \nmid (p_1, p_2, p_3, \dots, p_n + 1)$$

So $P \nmid N$

Thus p | N and $p \nmid N$, which is a contradiction.

Hence the supposition is false and the theorem is true.

PROOF BY CONTRAPOSITION:
A proof by contraposition is based on the logical equivalence between a statement and its contrapositive. Therefore, the implication p→ q can be proved by showing that its contrapositive ~ q → ~ p is true. The contrapositive is usually proved directly.
The method of proof by contrapositive may be summarized as:

1. Express the statement in the form if p then q.
2. Rewrite this statement in the contrapositive form
   if not q then not p.
3. Prove the contrapositive by a direct proof.

EXERCISE:
Prove that for all integers n, if n^2 is even then n is even.

PROOF:
The contrapositive of the given statement is:
"if n is not even (odd) then n^2 is not even (odd)"
We prove this contrapositive statement directly.
Suppose n is odd. Then n = 2k + 1 for some k € Z
Now $n^2 = (2k+1)^2 = 4k^2 + 4k + 1$
$= 2\cdot(2k^2 + 2k) + 1$
$= 2\cdot r + 1$ where $r = 2k^2 + 2k$ € Z
Hence n^2 is odd. Thus the contrapositive statement is true and so the given statement is true.

EXERCISE:
Prove that if 3n + 2 is odd, then n is odd.

PROOF:
The contrapositive of the given conditional statement is
" if n is even then 3n + 2 is even"
Suppose n is even, then

$n = 2k$ for some $k \in Z$

Now $3n + 2 = 3(2k) + 2$

$= 2. (3k + 1)$

$= 2.r$ where $r = (3k + 1) \in Z$

Hence $3n + 2$ is even. We conclude that the given statement is true since its contrapositive is true.

EXERCISE:

Prove that if n is an integer and $n^3 + 5$ is odd, then n is even.

PROOF:

Suppose n is an odd integer. Since, a product of two odd integers is odd, therefore $n^2 = n.n$ is odd; and $n^3 = n^2.n$ is odd.

Since a sum of two odd integers is even therefore $n^2 + 5$ is even.

Thus we have prove that if n is odd then $n^3 + 5$ is even.

Since this is the contrapositive of the given conditional statement, so the given statement is true.

EXERCISE:

Prove that if n^2 is not divisible by 25, then n is not divisible by 5.

SOLUTION:

The contra positive statement is:

"if n is divisible by 5, then n^2 is divisible by 25"

Suppose n is divisible by 5. Then by definition of divisibility

$n = 5 \cdot k$ for some integer k

Squaring both sides

$n^2 = 25 \cdot k^2$ where $k^2 \in Z$

n^2 is divisible by 25

.

EXERCISE:

Prove that if $|x| > 1$ then $x > 1$ or $x < -1$ for all $x \in R$.

PROOF:

The contrapositive statement is:

if $x \leq 1$ and $x \geq -1$ then $|x| \leq 1$ for $x \in R$.

Suppose that $x \leq 1$ and $x \geq -1$

$\Rightarrow$ $x \leq 1$ and $x \geq -1$

$\Rightarrow$ $-1 \leq x \leq 1$

and so

$|x| \leq 1$

Equivalently $|x| > 1$

EXERCISE:

Prove the statement by contraposition:

For all integers m and n, if m + n is even then m and n are both even or m and n are both odd.

PROOF:

The contrapositive statement is:

"For all integers m and n, if m and n are not both even and m and n are not both odd, then m + n is not even."
Or more simply,
"For all integers m and n, if one of m and n is even and the other is odd, then m + n is odd"
Suppose m is even and n is odd. Then

$m = 2p$ for some integer p

and $n = 2q + 1$ for some integer q

Now $m + n = (2p) + (2q + 1)$

$= 2\cdot(p+q) + 1$

$= 2\cdot r + 1$ where $r = p+q$ is an integer

Hence m + n is odd.
Similarly, taking m as odd and n even, we again arrive at the result that m + n is odd.
Thus, the contrapositive statement is true. Since an implication is logically equivalent to its contrapositive so the given implication is true.

Lecture No.27 Algorithm

PRE- AND POST-CONDITIONS OF AN ALGORITHM
LOOP INVARIANTS
LOOP INVARIANT THEOREM

ALGORITHM:

The word "algorithm" refers to a step-by-step method for performing some action. A computer program is, similarly, a set of instructions that are executed step-by-step for performing some specific task. Algorithm, however, is a more general term in that the term program refers to a particular programming language.

INFORMATION ABOUT ALGORITHM:

The following information is generally included when describing algorithms formally:

1.The name of the algorithm, together with a list of input and output variables.
2.A brief description of how the algorithm works.
3.The input variable names, labeled by data type.
4.The statements that make the body of the algorithm, with explanatory comments.
5.The output variable names, labeled by data type.
6.An end statement.

THE DIVISION ALGORITHM:

THEOREM (Quotient-Remainder Theorem):

Given any integer n and a positive integer d, there exist unique integers q and r such that $n = d \cdot q + r$ and $0 \le r < d$.

Example:

a) $n = 54, d = 4$ $54 = 4 \cdot 13 + 2;$ hence $q = 13, r = 2$
b) $n = -54, d = 4$ $-54 = 4 \cdot (-14) + 2;$ hence $q = -14, r = 2$
c) $n = 54, d = 70$ $54 = 70 \cdot 0 + 54;$ hence $q = 0, r = 54$

ALGORITHM (DIVISION)

{Given a nonnegative integer a and a positive integer d, the aim of the algorithm is to find integers q and r that satisfy the conditions $a = d \cdot q + r$ and $0 \le r < d$.
This is done by subtracting d repeatedly from a until the result is less than d but is still nonnegative.
The total number of d's that are subtracted is the quotient q. The quantity $a - d \cdot q$ equals the remainder r.}
Input: a {a nonnegative integer}, d {a positive integer}
Algorithm body: $r := a$, $q := 0$
{Repeatedly subtract d from r until a number less than d is obtained. Add 1 to d each time d is subtracted.}

while $(r \ge d)$
$r := r - d$ $q := q + 1$
end while
Output: q, r
end Algorithm (Division)

TRACING THE DIVISION ALGORITHM:

Example:

Trace the action of the Division Algorithm on the input variables $a = 54$ and $d = 11$

Solution

Iteration Number

Variable	0	1	2	3	4
a	54				
d	11				
r	54	43	32	21	10
q	0	1	2	3	4

PREDICATE:

Consider the sentence
"Aslam is a student at the Virtual University."
let *P* stand for the words
"is a student at the Virtual University"
and let *Q* stand for the words
"is a student at."
Then both *P* and *Q* are *predicate symbols*.
The sentences "x is a student at the Virtual University" and "x is a student at y" are symbolized as $P(x)$ and $Q(x, y)$, where x and y are predicate variables and take values in appropriate sets. When concrete values are substituted in place of predicate variables, a statement results.

DEFINITION:

A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables.
The domain of a predicate variable is the set of all values that may be substituted in place of the variable.

PRE-CONDITIONS AND POST-CONDITIONS:
Consider an algorithm that is designed to produce a certain final state from a given state. Both the initial and final states can be expressed as predicates involving the input and output variables.
Often the predicate describing the initial state is called the **pre-condition of the algorithm** and the predicate describing the final state is called the **post-condition of the algorithm**.

EXAMPLE:
1. Algorithm to compute a product of two nonnegative integers
pre-condition: The input variables m and n are nonnegative integers.
pot-condition: The output variable p equals $m \cdot n$.
2. Algorithm to find the quotient and remainder of the division of one positive integer by another
pre-condition: The input variables a and b are positive integers.
pot-condition: The output variable q and r are positive integers such that
$a = b \cdot q + r$ and $0 \le r < b$.

3. Algorithm to sort a one-dimensional array of real numbers
Pre-condition: The input variable $A[1], A[2], \ldots A[n]$ is a one-dimensional array of real numbers.
post-condition:The input variable $B[1], B[2], \ldots B[n]$ is a one-dimensional array of real numbers with same elements as $A[1], A[2], \ldots A[n]$ but with the property that $B[i] \le B[j]$ whenever $i \le j$.

THE DIVISION ALGORITHM:
[pre-condition: a is a nonnegative integer and
d is a positive integer, $r = a$, and $q = 0$]
while $(r \ge d)$
1. $r := r - d$
2. $q := q + 1$

end while
[post-condition: q and r are nonnegative integers
with the property that $a = q \cdot d + r$ and $0 \le r < d$.]

LOOP INVARIANTS:
The method of loop invariants is used to prove correctness of a loop with respect to certain pre and post-conditions. It is based on the principle of mathematical induction.
[pre-condition for loop]
while (G)
[Statements in body of loop. None contain branching statements that lead outside the loop.]
end while[post-condition for loop]

DEFINITION:
A loop is defined as **correct with respect to its pre- and post-conditions** if, and only if, whenever the algorithm variables satisfy the pre-condition for the loop and the loop is executed, then the algorithm variables satisfy the post-condition of the loop.

THEOREM:

Let a **while** loop with guard *G* be given, together with pre- and post conditions that are predicates in the algorithm variables.
Also let a predicate *I*(*n*), called the **loop invariant**, be given. If the following four properties are true, then the loop is correct with respect to its pre- and post-conditions.
I.Basis Property: The pre-condition for the loop implies that *I*(0) is true before the first iteration of the loop.
II.Inductive property: If the guard *G* and the loop invariant *I*(*k*) are both true for an integer $k \geq 0$ before an iteration of the loop, then *I*(*k* + 1) is true after iteration of the loop.
III.Eventual Falsity of Guard: After a finite number of iterations of the loop, the guard becomes false.
IV.Correctness of the Post-Condition: If *N* is the least number of iterations after which *G* is false and *I*(*N*) is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.

PROOF:

Let I(n) be a predicate that satisfies properties I-IV of the loop invariant theorem.
Properties I and II establish that:
For all integers $n \geq 0$, if the while loop iterates n times, then I(n) is true.
Property III indicates that the guard G becomes false after a finite number N of iterations.
Property IV concludes that the values of the algorithm variables are as specified by the post-condition of the loop.

Lecture No.28 Division algorithm

CORRECTNESS OF:
LOOP TO COMPUTE A PRODUCT
THE DIVISION ALGORITHM
THE EUCLIDEAN ALGORITHM

A LOOP TO COMPUTE A PRODUCT:

[pre-condition: m is a nonnegative integer,
x is a real number, $i = 0$, and product = 0.]
while (i # m)
1. product := product + x
2. $i := i + 1$

end while
[post-condition: product = $m \cdot x$]

PROOF:

Let the loop invariant be
$I(n)$: $i = n$ and product $= n \cdot x$
The guard condition G of the while loop is
G: $i \# m$

I.Basis Property:

[$I(0)$ is true before the first iteration of the loop.]
$I(0)$: $i = 0$ and product $= 0 \cdot x = 0$
Which is true before the first iteration of the loop.

II.Inductive property:

[If the guard G and the loop invariant $I(k)$ are both true before a loop iteration (where $k \geq 0$), then $I(k + 1)$ is true after the loop iteration.]
Before execution of statement 1,

$$product_{old} = k \cdot x.$$

Thus the execution of statement 1 has the following effect:
$product_{new} = product_{old} + x = k \cdot x + x = (k + 1) \cdot x$
Similarly, before statement 2 is executed,
$i_{old} = k$,
So after execution of statement 2,
$i_{new} = i_{old} + 1 = k + 1$.
Hence after the loop iteration, the statement $I(k + 1)$ (i.e., $i = k + 1$ and product = $(k + 1) \cdot x$) is true. This is what we needed to show.

III.Eventual Falsity of Guard:

[After a finite number of iterations of the loop, the guard becomes false.]

IV.Correctness of the Post-Condition:

[If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.]

THE DIVISION ALGORITHM:

[pre-condition: a is a nonnegative integer and
d is a positive integer, $r = a$, and $q = 0$]

while ($r \geq d$)
1. $r := r – d$
2. $q := q + 1$

end while

[post-condition: q and r are nonnegative integers
with the property that $a = q \cdot d + r$ and $0 \leq r < d$.]

PROOF:

Let the loop invariant be
$I(n)$: $r = a - n \cdot d$ and $n = q$.

The guard of the **while** loop is
G: $r \geq d$

I.Basis Property:

[$I(0)$ is true before the first iteration of the loop.]
$I(0)$: $r = a - 0 \cdot d = a$ and $0 = q$.

II.Inductive property:

[If the guard G and the loop invariant $I(k)$ are both true before a loop iteration (where $k \geq 0$), then $I(k + 1)$ is true after the loop iteration.]

$I(k)$: $r = a - k \cdot d \geq 0$ and $k = q$

$I(k + 1)$: $r = a - (k + 1) \cdot d \geq 0$ and $k + 1 = q$

$$
\begin{aligned}
r_{\text{new}} &= r - d \\
&= a - k \cdot d - d \\
&= a - (k + 1) \cdot d \\
q &= q + 1 \\
&= k + 1
\end{aligned}
$$

also

$$
\begin{aligned}
r_{\text{new}} &= r - d \\
&\geq d - d = 0 \quad \text{(since } r \geq 0)
\end{aligned}
$$

Hence $I(k + 1)$ is true.

III.Eventual Falsity of Guard:

[After a finite number of iterations of the loop, the guard becomes false.]

IV.Correctness of the Post-Condition:

[If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.]

G is false and I(N) is true.

That is, $r \geq d$ and $r = a - N \cdot d \geq 0$ and $N = q$.

or $\quad r = a - q \cdot d$

or $a = q \cdot d + r$

Also combining the two inequalities involving r we get

$0 \le r < d$

THE EUCLIDEAN ALGORITHM:

The greatest common divisor (gcd) of two integers a and b is the largest integer that divides both a and b. For example, the gcd of 12 and 30 is 6.

The Euclidean algorithm takes integers A and B with $A > B \geq 0$ and compute their greatest common divisor.

HAND CALCULATION OF gcd:

Use the Euclidean algorithm to find gcd(330, 156)

SOLUTION:

$$\begin{array}{r|r} & 2 \\ \cline{2-2} 156 & 330 \\ & 312 \\ \cline{2-2} & 18 \end{array} \qquad \begin{array}{r|r} & 8 \\ \cline{2-2} 18 & 156 \\ & 144 \\ \cline{2-2} & 12 \end{array}$$

$$\begin{array}{r|r} & 1 \\ \cline{2-2} 12 & 18 \\ & 12 \\ \cline{2-2} & 6 \end{array} \qquad \begin{array}{r|r} & 2 \\ \cline{2-2} 6 & 12 \\ & 12 \\ \cline{2-2} & 0 \end{array}$$

Hence gcd(330, 156) = 6

EXAMPLE:

Use the Euclidean algorithm to find gcd(330, 156)

Solution:

1.Divide 330 by 156:

This gives $330 = 156 \cdot 2 + 18$

2.Divide 156 by 18:

This gives $156 = 18 \cdot 8 + 12$

3.Divide 18 by 12:

This gives $18 = 12 \cdot 1 + 6$

4.Divide 12 by 6:

This gives $12 = 6 \cdot 2 + 0$

Hence gcd(330, 156) = 6.

LEMMA:

If a and b are any integers with $b \neq 0$ and q and r are nonnegative integers such that $a = q \cdot d + r$

then
gcd(a, b) = gcd(b, r)
[pre-condition: A and B are integers with
$A > B \geq 0$, $a = A$, b = B, $r = B$.]

while ($b \neq 0$)
1. $r := a \bmod b$
2. $a := b$
3. b := r

end while[post-condition: a = gcd(A, B)]

PROOF:

Let the **loop invariant** be
$I(n)$: gcd(a, b) = gcd(A, B) and $0 \leq b < a$.
The guard of the **while** loop is
G: $b \neq 0$

I.Basis Property:

[$I(0)$ is true before the first iteration of the loop.]
$I(0)$: gcd(a, b) = gcd(A, B) and $0 \leq b < a$.
According to the precondition,
a = A, b = B, r = B, and $0 \leq B < A$.
Hence $I(0)$ is true before the first iteration of the loop.

II.Inductive property:

[If the guard G and the loop invariant $I(k)$ are both true before a loop iteration (where k $\geq$ 0), then $I(k + 1)$ is true after the loop iteration.]
Since $I(k)$ is true before execution of the loop we have,
$\gcd(a_{old}, b_{old}) = \gcd(A, B)$ and $0 \leq b_{old} < a_{old}$
After execution of statement 1,
$r_{new} = a_{old} \bmod b_{old}$Thus,
$a_{old} = b_{old} \cdot q + r_{new}$ for some integer q
with,
$0 \leq r_{new} < b_{old}$.
But
$\gcd(a_{old}, b_{old}) = \gcd(b_{old}, r_{old})$
and we have,
$\gcd(b_{old}, r_{new}) = \gcd(A, B)$
When statements 2 and 3 are executed,
$a_{new} = b_{old}$ and $b_{new} = r_{new}$
It follows that
$\gcd(a_{new}, b_{new}) = \gcd(A, B)$
Also,
$0 \leq r_{new} < b_{old}$
becomes
$0 \leq b_{new} < a_{new}$
Hence $I(k + 1)$ is true.

III.Eventual Falsity of Guard:
[After a finite number of iterations of the loop, the guard becomes false.]

IV.Correctness of the Post-Condition:
[If N is the least number of iterations after which G is false and $I(N)$ is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.]

Lecture No.29 Combinatorics

COMBINATORICS
THE SUM RULE
THE PRODUCT RULE

COMBINATORICS:
Combinatorics is the mathematics of counting and arranging objects. Counting of objects with certain properties (enumeration) is required to solve many different types of problem

For example, counting is used to:

1) Determine number of ordered or unordered arrangement of objects.
2) Generate all the arrangements of a specified kind which is important in computer simulations.
3) Compute probabilities of events.
4) Analyze the chance of winning games, lotteries etc.
5) Determine the complexity of algorithms.

THE SUM RULE:
If one event can occur in n_1 ways, a second event can occur in n_2 (different) ways, then the total number of ways in which exactly one of the events (i.e., first or second) can occur is $n_1 + n_2$.

EXAMPLE:
Suppose there are 7 different optional courses in Computer Science and 3 different optional courses in Mathematics. Then there are $7 + 3 = 10$ choices for a student who wants to take one optional course.

EXERCISE:
A student can choose a computer project from one of the three lists. The three lists contain 23, 15 and 19 possible projects, respectively. How many possible projects are there to choose from?

SOLUTION:
The student can choose a project from the first list in 23 ways, from the second list in 15 ways, and from the third list in 19 ways. Hence, there are
$23 + 15 + 19 = 57$ projects to choose from.

GENERALIZED SUM RULE

If one event can occur in n_1 ways,
a second event can occur in n_2 ways,
a third event can occur in n_3 ways,
……………………………………..

then there are
$n_1 + n_2 + n_3 + \ldots$ ways in which exactly one of the events can occur.

SUM RULE IN TERMS OF SETS:

If $A_1, A_2, \ldots, A_m$ are finite disjoint sets, then the number of elements in the union of these sets is the sum of the number of elements in them.

If $n(A_i)$ denotes the number of elements in set A_i for $i = 1, 2, \ldots, m$, then

$$n(A_1 \cup A_2 \cup \ldots \cup A_m) = n(A_1) + n(A_2) + \ldots + n(A_m)$$

where $A_i \cap A_j = \phi$ if $i \neq j$

THE PRODUCT RULE:

If one event can occur in n_1 ways and if for each of these n_1 ways, a second event can occur in n_2 ways, then the total number of ways in which both events occur is $n_1 \cdot n_2$.

EXAMPLE:

Suppose there are 7 different optional courses in Computer Science and 3 different optional courses in Mathematics. A student who wants to take one optional course of each subject, there are $7 \times 3 = 21$ choices.

EXAMPLE:

The chairs of an auditorium are to be labeled with two characters, a letter followed by a digit. What is the largest number of chairs that can be labeled differently?

SOLUTION:

The procedure of labeling a chair consists of two events, namely,

1. Assigning one of the 26 letters: A, B, C, …, Z and
2. Assigning one of the 10 digits: 0, 1, 2, …, 9

By product rule, there are $26 \times 10 = 260$ different ways that a chair can be labeled by both a letter and a digit.

GENERALIZED PRODUCT RULE:

If some event can occur in n_1 different ways, and if, following this event, a second event can occur in n_2 different ways, and following this second event, a third event can occur in n_3 different ways, …, then the number of ways all the events can occur in the order indicated is $n_1 \cdot n_2 \cdot n_3 \cdot \ldots$

PRODUCT RULE IN TERMS OF SETS:

If $A_1, A_2, \ldots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements in each set.
If $n(A_i)$ denotes the number of elements in set A_i, then

$$n(A_1 \times A_2 \times \ldots \times A_m) = n(A_1) \cdot n(A_2) \cdot \ldots \cdot n(A_{m,})$$

EXERCISE:

Find the number n of ways that an organization consisting of 15 members can elect a president, treasurer, and secretary. (assuming no person is elected to more than one position)

SOLUTION:

The president can be elected in 15 different ways; following this, the treasurer can be elected in 14 different ways; and following this, the secretary can be elected in 13 different ways.

Thus, by product rule, there are

$$n = 15 \times 14 \times 13 = 2730$$

different ways in which the organization can elect the officers.

EXERCISE:

There are four bus lines between A and B; and three bus lines between B and C. Find the number of ways a person can travel:

(a) By bus from A to C by way of B;

(b) Round trip by bus from A to C by way of B;

(c) Round trip by bus from A to C by way of B, if the person does not want to use a bus line more than once.

SOLUTION:

(a) There are 4 ways to go from A to B and 3 ways to go from B to C; hence there are $4 \times 3 = 12$ ways to go from A to C by way of B.

(b) The person will travel from A to B to C to B to A for the round trip.

i.e $(A \rightarrow B \rightarrow C \rightarrow B \rightarrow A)$

The person can travel 4 ways from A to B and 3 way from B to C and back.

i.e., $A \xrightarrow{4} B \xrightarrow{3} C \xrightarrow{3} B \xrightarrow{4} A$

Thus there are $4 \times 3 \times 3 \times 4 = 144$ ways to travel the round trip.

(c) The person can travel 4 ways from A to B and 3 ways from B to C, but only 2 ways from C to B and 3 ways from B to A, since bus line cannot be used more than once. Thus

i.e., $A \xrightarrow{4} B \xrightarrow{3} C \xrightarrow{2} B \xrightarrow{3} A$

Hence there are $4 \times 3 \times 2 \times 3 = 72$ ways to travel the round trip without using a bus line more than once.

EXERCISE:

A bit string is a sequence of 0's and 1's.How many bit string are there of length 4?

SOLUTION:

Each bit (binary digit) is either 0 or 1.

Hence, there are 2 ways to choose each bit. Since we have to choose four bits therefore, the product rule shows, there are a total of $2 \times 2 \times 2 \times 2 = 2^4 = 16$ different bit strings of length four.

EXERCISE:

How many bit strings of length 8

(i) begin with a 1? (ii) begin and end with a 1?

SOLUTION:

(i) If the first bit (left most bit) is a 1, then it can be filled in only one way. Each of the remaining seven positions in the bit string can be filled in 2 ways (i.e., either by 0 or 1).

Hence, there are $1 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 = 128$

different bit strings of length 8 that begin with a 1.

(ii) If the first and last bit in an 8 bit string is a 1, then only the intermediate six bits can be filled in 2 ways, i.e. by a 0 or 1. Hence there are $1 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 1 = 2^6 = 64$ different bit strings of length 8 that begin and end with a 1.

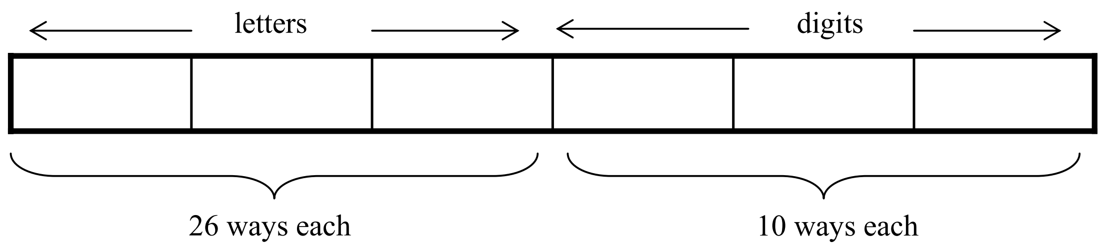
EXERCISE:

Suppose that an automobile license plate has three letters followed by three digits.

(a) How many different license plates are possible?

SOLUTION:

Each of the three letters can be written in 26 different ways, and each of the three digits can be written in 10 different ways.



Hence, by the product rule, there is a total of

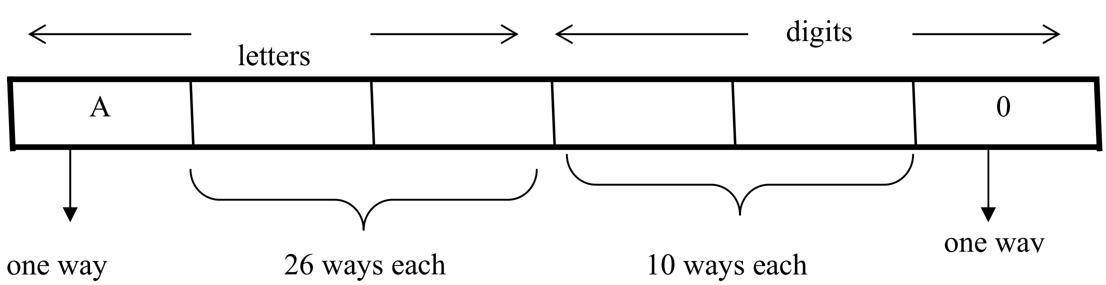
$26 \times 26 \times 26 \times 10 \times 10 \times 10 = 17,576,000$

different license plates possible.

(b) How many license plates could begin with A and end on 0?

SOLUTION:

The first and last place can be filled in one way only, while each of second and third place can be filled in 26 ways and each of fourth and fifth place can be filled in 10 ways.

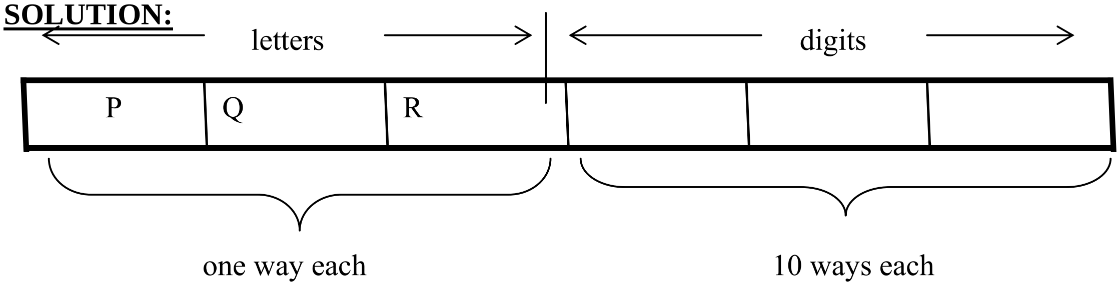


Number of license plates that begin with A and end in 0 are

$$1 \times 26 \times 26 \times 10 \times 10 \times 1 = 67600$$

(c) How many license plates begin with PQR

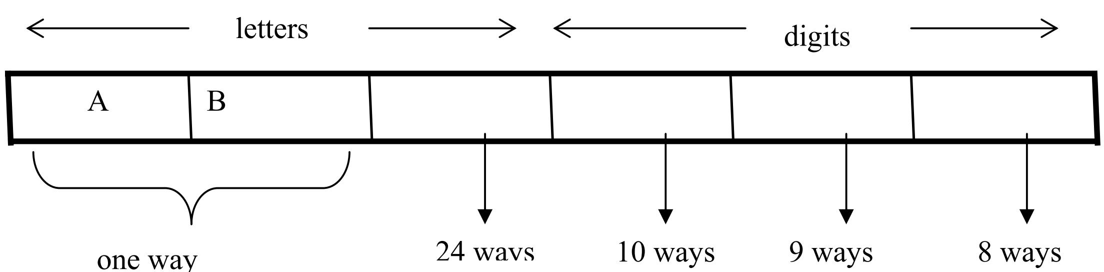
SOLUTION:



Number of license plates that begin with PQR are
$1 \times 1 \times 1 \times 10 \times 10 \times 10 = 1000$

(d) How many license plates are possible in which all the letters and digits are distinct?
SOLUTION:
The first letter place can be filled in 26 ways. Since, the second letter place should contain a different letter than the first, so it can be filled in 25 ways. Similarly, the third letter place can be filled in 24 ways.And the digits can be respectively filled in 10, 9, and 8 ways.
Hence; number of license plates in which all the letters and digits are distinct are
$26 \times 25 \times 24 \times 10 \times 9 \times 8 = 11, 232, 000$

(e) How many license plates could begin with AB and have all three letters and digits distinct.
SOLUTION:



The first two letters places are fixed (to be filled with A and B), so there is only one way to fill them. The third letter place should contain a letter different from A & B, so there are 24 ways to fill it.
The three digit positions can be filled in 10 and 8 ways to have distinct digits.
Hence, desired number of license plates are
$1 \times 1 \times 24 \times 10 \times 9 \times 8 = 17280$

EXERCISE:
A variable name in a programming language must be either a letter or a letter followed by a digit. How many different variable names are possible?

SOLUTION:
First consider variable names one character in length. Since such names consist of a single letter, there are 26 variable names of length 1.

Next, consider variable names two characters in length. Since the first character is a letter, there are 26 ways to choose it. The second character is a digit, there are 10 ways to choose it. Hence, to construct variable name of two characters in length, there are

$26\times10 = 260$ ways.

Finally, by sum rule, there are $26 + 260 = 286$ possible variable names in the programming language.

EXERCISE:

(a) How many bit strings consist of from one through four digits?
(b) How many bit strings consist of from five through eight digits?

SOLUTION:

(a) Number of bit strings consisting of 1 digit = 2
Number of bit strings consisting of 2 digits = $2\cdot2 = 2^2$
Number of bit strings consisting of 3 digits = $2\cdot2\cdot2 = 2^3$
Number of bit strings consisting of 4 digits = $2\cdot2\cdot2\cdot2 = 2^4$

Hence by sum rule, the total number of bit strings consisting of one through four digit is

$2+2^2 +2^3 +2^4 = 2 + 4 + 8 + 16 = 30$

(b) Number of bit strings of 5 digits = 2^5
Number of bit strings of 6 digits = 2^6
Number of bit strings of 7 digits = 2^7
Number of bit strings of 8 digits = 2^8

Hence, by sum rule, the total number of bit strings consisting of five through eight digit is

$2^5 + 2^6 + 2^7 + 2^8 = 480$

EXERCISE:

How many three-digit integers are divisible by 5?

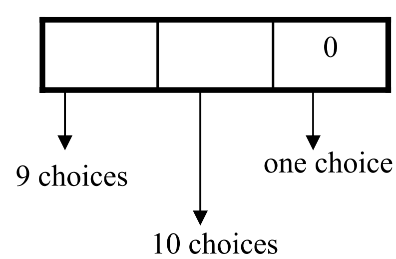
SOLUTION:

Integers that are divisible by 5, end either in 5 or in 0.

CASE-I (Integers that end in 0)

There are nine choices for the left-most digit (the digits 1 through 9) and ten choices for the middle digit.(the digits 0 through 9) Hence, total number of 3 digit integers that end in 0 is

$9 \times 10 \times 1 = 90$



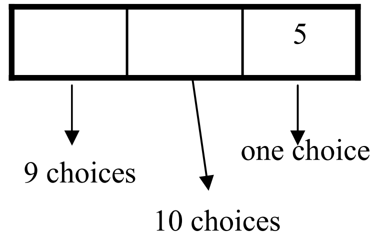
CASE-II (Integer that end in 5)

There are nine choices for the left-most digit and ten choices for the middle digit

Hence, total number of 3 digit integers that end in 5 is

$9 \times 10 \times 1 = 90$

Finally, by sum rule, the number of 3 digit integers that are divisible by 5 is
90 + 90 = 180



EXERCISE:

A computer access code word consists of from one to three letters of English alphabets with repetitions allowed. How many different code words are possible.

SOLUTION:

Number of code words of length 1 = 26^1
Number of code words of length 2 = 26^2
Number of code words of length 3 = 26^3

Hence, the total number of code words = $26^1 + 26^2 + 26^3$
= 18,278

NUMBER OF ITERATIONS OF A NESTED LOOP:

Determine how many times the inner loop will be iterated when the following algorithm is implemented and run

```
        for     i: = 1 to 4
for     j : = 1 to 3
        [Statement in body of inner loop.
                                None contain branching statements
         that lead out of the inner loop.]
  next j
  next i
```

SOLUTION:

The outer loop is iterated four times, and during each iteration of the outer loop, there are three iterations of the inner loop. Hence, by product rules the total number of iterations of inner loop is $4 \cdot 3 = 12$

EXERCISE:

Determine how many times the inner loop will be iterated when the following algorithm is implemented and run.

```
        for     i = 5 to 50
for     j: = 10 to 20
        [Statement in body of inner loop.
                                None contain branching statements
        that lead out of the inner loop.]
  next j
  next i
```

SOLUTION:

The outer loop is iterated 50 - 5 + 1 = 46 times and during each iteration of the outer loop there are 20 - 10 + 1 = 11 iterations of the inner loop. Hence by product rule, the total number of iterations of the inner loop is $46 \times 11 = 506$

EXERCISE:

Determine how many times the inner loop will be iterated when the following algorithm is implemented and run.

```
for     i: = 1 to 4
for j: = 1 to i
    [Statements in body of inner loop.
        None contain branching statements
    that lead outside the loop.]
 next j
 next i
```

SOLUTION:

The outer loop is iterated 4 times, but during each iteration of the outer loop, the inner loop iterates different number of times.
For first iteration of outer loop, inner loop iterates 1 times.
For second iteration of outer loop, inner loop iterates 2 times.
For third iteration of outer loop, inner loop iterates 3 times.
For fourth iteration of outer loop, inner loop iterates 4 times.
Hence, total number of iterations of inner loop = 1 + 2 + 3 + 4 = 10

Lecture No.30 Permutations

FACTORIAL
K-SAMPLE
K-PERMUTATION

FACTORIAL OF A POSITIVE INTEGER:

For each positive integer n, its factorial is defined to be the product of all the integers from 1 to n and is denoted n!. Thus $n! = n(n - 1)(n - 2) \ldots 3\cdot 2\cdot 1$

In addition, we define

$$0! = 1$$

REMARK:

n! can be recursively defined as

Base: $0! = 1$

Recursion $n! = n(n - 1)!$ for each positive integer n.

EXERCISE:

Compute each of the following

(i) $\dfrac{7!}{5!}$ (ii) $(-2)!$

(iii) $\dfrac{(n+1)!}{n!}$ (iv) $\dfrac{(n-1)!}{(n+1)!}$

SOLUTION:

$(i)\quad \dfrac{7!}{5!} = \dfrac{7\cdot 6\cdot 5!}{5!} = 7\cdot 6 = 42$

$(ii)\quad (-2)!$ is not defined

$(iii)\quad \dfrac{(n+1)!}{n!} = \dfrac{(n+1)n!}{n!} = n+1$

$(iv)\quad \dfrac{(n-1)!}{(n+1)!} = \dfrac{(n-1)!}{(n+1)\cdot n\cdot (n-1)!} = \dfrac{1}{(n+1)n} = \dfrac{1}{n^2+n}$

EXERCISE:

Write in terms of factorials.

(i) $25\cdot 24\cdot 23\cdot 22$ (ii) $n(n-1)(n-2) \ldots (n - r + 1)$

(iii) $\dfrac{n(n-1)(n-2)\cdots(n-r+1)}{1\cdot 2\cdot 3\cdots(r-1)\cdot r}$

SOLUTION:

$(i)\quad 25\cdot 24\cdot 23\cdot 22 = \dfrac{25\cdot 24\cdot 23\cdot 22\cdot 21!}{21!} = \dfrac{25!}{21!}$

$$(ii)\quad n(n-1)(n-2)\cdots(n-r+1) = \frac{n(n-1)(n-2)\cdots(n-r+1)(n-r)!}{(n-r)!}$$

$$= \frac{n!}{(n-r)!}$$

$$(iii) \quad \frac{n(n-1)(n-2)\cdots(n-r+1)}{1\cdot 2\cdot 3\cdots(r-1)\cdot r} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$$

$$= \frac{n(n-1)(n-2)\cdots(n-r+1)(n-r)!}{r!(n-r)!}$$

$$= \frac{n!}{r!(n-r)!}$$

COUNTING FORMULAS:

From a given set of n distinct elements, one can choose k elements in different ways. The number of selections of elements varies according as:
(i)elements may or may not be repeated.
(ii)the order of elements may or may not matter.
These two conditions therefore lead us to four counting methods summarized in the following table.

	ORDER MATTERS	ORDER DOES NOT MATTER
REPETITION ALLOWED	k-sample	k-selection
REPETITION NOT ALLOWED	k-permutation	k-combination

K-SAMPLE:

A k-sample of a set of n elements is a choice of k elements taken from the set of n elements such that the order of elements matters and elements can be repeated.

REMARK:

With k-sample, repetition of elements is allowed, therefore, k need not be less than or equal to n. i.e. k is independent of n.

FORMULA FOR K-SAMPLE:

Suppose there are n distinct elements and we draw a k-sample from it. The first element of the k-sample can be drawn in n ways. Since, repetition of elements is allowed, so the second element can also be drawn in n ways.
Similarly each of third, fourth, …, k-th element can be drawn in n ways.
Hence, by product rule, the total number of ways in which a k-sample can be drawn from n distinct elements is

$n \cdot n \cdot n \cdot \ldots \cdot n$ (k-times)
$= n^k$

EXERCISE:

How many possible outcomes are there when a fair coin is tossed three times.

SOLUTION:

Each time a coin is tossed it's outcome is either a head (H) or a tail (T). Hence in successive tosses, H and T are repeated. Also the order in which they appear is important. Accordingly, the problem is of 3-samples from a set of two elements H and T. [k = 3, n = 2]

Hence number of samples $= n^k$
$= 2^3 = 8$

These 8-samples may be listed as:
HHH, HHT, HTH, THH, HTT, THT, TTH, TTT

EXERCISE:

Suppose repetition of digits is permitted.

(a) How many three-digit numbers can be formed from the six digits 2, 3, 4, 5, 7 and 9

SOLUTION:

Given distinct elements = n = 6
Digits to be chosen = k = 3

While forming numbers, order of digits is important. Also digits may be repeated. Hence, this is the case of 3-sample from 6 elements.

Number of 3-digit numbers $= n^k = 6^3 = 216$

(b) How many of these numbers are less than 400?

SOLUTION:

From the given six digits 2, 3, 4, 5, 7 and 9, a three-digit number would be less than 400 if and only if its first digit is either 2 or 3.

The next two digits positions may be filled with any one of the six digits.

Hence, by product rule, there are

$$2\cdot 6 \cdot 6 = 72$$

three-digit numbers less than 400.

(c) How many are even?

SOLUTION:

A number is even if its right most digit is even. Thus, a 3-digit number formed by the digits 2, 3, 4, 5, 7 and 9 is even if it last digit is 2 or 4. Thus the last digit position may be filled in 2 ways only while each of the first two positions may be filled in 6 ways.

Hence, there are

$$6 \cdot 6 \cdot 2 = 72$$

3-digit even numbers.

(d) How many are odd?

SOLUTION:

A number is odd if its right most digit is odd. Thus, a 3-digit number formed by the digits 2, 3, 4, 5, 7 and 9 is odd if its last digit is one of 3, 5, 7, 9. Thus, the last digit position may be filled in 4 ways, while each of the first two positions may be filled in 6 ways.

Hence, there are $6 \cdot 6 \cdot 4 = 144$
3-digit odd numbers.

(e) How many are multiples of 5?

SOLUTION:

A number is a multiple of 5 if its right most digit is either 0 or 5. Thus, a 3-digit number formed by the digits 2, 3, 4, 5, 7 and 9 is multiple of 5 if its last digit is 5. Thus, the last digit position may be filled in only one way, while each of the first two positions may be filled in 6 ways.

Hence, there are $6 \cdot 6 \cdot 1 = 36$

3-digit numbers that are multiple of 5.

EXERCISE:

A box contains 10 different colored light bulbs. Find the number of ordered samples of size 3 with replacement.

SOLUTION:

Number of light bulbs = n = 10

Bulbs to be drawn = k = 3

Since bulbs are drawn with replacement, so repetition is allowed. Also while drawing a sample, order of elements in the sample is important.

Hence number of samples of size 3 $= n^k$

$= 10^3$

$= 1000$

EXERCISE:

A multiple choice test contains 10 questions; there are 4 possible answers for each question.

(a) How many ways can a student answer the questions on the test if every question is answered?

(b) How many ways can a student answer the questions on the test if the student can leave answers blank?

SOLUTION:

(a) Each question can be answered in 4 ways. Suppose answers are labeled as A, B, C, D. Since label A may be used as the answer of more than one question. So repetition is allowed. Also the order in which A, B, C, D are choosed as answers for 10 questions is important. Hence, this is the one of k-sample, in which

n = no. of distinct labels = 4

k = no. of labels selected for answering = 10

$\therefore$ No. of ways to answer 10 questions $= n^k$

$= 4^{10}$

$= 1048576$

(b) If the student can leave answers blank, then in addition to the four answers, a fifth option to leave answer blank is possible. Hence, in such case

n = 5

and k = 10 (as before)

$\therefore$ No. of possible answers$= n^k$

$= 5^{10}$

$= 9765625$

k-PERMUTATION:

A k-permutation of a set of n elements is a selection of k elements taken from the set of n elements such that the order of elements matters but repetition of the elements is not allowed. The number of k-permutations of a set of n elements is denoted P(n, k) or ${}^{n}p_{k}$.

REMARK:

1. With k-permutation, repetition of elements is not allowed, therefore $k \le n$.
2. The wording "number of permutations of a set with n elements" means that all n elements are to be permuted, so that k = n.

FORMULA FOR k-PERMUTATION:

Suppose a set of n elements is given. Formation of a k-permutation means that we have an ordered selection of k elements out of n, where elements cannot be repeated.

1st element can be selected in n ways
2nd element can be selected in (n-1) ways
3rd element can be selected in (n-2) ways
………………………………….
kth element can be selected in (n-(k-1)) ways
Hence, by product rule, the number of ways to form a k-permutation is

$$\begin{aligned} P(n,k) &= n\cdot(n-1)\cdot(n-2)\cdots(n-(k-1)) \\ &= n\cdot(n-1)\cdot(n-2)\cdots(n-k+1) \\ &= \frac{[n\cdot(n-1)\cdot(n-2)\cdots(n-k+1)][(n-k)(n-k-1)\cdots 3\cdot 2\cdot 1]}{[(n-k)(n-k-1)\cdots 3\cdot 2\cdot 1]} \\ &= \frac{n!}{(n-k)!} \end{aligned}$$

EXERCISE:

How many 2-permutation are there of {W, X, Y, Z}? Write them all.

SOLUTION:

Number of 2-permutation of 4 elements is

$$\begin{aligned} P(4,2) = {}^{4}p_{2} &= \frac{4!}{(4-2)!} \\ &= \frac{4\cdot 3\cdot 2!}{2!} \\ &= 4\cdot 3 = 12 \end{aligned}$$

These 12 permutations are:

WX, WY, WZ,
XW, XY, XZ,
YW, YX, YZ,
ZW, ZX, ZY.

EXERCISE:

Find (a) P(8, 3) (b) P(8,8)
(c) P(8,1) (d) P(6,8)

SOLUTION:

$$(a)\qquad P(8,3) = \frac{8!}{(8-3)!} = 8\cdot 7\cdot 6 = 336$$

$$(b)\qquad P(8,8) = \frac{8!}{(8-8)!} = \frac{8!}{0!} = 8! = 40320 \qquad (as\quad 0! = 1)$$

$$(c)\qquad P(8,1) = \frac{8!}{(8-1)!} = \frac{8\cdot 7!}{7!} = 8$$

(d) $P(6,8)$ is not defined, since the second integer cannot exceed the first integer.

EXERCISE:

Find n if

(a) P(n,2) = 72 **(b)** P(n,4) = 42 P (n, 2)

SOLUTION:

(a) Given P(n,2) = 72

$\Rightarrow$ $n \cdot (n-1) = 72$ (by using the definition of permutation)

$\Rightarrow$ $n^2 - n = 72$

$\Rightarrow$ $n^2 - n - 72 = 0$

$\Rightarrow$ $n = 9, -8$

Since n must be positive, so the only acceptable value of n is 9.

(b) Given P(n,4) = 42P(n,2)

$\Rightarrow$ $n(n-1)(n-2)(n-3) = 42\, n(n-1)$ (by using the definition of permutation)

$\Rightarrow$ $(n-2)(n-3) = 42$ if $n \neq 0$, $n \neq 1$

$\Rightarrow$ $n^2 - 5n + 6 = 42$

$\Rightarrow$ $n^2 - 5n - 36 = 0$

$\Rightarrow$ $(n-9)(n+4) = 0$

$\Rightarrow$ $n = 9, -4$

Since n must be positive, the only answer is n = 9

EXERCISE:

Prove that for all integers $n \geq 3$

$P(n + 1, 3) - P(n, 3) = 3P(n, 2)$

SOLUTION:

Suppose n is an integer greater than or equal to 3

Now L.H.S $= P(n + 1, 3) - P(n, 3)$

$= (n + 1)(n)(n-1) - n(n - 1)(n - 2)$

$= n(n - 1)[(n + 1) - (n - 2)]$

$= n(n - 1)[n + 1 - n + 2]$

$= 3n(n - 1)$

R.H.S $= 3P(n, 2)$

$= 3\cdot n(n-1)$

Thus L.H.S = R.H.S. Hence the result.

EXERCISE:

(a) How many ways can five of the letters of the word ALGORITHM be selected and written in a row?

(b) How many ways can five of the letters of the word ALGORITHM be selected and written in a row if the first two letters must be TH?

SOLUTION:

(a) The answer equals the number of 5-permutation of a set of 9 elements and

$$P(9,5) = \frac{9!}{(9-5)!} = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15120$$

(b) Since the first two letters must be TH hence we need to choose the remaining three letters out of the left 9 - 2 = 7 alphabets.
Hence, the answer is the number of 3-permutations of a set of seven elements which is

$$P(7,3) = \frac{7!}{(7-3)!} = 7 \cdot 6 \cdot 5 = 210$$

EXERCISE:

Find the number of ways that a party of seven persons can arrange themselves in a row of seven chairs.

SOLUTION:

The seven persons can arrange themselves in a row in P(7,7) ways.
Now

$$P(7,7) = \frac{7!}{(7-7)!} = \frac{7!}{0!} = 7!$$

EXERCISE:

A debating team consists of three boys and two girls. Find the number n of ways they can sit in a row if the boys and girls are each to sit together.

SOLUTION:

There are two ways to distribute them according to sex: BBBGG or GGBBB.
In each case
the boys can sit in a row in P(3,3) = 3! = 6 ways, and
the girls can sit in
P(2,2) = 2! = 2 ways and
Every row consist of boy and girl which is = 2!=2
Thus
The total number of ways=n = 2 · 3! · 2!
= 2 ·6 ·2 = 24

EXERCISE:

Find the number n of ways that five large books, four medium sized book, and three small books can be placed on a shelf so that all books of the same size are together.

SOLUTION:

In each case, the large books can be arranged among themselves in P(5,5)= 5! ways, the medium sized books in P(4,4) = 4! ways, and the small books in P(3,3) = 3! ways.

The three blocks of books can be arranged on the shelf in P(3,3) = 3! ways.

Thus

$$n = 3! \cdot 5! \cdot 4! \cdot 3!$$
$$= 103680$$

Lecture No.31 Combinations

K-COMBINATIONS
K-SELECTIONS

K-COMBINATIONS:

With a *k*-combinations the order in which the elements are selected does not matter and the elements cannot repeat.

DEFINITION:

A *k*-combination of a set of *n* elements is a choice of *k* elements taken from the set of *n* elements such that the order of the elements does not matter and elements can't be repeated.
The symbol $C(n, k)$ denotes the number of *k*-combinations that can be chosen from a set of *n* elements.

NOTE:

k-combinations are also written ${}^{n}C_{k}$ as or $\binom{n}{k}$

REMARK:

With *k*-combinations of a set of *n* elements, repetition of elements is not allowed, therefore, *k* must be less than or equal to *n*, i.e., $k \le n$.

EXAMPLE:

Let $X = \{a, b, c\}$. Then 2-combinations of the 3 elements of the set *X* are:
$\{a, b\}$, $\{a, c\}$, and $\{b, c\}$. Hence $C(3,2) = 3$.

EXERCISE:

Let $X = \{a, b, c, d, e\}$.
List all 3-combinations of the 5 elements of the set *X*, and hence find the value of $C(5,3)$.

SOLUTION:

Then 3-combinations of the 5 elements of the set *X* are:
$\{a, b, c\}$, $\{a, b, d\}$, $\{a, b, e\}$, $\{a, c, d\}$, $\{a, c, e\}$,
$\{a, d, e\}$, $\{b, c, d\}$, $\{b, c, e\}$, $\{b, d, e\}$, $\{c, d, e\}$
Hence $C(5, 3) = 10$

PERMUTATIONS AND COMBINATIONS:

EXAMPLE:

Let $X = \{A, B, C, D\}$.
The 3-combinations of *X* are:
$\{A, B, C\}$, $\{A, B, D\}$, $\{A, C, D\}$, $\{B, C, D\}$
Hence $C(4, 3) = 4$

The 3-permutations of *X* can be obtained from 3-combinations of X as follows.
ABC, *ACB*, *BAC*, *BCA*, *CAB*, *CBA*
ABD, *ADB*, *BAD*, *BDA*, *DAB*, *DBA*
ACD, *ADC*, *CAD*, *CDA*, *DAC*, *DCA*

BCD, *BDC*, *CBD*, *CDB*, *DBC*, *DCB*
So that $P(4, 3) = 24 = 4 \cdot 6 = 4 \cdot 3!$
Clearly $P(4, 3) = C(4, 3) \cdot 3!$
In general we have, $P(n, k) = C(n, k) \cdot k!$
In general we have,

$$P(n, k) = C(n, k) \cdot k!$$

or
$$C(n,k) = \frac{P(n,k)}{k!}$$

But we know that $P(n,k) = \dfrac{n!}{(n-k)!}$

Hence,
$$C(n,k) = \frac{n!}{(n-k)!k!}$$

COMPUTING *C*(*n*, *k*)

EXAMPLE:

Compute $C(9, 6)$.

SOLUTION:
$$\begin{aligned} C(9,6) &= \frac{9!}{(9-6)!6!} \\ &= \frac{9\cdot 8\cdot 7\cdot 6!}{3!\cdot 6!} \\ &= \frac{9\cdot 8\cdot 7}{3\cdot 2} \\ &= 84 \end{aligned}$$

SOME IMPORTANT RESULTS

(a) $C(n, 0) = 1$
(b) $C(n, n) = 1$
(c) $C(n, 1) = n$
(d) $C(n, 2) = n(n-1)/2$
(e) $C(n, k) = C(n, n-k)$
(f) $C(n, k) + C(n, k+1) = C(n+1, k+1)$

EXERCISE:

A student is to answer eight out of ten questions on an exam.
(a) Find the number *m* of ways that the student can choose the eight questions
(b) Find the number *m* of ways that the student can choose the eight questions, if the first three questions are compulsory.

SOLUTION:

(a) The eight questions can be answered in $m = C(10, 8) = 45$ ways.
(b) The eight questions can be answered in $m = C(7, 5) = 21$ ways.

EXERCISE:

An examination paper consists of 5 questions in section A and 5 questions in section B. A total of 8 questions must be answered. In how many ways can a student select the questions if he is to answer at least 4 questions from section A.

SOLUTION:

There are two possibilities:

(a) The student answers 4 questions from section A and 4 questions from section B.The number of ways 8 questions can be answered taking 4 questions from section A and 4 questions from section B are

$$C(5, 4) \cdot C(5, 4) = 5 \cdot 5 = 25.$$

(b) The student answers 5 questions from section A and 3 questions from section B.The number of ways 8 questions can be answered taking 5 questions from section A and 3 questions from section B are

$$C(5, 5) \cdot C(5, 3) = 1 \cdot 10 = 10.$$

Thus there will be a total of 25 + 10 = 35 choices.

EXERCISE:

A computer programming team has 14 members.

(a) How many ways can a group of seven be chosen to work on a project?

(b) Suppose eight team members are women and six are men

(i) How many groups of seven can be chosen that contain four women and three men

(ii) How many groups of seven can be chosen that contain at least one man?

(iii)How many groups of seven can be chosen that contain at most three women?

(c) Suppose two team members refuse to work together on projects. How many groups of seven can be chosen to work on a project?

(d) Suppose two team members insist on either working together or not at all on projects. How many groups of seven can be chosen to work on a project?

(e) How many ways a group of 7 be chosen to work on a project?

SOLUTION:

(a) Number of committees of 7

$$C(14,7) = \frac{14!}{(14-7)! \cdot 7!}$$

$$= \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}$$

$$= 3432$$

(b) Suppose eight team members are women and six are men

(i) How many groups of seven can be chosen that contain four women and three men?

SOLUTION:

Number of groups of seven that contain four women and three men.

$$C(8,4)\cdot C(6,3)=\frac{8!}{(8-4)!\cdot 4!}\cdot\frac{6!}{(6-3)!\cdot 3!}$$
$$=\frac{8\cdot 7\cdot 6\cdot 5}{4!}\cdot\frac{6\cdot 5\cdot 4}{3!}$$
$$=\frac{8\cdot 7\cdot 6\cdot 5}{4\cdot 3\cdot 2}\cdot\frac{6\cdot 5\cdot 4}{3\cdot 2}$$
$$=70\cdot 20=1400$$

(b) Suppose eight team members are women and six are men

(ii) How many groups of seven can be chosen that contain at least one man?

SOLUTION:

Total number of groups of seven

$$C(14,7)=\frac{14!}{(14-7)!\cdot 7!}$$
$$=\frac{14\cdot 13\cdot 12\cdot 11\cdot 10\cdot 9\cdot 8}{7\cdot 6\cdot 5\cdot 4\cdot 3\cdot 2}$$
$$=3432$$

Number of groups of seven that contain no men

$$C(8,7)=\frac{8!}{(8-7)!\cdot 7!}$$
$$=8$$

Hence, the number of groups of seven that contain at least one man

$C(14,7) - C(8, 7) = 3432 - 8 = 3424$

(b)Suppose eight team members are women and six are men

(iii) How many groups of seven can be chosen that contain atmost three women?

SOLUTION:

Number of groups of seven that contain no women $= 0$

Number of groups of seven that contain one woman $= C(8,1) \cdot C(6,6)$

$= 8 \cdot 1 = 8$

Number of groups of seven that contain two women $= C(8,2) \cdot C(6,5)$

$= 28 \cdot 6 = 168$

Number of groups of seven that contain three women $= C(8,3) \cdot C(6,4)$

$= 56 \cdot 15 = 840$

Hence the number of groups of seven that contain at most three women

$= 0 + 8 + 168 + 840 = 1016$

(c) Suppose two team members refuse to work together on projects. How many groups of seven can be chosen to work on a project?

SOLUTION:

Call the members who refuse to work together *A* and *B*.

Number of groups of seven that contain neither *A* nor *B*

$$C(12,7) = \frac{12!}{(12-7)!\cdot 7!}$$
$$= 792$$

Number of groups of seven that contain *A* but not *B*

C(12, 6) = 924

Number of groups of seven that contain *B* but not *A*

C(12,6) = 924

Hence the required number of groups are

C(12,7) + C(12,6) + C(12, 6)
= 792 + 924 + 924
= 2640

(d) Suppose two team members insist on either working together or not at all on projects. How many groups of seven can be chosen to work on a project?

SOLUTION:

Call the members who insist on working together *C* and *D*.

Number of groups of seven containing neither *C* nor *D*

C(12, 7) = 792

Number of groups of seven that contain both *C* and *D*

C(12, 5) = 792

Hence the required number

= C(12, 7) + C(12, 5)
= 792 + 792 = 1584

EXERCISE:

(a) How many 16-bit strings contain exactly 9 1's?
(b)How many 16-bit strings contain at least one 1?

SOLUTION:

(a) 16-bit strings that contain exactly 9 1's= $C(16,9) = \frac{16!}{(16-9)!9!} = 11440$

(b) Total no. of 16-bit strings = 2^{16}

Hence number of 16-bit strings that contain at least one 1

$2^{16} - 1 = 65536 - 1$
$= 65535$

***K*-SELECTIONS:**

k-selections are similar to *k*-combinations in that the order in which the elements are selected does not matter, but in this case repetitions can occur.

DEFINITION:

A *k*-selection of a set of *n* elements is a choice of *k* elements taken from a set of *n* elements such that the order of elements does not matter and elements can be repeated.

REMARK:

1. *k*-selections are also called *k*-combinations with repetition allowed or multisets of size *k*.

2. With k-selections of a set of n elements repetition of elements is allowed. Therefore k need not to be less than or equal to n.

<u>THEOREM:</u>
The number of k-selections that can be selected from a set of n elements is

$$C(k+n-1, k) \text{ or } {}^{k+n-1}C_k$$

<u>EXERCISE:</u>
A camera shop stocks ten different types of batteries.

(a) How many ways can a total inventory of 30 batteries be distributed among the ten different types?

(b) Assuming that one of the types of batteries is A76, how many ways can a total inventory of 30 batteries be distributed among the 10 different types if the inventory must include at least four A76 batteries?

<u>SOLUTION:</u>
(a) $k = 30$
$n = 10$

The required number is

$$C(30 + 10 - 1, 30) = C(39, 30) = \frac{39!}{(39-30)!30!} = 211915132$$

(b) $k = 26$
$n = 10$

The required number is

$$C(26 + 10 - 1, 26) = C(35, 26) = \frac{35!}{(35-26)!26!} = 70607460$$

WHICH FORMULA TO USE?

	ORDER MATTERS	**ORDER DOES NOT MATTER**
REPETITION ALLOWED	k-sample n^k	k-selection C(n+k-1, k)
REPETITION NOT ALLOWED	k-permutation P(n, k)	k-combination C(n, k)

Lecture No.32 K-Combinations

ORDERED AND UNORDERED PARTITIONS
PERMUTATIONS WITH REPETITIONS

K-SELECTIONS:
k-selections are similar to *k*-combinations in that the order in which the elements are selected does not matter, but in this case repetitions can occur.

DEFINITION:
A *k*-selection of a set of *n* elements is a choice of *k* elements taken from a set of *n* elements such that the order of elements does not matter and elements can be repeated.

REMARK:

1. *k*-selections are also called *k*-combinations with repetition allowed or multisets of size *k*.
2. With *k*-selections of a set of *n* elements repetition of elements is allowed. Therefore *k* need not to be less than or equal to *n*.

THEOREM:
The number of *k*-selections that can be selected from a set of *n* elements is

$$C(k+n-1, k) \text{ or } {}^{k+n-1}c_k$$

EXERCISE:
A camera shop stocks ten different types of batteries.
(a) How many ways can a total inventory of 30 batteries be distributed among the ten different types?
(b) Assuming that one of the types of batteries is A76, how many ways can a total inventory of 30 batteries be distributed among the 10 different types if the inventory must include at least four A76 batteries?

SOLUTION:
(a) $k = 30$
$n = 10$

The required number is

$$C(30 + 10 - 1, 30) = C(39, 30)$$

$$= \frac{39!}{(39-30)!30!}$$

$$= 211915132$$

(b) $k = 26$
$n = 10$
The required number is

$$C(26+10-1, 26) = C(35, 26)$$
$$= \frac{35!}{(35-26)!26!}$$
$$= 70607460$$

WHICH FORMULA TO USE?

	ORDER MATTERS	**ORDER DOES NOT MATTER**
REPETITION ALLOWED	k-sample n^k	k-selection C(n+k-1, k)
REPETITION NOT ALLOWED	k-permutation P(n, k)	k-combination C(n, k)

ORDERED AND UNORDERED PARTITIONS:

An unordered partition of a finite set S is a collection $[A_1, A_2, \ldots, A_k]$ of disjoint (nonempty) subsets of S (called cells) whose union is S.
The partition is ordered if the order of the cells in the list counts.

EXAMPLE:

Let S = {1, 2, 3, …, 7}
The collections
$P_1 = [\{1,2\}, \{3,4,5\}, \{6,7\}]$
And $P_2 = [\{6,7\}, \{3,4,5\}, \{1,2\}]$
determine the same partition of S but are distinct ordered partitions.

EXAMPLE:

Suppose a box **B** contains seven marbles numbered 1 through 7. Find the number m of ways of drawing from **B** firstly two marbles, then three marbles and lastly the remaining two marbles.

SOLUTION:

The number of ways of drawing 2 marbles from 7 = C(7, 2)
Following this, there are five marbles left in B.
The number of ways of drawing 3 marbles from 5 = C(5, 3)
Finally, there are two marbles left in B.
The number of way of drawing 2 marbles from 2 = C(2, 2)

Thus
$$m = \binom{7}{2}\binom{5}{3}\binom{2}{2}$$
$$= \frac{7!}{2!5!} \cdot \frac{5!}{2!3!} \cdot \frac{2!}{2!0!}$$
$$= \frac{7!}{2!3!2!} = 210$$

THEOREM:
Let S contain n elements and let $n_1, n_2, \ldots, n_k$ be positive integers with

$$n_1+n_2+\ldots+n_k = n.$$

Then there exist $\dfrac{n!}{n_1!n_2!n_3!\cdots n_k!}$

different ordered partitions of S of the form $[A_1, A_2, \ldots, A_k]$, where

A_1 contains n_1 elements

A_2 contains n_2 elements

A_3 contains n_3 elements

………………………..

A_k contains n_k elements

REMARK:
To find the number of unordered partitions, we have to count the ordered partitions and then divide it by suitable number to erase the order in partitions.

EXERCISE:
Find the number m of ways that nine toys can be divided among four children if the youngest child is to receive three toys and each of the others two toys.

SOLUTION:
We find the number m of ordered partitions of the nine toys into four cells containing 3, 2, 2 and 2 toys respectively.
Hence

$$\begin{aligned} m &= \frac{9!}{3!2!2!2!} \\ &= 2520 \end{aligned}$$

EXERCISE:
How many ways can 12 students be divided into 3 groups with 4 students in each group so that

(i) one group studies English, one History and one Mathematics.

(ii) all the groups study Mathematics.

SOLUTION:
(i) Since each group studies a different subject, so we seek the number of ordered partitions of the 12 students into cells containing 4 students each. Hence there are

$\dfrac{12!}{4!4!4!} = 34,650$ such partitions

(ii) When all the groups study the same subject, then order doesn't matter.
Now each partition $\{G_1, G_2, G_3\}$ of the students can be arranged in 3! ways as an ordered partition, hence there are

$$\frac{12!}{4!4!4!} \times \frac{1}{3!}$$

unordered partitions.

EXERCISE:

How many ways can 8 students be divided into two teams containing
(i) five and three students respectively.
(ii) four students each.

SOLUTION:

(i) The two teams (cells) contain different number of students; so the number of unordered partitions equals the number of ordered partitions, which is

$$\frac{8!}{5!3!} = 56$$

(ii) Since the teams are not labeled, so we have to find the number of unordered partitions of 8 students in groups of 4.
Firstly, note, there are $\frac{8!}{4!4!} = 70$ ordered partitions into two cells containing four students each.
Since each unordered partition determine 2! = 2 ordered partitions, there are

$$\frac{70}{2} = 35$$

unordered partitions

EXERCISE:

Find the number m of ways that a class X with ten students can be partitions into four teams A_1, A_2, B_1 and B_2 where A_1 and A_2 contain two students each and B_1 and B_2 contain three students each.

SOLUTION:

There are $\frac{10!}{2!2!3!3!} = 25,200$ ordered partitions of X into four cells containing 2, 2, 3 and 3 students respectively.
However, each unordered partition $[A_1, A_2, B_1, B_2]$ of X determines
$2!\cdot 2! = 4$ ordered partitions of X.
Thus,

$$m = \frac{25,200}{4} = 6300$$

EXERCISE:

Suppose 20 people are divided in 6 (numbered) committees so that 3 people each serve on committee C_1 and C_2, 4 people each on committees C_3 and C_4, 2 people on committee C_5 and 4 people on committee C_6. How many possible arrangements are there?

SOLUTION:

We are asked to count labeled group - the committee numbers labeled the group.So this is a problem of ordered partition. Now, the number of ordered partitions of 20 people into the specified committees is

$$\frac{20!}{3!3!4!4!2!4!} = 2444321880000$$

EXERCISE:

If 20 people are divided into teams of size 3, 3, 4, 4, 2, 4, find the number of possible arrangements.

SOLUTION:

Here, we are asked to count unlabeled groups. Accordingly, this is the case of ordered partitions.

Now number of ordered partitions $= \dfrac{20!}{3!3!4!4!2!4!} \times \dfrac{1}{3!2!}$

$= 203693490000$

GENERALIZED PERMUTATION or PERMUTATIONS WITH REPETITIONS:

The number of permutations of n elements of which n_1 are alike, n_2 are alike, …, n_k are alike is $\dfrac{n!}{n_1!n_2!\cdots n_k!}$

REMARK:

The number $\dfrac{n!}{n_1!n_2!\cdots n_k!}$ is often called a multinomial coefficient, and is denoted by the symbol.

$$\binom{n}{n_1, n_2, \cdots, n_k}$$

EXERCISE:

Find the number of distinct permutations that can be formed using the letters of the word "BENZENE".

SOLUTION:

The word "BENZENE" contains seven letters of which three are alike (the 3 E's) and two are alike (the 2 N's)

Hence, the number of distinct permutations are: $\dfrac{7!}{3!2!} = 420$

EXERCISE:

How many different signals each consisting of six flags hung in a vertical line, can be formed from four identical red flags and two identical blue flags?

SOLUTION:

We seek the number of permutations of 6 elements of which 4 are alike and 2 are alike.

There are $\dfrac{6!}{4!2!} = 15$ different signals.

EXERCISE:

(i) Find the number of "***words***" that can be formed of the letters of the word **ELEVEN.**

(ii) Find, if the *words* are to begin with L.

(iii) Find, if the *words* are to begin and end in E.

(iv) Find, if the *words* are to begin with E and end in N.

SOLUTION:

(i) There are six letters of which three are E; hence required number of "*words*" are $\dfrac{6!}{3!} = 120$

(ii) If the first letter is L, then there are five positions left to fill where three are E; hence required number of *words* are $\frac{5!}{3!}=20$

(iii) If the *words* are to begin and end in E, then there are only four positions to fill with four distinct letters.

Hence required number of words = 4! = 24

(iv) If the *words* are to begin with E and end in N, then there are four positions left to fill where two are E.

Hence required number of words = $\frac{4!}{2!}=12$

EXERCISE:

(i)Find the number of permutations that can be formed from all the letters of the word **BASEBALL**

(ii)Find, if the two B's are to be next to each other.

(iii)Find, if the words are to begin and end in a vowel.

SOLUTION:

(i) There are eight letter of which two are B, two are A, and two are L. Thus,

$$\text{Number of permutations} = \frac{8!}{2!2!2!}$$
$$= 5040$$

(ii) Consider the two B's as one letter. Then there are seven letters of which two are A and two are L. Hence,

$$\text{Number of permutations} = \frac{7!}{2!2!}$$
$$= 1260$$

(iii) There are three possibilities, the words begin and end in A, the words begin in A and end in E, or the words begin in E and end in A.
In each case there are six positions left to fill where two are B and two are L. Hence,

$$\text{Number of permutations} = 3\frac{6!}{2!2!} = 540$$

Lecture No.33 Tree Diagram

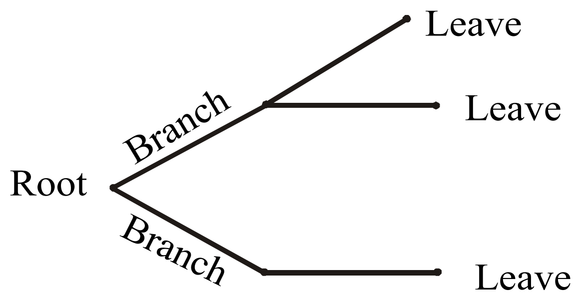
TREE DIAGRAM
INCLUSION - EXCLUSION PRINCIPLE

TREE DIAGRAM:

A tree diagram is a useful tool to list all the logical possibilities of a sequence of events where each event can occur in a finite number of ways.

A tree consists of a root, a number of branches leaving the root, and possible additional branches leaving the end points of other branches. To use trees in counting problems, we use a branch to represent each possible choice. The possible outcomes are represented by the leaves (end points of branches).

A tree is normally constructed from left to right.



A TREE STRUCTURE

EXAMPLE:

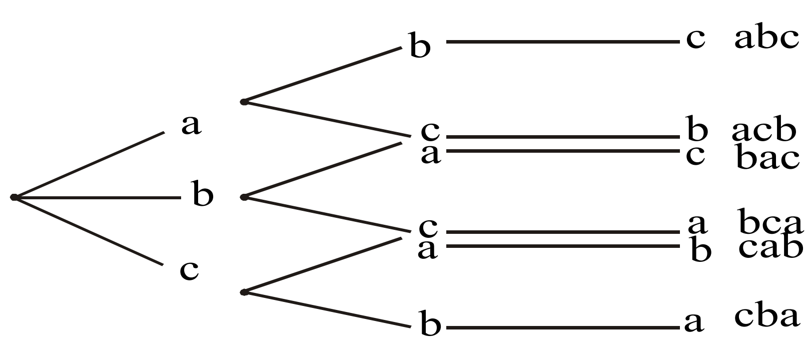
Find the permutations of {a, b, c}

SOLUTION:

The number of permutations of 3 elements is

$$P(3,3) = \frac{3!}{(3-3)!} = 3! = 6$$

We find the six permutations by constructing the appropriate tree diagram. The six permutations are listed on the right of the diagram.



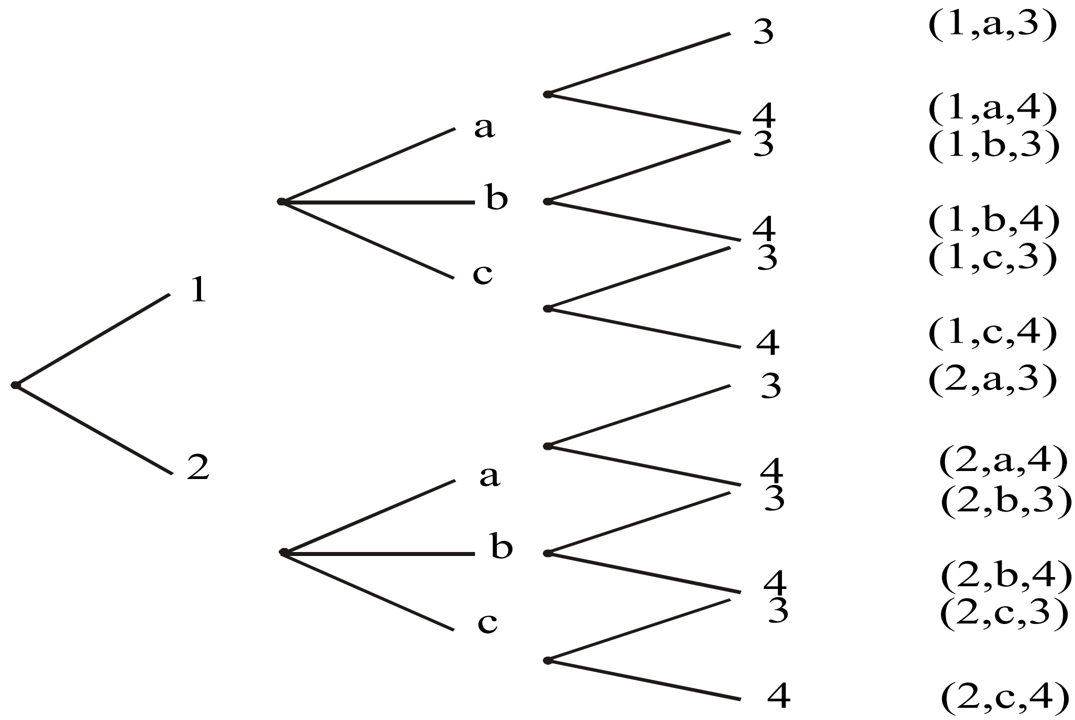
EXERCISE:

Find the product set A × B × C, where

A = {1,2}, B = {a,b,c}, and C = {3,4} by constructing the appropriate tree diagram.

SOLUTION:

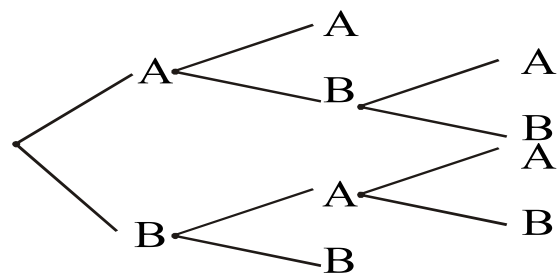
The required diagram is shown next. Each path from the beginning of the tree to the end point designates an element of A × B × C which is listed to the right of the tree.



EXERCISE:

Teams A and B play in a tournament. The team that first wins two games wins the tournament. Find the number of possible ways in which the tournament can occur.

SOLUTION: We construct the appropriate tree diagram.



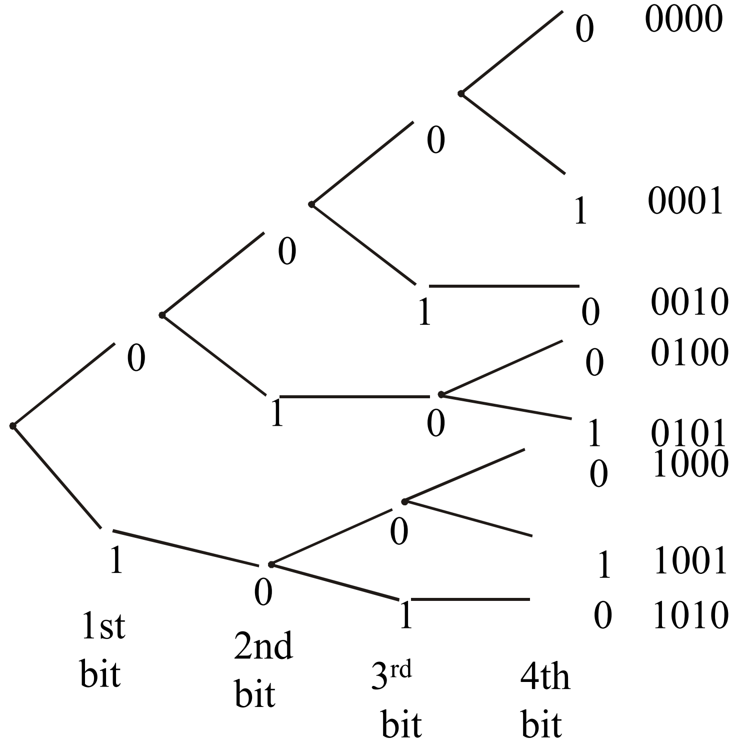
The tournament can occur in 6 ways: AA, ABA, ABB, BAA, BAB, BB

EXERCISE:

How many bit strings of length four do not have two consecutive 1's?

SOLUTION:

The following tree diagrams displays all bit strings of length four without two consecutive 1's. Clearly, there are 8 bit strings.



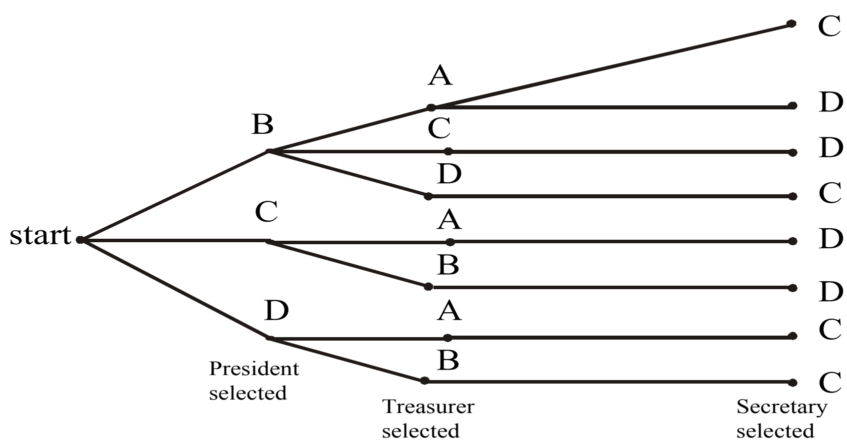
EXERCISE:

Three officers, a president, a treasurer, and a secretary are to be chosen from among four possible: A, B, C and D. Suppose that A cannot be president and either C or D must be secretary.
How many ways can the officers be chosen?

SOLUTION:

We construct the possibility tree to see all the possible choices.

From the tree given below, see that there are only eight ways possible to choose the offices under given conditions.

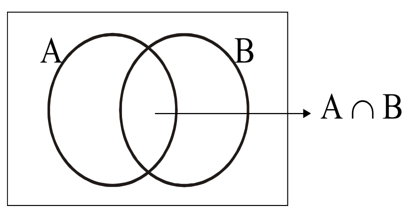


THE INCLUSION-EXCLUSION PRINCIPLE:

1. If A and B are disjoint finite sets, then
   $n(A\cup B) = n(A) + n(B)$
2. If A and B are finite sets, then
   $n(A\cup B) = n(A) + n(B) - n(A\cap B)$

REMARK:

Statement 1 follows from the sum rule
Statement 2 follows from the diagram



In counting the elements of $A \cup B$, we count the elements in A and count the elements in B. There are n(A) in A and n(B) in B. However, the elements in $A \cap B$ were counted twice. Thus we subtract $n(A \cap B)$ from $n(A) + n(B)$ to get $n(A \cup B)$.
Hence,

$$n(A\cup B) = n(A) + n(B) - n(A\cap B)$$

EXAMPLE:

There are 15 girls students and 25 boys students in a class. How many students are there in total?

SOLUTION:

Let G be the set of girl students and B be the set of boy students.

Then $n(G) = 15;\quad n(B) = 25$
and $n(G \cup B) = ?$

Since, the sets of boy and girl students are disjoint; here total number of students are

$$\begin{aligned} n(G \cup B) &= n(G) + n(B) \\ &= 15 + 25 \end{aligned}$$

$= 40$

EXERCISE:

Among 200 people, 150 either swim or jog or both. If 85 swim and 60 swim and jog, how many jog?

SOLUTION:

Let U be the set of people considered. Suppose S be the set of people who swim and J be the set of people who jog. Then given $n(U) = 200$; $n(S \cup J) = 150$

$n(S) = 85$; $n(S \cap J) = 60$ and $n(J) = ?$

By inclusion - exclusion principle,

$$n(S \cup J) = n(S) + n(J) - n(S \cap J)$$
$$150 = 85 + n(J) - 60$$
$$\Rightarrow \quad n(J) = 150 - 85 + 60$$
$$= 125$$

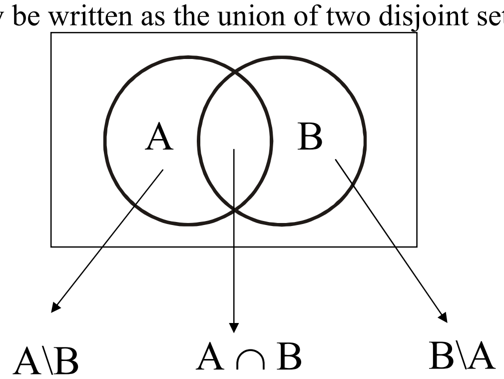
Hence 125 people jog.

EXERCISE:

Suppose A and B are finite sets. Show that

$n(A \backslash B) = n(A) - n(A \cap B)$**SOLUTION:**

Set A may be written as the union of two disjoint sets $A\backslash B$ and $A \cap B$.



i.e., $A = (A\backslash B) \cup (A \cap B)$

Hence, by inclusion exclusion principle (for disjoint sets)

$$n(A) = n(A\backslash B) + n(A \cap B)$$
$$\Rightarrow \quad n(A\backslash B) = n(A) - n(A \cap B)$$

REMARK:

$n(A') = n(U\backslash A)$ where U is the universal set

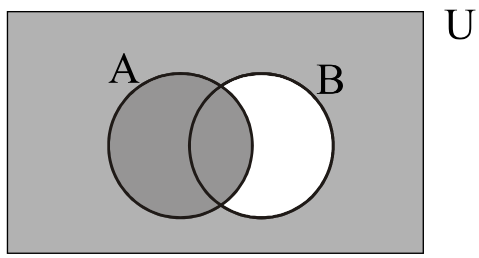
$= n(U) - n(U \cap A)$

$= n(U) - n(A)$

EXERCISE:

Let A and B be subsets of U with $n(A) = 10$, $n(B) = 15$, $n(A')=12$, and $n(A \cap B) = 8$. Find $n(A \cup B')$.

SOLUTION:



From the diagram $A \cup B' = U \setminus (B \setminus A)$
Hence

$$n(A \cup B') = n(U \setminus (B \setminus A))$$
$$= n(U) - n(B \setminus A) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Now $U = A \cup A'$ where A & A' are disjoint sets

$\Rightarrow$ $n(U) = n(A) + n(A')$
$= 10 + 12$
$= 22$

Also

$n(B \setminus A) = n(B) - n(A \cap B)$
$= 15 - 8$
$= 7$

Substituting values in (1) we get

$n(A \cup B') = n(U) - n(B \setminus A)$
$= 22 - 7$
$= 15$ Ans.

EXERCISE:

Let A and B are subset of U with $n(U) = 100$, $n(A) = 50$, $n(B)= 60$, and $n((A\cup B)') = 20$. Find $n(A\cap B)$

SOLUTION:

Since $(A \cup B)' = U \setminus (A \cup B)$

$\Rightarrow$ $n((A \cup B)') = n(U) - n(A \cup B)$
$\Rightarrow$ $20 = 100 - n(A \cup B)$
$\Rightarrow$ $n(A \cup B) = 100 - 20 = 80$

Now, by inclusion - exclusion principle

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$\Rightarrow$ $80 = 50 + 60 - n(A \cap B)$
$\Rightarrow$ $n(A \cap B) = 50 + 60 - 80 = 30$

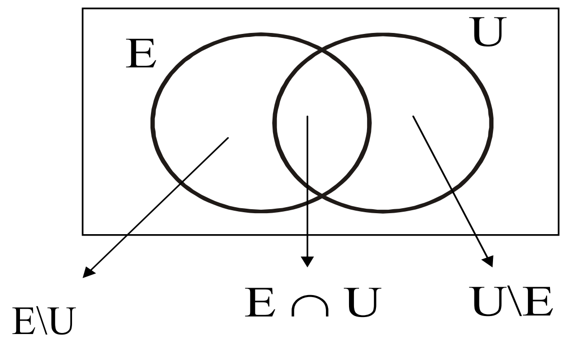
EXERCISE:

Suppose 18 people read English newspaper (E) or Urdu newspaper (U) or both. Given 5 people read only English newspaper and 7 read both, find the number "r" of people who read only Urdu newspaper.

SOLUTION:

Given $n(E \cup U) = 18$ $n(E\setminus U) = 5$, $n(E \cap U) = 7$
$r = n(U \setminus E) = ?$

From the diagram



$E \cup U = (E\backslash U) \cup (E \cap U) \cup (U\backslash E)$
and the union is disjoint. Therefore,
$n(E \cup U) = n(E\backslash U) + n(E \cap U) + n(U\backslash E)$

$\Rightarrow \quad 18 = 5 + 7 + r$

$\Rightarrow \quad r = 18 - 5 - 7$

$\Rightarrow \quad r = 6$ Ans.

EXERCISE:

Fifty people are interviewed about their food preferences. 20 of them like Chinese food, 32 like fast food, and 12 like neither Chinese nor fast food. How many like Chinese but not fast food?

SOLUTION:

Let U denote the set of people interviewed and C and F denotes the sets of people who like Chinese food and fast food respectively.
Now given

$n(U) = 50, \qquad n(C) = 20$

$n(F) = 32, \qquad n((C \cup F')) = 12$

To find $n(C \cap F') = n(C\backslash F)$

Since $n((C \cup F)') = n(U) - n(C \cup F)$

$\Rightarrow \quad 12 = 50 - n(C \cup F)$

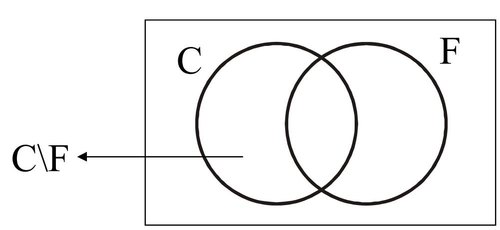
$\Rightarrow \quad n(C \cup F) = 50 - 12 = 38$

Next

$\Rightarrow \quad n(C \cup F) = n(C\backslash F) + n(F)$

$\Rightarrow \quad 38 = n(C\backslash F) + 32$

$\Rightarrow \quad n(C\backslash F) = 38 - 32 = 6$



Lecture No.34 Inclusion-Exclusion Principle

INCLUSION-EXCLUSION PRINCIPLE
PIGEONHOLE PRINCIPLE

EXERCISE:

(a) How many integers from 1 through 1000 are multiples of 3 or multiples of 5?
(b) How many integers from 1 through 1000 are neither multiples of 3 nor multiples of 5?

SOLUTION:

(a) Let A and B denotes the set of integers from 1 through 1000 that are multiples of 3 and 5 respectively.
Then $A \cap B$ contains integers that are multiples of 3 and 5 both i.e., multiples of 15.
Now

$$n(A) = \left\lfloor \frac{1000}{3} \right\rfloor = 333 \quad \text{and} \quad n(B) = \left\lfloor \frac{1000}{5} \right\rfloor = 200$$

and

$$n(A \cap B) = \left\lfloor \frac{1000}{15} \right\rfloor = 66$$

Hence by the inclusion - exclusion principle

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 333 + 200 - 66 \\ &= 467 \end{aligned}$$

(b) The set $(A \cup B)$ contains those integers that are either multiples of 3 or multiples of 5. Now

$$\begin{aligned} n((A \cup B)') &= n(U) - n(A \cup B) \\ &= 1000 - 467 \\ &= 533 \end{aligned}$$

where the universal set U contain integers 1 through 1000.

INCLUSION-EXCLUSION PRINCIPLE FOR 3 AND 4 SETS:

If A, B, C and D are finite sets, then

1. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

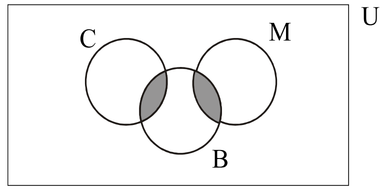
2. $$\begin{aligned} n(A \cup B \cup C \cup D) &= n(A) + n(B) + n(C) + n(D) - n(A \cap B) - n(A \cap C) - n(A \cap D) \\ &\quad - n(B \cap C) - n(B \cap D) - n(C \cap D) + n(A \cap B \cap C) + n(A \cap B \cap D) \\ &\quad + n(A \cap C \cap D) + n(B \cap C \cap D) - n(A \cap B \cap C \cap D) \end{aligned}$$

EXERCISE:

A survey of 100 college students gave the following data:
8 owned a car (C)
20 owned a motorcycle (M)
48 owned a bicycle (B)
38 owned neither a car nor a motorcycle nor a bicycle
No student who owned a car, owned a motorcycle

How many students owned a bicycle and either a car or a motorcycle?

SOLUTION:



No. of elements in the shaded region to be determined
Let U represents the universal set of 100 college students. Now given that

$n(U) = 100;$ $\quad n(C) = 8$
$n(M) = 20;$ $\quad n(B) = 48$
$n((C\cup M \cup B)') = 38;$ $\quad n(C \cap M) = 0$
and $\quad n(B \cap C) + n(B \cap M) = ?$

Firstly note $n((C \cup M \cup B)') = n(U) - n(C \cup M \cup B)$

$\Rightarrow \quad 38 = 100 - n(C \cup M \cup B)$

$\Rightarrow \quad n(C \cup M \cup B) = 100 - 38 = 62$

Now by inclusion - exclusion principle

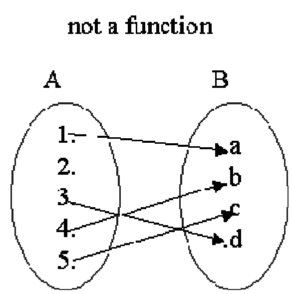
$$n(C \cup M \cup B) = n(C) + n(M) + n(B) - n(C\cap M) - n(C \cap B) - n(M \cap B) + n(C \cap M \cap B)$$

$$\Rightarrow \quad 62 = 8 + 20 + 48 - 0 - n(C \cap B) - n(M \cap B) - 0$$
$$(\therefore n(C \cap B) = 0)$$

$$\Rightarrow n(C \cap B) + n(M \cap B) = 8 + 20 + 48 - 62$$
$$= 76 - 62$$
$$= 14$$

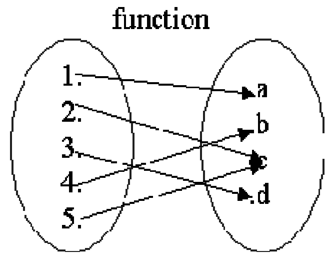
Hence, there are 14 students, who owned a bicycle and either a car or a motorcycle.

REVISION OF FUNCTIONS:

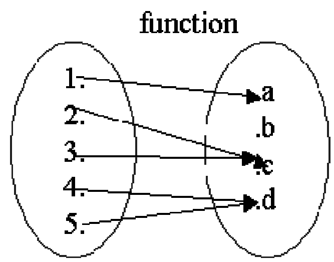


Clearly the above relation is not a function because 2 does not have any image under this relation. Note that if want to made it relation we have to map the 2 into some element of B which is also the image of some element of A.

Now,



The above relation is a function because it satisfy the conditions of functions(as each element of 1st set have the images in 2nd set). The following is a function.



The above relation is a function because it satisfy the conditions of functions(as each element of 1st set have the images in 2nd set). Therefore the above is also a function.

PIGEONHOLE PRINCIPLE

A function from a set of $k + 1$ or more elements to a set of k elements must have at least two elements in the domain that have the same image in the co-domain.

If $k + 1$ or more pigeons fly into k pigeonholes then at least one pigeonhole must contain two or more pigeons.

EXAMPLES:

1. Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

2. In any set of 27 English words, there must be at least two that begin with the same letter, since there are 26 letters in the English alphabet.

EXERCISE:

What is the minimum number of students in a class to be sure that two of them are born in the same month?

SOLUTION:

There are 12 (= n) months in a year. The pigeonhole principle shows that among any 13 (= $n + 1$) or more students there must be at least two students who are born in the same month.

EXERCISE:

Given any set of seven integers, must there be two that have the same remainder when divided by 6?

SOLUTION:

The set of possible remainders that can be obtained when an integer is divided by six is {0, 1, 2, 3, 4, 5}. This set has 6 elements. Thus by the pigeonhole principle if 7 = 6 + 1 integers are each divided by six, then at least two of them must have the same remainder.

EXERCISE:

How many integers from 1 through 100 must you pick in order to be sure of getting one that is divisible by 5?

SOLUTION:

There are 20 integers from 1 through 100 that are divisible by 5. Hence there are eighty integers from 1 through 100 that are not divisible by 5. Thus by the pigeonhole principle 81 = 80 + 1 integers from 1 though 100 must be picked in order to be sure of getting one that is divisible by 5.

EXERCISE:

Let A = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Suppose six integers are chosen from A. Must there be two integers whose sum is 11.

SOLUTION:

The set A can be partitioned into five subsets:
{1, 10}, {2, 9}, {3, 8}, {4, 7}, and {5, 6}
each consisting of two integers whose sum is 11.
These 5 subsets can be considered as 5 pigeonholes.
If 6 = (5 + 1) integers are selected from A, then by the pigeonhole principle at least two must be from one of the five subsets. But then the sum of these two integers is 11.

GENERALIZED PIGEONHOLE PRINCIPLE:

A function from a set of $n \cdot k + 1$ or more elements to a set of n elements must have at least $k + 1$ elements in the domain that have the same image in the co-domain.
If $n \cdot k + 1$ or more pigeons fly into n pigeonholes then at least one pigeonhole must contain $k + 1$ or more pigeons.

EXERCISE:

Suppose a laundry bag contains many red, white, and blue socks. Find the minimum number of socks that one needs to choose in order to get two pairs (four socks) of the same colour.

SOLUTION:

Here there are n = 3 colours (pigeonholes) and k + 1 = 4 or k = 3. Thus among any $n \cdot k + 1 = 3 \cdot 3 + 1 = 10$ socks (pigeons), at least four have the same colour.

DEFINITION:

1. Given any real number x, **the floor of *x***, denoted $\lfloor x \rfloor$, is the largest integer smaller than or equal to x.

2. Given any real number x, **the ceiling of *x***, denoted $\lceil x \rceil$, is the smallest integer greater than or equal to x.

EXAMPLE:

Compute $\lfloor x \rfloor$ and $\lceil x \rceil$ for each of the following values of x.

a. 25/4 **b.** 0.999 **c.** –2.01

SOLUTION:

a. $\lfloor 25/4 \rfloor = \lfloor 6 + 1/4 \rfloor = 6$

$\lceil 25/4 \rceil = \lceil 6 + 1/4 \rceil = 6 + 1 = 7$

b. $\lfloor 0.999 \rfloor = \lfloor 0 + 0.999 \rfloor = 0$

$\lceil 0.999 \rceil = \lceil 0 + 0.999 \rceil = 0 + 1 = 1$

c. $\lfloor -2.01 \rfloor = \lfloor -3 + 0.99 \rfloor = -3$

$\lceil -2.01 \rceil = \lceil -3 + 0.999 \rceil = -3 + 1 = -2$

EXERCISE:

What is the smallest integer N such that

a. $\lceil N/7 \rceil = 5$ b. $\lceil N/9 \rceil = 6$

SOLUTION:

a. $N = 7 \cdot (5 - 1) + 1 = 7 \cdot 4 + 1 = 29$

b. $N = 9 \cdot (6 - 1) + 1 = 9 \cdot 5 + 1 = 46$

PIGEONHOLE PRINCIPLE:

If N pigeons fly into k pigeonholes then at least one pigeonhole must contain $\lceil N/k \rceil$ or more pigeons.

EXAMPLE:

Among 100 people there are at least $\lceil 100/12 \rceil = \lceil 8 + 1/3 \rceil = 9$ who were born in the same month.

EXERCISE:

What is the minimum number of students required in a Discrete Mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F.

SOLUTION:

The minimum number of students needed to guarantee that at least six students receive the same grade is the smallest integer N such that $\lceil N/5 \rceil = 6$.

The smallest such integer is $N = 5(6-1)+1=5 \cdot 5 + 1 = 26$.

Thus 26 is the minimum number of students needed to be sure that at least 6 students will receive the same grades.

Lecture No.35 Probability

INTRODUCTION TO PROBABILITY

INTRODUCTION:

Combinatorics and probability theory share common origins. The theory of probability was first developed in the seventeenth century when certain gambling games were analyzed by the French mathematician Blaise Pascal. It was in these studies that Pascal discovered various properties of the binomial coefficients. In the eighteenth century, the French mathematician Laplace, who also studied gambling, gave definition of the probability as the number of successful outcomes divided by the number of total outcomes.

DEFINITIONS:

An **experiment** is a procedure that yields a given set of possible outcomes.

The **sample space** of the experiment is the set of possible outcomes.

An **event** is a subset of the sample space.

EXAMPLE:

When a die is tossed the sample space S of the experiment have the following six outcomes. $S = \{1, 2, 3, 4, 5, 6\}$

Let E_1 be the event that a 6 occurs,
E_2 be the event that an even number occurs,
E_3 be the event that an odd number occurs,
E_4 be the event that a prime number occurs,
E_5 be the event that a number less than 5 occurs, and
E_6 be the event that a number greater than 6 occurs.

Then

$E_1 = \{6\}$ $\qquad$ $E_2 = \{2, 4, 6\}$
$E_3 = \{1, 3, 5\}$ $\qquad$ $E_4 = \{2, 3, 5\}$
$E_5 = \{1, 2, 3, 4\}$ $\qquad$ $E_6 = \Phi$

EXAMPLE:

When a pair of dice is tossed, the sample space S of the experiment has the following thirty-six outcomes

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)$$
$$(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)$$
$$(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)$$
$$(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)$$
$$(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)$$
$$(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

or more compactly,

$$\{11, 12, 13, 14, 15, 16, 21, 22, 23, 24, 25, 26,$$
$$31, 32, 33, 34, 35, 36, 41, 42, 43, 44, 45, 46,$$
$$51, 52, 53, 54, 55, 56, 61, 62, 63, 64, 65, 66\}$$

Let E be the event in which the sum of the numbers is ten.
Then
$E = \{(4, 6), (5, 5), (6, 4)\}$

DEFINITION:

Let S be a finite sample space such that all the outcomes are equally likely to occur.
The **probability** of an event E, which is a subset of sample space S, is

$$P(E) = \frac{\text{the number of outcomes in } E}{\text{the numbr of total outcomes in } S} = \frac{n(E)}{n(S)}$$

REMARK:

Since $\varphi \subseteq E \subseteq S$ therefore, $0 \leq n(E) \leq n(S)$. It follows that the probability of an event is always between 0 and 1. (Since $n(\phi) = 0$, $n(S) = 1$)

EXAMPLE:

What is the probability of getting a number greater than 4 when a dice is tossed?

SOLUTION:

When a dice is rolled its sample space is $S=\{1,2,3,4,5,6\}$
Let E be the event that a number greater than 4 occurs. Then $E = \{5, 6\}$
Hence,

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

EXAMPLE:

What is the probability of getting a total of eight or nine when a pair of dice is tossed?

SOLUTION:

When a pair of dice is tossed,its sample space S has the 36 outcomes which are as fellows:

$$\begin{aligned} S = \{&(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)\\ &(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)\\ &(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\\ &(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\\ &(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\\ &(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\} \end{aligned}$$

Let E be the event that the sum of the numbers is eight or nine. Then
$E = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (3, 6), (4, 5), (5, 4), (6, 3)\}$
Hence,

$$P(E) = \frac{n(E)}{n(S)} = \frac{9}{36} = \frac{1}{4}$$

EXAMPLE:

An urn contains four red and five blue balls. What is the probability that a ball chosen from the urn is blue?

SOLUTION:

Since there are four red balls and five blue balls so if we take out one ball from the urn then there is possibility that it may be one of from four red and one of from five blue balls hence there are total of nine possibilities. Thus we have
The total number of possible outcomes = 4 + 5 = 9
Now our favourable event is that we get the blue ball when we choose a ball from the urn. So we have

The total number of favorable outcomes = 5

Now we have Favorable outcomes 5 and our sample space has total outcomes 9 .Thus we have

The probability that a ball chosen = 5/9

EXERCISE:

Two cards are drawn at random from an ordinary pack of 52 cards. Find the probability **p** that (i) both are spades, (ii) one is a spade and one is a heart.

SOLUTION:

There are $\binom{52}{2} = 1326$ ways to draw 2 cards from 52 card

(i) There are $\binom{13}{2} = 78$ ways to draw 2 spades from 13 spades (as spades are 13 in 52 cards); hence

$$p = \frac{\text{number of ways 2 spades can be drawn}}{\text{number of ways 2 cards can be drawn}} = \frac{78}{1326} = \frac{1}{17}$$

ii) Since there are 13 spades and 13 hearts, there are $\binom{13}{1}\binom{13}{1} = 13.13 = 169$ ways to draw a spade and a heart; hence $p = \frac{169}{1326} = \frac{13}{102}$

EXAMPLE:

In a lottery, players win the first prize when they pick three digits that match, in the correct order, three digits kept secret. A second prize is won if only two digits match. What is the probability of winning (a) the first prize, (b) the second prize?

SOLUTION:

Using the product rule, there are 10^3 = 1000 ways to choose three digits.

(a) There is only one way to choose all three digits correctly. Hence the probability that a player wins the first prize is 1/1000 = 0.001.

(b) There are three possible cases:

(i)The first digit is incorrect and the other two digits are correct
(ii)The second digit is incorrect and the other two digits are correct
(iii)The thirds digit is incorrect and the other digits are correct

To count the number of successes with the first digit incorrect, note that there are nine choices for the first digit to be incorrect, and one each for the other two digits to be correct. Hence, there are nine ways to choose three digits where the first digit is incorrect, but the other two are correct. Similarly, there are nine ways for the other two cases. Hence, there are 9 + 9 + 9 = 27 ways to choose three digits with two of the three digits correct.

It follows that the probability that a player wins the second prize is 27/1000 = 0.027.

EXAMPLE:
What is the probability that a hand of five cards contains four cards of one kind?

SOLUTION:

(i) For determining the favorable outcomes we note that
The number of ways to pick one kind = $C(13, 1)$
The number of ways to pick the four of this kind out of the four of this kind in the deck = $C(4, 4)$

The number of ways to pick the fifth card from the remaining 48 cards = $C(48, 1)$
Hence, using the product rule the number of hands of five cards with four cards of one kind = $C(13, 1) \times C(4, 4) \times C(48, 1)$

$$\frac{C(13,1)\cdot C(4,4)\cdot C(48,1)}{C(52,5)} = \frac{13\cdot 1\cdot 48}{2,598,960} \approx 0.0024$$

(ii) The total number of different hands of five cards = $C(52, 5)$.

From (i) and (ii) it follows that the probability that a hand of five cards contains four cards of one kind is

EXAMPLE:
Find the probability that a hand of five cards contains three cards of one kind and two of another kind.

SOLUTION:
(i) For determining the favorable outcomes we note that
The number of ways to pick two kinds = $C(13, 2)$
The number of ways to pick three out four of the first kind = $C(4, 3)$
The number of ways to pick two out four of the second kind = $C(4, 2)$
Hence, using the product rule the number of hands of five cards with three cards of one kind and two of another kind = $C(13, 2) \times C(4, 3) \times C(4, 2)$

(ii) The total number of different hands of five cards = $C(52, 5)$.

From (i) and (ii) it follows that the probability that a hand of five cards contains three cards of one kind and two of another kind is

$$\frac{C(13,2)\cdot C(4,3)\cdot C(4,2)}{C(52,5)} = \frac{3744}{2,598,960} \approx 0.0014$$

EXAMPLE:

What is the probability that a randomly chosen positive two-digit number is a multiple of 6?

SOLUTION:

1. There are $\left\lfloor \frac{99}{6} \right\rfloor = \left\lfloor 16 + \frac{1}{2} \right\rfloor = 16$ positive integers from 1 to 99 that are divisible by 6. Out of these 16 – 1 = 15 are two-digit numbers(as 6 is a multiple of 6 but not a two-digit number).

2. There are 99 – 9 = 90 positive two-digit numbers in all.

Hence, the probability that a randomly chosen positive two-digit number is a multiple of 6 = 15/90 = 1/6 ≈ 0.166667

DEFINITION:

Let E be an event in a sample space S, the complement of E is the event that occurs if E does not occur. It is denoted by E^c. Note that $E^c = S\backslash E$

EXAMPLE:

Let E be the event that an even number occurs when a die is tossed. Then E^c is the event that an odd number occurs.

THEOREM:

Let E be an event in a sample space S. The probability of the complementary event E^c of E is given by

$$P(E^c) = 1 - P(E).$$

EXAMPLE:

Let 2 items be chosen at random from a lot containing 12 items of which 4 are defective. What is the probability that (i) none of the items chosen are defective, (ii) at least one item is defective?

SOLUTION:

The number of ways 2 items can be chosen from 12 items = $C(12, 2) = 66$.

(i) Let A be the event that none of the items chosen are defective.
The number of favorable outcomes for A = The number of ways 2 items can be chosen from 8 non-defective items = $C(8, 2) = 28$.
Hence, P(A) = 28/66 = 14/33.

(ii) Let B be the event that at least one item chosen is defective.

Then clearly, $B = A^c$
It follows that
$P(B) = P(A^c)$
$=1 - P(A) = 1 - 14/33 = 19/33.$

EXERCISE:

Three light bulbs are chosen at random from 15 bulbs of which 5 are defective. Find the probability p that (i) none is defective, (ii) exactly one is defective, (iii) at least one is defective.

SOLUTION:

There are $\binom{15}{3} = 455$ ways to choose 3 bulbs from the 15 bulbs.

(i) Since there are 15 - 5 = 10 non-defective bulbs, there are $\binom{10}{3} = 120$ ways to choose 3 non-defective bulbs.

Thus

$$p = \frac{120}{455} = \frac{24}{91}$$

(ii) There are 5 defective bulbs and $\binom{10}{2} = 45$ different pairs of non-defective bulbs; hence there are $\binom{5}{1}\binom{10}{2} = 5.45 = 225$ ways to choose 3 bulbs of which one is defective.

Thus

$$P = \frac{225}{455} = \frac{45}{91}.$$

(iii) The event that at least one is defective is the complement of the event that none are defective which has by (i), probability $\frac{24}{91}$.

Hence $p(atleast\ one\ is\ defective) = 1 - p(none\ is\ defective) = 1 - \frac{24}{91} = \frac{67}{91}$

Lecture No.36 Laws of Probability

ADDITION LAW OF PROBABILITY

THEOREM:
If A and B are two disjoint (mutually exclusive) events of a sample space S, then

$$P(A \cup B) = P(A) + P(B)$$

In words, the probability of the happening of an event A or an event B or both is equal to the sum of the probabilities of event A and event B provided the events have nothing in common.

PROOF:
By inclusion - exclusion principle for mutually disjoint sets,

$$n(A \cup B) = n(A) + n(B)$$

Dividing both sides by n(S), we get

$$\frac{n(A \cup B)}{n(S)} = \frac{n(A) + n(B)}{n(S)}$$

$$= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)}$$

$$\Rightarrow \quad P(A \cup B) = p(A) \quad + P(B)$$

EXAMPLE:
Suppose a die is rolled. Let A be the event that 1 appears & B be the event that some even number appears on the die. Then

S = {1, 2, 3, 4, 5, 6}, A = {1} & B = {2, 4, 6}

Clearly A & B are disjoint events and

$$P(A) = \frac{1}{6}, \qquad P(B) = \frac{3}{6}$$

Hence the probability that a 1 appears or some even number appears is given by

$$P(A \cup B) = P(A) + P(B)$$

$$= \frac{1}{6} + \frac{3}{6}$$

$$= \frac{4}{6} = \frac{2}{3} \qquad Ans.$$

EXERCISE:
A bag contains 6 white, 5 black and 4 red balls. Find the probability of getting a white or a black ball in a single draw.

SOLUTION:
Let A be the event of getting a white ball and B be the event of getting a black ball.

Total number of balls = 6 + 5 + 4 = 15

Since the two events are disjoint (mutually exclusive),therefore

$$P(A) = \frac{6}{15}, \qquad P(B) = \frac{5}{15}$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= \frac{6}{15} + \frac{5}{15} \\ &= \frac{11}{15} \qquad \textit{Ans} \end{aligned}$$

EXERCISE:

A pair of dice is thrown. Find the probability of getting a total of 5 or 11.

SOLUTION:

When two dice are thrown, the sample space has 6 * 6 = 36 outcomes. Let A be the event that a total of 5 occurs and B be the event that a total of 11 occurs.
Then
A = {(1,4), (2,3), (3,2), (4,1)} and B = {(5,6), (6,5)}
Clearly, the events A and B are disjoint (mutually exclusive) with probabilities given by

Now by using the sum Rule for Mutually Exclusive events we get

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= \frac{4}{36} + \frac{2}{36} = \frac{6}{36} = \frac{1}{6} \qquad \textit{Ans} \end{aligned}$$

EXERCISE:

For any two event A and B of a sample space S. Prove that
$P(A\backslash B) = P(A\cap B') = P(A) - P(A \cap B)$

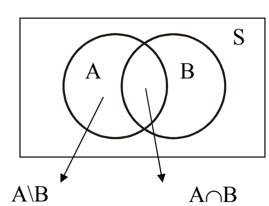
SOLUTION:

The event A can be written as the union of two disjoint events $A\backslash B$ and $A \cap B$. i.e. $A = (A\backslash B) \cup (A\cap B)$

Hence, by addition law of probability

$P(A) = P(A\backslash B) + P(A \cap B)$

$\Rightarrow P(A\backslash B) = P(A) - P(A \cap B)$



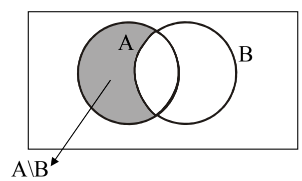
GENERAL ADDITION LAW OF PROBABILITY

THEOREM

If A and B are any two events of a sample space S, then
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

PROOF:

The event $A \cup B$ may be written as the union of two disjoint events A\B and B.
i.e., $A \cup B = (A \backslash B) \cup B$
Hence, by addition law of probability (for disjoint events)
$P(A \cup B) = P(A \backslash B) + P(B)$
$= [P(A) - P(A \cap B)] + P(B)$
$= P(A) + P(B) - P(A \cap B)$ (proved)



REMARK:

By inclusion - exclusion principle
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ (where A and B are finite)

Dividing both sides by n(S) and denoting the ratios as respective probabilities we get the

Generalized Addition Law of probability.
i.e $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

EXERCISE:

Let A and B be events in a sample space S, and let
$P(A) = 0.65$, $P(B) = 0.30$ and $P(A \cap B) = 0.15$
Determine the probability of the events

(a) $A \cap B'$ (b) $A \cup B$ (c) $A' \cap B'$

SOLUTION:

(a) As we know that
$P(A \cap B') = P(A) - P(A \cap B)$ (as $A-B=A \cap B'$)
$= 0.65 - 0.15$
$= 0.50$

(b) By addition Law of probability
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (as $A \cap B \neq \phi$)
$= 0.65 + 0.30 - 0.15$
$= 0.80$

(c) By DeMorgan's Law

$$A' \cap B' = (A \cup B)'$$

$$\therefore P(A' \cap B') = P(A \cup B)'$$
$$= 1 - P(A \cup B)$$
$$= 1 - 0.80$$
$$= 0.20 \quad \text{Ans.}$$

EXERCISE:

Let A, B, C and D be events which form a partition of a sample space S. If P(A) = P(B), P(C) = 2 P(A) and P(D) = 2 P(C). Determine each of the following probabilities.

(a) P(A) (b) $P(A \cup B)$ (c) $P(A \cup C \cup D)$

SOLUTION:

(a) Since A, B, C and D form a partition of S, therefore

$S = A \cup B \cup C \cup D$ and A, B, C, D are pair wise disjoint. Hence, by addition law of probability.

$$P(S) = P(A) + P(B) + P(C) + P(D)$$
$$\Rightarrow 1 = P(A) + P(A) + 2P(A) + 2P(C)$$
$$\Rightarrow 1 = 4P(A) + 2(2P(A))$$
$$\Rightarrow 1 = 8P(A)$$
$$\Rightarrow P(A) = \frac{1}{8}$$

(b)
$$P(A \cup B) = P(A) + P(B)$$
$$= P(A) + P(A) \quad [\therefore P(B) = P(A)]$$
$$= 2P(A)$$
$$= 2\left(\frac{1}{8}\right) = \frac{1}{4}$$

(c)
$$P(A \cup C \cup D) = P(A) + P(C) + P(D)$$
$$= P(A) + 2P(A) + 2(2P(A)) \quad [\therefore P(C) = 2P(A) \;\&\; P(D) = 2P(C)]$$
$$= 7P(A)$$
$$= 7\left(\frac{1}{8}\right) = \frac{7}{8} \quad \text{Ans.}$$

EXERCISE:

A card is drawn from a well-shuffled pack of playing card. What is the probability that it is either a spade or an ace?

SOLUTION:

Let A be the event of drawing a spade and B be the event of drawing an ace. Now A and B are not disjoint events. $A \cap B$ represents the event of drawing an ace of spades.

Now

$$P(A) = \frac{13}{52}; \qquad P(B) = \frac{4}{52}$$

$$P(A \cap B) = \frac{1}{52}$$

$$\therefore \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52}$$

$$= \frac{16}{52} = \frac{4}{13}$$

EXERCISE:

A class contains 10 boys and 20 girls of which half the boys and half the girls have brown eyes. Find the probability that a student chosen at random is a boy or has brown eyes.

SOLUTION:

Let A be the event that a boy is chosen and B be the event that a student with brown eyes is chosen.Then A and B are not disjoint events. Infact, A∩B represents the event that a boy with brown eyes is chosen.

$$P(A) = \frac{10}{10+20} = \frac{10}{30}$$

$$P(B) = \frac{5+10}{10+20} = \frac{15}{30}$$

and

$$P(A \cap B) = \frac{5}{10+20} = \frac{5}{30} \quad (as\, some\, boys\, also\, have\, brown\, eyes)$$

$$\therefore \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{10}{30} + \frac{15}{30} - \frac{5}{30}$$

$$= \frac{20}{30} = \frac{2}{3} \qquad Ans.$$

EXERCISE:

An integer is chosen at random from the first 100 positive integers. What is the probability that the integer chosen is divisible by 6 or by 8?

SOLUTION:

Let A be the event that the integer chosen is divisible by 6, and B be the event that the integer chosen is divisible by 8.

A∩B is the event that the integer is divisible by both 6 and 8 (i.e. as their L.C.M. is 24)

Now

$$n(A)=\left\lfloor\frac{100}{6}\right\rfloor=16;\qquad n(B)=\left\lfloor\frac{100}{8}\right\rfloor=12$$

and

$$n(A\cap B)=\left\lfloor\frac{100}{24}\right\rfloor=4$$

Hence $P(A\cup B)$ $=P(A)+P(B)-P(A\cap B)$

$$=\frac{16}{100}+\frac{12}{100}-\frac{4}{100}$$

$$=\frac{24}{100}=\frac{6}{25}\qquad Ans$$

OR

Let A denote the event that the integer chosen is divisible by 6, and B denote the event that the integer chosen is divisible by 8 i.e

A={6,12,18,24,…,90,96} $\Rightarrow$ n(A)=16 $\Rightarrow P(A)=\frac{16}{100}$

B={8,16,24,40,…,88,96} $\Rightarrow$ n(B)=12 $\Rightarrow P(B)=\frac{12}{100}$

$A\cap B$={24,48,72,96} $\Rightarrow$ $n(A\cap B)$=12 $\Rightarrow P(A\cap B)=\frac{4}{100}$

EXERCISE:

A student attends mathematics class with probability 0.7 skips accounting class with probability 0.4, and attends both with probability 0.5. Find the probability that

(1) he attends at least one class
(2) he attends exactly one class

SOLUTION:

(1) Let A be the event that the student attends mathematics class and B be the event that the student attends accounting class.

Then given

P(A) = 0.7; P(B) = 1 - 0.4 = 0.6

And $P(A\cap B)$ = 0.5, $P(A\cup B)$ = ?

By addition law of probability

$P(A \cup B) = P(A) + P(B) - P(A\cap B)$
$= 0.7 + 0.6 - 0.5$
$= 0.8$

(2) Students can attend exactly one class in two ways

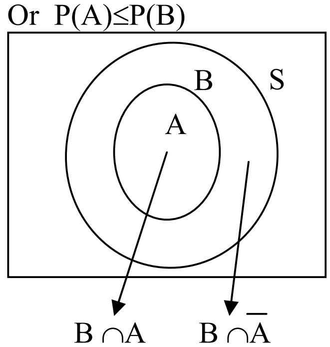
(a)He attends mathematics class but not accounting i.e., event $A \cap B^c$ or
(b)He does not attend mathematics class and attends accounting class i.e., event $A^c \cap B$

Since the two event $A\cap B^c$ and $A^c \cap B$ are disjoint, hence required probability is

$P(A \cap B^c) + P(A^c \cap B)$
Now
$P(A \cap B^c) = P(A \backslash B)$
$= P(A) - P(A \cap B)$
$= 0.7 - 0.5$
$= 0.2$
and
$P(A^c \cap B) = P(B \backslash A)$
$= P(B) - P(A \cap B)$
$= 0.6 - 0.5$
$= 0.1$
Hence required probability is
$P(A \cap B^c) + P(A^c \cap B) = 0.2 + 0.1 = 0.3$

PROBABILITY OF SUB EVENT
THEOREM:
If A and B are two events such that $A \subseteq B$, then $P(A) \le P(B)$

PROOF:
Suppose $A \subseteq B$. The event B may be written as the union of disjoint events $B \cap A$ and $B \cap \overline{A}$
i.e., $B = (B \cap A) \cup (B \cap \overline{A})$
But $B \cap A = A$ (as $A \subseteq B$)
So $B = A \cup (B \cap \overline{A})$
$\therefore P(B) = P(A) + P(B \cap \overline{A})$
But $P(B \cap \overline{A}) \ge 0$
Hence $P(B) \ge P(A)$
Or $P(A) \le P(B)$



EXERCISE:
Let A and B be subsets of a sample space S with $P(A) = 0.7$ and $P(B) = 0.5$. What are the maximum and minimum possible values of $P(A \cup B)$.
SOLUTION:
By addition law of probabilities
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 0.7 + 0.5 - P(A \cap B)$

$= 1.2 - P(A \cap B)$

Since probability of any event is always less than or equal to 1, therefore

max $P(A \cup B) = 1$, for which $P(A \cap B) = 0.2$

Next to find the minimum value, we note

$A \cap B \subseteq B$

$\Rightarrow P(A \cap B) \leq P(B) = 0.5$

Thus for min $P(A \cup B)$ we take maximum possible value of $P(A \cap B)$ which is 0.5. Hence

min $P(A \cup B) = 1.2 -$ max $P(A \cap B)$

$= 1.2 - 0.5$

$= 0.7$ is the required minimum value.

ADDITION LAW OF PROBABILITY FOR THREE EVENTS:

If A, B and C are any three events, then

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

REMARK:

If A, B, C are mutually disjoint events, then $P(A \cup B \cup C) = P(A) + P(B) + P(C)$

EXERCISE:

Three newspapers A, B, C are published in a city and a survey of readers indicates the following:

20% read A, 16% read B, 14% read C

8% read both A and B, 5% read both A and C

4% read both B and C, 2% read all the three

For a person chosen at random, find the probability that he reads none of the papers.

SOLUTION:

Given

$$P(A) = 20\% = \frac{20}{100} = 0.2; \qquad P(B) = 16\% = \frac{16}{100} = 0.16$$

$$P(C) = 14\% = \frac{14}{100} = 0.14; \qquad P(A \cap B) = 8\% = \frac{8}{100} = 0.08$$

$$P(A \cap C) = 5\% = \frac{5}{100} = 0.05; P(B \cap C) = 4\% = \frac{4}{100} = 0.04$$

and

$$P(A \cap B \cap C) = 2\% = \frac{2}{100} = 0.02$$

Now the probability that person reads A or B or C = $P(A \cup B \cup C)$

$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

$$= \frac{20}{100} + \frac{16}{100} + \frac{14}{100} - \frac{8}{100} - \frac{5}{100} - \frac{4}{100} + \frac{2}{100}$$

$$= \frac{35}{100}$$

Hence, the probability that he reads none of the papers

$$= P((A \cup B \cup C)^c)$$
$$= 1 - P(A \cup B \cup C)$$
$$= 1 - \frac{35}{100}$$
$$= \frac{65}{100}$$
$$= 65\%$$

EXERCISE:

Let A, B and C be events in a sample space S, with $A \cup B \cup C = S$, $A \cap (B \cup C) = \phi$, $P(A) = 0.2$, $P(B) = 0.5$ and $P(C) = 0.7$. Find $P(A^C)$, $P(B \cup C)$, $P(B \cap C)$.

SOLUTION:

$P(A^c) = 1 - P(A)$
$= 1 - 0.2$
$P(A^c) = 0.8$

Next, given that the events A and $B \cup C$ are disjoint, since $A \cap (B \cup C) = \phi$, therefore

$P(A \cup (B \cup C)) = P(A) + P(B \cup C)$
$\Rightarrow \quad P(S) = 0.2 + P(B \cup C)$
$\Rightarrow \quad 1 = 0.2 + P(B \cup C)$
$\Rightarrow \quad P(B \cup C) = 1 - 0.2 = 0.8$

Finally, by addition law of probability
$P(B \cup C) = P(B) + P(C) - P(B \cap C)$
$\Rightarrow \quad 0.8 = 0.5 + 0.7 - P(B \cap C)$
$\Rightarrow P(B \cap C) = 0.4$ is the required probability.

Lecture# 37 Conditional probability

CONDITIONAL PROBABILITY
MULTIPLICATION THEOREM
INDEPENDENT EVENTS

EXAMPLE:

a. What is the probability of getting a 2 when a dice is tossed?
b. An even number appears on tossing a die.
   (i) What is the probability that the number is 2?
   (ii) What is the probability that the number is 3?

SOLUTION:

When a dice is tossed, the sample space is S={1,2,3,4,5,6}
n(S)=6

a. Let "A" denote the event of getting a 2 i.e A={ 2 }, n(A)=1

$$P(\text{ 2 appears when the die is tossed}) = \frac{n(A)}{n(S)} = \frac{1}{6}$$

b. (i) Let "S_1" denote the total number of even numbers from a sample space S,when a dice is tossed (i.e $S_1 \subseteq S$)

S_1={ 2,4,6 }, $n(S_1)$=3

Let "B" denote the event of getting a 2 from total number of even number i.e B={ 2 }
. n(B) = 1

$$P(\text{2 appears; given that the number is even}) = P(B) = \frac{n(B)}{n(S_1)} = \frac{1}{3}$$

(ii) Let "C" denote the event of getting a 3 in S_1(among the even numbers)i.e C={ }
n(C)=0

$$P(\text{3 appears; given that the number is even}) = P(C) = \frac{n(C)}{n(S_1)} = \frac{0}{3} = 0$$

EXAMPLE:

Suppose that an urn contains 3 red balls, 2 blue balls, and 4 white balls, and that a ball is selected at random.
Let E be the event that the ball selected is red.
Then P(E) = 3/9 (as there are 3 red balls out of total 9 balls)
Let F be the event that the ball selected is not white.
Then the probability of E if it is already known that the selected ball is not white would be
P(red ball selected; given that the selected ball is not white) = 3/5 (as we count no white ball so there are total 9 balls(i.e 2 blue and 3 red balls))
This is called the conditional probability of E given F and is denoted by P(E|F).

DEFINITION:

Let E and F be two events in the sample space of an experiment with P(F) ≠ 0. The **conditional probability** of E given F, denoted by P(E|F), is defined as

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

EXAMPLE:

Let A and B be events of an experiment such that
P(B) = 1/4 and P(A∩B) = 1/6.

What is the conditional probability P(A|B)?

SOLUTION:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{1/4} = \frac{4}{6} = \frac{2}{3}$$

EXERCISE:

Let A and B be events with $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$
Find

(i) $P(A|B)$ (ii) $P(B|A)$

(iii) $P(A \cup B)$ (iv) $P(A^c|B^c)$

SOLUTION:

Using the formula of the conditional Probability we can write

(i) $P(A|B)$
$$= \frac{P(A \cap B)}{P(B)}$$
$$= \frac{1/4}{1/3} = \frac{3}{4}$$

(ii) $P(B|A)$
$$= \frac{P(B \cap A)}{P(A)}$$
$$= \frac{1/4}{1/2} = \frac{2}{4} = \frac{1}{2} \quad (\text{As } P(B \cap A) = P(A \cap B) = 1/4 \ )$$

(iii) $P(A \cup B)$
$$= P(A) + P(B) - P(A \cap B)$$
$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{4}$$
$$= \frac{7}{12}$$

$$(iv)\qquad P(A^c \mid B^c) = \frac{P(A^c \cap B^c)}{P(B^c)}$$

$$= \frac{P((A \cup B)^c)}{P(B^c)} \quad \text{(By using DeMorgan's Law)}$$

$$= \frac{1 - P(A \cup B)}{1 - P(B)} \qquad [P(E^c) = 1 - P(E)]$$

$$= \frac{1 - 7/12}{1 - 1/3} = \frac{5/12}{2/3} = \frac{5}{12} \times \frac{3}{2} = \frac{5}{8}$$

EXERCISE:

Find P(B|A) if

(i) A is a subset of B

(ii) A and B are mutually exclusive

SOLUTION:

(i) When $A \subseteq B$, then $B \cap A = A$ (As $A \cap A \subseteq B \cap A$ □□ $\subseteq B \cap A$ ……………..(i)
also we know that $B \cap A \subseteq A$ ……………..(ii) , From (i) and (ii) clearly $B \cap A = A$)

$$\therefore \quad P(B \mid A) = \frac{P(B \cap A)}{P(A)} \quad (as\, B \cap A = A \Rightarrow P(B \cap A) = P(A))$$

$$= \frac{P(A)}{P(A)} = 1$$

(ii) When A and B are mutually exclusive, then $B \cap A = \varnothing$

$$\therefore \qquad P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

$$= \frac{P(\phi)}{P(A)} \qquad (Since\ B \cap A = \phi \Rightarrow P(B \cap A) = 0)$$

$$= \frac{0}{P(A)} = 0 \qquad (\ as\ P(\phi) = 0\ )$$

EXAMPLE:

Suppose that an urn contains **three** red balls marked 1, 2, 3, one blue ball marked 4, and **four** white balls marked 5, 6, 7, 8.

A ball is selected at random and its color and number noted.

(i) What is the probability that it is red?

(ii) What is the probability that it has an even number marked on it?

(iii) What is the probability that it is red, if it is known that the ball selected has an even number marked on it?

(iv) What is the probability that it has an even number marked on it, if it is known that the ball selected is red?

SOLUTION:

Let E be the event that the ball selected is red .

(i) P(E) = 3/8

let F be the event that the ball selected has an even number marked on it.

(ii) P(F) = 4/8 (as there are four even numbers 2,4,6 & 8 out of total eight numbers).

(iii) $E\cap F$ is the event that the ball selected is red and has an even number marked on it.
Clearly $P(E\cap F)$ =1/8 (as there is only one ball which is red and marked an even number "2" out of total eight balls).

Hence,

P(Selected ball is red, given that the ball selected has an even number marked on it.) = P(E|F)

$$=\frac{P(E\cap F)}{P(F)}=\frac{1/8}{4/8}=1/4$$

(iv) P(Selected ball has an even number marked on it, given that the ball selected is red) = P(F|E)

$$=\frac{P(E\cap F)}{P(E)}=\frac{1/8}{3/8}=\frac{1}{3}$$

EXAMPLE:

Let a pair of dice be tossed. If the sum is 7, find the probability that one of the dice is 2.

SOLUTION:

Let E be the event that a 2 appears on at least one of the two dice, and F be the event that the sum is 7.

Then

E = {(1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6) ,(3, 2) (4, 2), (5, 2), ,(6, 2)}

F = {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}

$E\cap F$ = {(2, 5), (5, 2)}

P(F) = 6/36 and $P(E\cap F)$ = 2/36.

Hence,

P(Probability that one of the dice is 2, given that the sum is 7)

= P(E|F)

$$=\frac{P(E\cap F)}{P(E)}=\frac{2/36}{6/36}=\frac{1}{3}$$

EXAMPLE:

A man visits a family who has two children. One of the children, a boy, comes into the room.

Find the probability that the other child is also a boy if

(i) The other child is known to be elder,

(ii) Nothing is known about the other child.

SOLUTION:

The sample space of the experiment is S = {*bb*, *bg*, *gb*, *gg*}
(The outcome *bg* specifies that younger is a boy and elder is a girl, etc.)
Let A be the event that both the children are boys.
Then, A = {*bb*}.
(i) Let B be the event that the younger is a boy. Then, B = {*bb*, *bg*},and $A \cap B$ = {*bb*}.
Hence, the required probability is
P(Probability that the other child is also a boy, given that the other child is elder) =

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{2/4} = \frac{1}{2}$$

(ii) Let C be the event that one of the children is a boy.
Then C = {*bb*, *bg*, *gb*}, and $A \cap C$ = {*bb*}.
Hence, the required probability is
P(Probability that both the children are boys, given that one of the children is a boy)
= P(A|C)

$$= \frac{P(A \cap C)}{P(C)} = \frac{1/4}{3/4} = \frac{1}{3}$$

MULTIPLICATION THEOREM

Let E and F be two events in the sample space of an experiment, then
$P(E \cap F) = P(F)P(E \mid F)$
Or $P(E \cap F) = P(E)P(F \mid E)$

Let $E_1, E_2, \ldots, E_n$ be events in the sample space of an experiment, then
$P(E_1 \cap E_2 \cap \ldots \cap E_n) = P(E_1)P(E_2 \mid E_1)\, P(E_3 \mid E_1 \cap E_2) \ldots P(E_n \mid E_1 \cap E_2 \cap \ldots \cap E_n)$

EXAMPLE:

A lot contains 12 items of which 4 are defective. Three items are drawn at random from the lot one after the other. What is the probability that all three are non-defective?

SOLUTION:

Let A_1 be the event that the first item is not defective.
Let A_2 be the event that the second item is not defective.
Let A_3 be the event that the third item is not defective.
Then $P(A_1) = 8/12$, $P(A_2|A_1) = 7/11$, and $P(A_3|A_1 \cap A_2) = 6/10$
Hence, by multiplication theorem, the probability that all three are non-defective is
$P(A_1 \cap A_2 \cap A_3) = P(A_1)\, P(A_2|A_1)\, P(A_3|A_1 \cap A_2)$

$$= \frac{8}{12} \cdot \frac{7}{11} \cdot \frac{6}{10} = \frac{14}{55}$$

INDEPENDENCE:

An event A is said to be independent of an event B if the probability that A occurs is not influenced by whether B has or has not occurred. That is, P(A|B) = P(A).

It follows then from the Multiplication Theorem that,
$P(A \cap B) = P(B)P(A|B) = P(B)P(A)$
We also know that,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$
$$= \frac{P(A)P(B)}{P(A)} \qquad \textit{Because } P(A \cap B) = P(A)P(B) \text{ ,due to independence}$$
$$= P(B)$$

EXAMPLE:

Let A be the event that a randomly generated bit string of length four begins with a 1 and let B be the event that a randomly generated bit string of length four contains an even number of 0s.
Are A and B independent events?

SOLUTION:

A = {1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111}
B = {0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111}

Since there are 16 bit strings of length four, we have
P(A) = 8/16 = 1/2, P(B) = 8/16 = 1/2
Also,
$A \cap B$ = {1001, 1010, 1100, 1111} so that $P(A \cap B)$ = 4/16 =1/4
We note that,

$$P(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B)$$

Hence A and B are independent events.

EXAMPLE:

Let a fair coin be tossed three times. Let A be the event that first toss is heads, B be the even that the second toss is a heads, and C be the event that exactly two heads are tossed in a row. Examine pair wise independence of the three events.

SOLUTION:

The sample space of the experiment is
S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT} and the events are
A = {HHH, HHT, HTH, HTT}
B = {HHH, HHT, THH, THT}
C = {HHT, THH}
$A \cap B$ = {HHH, HHT}, $A \cap C$ = {HHT}, $B \cap C$ = {HHT, THH}

It follows that
P(A) = 4/8=1/2
P(B) = 4/8= 1/2
P(C) = 2/8=1/4
and
$P(A \cap B)$=2/8=1/4
$P(A \cap C)$= 1/8

$P(B \cap C) = 2/8 = 1/4$
Note that,

$$P(A)P(B) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4} = P(A \cap B), \text{ so that A and B are independent.}$$

$$P(A)P(C) = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8} = P(A \cap C), \text{ so that A and C are independent.}$$

$$P(B)P(C) = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8} \neq P(B \cap C), \text{ so that B and C are dependent.}$$

EXAMPLE:

The probability that A hits a target is 1/3 and the probability that B hits the target is 2/5. What is the probability that target will be hit if A and B each shoot at the target?

SOLUTION:

It is clear from the nature of the experiment that the two events are independent.
Hence,
$P(A \cap B) = P(A)P(B)$
It follows that,

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A)P(B) \quad \text{(due to independence)} \\ &= \frac{1}{3} + \frac{2}{5} - \frac{1}{3}\cdot\frac{2}{5} \\ &= \frac{3}{5} \end{aligned}$$

Lecture# 38 Random variable

RANDOM VARIABLE
PROBABILITY DISTRIBUTION
EXPECTATION AND VARIANCE

INTRODUCTION:

Suppose S is the sample space of some experiment. The outcomes of the experiment, or the points in S, need not be numbers. For example in tossing a coin, the outcomes are H (heads) or T(tails), and in tossing a pair of dice the outcomes are pairs of integers.

However, we frequently wish to assign a specific number to each outcome of the experiment. For example, in coin tossing, it may be convenient to assign 1 to H and 0 to T; or in the tossing of a pair of dice, we may want to assign the sum of the two integers to the outcome. Such an assignment of numerical values is called a random variable.

RANDOM VARIABLE:

A random variable X is a rule that assigns a numerical value to each outcome in a sample Space S. **OR** It is a function which maps each outcome of the sample space into the set of real numbers.

We shall let X(S) denote the set of numbers assigned by a random variable X, and refer to X(S) as the range space.
In formal terminology, X is a function from S(sample space) to the set of real numbers R, and X(S) is the range of X.

REMARK:

1. A random variable is also called a chance variable, or a stochastic variable(not called simply a variable, because it is a function).
2. Random variables are usually denoted by capital letters such as X, Y, Z; and the values taken by them are represented by the corresponding small letters.

EXAMPLE:

A pair of fair dice is tossed. The sample space S consists of the 36 ordered pairs i.e

$$S = \{(1,1), (1,2), (1,3), \dots, (6,6)\}$$

Let X assign to each point in S the sum of the numbers; then X is a random variable with range space i.e

$$X(S) = \{2,3,4,5,6,7,8,9,10,11,12\}$$

Let Y assign to each point in S the maximum of the two numbers in the outcomes; then Y is a random variable with range space.

$$Y(S) = \{1,2,3,4,5,6\}$$

PROBABILITY DISTRIBUTION OF A RANDOM VARIABLE:

Let $X(S) = \{x_1, x_2, \dots, x_n\}$ be the range space of a random variable X defined on a finite sample space S.
Define a function f on X(S) as follows:

$f(\mathbf{x_i}) = P(X = \mathbf{x_i})$
$= \text{sum of probabilities of points in S whose image is } \mathbf{x_i}.$

This function f is called the probability distribution or the probability function of X. The probability distribution f of X is usually given in the form of a table.

x_1	x_2	…	x_n
$f(x_1)$	$f(x_2)$	…	$f(x_n)$

The distribution f satisfies the conditions.

$(i) \quad f(x_i) \geq 0$ and $(ii) \quad \sum_{i=1}^{n} f(x_i) = 1$

EXAMPLE:

A pair of fair dice is tossed. Let X assign to each point (a, b) in S = {(1,1), (1,2), …, (6,6)}, the sum of its number, i.e., X (a,b) = a+b. Compute the distribution f of X.

SOLUTION:

X is clearly a random variable with range space
X(S) = {2,3,4,5,6,7,8,9,10, 11, 12}
(because X (a,b) = a+b ⇒ X (1,1) = 1+1=2, X (1,2) = 1+2=3, X (1,3) = 1+3=4 etc).

The distribution f of X may be computed as:

$$f(2) = P(X=2)=P(\{(1,1)\}) = \frac{1}{36}$$

$$f(3) = P(X=3) = P(\{(1,2), (2,1)\}) = \frac{2}{36}$$

$$f(4) = P(X=4) = P(\{(1,3), (2,2),(3,1)\}) = \frac{3}{36}$$

$$f(5) = P(X=5) = P(\{(1,4), (2,3),(3,2),(4,1)\}) = \frac{4}{36}$$

$$f(6) = P(X=6) = P(\{(1,5),(2,4),(3,3),(4,2),(5,1)\}) = \frac{5}{36}$$

$$f(7) = P(X=7) = P(\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,2)\}) = \frac{6}{36}$$

$$f(8) = P(X=8) = P(\{(2,6),(3,5),(4,4),(5,3),(6,2)\}) = \frac{5}{36}$$

$$f(9) = P(X=9) = P(\{(3,6),(4,5),(5,4),(6,3)\}) = \frac{4}{36}$$

$$f(10) = P(X=10) = P(\{(4,6),(5,5),(6,4)\}) = \frac{3}{36}$$

$$f(11) = P(X = 11) = P(\{(5,6),(6,5)\}) = \frac{2}{36}$$

$$f(12) = P(X = 12) = P(\{(6,6)\}) = \frac{1}{36}$$

The distribution of X consists of the points in X(S) with their respective probabilities.

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

EXAMPLE:

A box contains 12 items of which three are defective. A sample of three items is selected from the box.
If X denotes the number of defective items in the sample; find the distribution of X.

SOLUTION:

The sample space S consists of $\binom{12}{3} = 220$ that is 220 different samples of size 3.
The random variable X, denoting the number of defective items has the range space X(S) = {0,1,2,3}
There are $\binom{3}{0}\binom{9}{3} = 84$ samples of size 3 with no defective items;

hence

$$p_0 = P(X = 0) = \frac{84}{220}$$

There are $\binom{3}{1}\binom{9}{2} = 108$ samples of size 3 containing one defective item;
hence .

$$p_1 = P(X = 1) = \frac{108}{220}$$

There are $\binom{3}{2}\binom{9}{1} = 27$ samples of size 3 containing two defective items;

hence

$$p_2 = P(X = 2) = \frac{27}{220}$$

Finally, there is $\binom{3}{3}\binom{9}{0} = 27$, only one sample of size 3 containing three defective items;
hence

$$p_3 = P(X = 3) = \frac{1}{220}$$

The distribution of X follows:

x_i	0	1	2	3
p_i	84/220	108/220	27/220	1/220

EXPECTATION OF A RANDOM VARIABLE

Let X be a random variable with probability distribution

x_i	x_1	x_2	x_3	…	x_n
$f(x_i)$	$f(x_1)$	$f(x_2)$	$f(x_3)$	…	$f(x_n)$

The mean (denoted μ) or the expectation of X (written E(X)) is defined by
$\mu = E(X) = x_1 f(x_1) + x_2 f(x_2) + \ldots + x_n f(x_n)$

$$= \sum_{i=1}^{n} x_i f(x_i)$$

EXAMPLE:

What is the expectation of the number of heads when three fair coins are tossed?

SOLUTION:

The sample space of the experiment is:
S = {TTT, TTH, THT, HTT, THH, HTH, HHT, HHH}
Let the random variable X represents the number of heads (i.e 0,1,2,3) when three fair coins are tossed. Then X has the probability distribution.

x_i	$x_0=0$	$x_1=1$	$x_2=2$	$x_3=3$
$f(x_i)$	1/8	3/8	3/8	1/8

Hence, expectation of X is

$$E(X) = x_0\ f(x_0) + x_1 f(x_1) + x_2 f(x_2) + x_3 f(x_3)$$

$$= 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{12}{8} = 1 \cdot 5$$

EXERCISE:

A player tosses two fair coins. He wins Rs. 1 if one head appears, Rs.2 if two heads appear. On the other hand, he loses Rs.5 if no heads appear. Determine the expected value E of the game and if it is favourable to be player.

SOLUTION:

The sample space of the experiment is S = {HH, HT, TH, TT}
Now

$$P(\text{Two heads}) = P(HH) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

$$P(\text{One head}) = P(HT, TH) = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}$$

$$P(\text{No heads}) = P(TT) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

Thus, the probability of winning Rs.2 is $\frac{1}{4}$, of winning Rs 1 is $\frac{1}{2}$ and of losing Rs 5 is $\frac{1}{4}$

Hence,

$$E = 2(\frac{1}{4}) + 1(\frac{1}{2}) - 5(\frac{1}{4})$$

$$= -\frac{1}{4} = -0.25$$

Since, the expected value of the game is negative, so it is unfavorable to the player.

EXAMPLE:

A coin is weighted so that and $P(H) = \frac{3}{4}$ and $P(T) = \frac{1}{4}$

The coin is tossed three times.

Let X denotes the number of heads that appear.

(a) Find the distribution of X

(b) Find the expectation of E(X)

SOLUTION:

(a) The sample space is S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

The probabilities of the points in sample space are

$$p(HHH) = \frac{3}{4}\cdot\frac{3}{4}\cdot\frac{3}{4} = \frac{27}{64} \qquad p(HHT) = \frac{3}{4}\cdot\frac{3}{4}\cdot\frac{1}{4} = \frac{9}{64}$$

$$p(HTH) = \frac{3}{4}\cdot\frac{1}{4}\cdot\frac{3}{4} = \frac{9}{64} \qquad p(THH) = \frac{1}{4}\cdot\frac{3}{4}\cdot\frac{3}{4} = \frac{9}{64}$$

$$p(HTT) = \frac{3}{4}\cdot\frac{1}{4}\cdot\frac{1}{4} = \frac{3}{64} \qquad p(THT) = \frac{1}{4}\cdot\frac{3}{4}\cdot\frac{1}{4} = \frac{3}{64}$$

$$p(TTH) = \frac{1}{4}\cdot\frac{1}{4}\cdot\frac{3}{4} = \frac{3}{64} \qquad p(TTT) = \frac{1}{4}\cdot\frac{1}{4}\cdot\frac{1}{4} = \frac{1}{64}$$

The random variable X denoting the number of heads assumes the values 0,1,2,3 with the probabilities:

$$P(0) = P(TTT) = \frac{1}{64}$$
$$P(1) = P(HTT, THT, TTH) = \frac{3}{64} + \frac{3}{64} + \frac{3}{64} = \frac{9}{64}$$
$$P(2) = P(HHT, HTH, THH) = \frac{9}{64} + \frac{9}{64} + \frac{9}{64} = \frac{27}{64}$$
$$P(3) = P(HHH) = \frac{27}{64}$$

Hence, the distribution of X is

x_i	0	1	2	3
$P(x_i)$	1/64	9/64	27/64	27/64

(b) The expected value E(X) is obtained by multiplying each value of X by its probability and taking the sum.
The distribution of X is

x_i	0	1	2	3
$P(x_i)$	1/64	9/64	27/64	27/64

Hence

$$\begin{aligned} E(X) &= 0\left(\frac{1}{64}\right) + 1\left(\frac{9}{64}\right) + 2\left(\frac{27}{64}\right) + 3\left(\frac{27}{64}\right) \\ &= \frac{144}{64} \\ &= 2\cdot 25 \end{aligned}$$

VARIANCE AND STANDARD DEVIATION OF A RANDOM VARIABLE:

Let X be a random variable with mean μ and the probability distribution

x_1	x_2	x_3	…	x_n
$f(x_1)$	$f(x_2)$	$f(x_3)$	…	$f(x_n)$

The variance of X, measures the "spread" or "dispersion" of X from the mean μ and is denoted and defined as

$$\sigma_x^2 = \text{Var}(X) = \sum_{i=1}^{n}(x_i-\mu)^2 f(x_i)$$
$$= E((X-\mu)^2)$$
$$= E(X^2)-\mu^2$$
$$= \sum x_i^2 f(x_i)-\mu^2$$

The last expression is a more convenient form for computing Var(X).
The standard derivation of X, denoted by σ_X , is the non-negative square root of Var(X):

Where $\sigma_X = \sqrt{\text{Var}(X)}$

EXERCISE:

Find the expectation μ, variance σ^2 and standard deviation σ of the distribution given in the following table.

x_i	1	3	4	5
$f(x_i)$	0.4	0.1	0.2	0.3

SOLUTION:

$\mu = E(X) = \sum x_i f(x_i)$
$= 1(0.4) + 3(0.1) + 4(0.2) + 5(0.3)$
$= 0.4 + 0.3 + 0.8 + 1.5$
$= 3.0$

Next

$E(X^2) = \sum x_i^2 f(x_i)$
$= 1^2 (0.4) + 3^2 (0.1) + 4^2 (0.2) + 5^2 (0.3)$
$= 0.4 + 0.9 + 3.2 + 7.5$
$= 12.0$

Hence

$\sigma^2 = \text{Var}(X)= E(X^2) - \mu^2$
$= 12.0 - (3.0)^2 = 3.0$

and $\sigma = \sqrt{\text{Var}(X)} = \sqrt{3.0} \approx 1.7$

EXERCISE:

A pair of fair dice is thrown. Let X denote the maximum of the two numbers which appears.

(a) Find the distribution of X
(b) Find the μ, variance σ_x^2 = Var(X), and standard deviation σx of X

SOLUTION:

(a) The sample space S consist of the 36 pairs of integers (a,b) where a and b range from 1 to 6;
that is S = {(1,1), (1,2), …, (6,6)}
Since X assigns to each pair in S the larger of the two integers, the value of X are the integers from 1 to 6.

Note that:

$$f(1) = P(X = 1) = P(\{(1,1)\}) = \frac{1}{36}$$

$$f(2) = P(X = 2) = P(\{(2,1),(2,2),(1,2)\}) = \frac{3}{36}$$

$$f(3) = P(X = 3) = P(\{(3,1),(3,2),(3,3),(2,3),(1,3)\}) = \frac{5}{36}$$

$$f(4) = P(X = 4) = P(\{(4,1),(4,2),(4,3),(4,4),(3,4),(2,4),(1,4)\}) = \frac{7}{36}$$

Similarly

$$f(5) = P(X = 5) = \{(1,5),(2,5),(3,5),(4,5),(5,1),(5,2),(5,3),(5,4),(5,5)\}$$
$$= \frac{9}{36}$$

$$f(6) = P(X = 6) = \{(1,6),(2,6),(3,6),(4,6),(5,6),(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$$
$$= \frac{11}{36}$$

Hence, the probability distribution of x is:

x_i	1	2	3	4	5	6
$f(x_i)$	1/36	3/36	5/36	7/36	9/36	11/36

(b) We find the expectation (mean) of X as

$$\mu = E(X) = \sum x_i\, f(x_i)$$
$$= 1\cdot\frac{1}{36} + 2\cdot\frac{1}{36} + 3\cdot\frac{5}{36} + 4\cdot\frac{7}{36} + 5\cdot\frac{9}{36} + 6\cdot\frac{11}{36}$$
$$= \frac{161}{36} \approx 4.5$$

Next

$$E(X^2) = \sum x_i^2 f(x_i)$$
$$= 1^2\cdot\frac{1}{36} + 2^2\cdot\frac{1}{36} + 3^2\cdot\frac{1}{36} + 4^2\cdot\frac{7}{36} + 5^2\cdot\frac{9}{36} + 6^2\cdot\frac{11}{36}$$
$$= \frac{791}{36} \approx 22.0$$

Then

$$\sigma_x^2 = \mathrm{Var}(X) = E(X^2) - \mu^2$$
$$= 22.0 - (4.5)^2$$
$$= 17.5$$

and

$$\sigma_x = \sqrt{17.5} \approx 1.3$$

Lecture# 39 Introduction to Graphs

INTRODUCTION TO GRAPHS

INTRODUCTION:

Graph theory plays an important role in several areas of computer science such as:

- switching theory and logical design
- artificial intelligence
- formal languages
- computer graphics
- operating systems
- compiler writing
- information organization and retrieval.

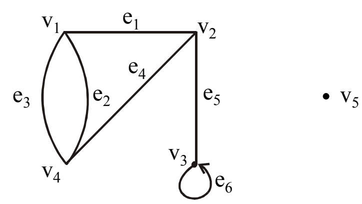
GRAPH:

A graph is a non-empty set of points called vertices and a set of line segments joining pairs of vertices called edges.

Formally, a graph G consists of two finite sets:

(i) A set V=V(G) of vertices (or points or nodes)

(ii)A set E=E(G) of edges; where each edge corresponds to a pair of vertices.

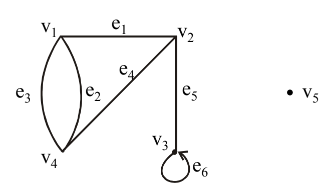


The graph G with

$V(G) = \{v_1, v_2, v_3, v_4, v_5\}$ and

$E(G) = \{e_1, e_2, e_3, e_4, e_5, e_6\}$

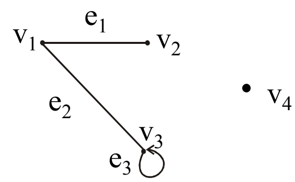
SOME TERMINOLOGY:



1. An edge connects either one or two vertices called its **endpoints** (edge e_1 connects vertices v_1 and v_2 described as $\{v_1, v_2\}$ i.e v_1 and v_2 are the endpoints of an edge e_1).
2. An edge with just one endpoint is called a **loop**. Thus a loop is an edge that connects a vertex to itself (e.g., edge e_6 makes a loop as it has only one endpoint v_3).
3. Two vertices that are connected by an edge are called **adjacent**; and a vertex that is an endpoint of a loop is said to be adjacent to itself.
4. An edge is said to be **incident** on each of its endpoints(i.e. e_1 is incident on v_1 and v_2).
5. A vertex on which no edges are incident is called **isolated** (e.g., v_5)
6. Two distinct edges with the same set of end points are said to be **parallel** (i.e. e_2 & e_3).

EXAMPLE:

Define the following graph formally by specifying its vertex set, its edge set, and a table giving the edge endpoint function.



SOLUTION:

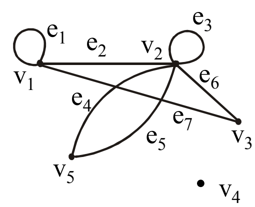
Vertex Set = $\{v_1, v_2, v_3, v_4\}$
Edge Set = $\{e_1, e_2, e_3\}$

Edge - endpoint function is:

Edge	Endpoint
e_1	$\{v_1, v_2\}$
e_2	$\{v_1, v_3\}$
e_3	$\{v_3\}$

EXAMPLE:

For the graph shown below
(i) find all edges that are incident on v_1;
(ii)find all vertices that are adjacent to v_3;
(iii)find all loops;
(iv)find all parallel edges;
(v)find all isolated vertices;



SOLUTION:

(i) v_1 is incident with edges e_1, e_2 and e_7
(ii) vertices adjacent to v_3 are v_1 and v_2
(iii) loops are e_1 and e_3
(iv) only edges e_4 and e_5 are parallel
(v) The only isolated vertex is v_4 in this Graph.

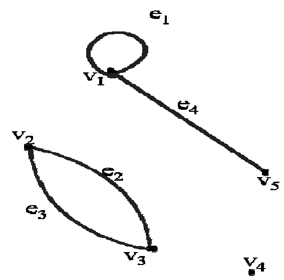
DRAWING PICTURE FOR A GRAPH:

Draw picture of Graph H having vertex set $\{v_1, v_2, v_3, v_4, v_5\}$ and edge set $\{e_1, e_2, e_3, e_4\}$ with edge endpoint function

Edge	Endpoint
e_1	$\{v_1\}$
e_2	$\{v_2,v_3\}$
e_3	$\{v_2,v_3\}$
e_4	$\{v_1,v_5\}$

SOLUTION:

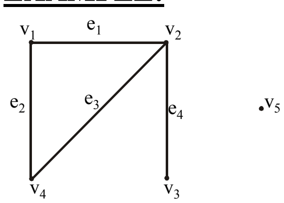
Given $V(H) = \{v_1, v_2, v_3, v_4, v_5\}$
and $E(H) = \{e_1, e_2, e_3, e_4\}$
with edge endpoint function



SIMPLE GRAPH

A simple graph is a graph that does not have any loop or parallel edges.

EXAMPLE:



It is a simple graph H
$V(H) = \{v_1, v_2, v_3, v_4, v_5\}$ & $E(H) = \{e_1, e_2, e_3, e_4\}$

EXERCISE:

Draw all simple graphs with the four vertices {u, v, w, x} and two edges, one of which is {u, v}.

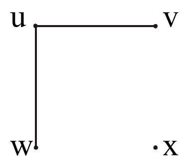
SOLUTION:

There are C(4,2) = 6 ways of choosing two vertices from 4 vertices. These edges may be listed as:

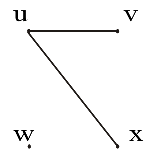
{u,v},{u,w},{u,x},{v,w}, {v,x},{w,x}

One edge of the graph is specified to be {u,v}, so any of the remaining five from this list may be chosen to be the second edge. This required graphs are:

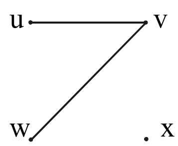
1.



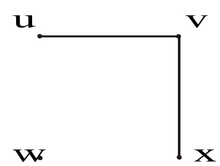
2.



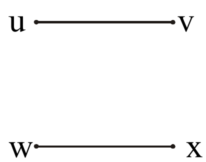
3.



4.



5.

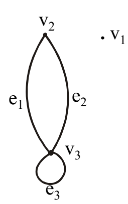


<u>DEGREE OF A VERTEX:</u>

Let G be a graph and ***v*** a vertex of G. The degree of ***v***, denoted **deg(*v*),** equals the number of edges that are incident on ***v***, with an edge that is a loop counted twice.
Note:(i)The total degree of G is the sum of the degrees of all the vertices of G.
(ii) The **degree** of a loop is counted twice.

<u>EXAMPLE:</u>

For the graph shown



deg (v_1) = 0, since v_1 is isolated vertex.
deg (v_2) = 2, since v_2 is incident on e_1 and e_2 .
deg (v_3) = 4, since v_3 is incident on e_1,e_2 and the loop e_3.
Total degree of G = deg(v_1) + deg(v_2) + deg(v_3)
= 0 + 2 + 4
= 6

<u>REMARK:</u>

The total degree of G, which is 6, equals twice the number of edges of G, which is 3.

<u>THE HANDSHAKING THEOREM:</u>

If G is any graph, then the sum of the degrees of all the vertices of G equals twice the number of edges of G.
Specifically, if the vertices of G are v_1, v_2, …, v_n, where n is a positive integer, then

the total degree of G $= \deg(v_1) + \deg(v_2) + \ldots + \deg(v_n)$
$= 2$. (the number of edges of G)

PROOF:

Each edge "e" of G connects its end points v_i and v_j. This edge, therefore contributes 1 to the degree of v_i and 1 to the degree of v_j.
If "e" is a loop, then it is counted twice in computing the degree of the vertex on which it is incident.
Accordingly, each edge of G contributes 2 to the total degree of G.
Thus,

the total degree of G = 2. (the number of edges of G)

COROLLARY:

The total degree of G is an even number

EXERCISE:

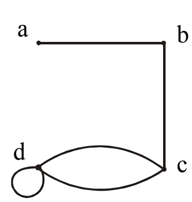
Draw a graph with the specified properties or explain why no such graph exists.
(i) Graph with four vertices of degrees 1, 2, 3 and 3
(ii) Graph with four vertices of degrees 1, 2, 3 and 4
(iii)Simple graph with four vertices of degrees 1, 2, 3 and 4

SOLUTION:

(i) Total degree of graph $= 1 + 2 + 3 + 3$
$= 9$ an odd integer
Since, the total degree of a graph is always even, hence no such graph is possible.
Note: As we know that "for any graph,the sum of the degrees of all the vertices of G equals twice the number of edges of G **or** the total degree of G is an even number".

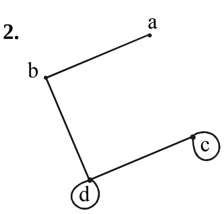
(ii) Two graphs with four vertices of degrees 1, 2, 3 & 4 are

1.



or

2.



The vertices a, b, c, d have degrees 1,2,3, and 4 respectively(i.e graph exists).

(iii) Suppose there was a simple graph with four vertices of degrees 1, 2, 3, and 4. Then the vertex of degree 4 would have to be connected by edges to four distinct vertices other than itself because of the assumption that the graph is simple (and hence has no loop or parallel edges.) This contradicts the assumption that the graph has four vertices in total. Hence there is no simple graph with four vertices of degrees 1, 2, 3, and 4, so simple graph is not possible in this case.

EXERCISE:

Suppose a graph has vertices of degrees 1, 1, 4, 4 and 6. How many edges does the graph have?

SOLUTION:

The total degree of graph = 1 + 1 + 4 + 4 + 6
= 16

Since, the total degree of graph = 2.(number of edges of graph) [by using Handshaking theorem]

$\Rightarrow$ 16 = 2.(number of edges of graph)

$$\Rightarrow \text{Number of edges of graph} = \frac{16}{2} = 8$$

EXERCISE:

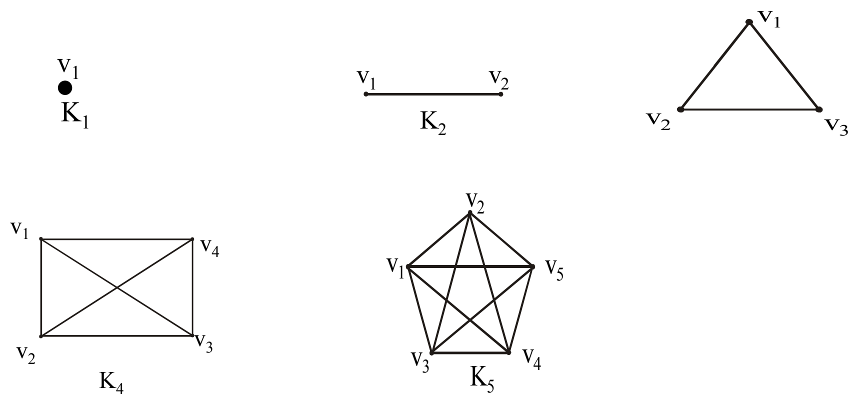
In a group of 15 people, is it possible for each person to have exactly 3 friends?

SOLUTION:

Suppose that in a group of 15 people, each person had exactly 3 friends. Then we could draw a graph representing each person by a vertex and connecting two vertices by an edge if the corresponding people were friends.
But such a graph would have 15 vertices each of degree 3, for a total degree of 45 (not even) which is not possible.
Hence, in a group of 15 people it is not possible for each to have exactly three friends.

COMPLETE GRAPH:

A complete graph on n vertices is a simple graph in which each vertex is connected to every other vertex and is denoted by K_n (K_n means that there are n vertices).
The following are complete graphs K_1, K_2, K_3, K_4 and K_5.



EXERCISE:

For the complete graph K_n, find
(i) the degree of each vertex
(ii)the total degrees
(iii)the number of edges

SOLUTION:

(i) Each vertex v is connected to the other (n-1) vertices in K_n; hence deg (v) = n - 1 for every v in K_n.

(ii) Each of the n vertices in K_n has degree n - 1; hence, the total degree in
$K_n = (n - 1) + (n - 1) + \ldots + (n - 1)$ n times
$= n (n - 1)$

(iii) Each pair of vertices in K_n determines an edge, and there are C(n, 2) ways of selecting two vertices out of n vertices. Hence,
Number of edges in K_n = C(n, 2)

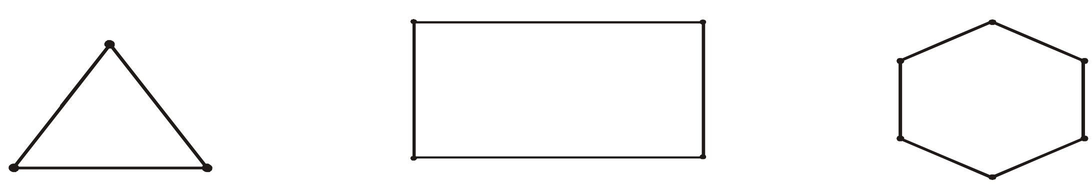
$$= \frac{n(n-1)}{2}$$

Alternatively,
The total degrees in graph K_n = 2 (number of edges in K_n)
$\Rightarrow$ n(n-1) = 2(number of edges in K_n)
$\Rightarrow$
Number of edges in K_n $= \dfrac{n(n-1)}{2}$

REGULAR GRAPH:
A graph G is regular of degree k or k-regular if every vertex of G has degree k.
In other words, a graph is regular if every vertex has the same degree.
Following are some regular graphs.

(i) 0-regular

(ii) 1-regular

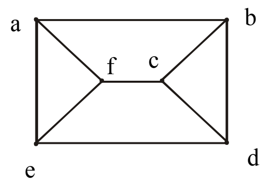


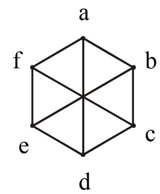
(iii) 2-regular

REMARK: The complete graph K_n is (n-1) regular.

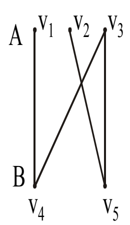
EXERCISE:
Draw two 3-regular graphs with six vertices.

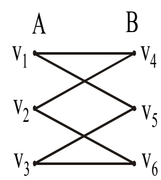
SOLUTION:





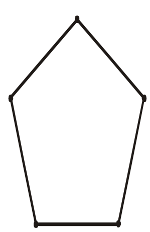
BIPARTITE GRAPH:
A bipartite graph G is a simple graph whose vertex set can be partitioned into two mutually disjoint non empty subsets A and B such that the vertices in A may be connected to vertices in B, but no vertices in A are connected to vertices in A and no vertices in B are connected to vertices in B.
The following are bipartite graphs

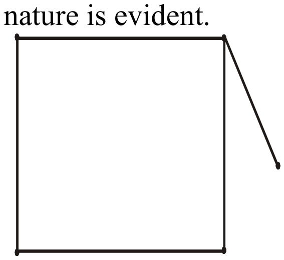




DETERMINING BIPARTITE GRAPHS:
The following labeling procedure determines whether a graph is bipartite or not.
1. Label any vertex **a**
2. Label all vertices adjacent to **a** with the label **b**.
3. Label all vertices that are adjacent to a vertex just labeled **b** with label **a**.
4. Repeat steps 2 and 3 until all vertices got a distinct label (a bipartite graph) or there is a conflict i.e., a vertex is labeled with a and b (not a bipartite graph).

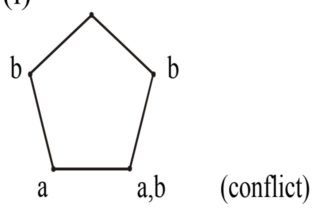
EXERCISE:
Find which of the following graphs are bipartite. Redraw the bipartite graph so that its bipartite nature is evident.





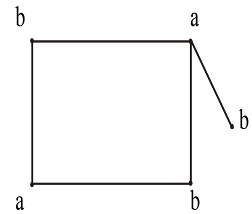
SOLUTION:

(i)

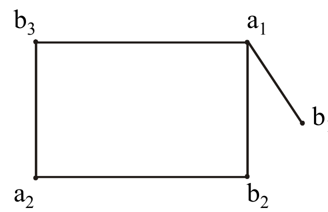


The graph is not bipartite.

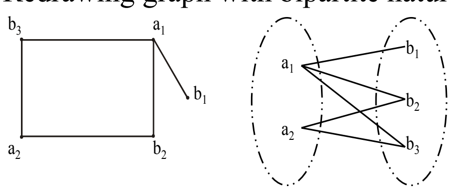
(ii)



By labeling procedure, each vertex gets a distinct label. Hence the graph is bipartite. To redraw the graph we mark labels a's as a_1, $a_2$and b's as b_1, b_2,

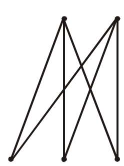


Redrawing graph with bipartite nature evident.

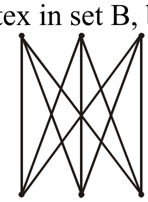


COMPLETE BIPARTITE GRAPH:

A complete bipartite graph on (m+n) vertices denoted $K_{m,n}$ is a simple graph whose vertex set can be partitioned into two mutually disjoint non empty subsets A and B containing m and n vertices respectively, such that each vertex in set A is connected (adjacent) to every vertex in set B, but the vertices within a set are not connected.



$K_{2,3}$

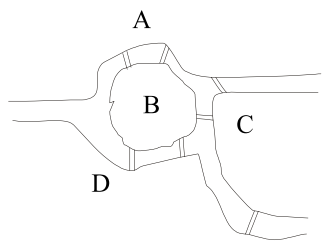


$K_{3,3}$

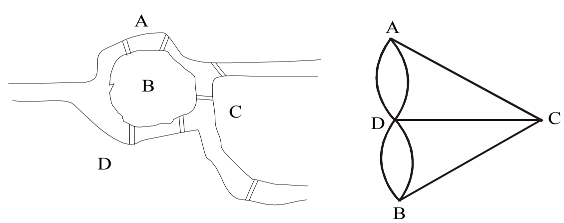
Lecture# 40 Paths and Circuits

PATHS AND CIRCUITS

KONIGSBERG BRIDGES PROBLEM



It is possible for a person to take a walk around town, starting and ending at the same location and crossing each of the seven bridges exactly once?



Is it possible to find a route through the graph that starts and ends at some vertex A, B, C or D and traverses each edge exactly once?

Equivalently:

Is it possible to trace this graph, starting and ending at the same point, without ever lifting your pencil from the paper?

DEFINITIONS:

Let G be a graph and let v and w be vertices in graph G.

1. WALK

A walk from v to w is a finite alternating sequence of adjacent vertices and edges of G.
Thus a walk has the form

$$v_0e_1v_1e_2 \ldots v_{n-1}e_nv_n$$

where the v's represent vertices, the e's represent edges $v_0=v$, $v_n=w$, and for all $i = 1, 2 \ldots n$, v_{i-1} and v_i are endpoints of e_i.

The trivial walk from v to v consists of the single vertex v.

2. CLOSED WALK

A closed walk is a walk that starts and ends at the same vertex.

3. CIRCUIT

A circuit is a closed walk that does not contain a repeated edge. Thus a circuit is a walk of the form

$$v_0e_1v_1e_2 \dots v_{n-1}\ e_n\ v_n$$

where $v_0 = v_n$ and all the e_i,s are distinct.

4. SIMPLE CIRCUIT

A simple circuit is a circuit that does not have any other repeated vertex except the first and last.
Thus a simple circuit is a walk of the form

$$v_0e_1v_1e_2 \dots v_{n-1}e_nv_n$$

where all the e_i,s are distinct and all the v_j,s are distinct except that $v_0 = v_n$

5. PATH

A path from v to w is a walk from v to w that does not contain a repeated edge. Thus a path from v to w is a walk of the form

$$v = v_0e_1v_1e_2 \dots v_{n-1}e_nv_n = w$$

where all the e_i,s are distinct (that is $e_i \neq e_k$ for any $i \neq k$).

6. SIMPLE PATH

A simple path from v to w is a path that does not contain a repeated vertex.
Thus a simple path is a walk of the form

$$v = v_0e_1v_1e_2 \dots v_{n-1}e_nv_n = w$$

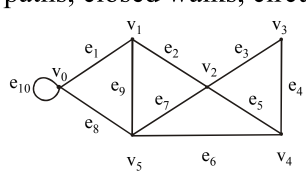
where all the e_i,s are distinct and all the v_j,s are also distinct (that is, $v_j \neq v_m$ for any $j \neq m$).

SUMMARY

	Repeated Edge	Repeated Vertex	Starts and Ends at Same Point
walk	allowed	Allowed	allowed
closed walk	allowed	Allowed	yes(means, where it starts also ends at that point)
circuit	no	Allowed	yes
simple circuit	no	first and last only	yes
path	no	Allowed	allowed
simple path	no	no	No

<u>**EXERCISE:**</u>

In the graph below, determine whether the following walks are paths, simple paths, closed walks, circuits, simple circuits, or are just walks.

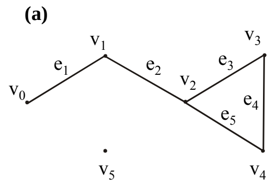


(a) $v_1e_2v_2e_3v_3e_4v_4e_5v_2e_2v_1e_1v_0$

(b) $v_1v_2v_3v_4v_5v_2$
(c) $v_4v_2v_3v_4v_5v_2v_4$
(d) $v_2v_1v_5v_2v_3v_4v_2$
(e) $v_0v_5v_2v_3v_4v_2v_1$
(f) $v_5v_4v_2v_1$

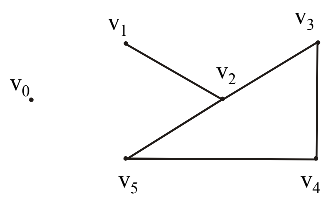
SOLUTION:

(a) $v_1e_2v_2e_3v_3e_4v_4e_5v_2e_2v_1e_1v_0$



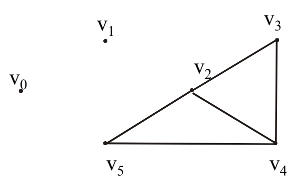
This graph starts at vertex v_1,then goes to v_2 along edge e_2,and moves continuously, at the end it goes from v_1 to v_0 along e_1.Note it that the vertex v_2 and the edge e_2 is repeated twice, and starting and ending, not at the same points.Hence The graph is just a walk.

(b) $v_1v_2v_3v_4v_5v_2$



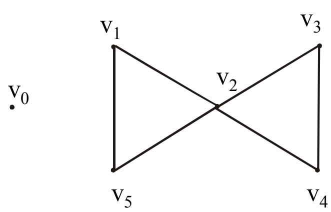
In this graph vertex v_2 is repeated twice.As no edge is repeated so the graph is a path.

(c) $v_4v_2v_3v_4v_5v_2v_4$



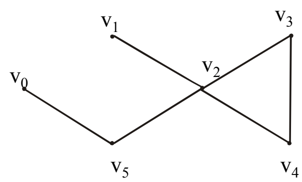
As vertices v_2 & v_4 are repeated and graph starts and ends at the same point v_4,also the edge(i.e. e_5)connecting v_2 & v_4 is repeated, so the graph is a closed walk.

(d) $v_2v_1v_5v_2v_3v_4v_2$



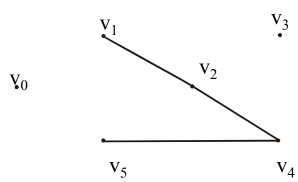
In this graph, vertex v_2 is repeated and the graph starts and end at the same vertex (i.e. at v_2) and no edge is repeated, hence the above graph is a circuit.

(e) $v_0v_5v_2v_3v_4v_2v_1$



Here vertex v_2 is repeated and no edge is repeated so the graph is a path.

(f) $v_5v_4v_2v_1$



Neither any vertex nor any edge is repeated so the graph is a simple path**.**

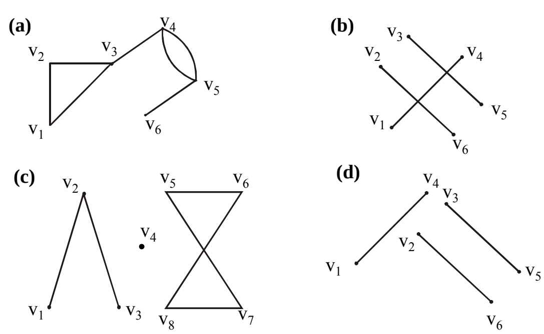
<u>CONNECTEDNESS:</u>

Let G be a graph. Two vertices *v* and *w* of G are connected if, and only if, there is a walk from *v* to *w*. The graph G is connected if, and only if, given any two vertices *v* and *w* in G, there is a walk from *v* to *w*. Symbolically:

G is connected $\Leftrightarrow$ $\forall$ vertices v, w $\in$ V (G), $\exists$ a walk from v to w:

<u>EXAMPLE:</u>

Which of the following graphs have a connectedness**?**



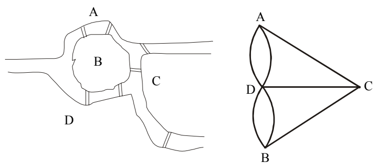
EULER CIRCUITS

<u>DEFINITION:</u>

Let G be a graph. An Euler circuit for G is a circuit that contains every vertex and every edge of G. That is, an Euler circuit for G is sequence of adjacent vertices and edges in G that starts and ends at the same vertex uses every vertex of G at least once, and used every edge of G exactly once.

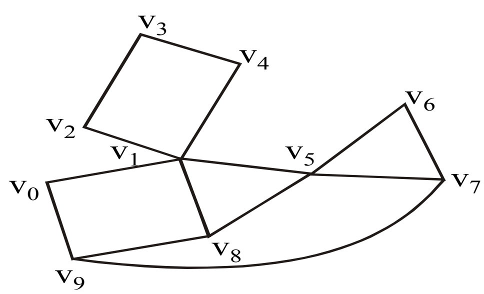
THEOREM:
A graph G has an Euler circuit if, and only if, G is connected and every vertex of G has an even degree.

KONIGSBERG BRIDGES PROBLEM



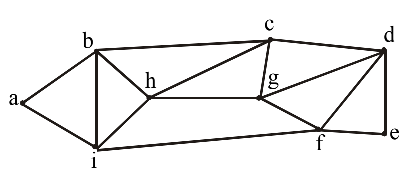
We try to solve Konigsberg bridges problem by Euler method.
Here deg(a)=3,deg(b)=3,deg(c)=3 and deg(d)=5 as the vertices have odd degree so there is no possibility of an Euler circuit.

EXERCISE:
Determine whether the following graph has an Euler circuit**.**



SOLUTION:
As deg (v_1) =5, an odd degree so the following graph has not an Euler circuit.

EXERCISE:
Determine whether the following graph has Euler circuit.



SOLUTION:
From above clearly deg(a)=2, deg(b)=4, deg(c)=4, deg(d)=4, deg(e)=2, deg(f)=4, deg(g)=4, deg(h)=4, deg(i)=4
Since the degree of each vertex is even, and the graph has Euler Circuit. One such circuit is:
a b c d e f g d f i h c g h b i a

EULER PATH

DEFINITION:

Let G be a graph and let v and w be two vertices of G. An Euler path from v to w is a sequence of adjacent edges and vertices that starts at v, end and w, passes through every vertex of G at least once, and traverses every edge of G exactly once.

COROLLARY

Let G be a graph and let v and w be two vertices of G. There is an Euler path from v to w if, and only if, G is connected, v and w have odd degree and all other vertices of G have even degree.

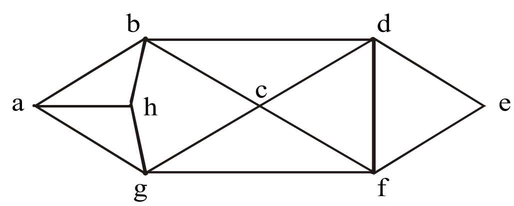
HAMILTONIAN CIRCUITS

DEFINITION:

Given a graph G, a Hamiltonian circuit for G is a simple circuit that includes every vertex of G. That is, a Hamiltonian circuit for G is a sequence of adjacent vertices and distinct edges in which every vertex of G appears exactly once.

EXERCISE:

Find Hamiltonian Circuit for the following graph.



SOLUTION:

The Hamiltonian Circuit for the following graph is:

a b d e f c g h a

Another Hamiltonian Circuit for the following graph could be:

a b c d e f g h a

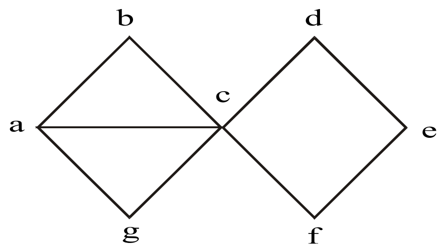
PROPOSITION:

If a graph G has a Hamiltonian circuit then G has a sub-graph H with the following properties:

1. H contains every vertex G
2. H is connected
3. H has the same number of edges as vertices
4. Every vertex of H has degree 2

EXERCISE:

Show that the following graph does not have a Hamiltonian circuit.



Here deg(c)=5,if we remove 3 edges from vertex c then deg(b)< 2 , deg(g)< 2
or deg(f)< 2,deg(d)< 2.

It means that this graph does not satisfy the desired properties as above, so the graph does not have a Hamiltonian circuit.

Lecture# 41 Matrix Representation of Graphs

MATRIX REPRESENTATIONS OF GRAPHS

MATRIX:

An m × n matrix A over a set S is a rectangular array of elements of S arranged into m rows and n columns:

$$A=\begin{array}{c} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \end{array} \begin{array}{l} \\ \\ \\ \leftarrow \text{ith row of A} \\ \\ \\ \end{array}$$

↑

jth column of A

Briefly, it is written as:

$$\mathbf{A = [a_{ij}]_{m\times n}}$$

EXAMPLE:

$$A=\begin{bmatrix} 4 & -2 & 0 & 6 \\ 2 & -3 & 1 & 9 \\ 0 & 7 & 5 & -1 \end{bmatrix}$$

A is a matrix having 3 rows and 4 columns. We call it a 3 × 4 matrix, or matrix of size 3 × 4(or we say that a matrix having an order 3 × 4).

Note it that

$a_{11} = 4$ (11 means 1st row and 1st column), $a_{12} = -2$ (12 means 1st row and 2nd column),

$a_{13} = 0$, $a_{14} = 6$

$a_{21} = 2$, $a_{22} = -3$, $a_{23} = 1$, $a_{24} = 9$ etc.

SQUARE MATRIX:

A matrix for which the number of rows and columns are equal is called a square matrix. A square matrix A with m rows and n columns (size m × n) but m=n (i.e of order n × n) has the form:

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1i} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2i} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ii} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{ni} & \cdots & a_{nn} \end{bmatrix}$$

Diagonal entries

Note:

The main diagonal of A consists of all the entries

$a_{11}, a_{22}, a_{33}, \ldots, a_{ii}, \ldots, a_{nn}$

TRANSPOSE OF A MATRIX:
The transpose of a matrix A of size m × n, is the matrix denoted by A^t of size n × m, obtained by writing the rows of A, in order, as columns.(**Or** we can say that transpose of a matrix means "write the rows instead of columns or write the columns instead of rows". Thus if

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad then \;\; A^t = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

EXAMPLE:

$$A = \begin{bmatrix} 4 & -2 & 0 & 6 \\ 2 & -3 & 1 & 9 \\ 0 & 7 & 5 & -1 \end{bmatrix}$$

Then

$$A^t = \begin{bmatrix} 4 & 2 & 0 \\ -2 & -3 & 7 \\ 0 & 1 & 5 \\ 6 & 9 & -1 \end{bmatrix}$$

SYMMETRIC MATRIX:
A square matrix A = $[a_{ij}]$ of size n × n is called symmetric if, and only if, $A^t = A$
i.e., for all i, j = 1, 2, …, n, $a_{ij} = a_{ji}$
EXAMPLE:

$$\text{Let} \quad A = \begin{bmatrix} 1 & 3 & 7 \\ 5 & 2 & 9 \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} 4 & 2 & 0 \\ 2 & -3 & 1 \\ 0 & 1 & 5 \end{bmatrix}$$

$$\text{Then} \quad A^t = \begin{bmatrix} 1 & 5 \\ 3 & 2 \\ 7 & 9 \end{bmatrix}, \quad \text{and} \quad B^t = \begin{bmatrix} 4 & 2 & 0 \\ 2 & -3 & 1 \\ 0 & 1 & 5 \end{bmatrix}$$

Note that $B^t = B$, so that B is a symmetric matrix.

MATRIX MULTIPLICATION:
Suppose A and B are two matrices such that the number of columns of A is equal to the number of rows of B, say A is an m × p matrix and B is a p × n matrix. Then the product of A and B, written AB, is the m × n matrix whose ijth entry is obtained by multiplying the elements of the ith row of A by the corresponding elements of the jth column of B and then adding;

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ip} \\ \vdots & \vdots & & \vdots \\ a_{mi} & a_{m2} & \cdots & a_{mp} \end{bmatrix} \begin{bmatrix} \mathrm{b}_{11} & \cdots & b_{1j} & \cdots & b_{1n} \\ \mathrm{b}_{21} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & & & & \\ \mathrm{b}_{\mathrm{p}1} & \cdots & b_{pj} & \cdots & b_{pn} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1j} & \cdots & c_{1n} \\ \vdots & & & & \\ c_{i1} & \cdots & c_{ij} & \cdots & c_{in} \\ \vdots & & & & \\ c_{m1} & \cdots & c_{mj} & \cdots & c_{mn} \end{bmatrix}$$

where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^{p} a_{ik}b_{kj}$$

REMARK:

If the number of columns of A is not equal to the number of rows of B, then the product AB is not defined.

EXAMPLE:

Find the product AB and BA of the matrices

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 2 & 0 & -4 \\ 3 & -2 & 6 \end{bmatrix}$$

SOLUTION:

Size of A is 2 × 2 and of B is 2 × 3, the product AB is defined as a 2 × 3 matrix. But BA is not defined, because no. of columns of B = 3 ≠ 2 = no. of rows of A.

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -4 \\ 3 & -2 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(2)+(3)(3) & (1)(0)+(3)(6) & (1)(-4)+(3)(6) \\ (2)(2)+(-1)(3) & (2)(0)+(-1)(-2) & (2)(-4)+(-2)(6) \end{bmatrix} = \begin{bmatrix} 11 & -6 & 14 \\ 1 & 2 & -14 \end{bmatrix}$$

EXERCISE:

Find AA^t and A^tA, where

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$$

SOLUTION:

A^t is obtained from A by rewriting the rows of A as columns:

i.e $$A^t = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix}$$

Now

$$AA^t = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 1+4+0 & 3-2+0 \\ 3-2+0 & 9+1+16 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 26 \end{bmatrix}$$

and
$$A^tA = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$$
$$= \begin{bmatrix} 1+9 & 2-3 & 0+12 \\ 2-3 & 4+1 & 0-4 \\ 0+12 & 0-4 & 0+16 \end{bmatrix}$$
$$= \begin{bmatrix} 10 & -1 & 12 \\ -1 & 5 & -4 \\ 12 & -4 & 16 \end{bmatrix}$$

ADJACENCY MATRIX OF A GRAPH:

Let G be a graph with ordered vertices $v_1, v_2, \ldots, v_n$. The adjacency matrix of G is the matrix $A = [a_{ij}]$ over the set of non-negative integers such that
a_{ij} = the number of edges connecting v_i and v_j for all i, j = 1, 2, …, n.

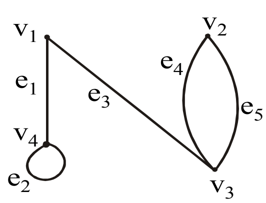
OR

The adjancy matrix say A= $[a_{ij}]$ is also defined as

$$a_{ij} = \begin{cases} 1 & if\ \{v_i, v_j\}\ is\ an\ edge\ of\ G \\ 0 & otherwise \end{cases}$$

EXAMPLE:

A graph with it's adjacency matrix is shown.



$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Note that the nonzero entries along the main diagonal of A indicate the presence of loops and entries larger than 1 correspond to parallel edges.
Also note A is a symmetric matrix.

EXERCISE:

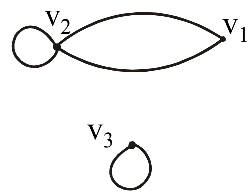
Find a graph that have the following adjacency matrix.

$$\begin{bmatrix} 0 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

SOLUTION:

Let the three vertices of the graph be named v_1, v_2 and v_3. We label the adjacency matrix across the top and down the left side with these vertices and draw the graph accordingly(as from v_1 to v_2 there is a value "2",it means that two parallel edges between v_1 and v_2 and same condition occurs between v_2 and v_1 and the value "1" represent the loops of v_2 and v_3).

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{bmatrix} 0 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

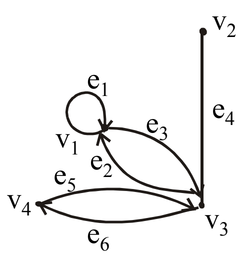


DIRECTED GRAPH:

A directed graph or digraph, consists of two finite sets: a set V(G) of vertices and a set D(G) of directed edges, where each edge is associated with an ordered pair of vertices called its end points.

If edge e is associated with the pair (v, w) of vertices, then e is said to be the directed edge from v to w and is represented by drawing an arrow from v to w.

EXAMPLE OF A DIGRAPH:



ADJACENCY MATRIX OF A DIRECTED GRAPH:

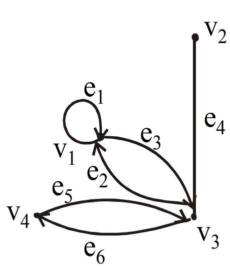
Let G be a graph with ordered vertices $v_1, v_2, \ldots, v_n$.

The adjacency matrix of G is the matrix $A = [a_{ij}]$ over the set of non-negative integers such that

a_{ij} = the number of arrows from v_i to v_j for all i, j = 1, 2, …, n.

EXAMPLE:

A directed graph with its adjacency matrix is shown



$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

is the adjacency matrix

EXERCISE:

Find directed graph that has the adjacency matrix

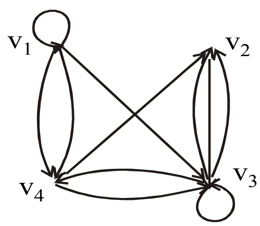
$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

SOLUTION:

The 4 × 4 adjacency matrix shows that the graph has 4 vertices say v_1, v_2, v_3 and v_4 labeled across the top and down the left side of the matrix.

$$A = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

A corresponding directed graph is



It means that a loop exists from v_1 and v_3 , two arrows go from v_1 to v_4 and two from v_3 and v_2 and one arrow go from $v_1$to v_3 , v_2 to v_3 , v_3 to v_4, v_4 to v_2 and v_3.

THEOREM

If G is a graph with vertices v_1, v_2, …,v_m and A is the adjacency matrix of G, then for each positive integer n,
the ijth entry of A^n = the number of walks of length n from v_i to v_j
for all integers i,j = 1, 2, …, n

PROBLEM:

Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$

be the adjacency matrix of a graph G with vertices v_1, v_2, and v_3. Find
(a) the number of walks of length 2 from v_2 to v_3
(b) the number of walks of length 3 from v_1 to v_3
Draw graph G and find the walks by visual inspection for (a)

SOLUTION:

(a)

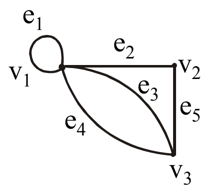
$$A^2 = AA = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 5 \end{bmatrix} \longrightarrow \text{it shows the entry (2,3) from } v_2 \text{ to } v_3$$

Hence, number of walks of length 2(means "multiply matrix A two times") from v_2 to v_3 = the entry at (2,3) of A^2 = 2

(b) $A^3 = AA^2 = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 6 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 15 & 9 & 15 \\ 9 & 5 & 8 \\ 15 & 8 & 8 \end{bmatrix}$ it shows the entry (1,3) from v_1 to v_3

Hence, number of walks of length 3 from v_1 to v_3 = the entry at (1,3) of $A^3 = 15$
Walks from v_2 to v_3 by visual inspection of graph is

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$



so in part (a)two Walks of length 2 from v_2 to v_3 are
(i) $v_2\ e_2\ v_1\ e_3\ v_3$ (by using the above theorem).
(ii) $v_2\ e_2\ v_1\ e_4\ v_3$

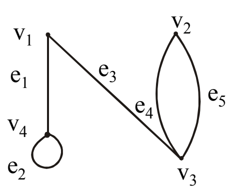
INCIDENCE MATRIX OF A SIMPLE GRAPH:

Let G be a graph with vertices $v_1, v_2, \ldots, v_n$ and edges $e_1, e_2, \ldots, e_n$. The incidence matrix of G is the matrix $M = [m_{ij}]$ of size n × m defined by

$$m_{ij} = \begin{cases} 1 & \text{if the vertex } v_i \text{ is incident on the edge } e_j \\ 0 & \text{otherwise} \end{cases}$$

EXAMPLE:

A graph with its incidence matrix is shown.



$$M = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

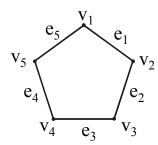
REMARK:

In the incidence matrix
1. Multiple edges are represented by columns with identical entries (in this matrix e_4 & e_5 are multiple edges).
2. Loops are represented using a column with exactly one entry equal to 1, corresponding to the vertex that is incident with this loop and other zeros (here e_2 is only a loop).

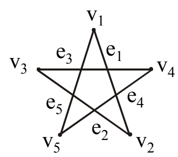
Lecture# 42 Isomorphism of graphs

ISOMORPHISM OF GRAPHS

Here we have a graph



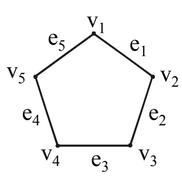
Which can also be defined as



Its vertices and edges can be written as:
$V(G) = \{v_1, v_2, v_3, v_4, v_5\}$, $E(G) = \{e_1, e_2, e_3, e_4, e_5\}$
Edge endpoint function is:

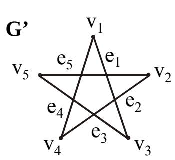
Edge	Endpoints
E_1	$\{v_1,v_2\}$
E_2	$\{v_2,v_3\}$
E_3	$\{v_3,v_4\}$
E_4	$\{v_4,v_5\}$
E_5	$\{v_5,v_1\}$

G



Another graph G' is

G'



Edge endpoint function of G is:

Edge	Endpoints
e_1	$\{v_1,v_2\}$
e_2	$\{v_2,v_3\}$
e_3	$\{v_3,v_4\}$
e_4	$\{v_4,v_5\}$
e_5	$\{v_5,v_1\}$

Edge endpoint function of G' is:

Edge	Endpoints
e_1	$\{v_1,v_3\}$
e_2	$\{v_2,v_4\}$
e_3	$\{v_3,v_5\}$
e_4	$\{v_1,v_4\}$
e_5	$\{v_2,v_5\}$

Two graphs (G and G') that are the same except for the labeling of their vertices are not considered different.

GRAPHS OF EDGE POINT FUNCTIONS

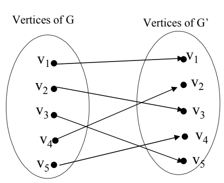
Edge point function of G is:

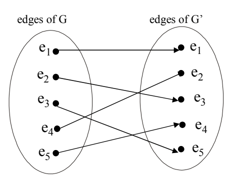
Edge	Endpoints
e_1	$\{v_1,v_2\}$
e_2	$\{v_2,v_3\}$
e_3	$\{v_3,v_4\}$
e_4	$\{v_4,v_5\}$
e_5	$\{v_5,v_1\}$

Edge point function of G' is:

Edge	Endpoints
e_1	$\{v_1,v_3\}$
e_2	$\{v_2,v_4\}$
e_3	$\{v_3,v_5\}$
e_4	$\{v_1,v_4\}$
e_5	$\{v_2,v_5\}$

Note it that the graphs G and G' are looking different because in G the end points of e_1 are v_1,v_2 but in G' are v_1, v_3 etc.
Buts G' is very similar to G ,if the vertices and edges of G' are relabeled by the function shown below, then G' becomes same as G:





It shows that if there is one-one correspondence between the vertices of G and G', then also one-one correspondence between the edges of G and G'.

ISOMORPHIC GRAPHS:
Let G and G' be graphs with vertex sets V(G) and V(G') and edge sets E(G) and E(G'), respectively.
G is isomorphic to G' if, and only if, there exist one-to-one correspondences g: $V(G) \to V(G')$ and h: $E(G) \to E(G')$ that preserve the edge-endpoint functions of G and G' in the sense that for all $v \in V(G)$ and $e \in E(G)$.
v is an endpoint of e $\Leftrightarrow$ g(v) is an endpoint of h(e).

EQUIVALENCE RELATION:
Graph isomorphism is an equivalence relation on the set of graphs.

1. Graphs isomorphism is Reflexive (It means that the graph should be isomorphic to itself).

2. Graphs isomorphism is Symmetric (It means that if G is isomorphic to G' then G' is also isomorphic to G).

3. Graphs isomorphism is Transitive (It means that if G is isomorphic to G' and G' is isomorphic to G'', then G is isomorphic to G'').

ISOMORPHIC INVARIANT:

A property P is called an isomorphic invariant if, and only if, given any graphs G and G', if G has property P and G' is isomorphic to G, then G' has property P.

THEOREM OF ISOMORPHIC INVARIANT:

Each of the following properties is an invariant for graph isomorphism, where n, m and k are all non-negative integers, if the graph:

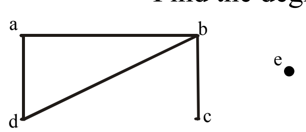
1. has n vertices.
2. has m edges.
3. has a vertex of degree k.
4. has m vertices of degree k.
5. has a circuit of length k.
6. has a simple circuit of length k.
7. has m simple circuits of length k.
8. is connected.
9. has an Euler circuit.
10. has a Hamiltonian circuit.

DEGREE SEQUENCE:

The degree sequence of a graph is the list of the degrees of its vertices in non-increasing order.

EXAMPLE:

Find the degree sequence of the following graph.



SOLUTION:

Degree of a = 2, Degree of b = 3, Degree of c = 1,
Degree of d = 2, Degree of e = 0

By definition, degree of the vertices of a given graph should be in decreasing (non-increasing) order.

Therefore Degree sequence is: 3, 2, 2, 1, 0

GRAPH ISOMORPHISM FOR SIMPLE GRAPHS:

If G and G' are simple graphs (means the "graphs which have no loops or parallel edges") then G is isomorphic to G' if, and only if, there exists a one-to-one correspondence (1-1 and onto function) g from the vertex set V (G) of G to the vertex set V (G') of G' that preserves the edge-endpoint functions of G and G' in the sense that for all vertices u and v of G,

{u, v} is an edge in G ⇔ {g(u), g(v)} is an edge in G'.

OR

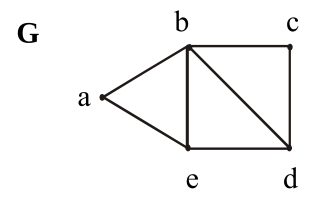
You can say that with the property of one-one correspondence, u and v are adjacent in graph G ⇔ if g (u) and g (v) are adjacent in G'.

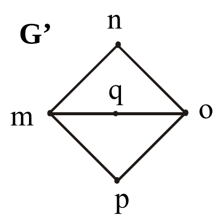
Note:

It should be noted that unfortunately, there is no efficient method for checking that whether two graphs are isomorphic(methods are there but take so much time in calculations).Despite that there is a simple condition. Two graphs are isomorphic if they have the same number of vertices(as there is a 1-1 correspondence between the vertices of both the graphs) and the same number of edges(also vertices should have the same degree.

EXERCISE:

Determine whether the graph G and G' given below are isomorphic.





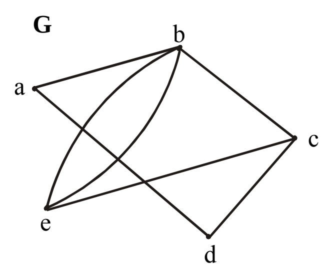
SOLUTION:

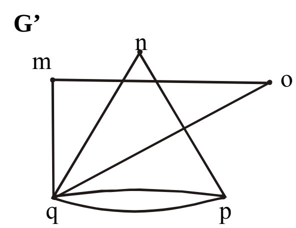
As both the graphs have the same number of vertices. But the graph G has 7 edges and the graph G' has only 6 edges. Therefore the two graphs are not isomorphic.

Note: As the edges of both the graphs G and G' are not same then how the one-one correspondence is possible ,that the reason the graphs G and G' are not isomorphic.

EXERCISE:

Determine whether the graph G and G' given below are isomorphic.



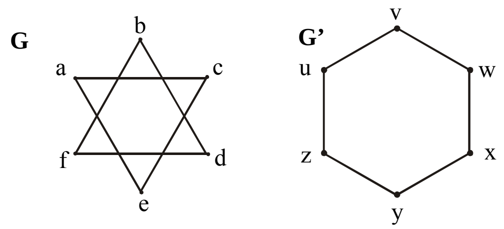


SOLUTION:

Both the graphs have 5 vertices and 7 edges. The vertex q of G' has degree 5. However G does not have any vertex of degree 5 (so one-one correspondence is not possible). Hence, the two graphs are not isomorphic.

EXERCISE:

Determine whether the graph G and G' given below are isomorphic.

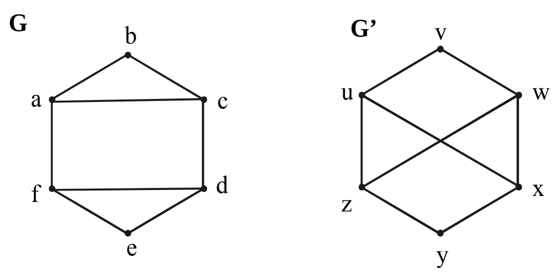


SOLUTION:

Clearly the vertices of both the graphs G and G' have the same degree (i.e "2") and having the same number of vertices and edges but isomorphism is not possible. As the graph G' is a connected graph but the graph G is not connected due to have two components (eca and bdf). Therefore the two graphs are non isomorphic.

EXERCISE:

Determine whether the graph G and G' given below are isomorphic.



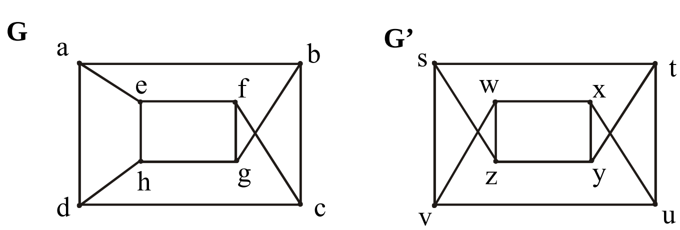
SOLUTION:

Clearly G has six vertices, G' also has six vertices. And the graph G has two simple circuits of length 3; one is abca and the other is defd. But G' does not have any simple circuit of length 3(as one simple circuit in G' is uxwv of length 4).Therefore the two graphs are non-isomorphic.

Note: A simple circuit is a circuit that does not have any other repeated vertex except the first and last.

EXERCISE:

Determine whether the graph G and G' given below are isomorphic.

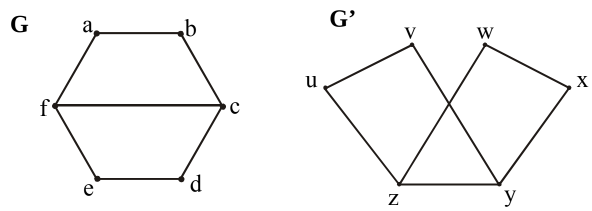


SOLUTION:

Both the graph G and G' have 8 vertices and 12 edges and both are also called regular graph(as each vertex has degree 3).The graph G has two simple circuits of length 5; abcfea(i.e starts and ends at a) and cdhgfc(i.e starts and ends at c). But G' does not have any simple circuit of length 5 (it has simple circuit tyxut,vwxuv of length 4 etc). Therefore the two graphs are non-isomorphic.

EXERCISE:

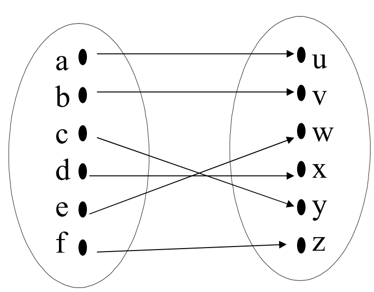
Determine whether the graph G and G' given below are isomorphic.



SOLUTION:

We note that all the isomorphism invariants seems to be true.
We shall prove that the graphs G and G' are isomorphic.
Here G has four vertices of degree "2" and two vertices of degree"3". Similar case in G'. Also G and G' have circuits of length 4.As a is adjacent to b and f in graph G.In graph G' u is adjacent to v and z. And as a and u has degree 2 so both are mapped. And b mapped with v, f mapped with z(as both have the same degree also a is adjacent to f and u is to z), and as we moves further we get the 1-1 correspondence.

Define a function f: V(G) →V(G') as follows.



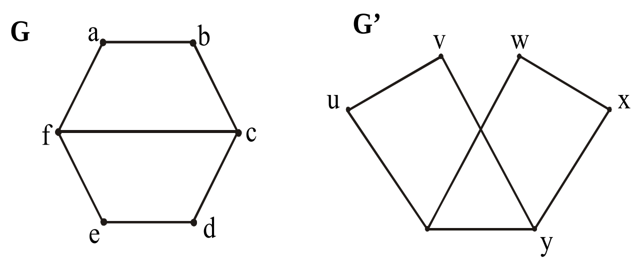
Clearly the above function is one and onto that is a bijective mapping. Note that I write the above mapping by keeping in mind the invariants of isomorphism as well as the fact that the mapping should preserve edge end point function. Also you should note that the mapping is not unique.

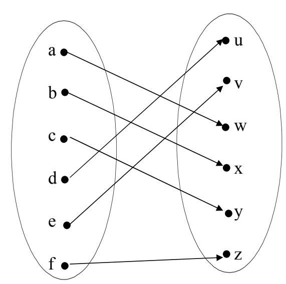
f is clearly a bijective function. The fact that f preserves the edge endpoint functions of G and G' is shown below.

Edges of G	Edges of G’
{a, b}	{u, v} = {g(a), g(b)}
{b, c}	{v, y} = {g(b), g(c)}
{c, d}	{y, x} = {g(c), g(d)}
{d, e}	{x, w} = {g(d), g(e)}
{e, f}	{w, z} = {g(e), g(f)}
{a, f}	{u, z} = {g(a), g(f)}
{c, f}	{y, z} = {g(c), g(f)}

ALTERNATIVE SOLUTION:

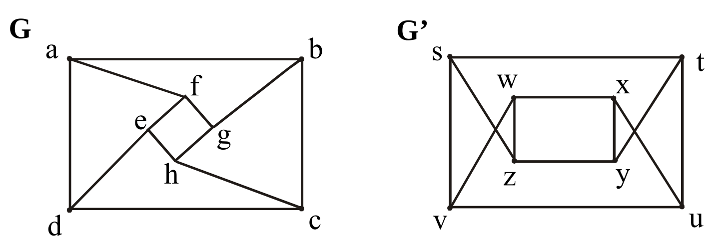
We shall prove that the graphs G and G’ are isomorphic.
Define a function f: V(G) →V(G’) as follows.





EXERCISE:

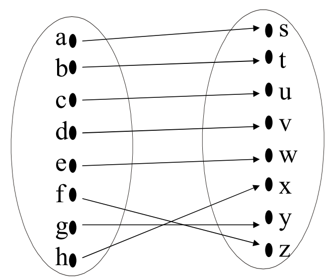
Determine whether the graph G and G’ given below are isomorphic.



SOLUTION:

We shall prove that the graphs G and G’ are isomorphic.
Clearly the isomorphism invariants seems to be true between G and G’.

Define a function f: V(G) →V(G') as follows.



f is clearly a bijective function(as it satisfies conditions the one-one and onto function clearly). The fact that f preserves the edge endpoint functions of G and G' is shown below.

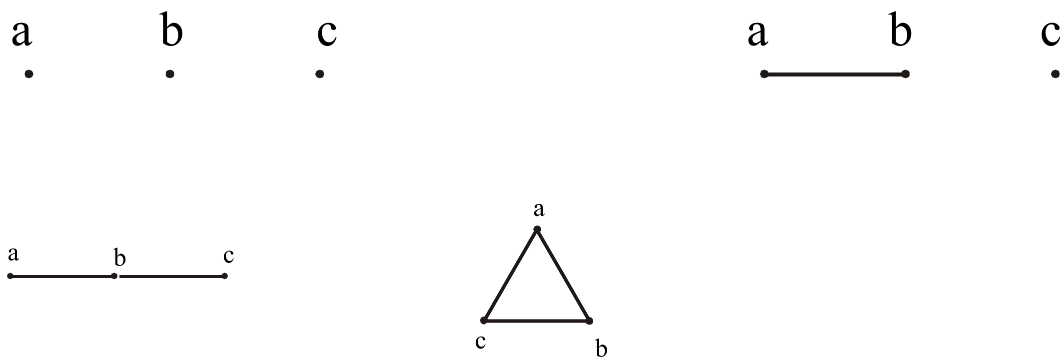
Edges of G	**Edges of G'**
{a, b}	{s, t} = {f(a), f(b)}
{b, c}	{t, u} = {f(b), f(c)}
{c, d}	{u, v} = {f(c), f(d)}
{a,d}	{s, v} = {f(a), f(d)}
{a, f}	{s, z} = {f(a), f(f)}
{b, g}	{t, y} = {f(b), f(g)}
{c, h}	{u, x} = {f(c), f(h)}
{d, e}	{v, w} = {f(d), f(e)}
{e, f}	{w, z} = {f(e), f(f)}
{f, g}	{z, y} = {f(f), f(g)}
{g, h}	{y, x} = {f(g), f(h)}
{h, e}	{x, w} = {f(h), f(e)}

EXERCISE:

Find all non isomorphic simple graphs with three vertices.

SOLUTION:

There are four simple graphs with three vertices as given below(which are non-isomorphic simple graphs).



EXERCISE:

Find all non isomorphic simple connected graphs with three vertices.

SOLUTION:

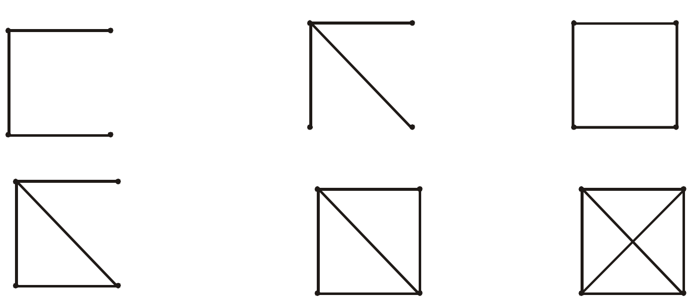
There are two simple connected graphs with three vertices as given below(which are non-isomorphic connected simple graphs).

EXERCISE:

Find all non isomorphic simple connected graphs with four vertices.

SOLUTION:

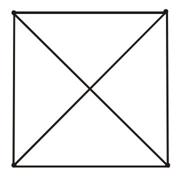
There are six simple connected graphs with four vertices as given below.



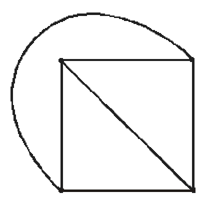
Lecture# 43 Planar graphs

PLANAR GRAPHS
GRAPH COLORING

In this lecture, we will study that whether any graph can be drawn in the plane (means "a flat surface") without crossing any edges.



It is a graph on 4 vertices and written as K_4. Each vertex is connected to every other vertex.
Note it that here edges are crossed. Also the above graph can also be drawn as



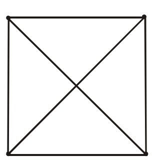
In this graph, note it that each vertex is connected to every other vertex, but no edge is crossed.

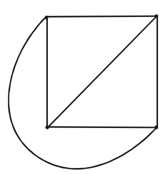
Note: The graphs shown above are complete graphs with four vertices (denoted by K_4).

DEFINITION:

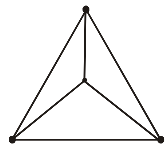
A graph is called planar if it can be drawn in the plane without any edge crossed (crossing means the intersection of lines). Such a drawing is called a plane drawing of the graph.
OR
You can say that a graph is called planar in which the graph crossing number is "0".

EXAMPLES:





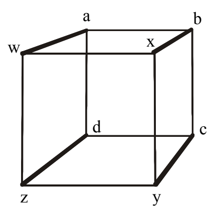
The graphs given above are planar .In the first figure edges are crossed but it can be redrawn in second figure where edges are not crossed, so called planar.



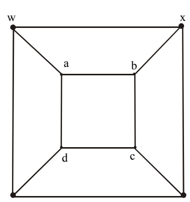
It is also a graph on 4 vertices (written as K_4) with no edge crossed, hence called planar.

Note: The graphs given above are also complete graphs (except second; are those where each vertex is connected to every other vertex) on 4 vertices and is written as K_4.
Note: Complete graphs are planar only for $n \leq 4$.

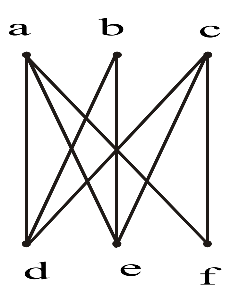
EXAMPLE:
Show that the graph below is planar.



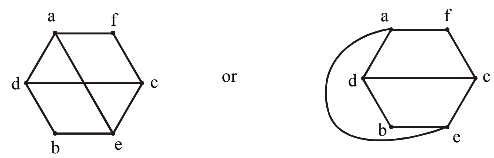
SOLUTION:
This graph has 8 vertices and 12 edges, and is called **3-cube** and is denoted **Q_3**.
The above representation includes many "edge crossing." A plane drawing of the graph in which no two edges cross is possible and shown below.



EXERCISE:
Determine whether the graph below is planar. If so, draw it so that no edges cross.



SOLUTION:
The graph given above is bipartite graph denoted by K_3. It also has a circuit afcebda. This graph can be re-drawn as

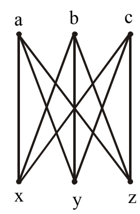


Hence the given graph is planar

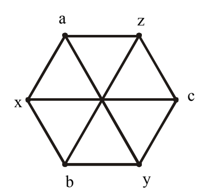
THEOREM:

Show that $K_{3,3}$ is not planar.

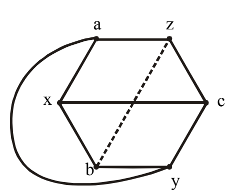
PROOF:



Clearly it is a complete bipartite graph (means bipartite graph, but the vertices within a set are not connected) denoted by $\mathbf{K_{3,3}}$. Now $\mathbf{K_{3,3}}$ can be re-drawn as



We re-draw the edge ***ay*** so that it does not cross any other edge like that.



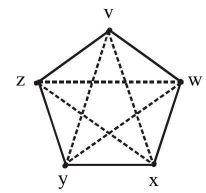
Note it that ***bz*** cannot be drawn without crossings. Hence, $K_{3,3}$ is not planar.
Similary if ay can be drawn inside(i.e drawn with crossing) and bz drawn outside, then same result exits.

THEOREM:

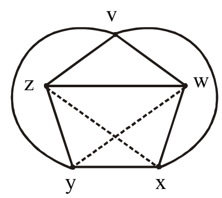
Show that K_5 is non-planar.

PROOF:

Graph K_5 (means a "complete graph" in which every vertex is connected to every other vertex) can be drawn as



To show that K_5 is non-planar, it can be re-drawn as

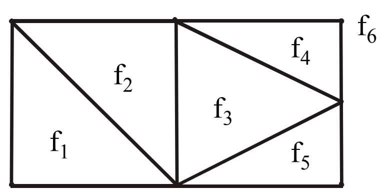


But still edges wy and zx contain the lines which crossed each other. Hence called non-planar.

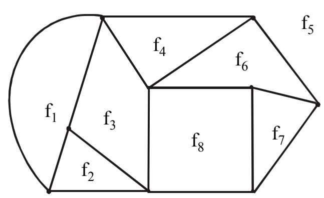
DEFINITION:

A plane drawing of a planar graph divides the plane into regions, including an unbounded region, called **faces**.

The unbounded region is called the **infinite face**.



Here we have 6 faces,7 vertices and 10 edges.f_6 is the unbounded region or called the infinite face because f_6 is outside of the graph.



In this graph, it has 8 faces,9 vertices and 14 edges. Here f_5 is the infinite face or unbounded region.

EULER'S FORMULA

THEOREM:

Let G be a connected planar simple graph with e edges and v vertices. Let f be the number of faces in a plane drawing of G. Then $\boldsymbol{f = e - v + 2}$

EXERCISE:

Suppose that a connected planar simple graph has 30 edges. If a plane drawing of this graph has 20 faces, how many vertices does this graph have?

SOLUTION:

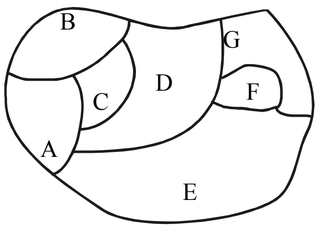
Given that $e = 30$, and $f = 20$. Substituting these values in the Euler's Formula $f = e - v + 2$, we get

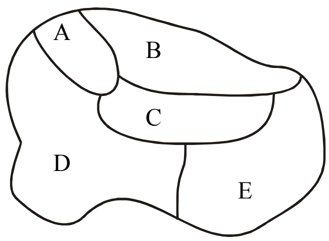
$$20 = 30 - v + 2$$

Hence,

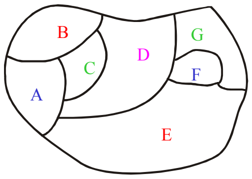
$$v = 30 - 20 + 2 = 12$$

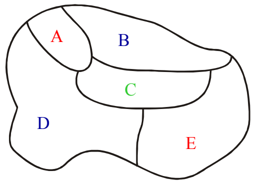
GRAPH COLORING





We also have to face many problems in the form of maps (maps like the parts of the world), which have generated many results in graph theory. Note it that in any graph, many regions are there, but two adjacent regions can't have the same color. And we have to choose a small number of color whenever possible.
Given two graphs above, our problem is to determine the least number of colors that can be used to color the map so that no adjacent regions have the same color.
In the first map given above, 4 colors are necessary, but three colors are not enough. In the second graph, 3 colors are necessary but 2 colors are not enough.



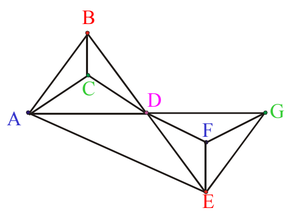


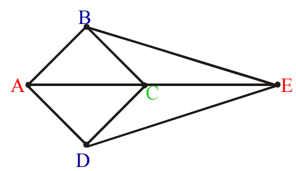
As in the 1^{st} graph, four colors (red, pink, green, blue) are used like that adjacent regions not have the same color. In 2^{nd} graph, three colors (red, blue, green) are used in the same manner.

HOW TO DRAW A GRAPH FROM A MAP:

1. Each map in the plane can be represented by a graph.
2. Each region is represented by a vertex (in 1^{st} map as there are 7 regions, so 7 vertices are used in drawing a graph, similarly we can see 2^{nd} map).

3. If the regions connected by these vertices have the common border, then edge connect two vertices.
4. Two regions that touch at only one point are not adjacent.
So apply these rules, we have (first graph drawn from first map given above, second graph from second map).

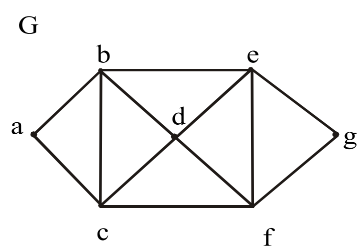


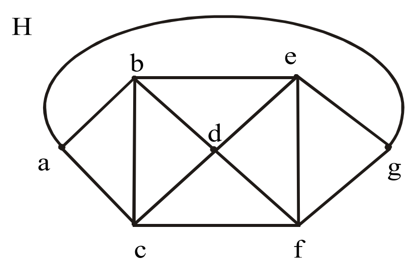


DEFINITION:

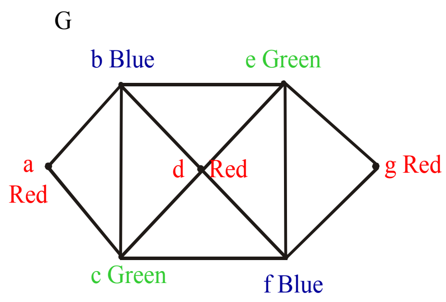
1. A **coloring** of a **simple graph** is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.
2. The **chromatic number** of a graph is the least (minimum) number of colors for coloring of this graph.

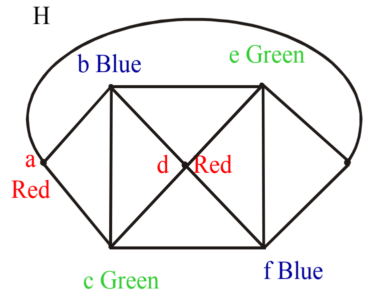
EXAMPLE:
What is the chromatic number of the graphs G and H shown below?





SOLUTION:
Clearly the chromatic number of G is 3 and chromatic number of H is 4(by using the above definition).
In graph G,
As vertices a, b and c are adjacent to each other so assigned different colors. So we assign red color to vertex a, blue to b and green to vertex c. Then no more colors we choose (due to above definition).Now vertex d must be colored red because it is adjacent to vertex b(with blue color) and c(with green color). And e must be colored green because it is adjacent to vertex b(blue color) and vertex d(red color). And f must be colored blue as it is adjacent to red and green color. At last,vertex g must be colored red as it is adjacent to green and blue color.
Same process is used in Graph H.



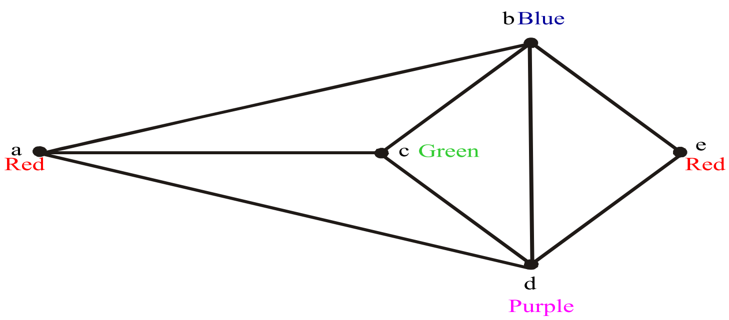


THE FOUR COLOR THEOREM:
The chromatic number of a simple planar graph is no greater than four.
APPLICATION OF GRAPH COLORING

EXAMPLE:
Suppose that a chemist wishes to store five chemicals *a, b, c, d* and *e* in various areas of a warehouse. Some of these chemicals react violently when in contact, and so must be kept in separate areas. In the following table, an asterisk indicates those pairs of chemicals that must be separated. How many areas are needed?

	a	*b*	*c*	*d*	*e*
a	–	*	*	*	–
b	*	–	*	*	*
c	*	*	–	*	–
d	*	*	–	–	*
e	–	*	*	*	–

SOLUTION:
We draw a graph whose vertices correspond to the five chemicals, with two vertices adjacent whenever the corresponding chemicals are to be kept apart.



Clearly the chromatic number is 4 and so four areas are needed.

Lecture# 44   Trees

TREES

<u>APPLICATION AREAS:</u>

Trees are used to solve problems in a wide variety of disciplines. In computer science trees are employed to

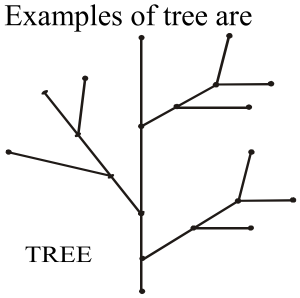
1) construct efficient algorithms for locating items in a list.
2) construct networks with the least expensive set of telephone lines linking distributed computers.
3) construct efficient codes for storing and transmitting data.
4) model procedures that are carried out using a sequence of decisions, which are valuable in the study of sorting algorithms.

<u>TREE:</u>

A tree is a connected graph that does not contain any non-trivial circuit. (i.e. it is circuit-free).

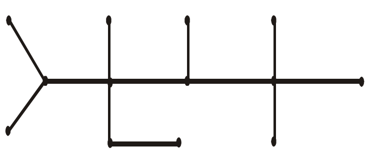
A trivial circuit is one that consists of a single vertex.

Examples of tree are



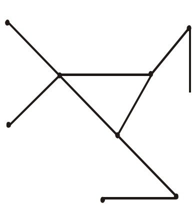
TREE

TREE

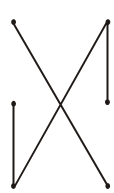


TREE

EXAMPLES OF NON TREES



(a) Graph with a circuit



(b) Disconnected graph

(c) Graph with a circuit

In graph (a), there exists circuit, so not a tree.
In graph (b), there exists no connectedness, so not a tree.
In graph (c), there exists a circuit (also due to loop), so not a tree (because trees have to be a circuit free).

SOME SPECIAL TREES

1. TRIVIAL TREE:

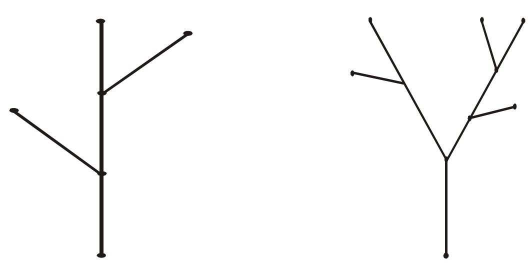
A graph that consists of a single vertex is called a trivial tree or degenerate tree.

2. EMPTY TREE

A tree that does not have any vertices or edges is called an empty tree.

3. FOREST

A graph is called a forest if, and only if, it is circuit-free.
OR "Any non-connected graph that contains no circuit is called a forest."
Hence, it clears that the connected components of a forest are trees.



A forest

As in both the graphs above, there exists no circuit, so called forest.

PROPERTIES OF TREES:

1. A tree with n vertices has n - 1 edges (where n ≥0).
2. Any connected graph with n vertices and n - 1 edges is a tree.
3. A tree has no non-trivial circuit; but if one new edge (but no new vertex) is added to it, then the resulting graph has exactly one non-trivial circuit.
4. A tree is connected, but if any edge is deleted from it, then the resulting graph is not connected.

5. Any tree that has more than one vertex has at least two vertices of degree 1.
6. A graph is a tree iff there is a unique path between any two of its vertices.

EXERCISE:

Explain why graphs with the given specification do not exist.
1. Tree, twelve vertices, fifteen edges.
2. Tree, five vertices, total degree 10.

SOLUTION:

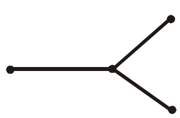
1. Any tree with 12 vertices will have 12 - 1 = 11 edges, not 15.
2. Any tree with 5 vertices will have 5 - 1 = 4 edges.
Since, total degree of graph = 2 (No. of edges)
= 2(4) = 8
Hence, a tree with 5 vertices would have a total degree 8, not 10.

EXERCISE:

Find all non-isomorphic trees with four vertices.

SOLUTION:

Any tree with four vertices has (4-1=3) three edges. Thus, the total degree of a tree with 4 vertices must be 6 [by using total degree=2(total number of edges)].
Also, every tree with more than one vertex has at least two vertices of degree 1, so the only possible combinations of degrees for the vertices of the trees are 1, 1, 1, 3 and 1, 1, 2, 2.
The corresponding trees (clearly non-isomorphic, by definition) are



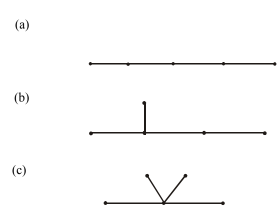
and

EXERCISE:

Find all non-isomorphic trees with five vertices.

SOLUTION:

There are three non-isomorphic trees with five vertices as shown (where every tree with five vertices has 5-1=4 edges).



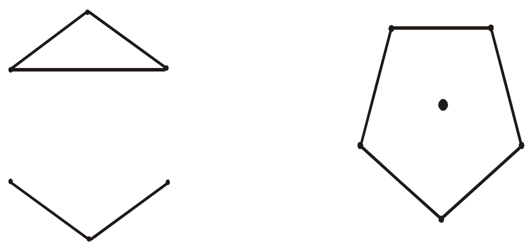
In part (a), tree has 2 vertices of degree '1' and 3 vertices of degree '2'.
In part (b), 3 vertices have degree '1', 1 has degree '2'and 1 vertex has degree '3'.
In part (c), possible combinations of degree are 1, 1, 1, 1, 4.

EXERCISE:

Draw a graph with six vertices, five edges that is not a tree.

SOLUTION:

Two such graphs are:



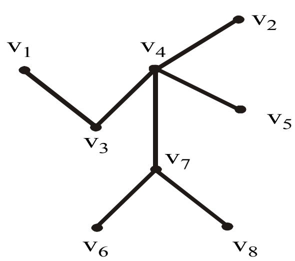
First graph is not a tree; because it is not connected also there exists a circuit.
Similarly, second graph not a tree.

DEFINITION:

A vertex of degree 1 in a tree is called a ***terminal vertex*** or a leaf and a vertex of degree greater than 1 in a tree is called an ***internal vertex*** or a branch vertex.

EXAMPLE:

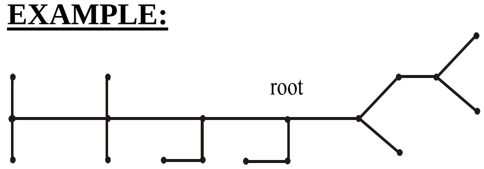
The terminal vertices of the tree are v_1, v_2, v_5, v_6 and v_8 and internal vertices are v_3, v_4, v_7.



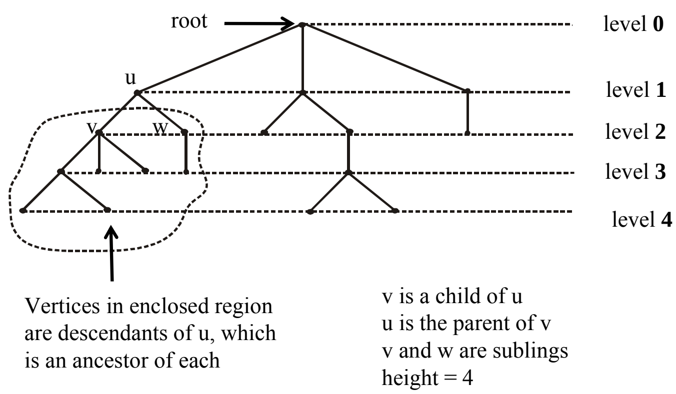
ROOTED TREE:

A **rooted tree** is a tree in which one vertex is distinguished from the others and is called the ***root.***
The **level** of a vertex is the number of edges along the unique path between it and the root.
The **height** of a rooted tree is the maximum level to any vertex of the tree.
The **children** of any internal vertex v are all those vertices that are adjacent to v and are one level farther away from the root than v.
If **w** is a **child** of **v**, then **v** is called the **parent** of **w**.
Two vertices that are both children of the same parent are called **siblings**.
Given vertices v and w, if v lies on the unique path between w and the root, then v is an **ancestor** of w and w is a **descendant** of v.

EXAMPLE:



We redraw the tree as and see what the relations are



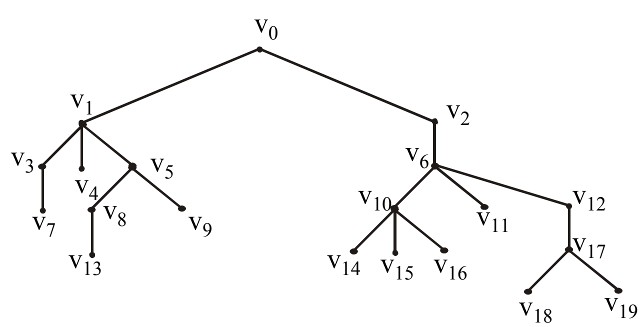
EXERCISE:

Consider the rooted tree shown below with root v_0

a. What is the level of v_8? b. What is the level of v_0?

c. What is the height of this tree? d. What are the children of v_{10}?

e. What are the siblings of v_1? f. What are the descendants of v_{12}?



SOLUTION:

As we know that "Level means

the total number of edges along the unique path between it and the root".

(a). As v_0 is the root so the level of v_8 (from the root v_0 along the unique path) is 3,because it covers the 3 edges.

(b).The level of v_0 is 0(as no edge cover from v_0 to v_0).

(c).The height of this tree is 5.

Note: As levels are 0, 1, 2, 3, 4, 5 but to find height we have to take the maximum level.

(d).The children of v_{10} are v_{14}, v_{15} and v_{16} .

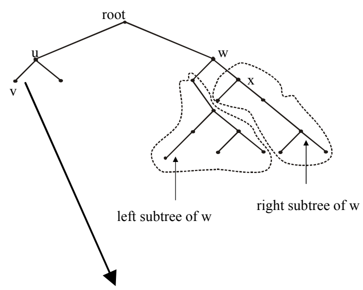
(e).The siblings of v_1 are v_3 , v_4 , and v_5.

(f).The descendants of v_{12} are v_{17}, v_{18}, and v_{19}.

BINARY TREE

A ***binary tree*** is a rooted tree in which every internal vertex has at most two children.
Every child in a binary tree is designated either a left child or a right child (but not both).
A ***full binary tree*** is a binary tree in which each internal vertex has exactly two children.

EXAMPLE:



v is the left child of u.

THEOREMS:

1. If k is a positive integer and T is a full binary tree with k internal vertices, then T has a total of 2k + 1 vertices and has k + 1 terminal vertices.

2. If T is a binary tree that has t terminal vertices and height h, then $t \leq 2^h$
Equivalently,

$$\log_2 t \leq h$$

Note: The maximum number of terminal vertices of a binary tree of height h is 2^h.

EXERCISE:

Explain why graphs with the given specification do not exist.
1. full binary tree, nine vertices, five internal vertices.
2. binary tree, height 4, eighteen terminal vertices.

SOLUTION:

1. Any full binary tree with five internal vertices has six terminal vertices, for a total of eleven vertices (according to 2(5) +1=11), not nine vertices in all.

OR

As total vertices = 2k+1=9
k=4(internal vertices)
but given internal vertices=5 ,which is a contradiction.
Thus there is no full binary tree with the given properties.

2. Any binary tree of height 4 has at most 2^4 = 16 terminal vertices.
Hence, there is no binary tree that has height 4 and eighteen terminal vertices.

EXERCISE:

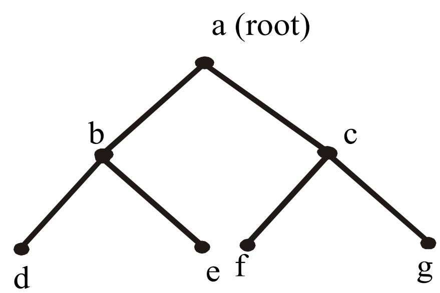
Draw a full binary tree with seven vertices.

SOLUTION:

Total vertices= 2k + 1 = 7 (by using the above theorem)

$\Rightarrow \quad k = 3$

Hence, total number of internal vertices (i.e. a vertex of degree greater than 1)=k=3
and total number of terminal vertices(i.e. a vertex of degree 1 in a tree) =k+1=3+1=4
Hence, a full binary tree with seven vertices is

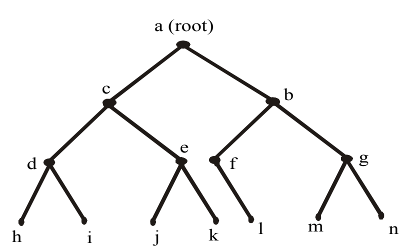


EXERCISE:

Draw a binary tree with height 3 and having seven terminal vertices.

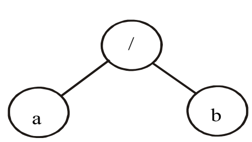
SOLUTION:

Given height=h=3
Any binary tree with height 3 has atmost 2^3=8 terminal vertices.
But here terminal vertices are 7
and Internal vertices=k=6 so binary tree exists and is as fellows:



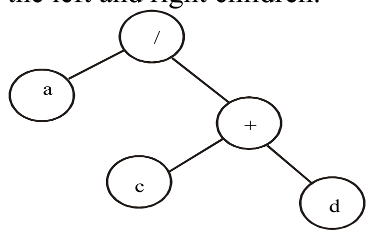
REPRESENTATION OF ALGEBRAIC EXPRESSIONS BY BINARY TREES

Binary trees are specially used in computer science to represent algebraic expression with Arbitrary nesting of balanced parentheses.



Binary tree for a/b

The above figure represents the expression a/b. Here the operator (/) is the root and b are the left and right children.



Binary tree for a/(c+d)

The second figure represents the expression a/(c+d).Here the operator (/) is the root. Here the terminal vertices are variables (here a, c and d), and the internal vertices are arithmetic operators (+ and /).

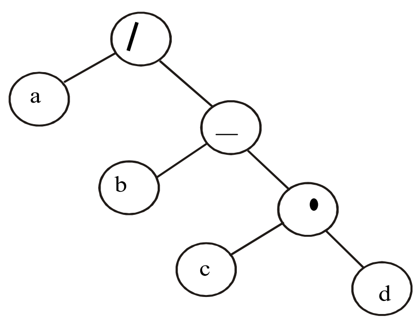
EXERCISE:

Draw a binary tree to represent the following expression

a/(b-c.d)

SOLUTION:

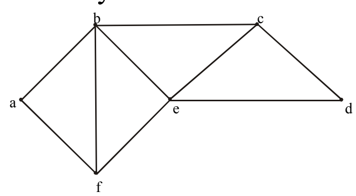
Note that the internal vertices are arithmetic operators, the terminal vertices are variables and the operator at each vertex acts on its left and right sub trees in left-right order.



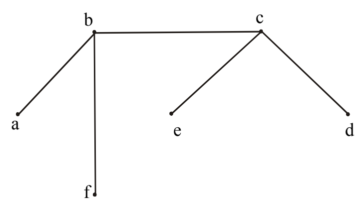
Lecture# 45 Spanning Trees

SPANNING TREES:

Suppose it is required to develop a system of roads between six major cities.
A survey of the area revealed that only the roads shown in the graph could be constructed.



For economic reasons, it is desired to construct the least possible number of roads to connect the six cities.
One such set of roads is



Note that the subgraph representing these roads is a tree, it is connected & circuit-free (six vertices and five edges)

SPANNING TREE:

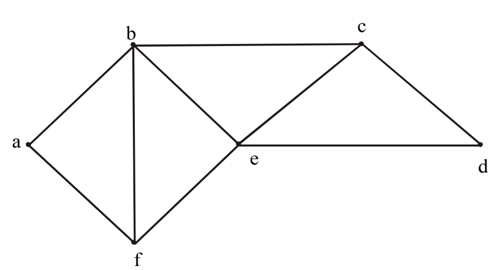
A spanning tree for a graph G is a subgraph of G that contains every vertex of G and is a tree.

REMARK:

1. Every connected graph has a spanning tree.
2. A graph may have more than one spanning trees.
3. Any two spanning trees for a graph have the same number of edges.
4. If a graph is a tree, then its only spanning tree is itself.

EXERCISE:

Find a spanning tree for the graph below:



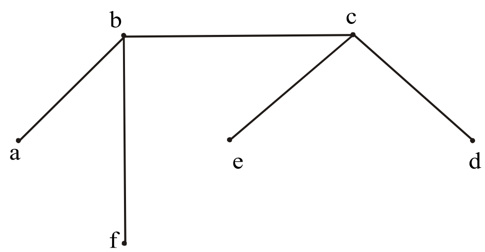
SOLUTION:

The graph has 6 vertices (a, b, c, d, e, f) & 9 edges so we must delete 9 - 6 + 1 = 4 edges (as we have studied in lecture 44 that a tree of vertices n has n-1 edges). We delete an edge in each cycle.

1. Delete af 2. Delete fe
3. Delete be 4. Delete ed

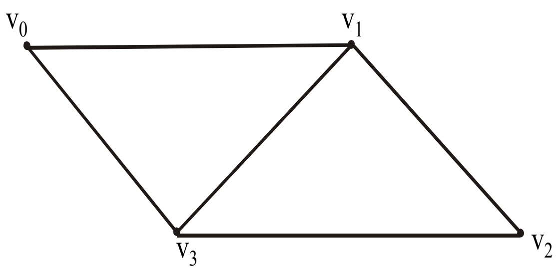
Note it that we can construct road from vertex a to b, but can't go from "a to e", also from "a to d" and from "a to c ", because there is no path available.

The associated spanning tree is



EXERCISE:

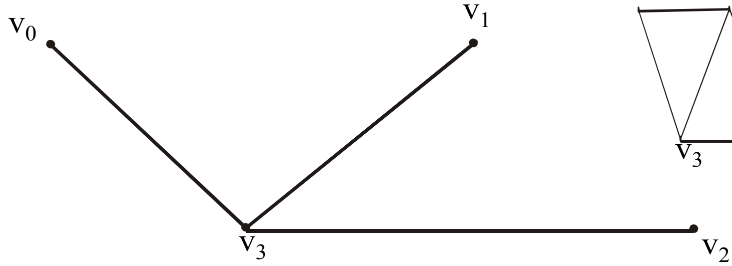
Find all the spanning trees of the graph given below.

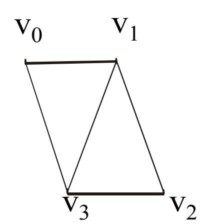


SOLUTION:

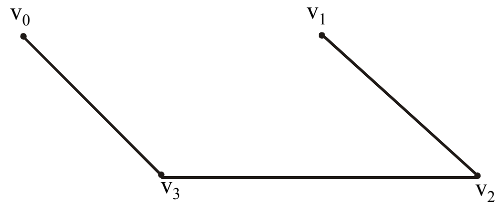
The graph has n = 4 vertices and e = 5 edges. So we must delete e - v + 1 = 5 - 4 + 1 = 2 edges from the cycles in the graph to obtain a spanning tree.

(1) Delete v_0v_1 & v_1v_2 to get

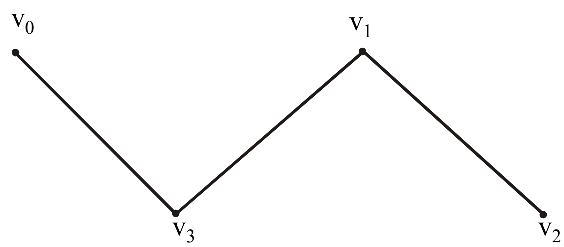


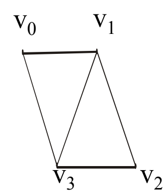


(2) Delete v_0v_1 & v_1v_3 to get

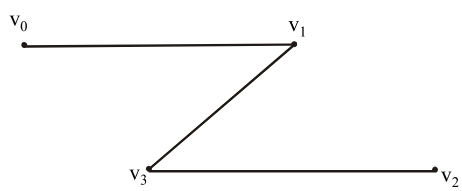


(3) Delete v_0v_1 & v_2v_3 to get

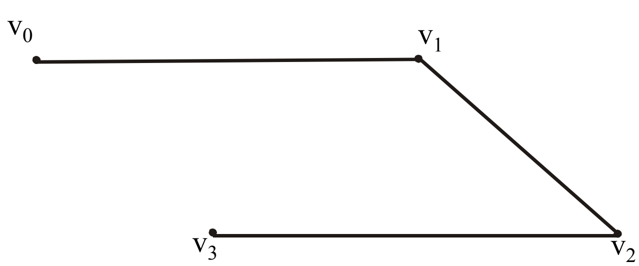


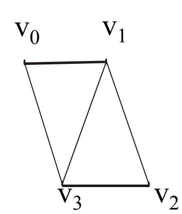


(4) Delete v_0v_3 & v_1v_2 to get

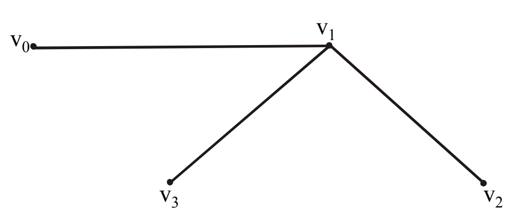


(5) Delete v_0v_3 & v_1v_3 to get

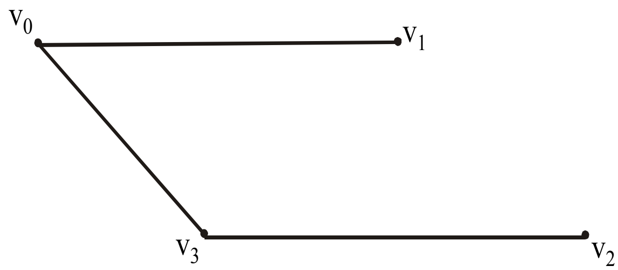


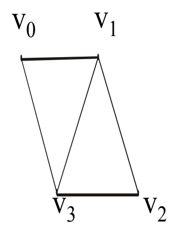


(6) Delete v_0v_3 & v_2v_3 to get

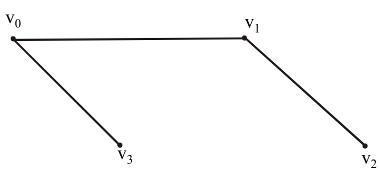


(7) Delete v_1v_3 & v_1v_2 to get





(8) Delete v_1v_3 & v_2v_3 to get



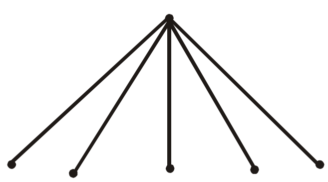
EXERCISE:

Find a spanning tree for each of the following graphs.

(a) $k_{1,5}$ (b) k_4

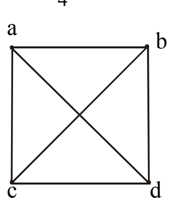
SOLUTION:

(a). $k_{1,5}$ represents a complete bipartite graph on (1,5) vertices, drawn below:



Clearly the graph itself is a tree (six vertices and five edges). Hence the graph is itself a spanning tree.

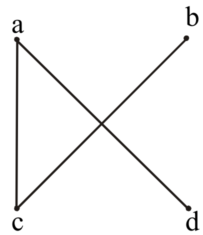
(b) k_4 represents a complete graph on four vertices.



Now
number of vertices = n = 4 and number of edges = e = 6
Hence we must remove

$$e - v + 1 = 6 - 4 + 1 = 3$$

edges to obtain a spanning tree.
Let ab, bd & cd edges are removed. The associated spanning tree is

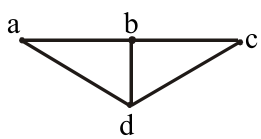


KIRCHHOFF'S THEOREM
OR MATRIX - TREE THEOREM

Let M be the matrix obtained from the adjacency matrix of a connected graph G by changing all 1's to -1's and replacing each diagonal 0 by the degree of the corresponding vertex. Then the number of spanning trees of G is equal to the value of any cofactor of M.

EXAMPLE:

Find the number of spanning trees of the graph G.



SOLUTION:

The adjacency matrix of G is

$$A(G)=\begin{array}{c c}
 & \begin{array}{cccc} a & b & c & d \end{array} \\
\begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}
\end{array}$$

The matrix specified in Kirchhoff's theorem is

$$M=\begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

Now cofactor of the element at (1,1) in M is

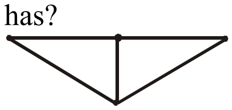
$$\begin{vmatrix} 3 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 3 \end{vmatrix}$$

Expanding by first row, we get

$$
\begin{aligned}
&= 3\begin{vmatrix} 2 & -1 \\ -1 & 3 \end{vmatrix} - (-1)\begin{vmatrix} -1 & -1 \\ -1 & 3 \end{vmatrix} + (-1)\begin{vmatrix} -1 & 2 \\ -1 & -1 \end{vmatrix} \\
&= 3(6-1) + (-3-1) + (-1)(1+2) \\
&= 15 - 4 - 3 = 8
\end{aligned}
$$

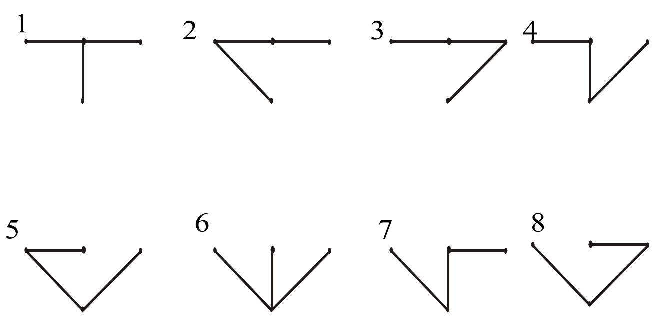
EXERCISE:

How many non-isomorphic spanning trees does the following simple graph has?



SOLUTION:

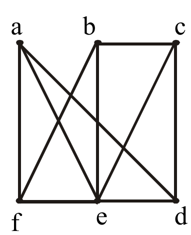
There are eight spanning tree of the graph



Clearly 1 & 6 are isomorphic, and 2, 3, 4, 5, 7, 8 are isomorphic. Hence there are only two non-isomorphic spanning trees of the given graph.

EXERCISE:

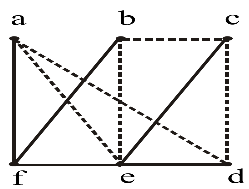
Suppose an oil company wants to build a series of pipelines between six storage facilities in order to be able to move oil from one storage facility to any of the other five. For environmental reasons it is not possible to build a pipeline between some pairs of storage facilities. The possible pipelines that can be build are.



Because the construction of a pipeline is very expensive, construct as few pipelines as possible.
(The company does not mind if oil has to be routed through one or more intermediate facilities)

SOLUTION:

The task is to find a set of edges which together with the incident vertices from a connected graph containing all the vertices and having no cycles. This will allow oil to go from any storage facility to any other without unnecessary building costs. Thus, a tree containing all the vertices of the graph is to be soughed. One selection of edges is



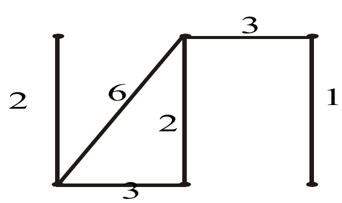
DEFINITION:

A WEIGHTED GRAPH is a graph for which each edge has an associated real number weight.

The sum of the weights of all the edges is the total weight of the graph.

EXAMPLE:

The figure shows a weighted graph



with total weight is $2 + 6 + 3 + 2 + 3 + 1 = 17$

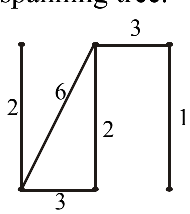
MINIMAL SPANNING TREE:

A minimal spanning tree for a weighted graph is a spanning tree that has the least possible total weight compared to all other spanning trees of the graph.

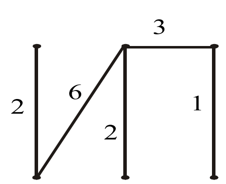
If G is a weighted graph and e is an edge of G then w(e) denotes the weight of e and w(G) denotes the total weight of G.

EXERCISE:

Find the three spanning trees of the weighted graph below. Also indicate the minimal spanning tree.

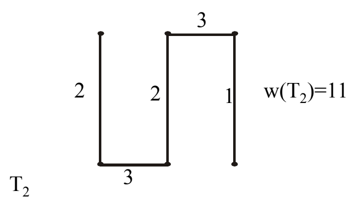


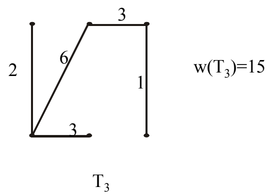
SOLUTION:



T_1

$w(T_1)=14$





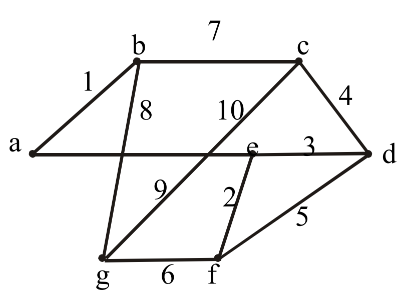
T_2 is the minimal spanning tree, since it has the minimum weight among the spanning trees.

KRUSKAL'S ALGORITHM:

Input: G [a weighted graph with n vertices]
Algorithm:
1. Initialize T(the minimal spanning tree of G) to have all the vertices of G and no edges.
2. Let E be the set of all edges of G and let m: = 0.
3. While (m < n - 1)
3a. Find an edge e in E of least weight.
3b. Delete e from E.
3c. If addition of e to the edge set of T does not produce a circuit then add e to the edge set of T and set m: = m + 1
end while
Output T
end Algorithm

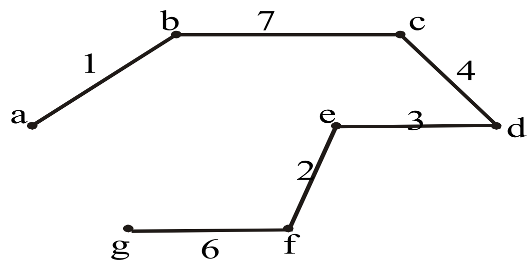
EXERCISE:

Use Kruskal's algorithm to find a minimal spanning tree for the graph below. Indicate the order in which edges are added to form the tree.



SOLUTION:

Minimal spanning tree:



Order of adding the edges:
{a, b}, {e, f}, {e, d}, {c, d}, {g, f}, {b, c}

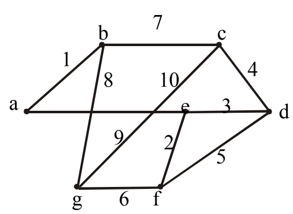
PRIM'S ALGORITHM:

Input: G[a weighted graph with n vertices]
Algorithm Body:
1. Pick a vertex v of G and let T be the graph with one vertex, v, and no edges.
2. Let V be the set of all vertices of G except v
3. for i: = 1 to n - 1
3a. Find an edge e of G such that
(1) e connects T to one of the vertices in V and
(2)e has the least weight of all edges connecting T to a vertex in V.
Let w be the end point of e that is in V.
3b. Add e and w to the edge and vertex sets of T and delete w from V.

next i
Output: T
end Algorithm

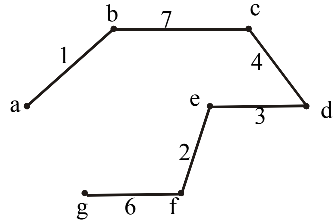
EXERCISE:

Use Prim's algorithm starting with vertex ***a*** to find a minimal spanning tree of the graph below. Indicate the order in which edges are added to form the tree.



SOLUTION:

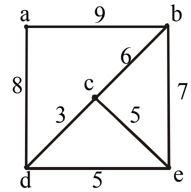
Minimal spanning tree is



Order of adding the edges:
{a, b}, {b, c}, {c, d}, {d, e}, {e, f}, {f, g}

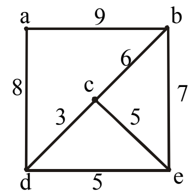
EXERCISE:

Find all minimal spanning trees that can be obtained using
(a) Kruskal's algorithm
(b) Prim's algorithm starting with vertex a



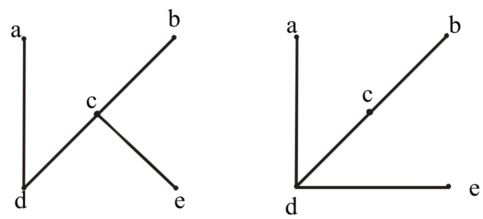
SOLUTION:

Given :

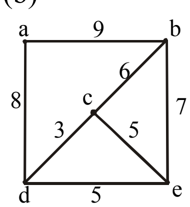


(a) When Kruskal's algorithm is applied, edges are added in one of the following two orders:
1. {c, d}, {c, e}, {c, b}, {d, a}
2. {c, d}, {d, e}, {c, b}, {d, a}

Thus, there are two distinct minimal spanning trees:

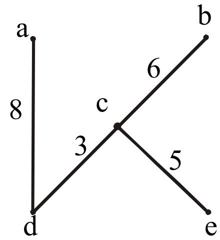


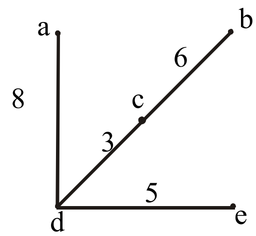
(b)



When Prim's algorithm is applied starting at a, edges are added in one of the following two orders:
1. {a, d}, {d, c}, {c, e}, {c, b}
2. {a, d}, {d, c}, {d, e}, {c, b}

Thus, the two distinct minimal spanning trees are:


a
b
c
d
e
8
6
3
5


a
b
c
d
e
8
6
3
5

Solution File Lecture No 1-3

Question No 1:

Let p be “I am intelligent” and let q be “I have a book”, Give a simple verbal sentence which describes each of the following statement:

(a) □ p
(b) $p \vee q$
(c) $q \wedge$ □ p

Solution:

(a) I am not intelligent
(b) I am intelligent or I have a book
(c) I have a book but I am not intelligent

Question No 2:

Construct a truth table for the compound proposition $[p \wedge (\sim p \rightarrow q)] \rightarrow q$

Solution:

p	**q**	**~p**	**~p →q**	**P^(~p→q)**	$[p \wedge (\sim p \rightarrow q)] \rightarrow q$
T	T	F	T	T	T
T	F	F	T	T	F
F	T	T	T	F	T
F	F	T	F	F	T

Question No 3:

Convert into logical form and then write converse, inverse and contra positive of the following statement.

“If I study, then I shall pass the test”.

Solution:

Inverse: If I do not study, then I shall not pass the test.

Converse: If I shall pass the test then Istudy.

Contrapositive: if I shall not pass the test then I do not study.

Question No 4:

Let p = "Ahmad eats apple", q = "Ahmad eats banana" and r = "Ahmad eats grapes". Write each of the following in symbolic form:

a) Ahmad eats apple or banana but not grapes.
b) It is not true that Ahmad eats apple but not grapes.
c) It is not true that Ahmad eats apples or bananas but not grapes.

Solution:

i) $(p \vee q) \wedge \sim r$

ii) $\sim (p \wedge \sim r)$

iii) $\sim ((p \vee q) \wedge \sim r)$

Question No 5:

Assume that for the truth values p = F and q = T. Show that the proposition $(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$ is true.

Solution:

p	q	~p	~q	$(p \wedge q)$	$(p \wedge \sim q)$	$\sim p \vee (p \wedge \sim q)$	$(p \wedge q) \vee (\sim p \vee (p \wedge \sim q))$
F	T	T	F	F	F	T	T

Hence the given proposition is true.

Question No 6:

Use truth table to verify the statement (how to prove it using laws)

$q \vee (p \wedge q) \equiv q \wedge (p \rightarrow q)$

Solution:

p	**q**	$(p\wedge q)$	$q \vee (p \wedge q)$	$(p \rightarrow q)$	$q\wedge (p \rightarrow q)$
T	T	T	T	T	T
T	F	F	F	F	F
F	T	F	T	T	T
F	F	F	F	T	F

Therefore from column 4&6, $q \vee (p \wedge q) \equiv q\wedge (p \rightarrow q)$.

Question No 7:

Do it by yourself

Question No 8:

Use De Morgan's laws to find the negation of

"Ahmad loves football and his brother loves cricket"

Solution:

Let p=Ahmed loves football

q=His brother loves cricket

$p \wedge q$ = Ahmed loves football and His brother loves cricket

Using De Morgan's law

$\sim (p \wedge q) = \sim p \vee \sim q$ =Ahmed does not love football or his brother does not love cricket

Question No 9:

Do it by yourself.

Lecture 4-6

Question # 1: Formulate the symbolic expression $\sim(p \vee q) \rightarrow r$ in words using

p =: Today is Holiday
q =: It is raining
r =: We should study.

Solution:

If it is not true that today is holiday or it is raining, then we should study.

Question # 2: Check the validity of the following argument form using truth table:

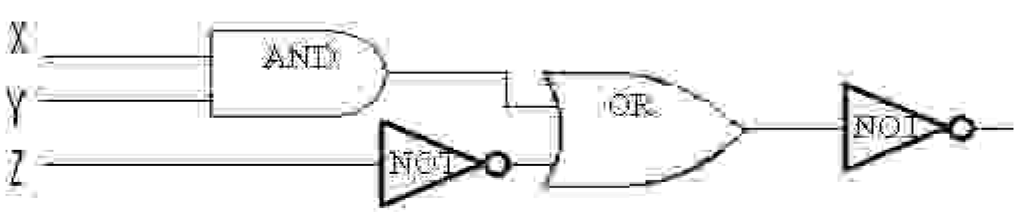
$$\begin{array}{l} p \rightarrow q \\ \underline{p \vee q} \\ \therefore p \end{array}$$

Solution:

p	q	$p \rightarrow q$	$p \vee q$	p
F	F	T	F	F
F	T	T	T	F
T	F	F	T	T
T	T	T	T	T

Row 2 and 4 are critical rows, in second row conclusion is false so it is an invalid argument.

Question # 3: Write the output of each gate in the following circuit



Solution:

X	Y	Z	$X \wedge Y$	$\sim Z$	$(X \wedge Y) \vee \sim Z$	$\sim((X \wedge Y) \vee \sim Z)$
0	0	0	0	1	1	0
0	0	1	0	0	0	1
0	1	0	0	1	1	0
0	1	1	0	0	0	1
1	0	0	0	1	1	0
1	0	1	0	0	0	1
1	1	0	1	1	1	0
1	1	1	1	0	1	0

Question # 4: Construct a truth table to determine whether the following statement is logically equivalent or not:

$(p \wedge q) \vee r$ and $p \wedge (q \vee r)$.

Solution:

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$	$q \vee r$	$p \wedge (q \vee r)$
F	F	F	F	F	F	F
F	F	T	F	T	T	F
F	T	F	F	F	T	F
F	T	T	F	T	T	F
T	F	F	F	F	F	F
T	F	T	F	T	T	T
T	T	F	T	T	T	T
T	T	T	T	T	T	T

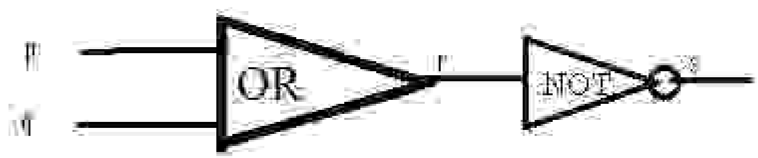
Columns 5 and 7 tells that they are not equivalent

Question # 5: By using the laws of logic verify the following logical equivalence: $\sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$.

Solution:

$$
\begin{aligned}
\sim\big((\sim p \wedge q) \vee (\sim p \wedge \sim q)\big) \vee (p \wedge q) &\equiv \sim\big(\sim p \wedge (q \vee \sim q)\big) \vee (p \wedge q) && \textit{Distributive Law} \\
&\equiv \sim(\sim p \wedge t) \vee (p \wedge q) && \textit{Negation Law} \\
&\equiv \sim(\sim p) \vee (p \wedge q) && \textit{Identity Law} \\
&\equiv p \vee (p \wedge q) && \textit{Double Negation Law} \\
&\equiv p && \textit{Absorption Law}
\end{aligned}
$$

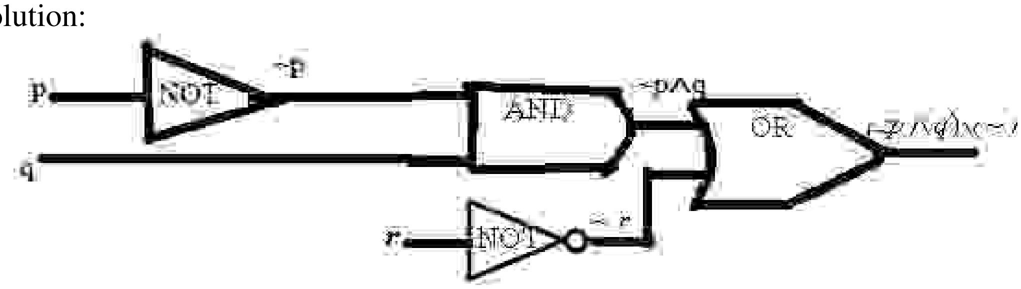
Question # 6: Construct the input/output table for ~ $(p \vee q)$



Solution:

Inputs			Output
p	q	$(p \vee q) = r$	$\sim (p \vee q) = s$
1	1	1	0
1	0	1	0
0	1	1	0
0	0	0	1

Question # 7: Construct circuit for the Boolean expression $(\sim p \wedge q) \vee \sim r$.
Solution:



Solution File of Practice Questions Lecture No 7-9

Solution No 1:

$$U = \{a, b, c, d, e, f, g\}$$
$$A = \{a, c, e, g\}$$
$$B = \{d, e, f, g\}$$
$$AUB = \{a, c, e, g\} U \{d, e, f, g\} = \{a, c, d, e, f, g\}$$
$$A \cap B = \{a, c, e, g\} \cap \{d, e, f, g\} = \{e, g\}$$
$$B - A = \{d, e, f, g\} - \{a, c, e, g\} = \{d, f\}$$
$$A^c = U - A = \{a, b, c, d, e, f, g\} - \{a, c, e, g\} = \{b, d, f\}$$

Solution No 2:

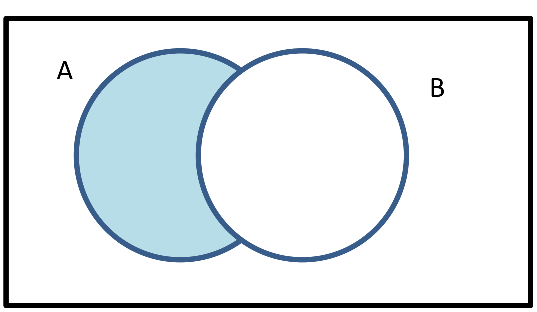
$$A = \{x \in Z \mid 0 < x \leq 2\}$$
$$B = \{x \in Z \mid 1 \leq x < 4\}$$
$$C = \{x \in Z \mid 3 \leq x < 9\}$$

$$AUB = \{x \in Z \mid 0 < x \leq 2\} U \{x \in Z \mid 1 \leq x < 4\} = \{x \in Z \mid 0 < x < 4\}$$
$$(AUB)^C = U - AUB = \{x \mid x \in Z^+\} - \{x \in Z \mid 0 < x < 4\} = \{x \in Z \mid \ \mathrm{x} \geq 4\}$$

$$A \cap B = \{x \in Z \mid 0 < x \leq 2\} \cap \{x \in Z \mid 1 \leq x < 4\} = \{x \in Z \mid 1 \leq x \leq 2\}$$
$$(A \cap B)^C = U - A \cap B = \{x \mid x \in Z^+\} - \{x \in Z \mid 1 \leq x \leq 2\} = \{x \in Z \mid \ \mathrm{x} \geq 3\}$$
$$A^C = U - A = \{x \mid x \in Z^+\} - \{x \in Z \mid 0 < x \leq 2\} = \{x \in Z \mid x \geq 3\}$$
$$B^c = U - B = \{x \mid x \in Z^+\} - \{x \in Z \mid 1 \leq x < 4\} = \{x \in Z \mid \mathrm{x} \geq 4\}$$
$$A^C U B^C = \{x \in Z \mid x \geq 3\} U \{x \in Z \mid \mathrm{x} \geq 4\} = \{x \in Z \mid x \geq 3\}$$
$$A^C \cap B^C = \{\{x \in Z \mid x \geq 3\} \cap \{x \in Z \mid \mathrm{x} \geq 4\} = \{x \in Z \mid x \geq 4\}$$
$$\mathrm{A} \cup C = \{x \in Z \mid 0 < x \leq 2\} \cup \{x \in Z \mid 3 \leq x < 9\} = \{x \in Z \mid 0 < x \leq 8\}$$

Solution No 3:

y will be in the shaded area



Solution No 4:

Do it by yourself.

Solution No 5:

$A = \{4, 8, 12, 16\}$

$\mathrm{B} = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$

$A \cap B = \{4, 8, 12, 16\}$

$\mathrm{B} - \mathrm{A} = \{2, 6, 10, 14, 18\}$

Practice Questions solution Files

Lecture No (10 – 12)

Solution No 1:

Let A = {1, 2, 3, 4, 5, 6, 7, 8}

$A_1 = \{1, 2, 3\}$
$A_2 = \{4, 5, 6\}$
$A_3 = \{7, 8\}$

To show that A_1, A_2 and A_3 is the partition of A, we have to show

1) $A_1 \cup A_2 \cup A_3 = A$
2) $A_1 \cap A_2 \cap A_3 = \varnothing$

$A_1 \cup A_2 \cup A_3 = \{1, 2, 3, 4, 5, 6, 7, 8\} = A$
$A_1 \cap A_2 \cap A_3 = \varnothing$

Hence {{1, 2, 3}, {4, 5, 6}, {7, 8}} is a partition of {1, 2, 3, 4, 5, 6, 7, 8}.

Solution No 2:

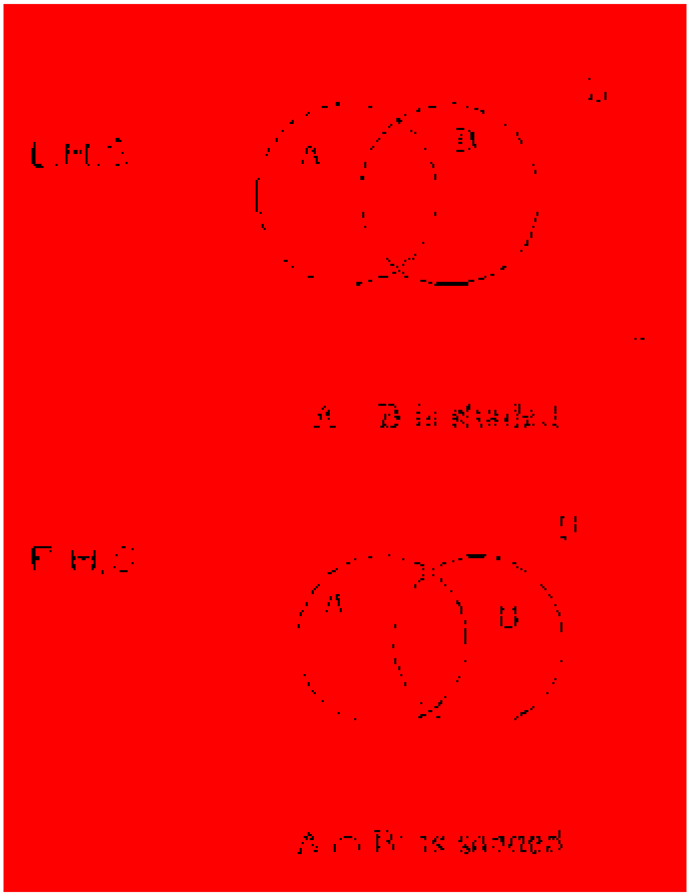
$$Domain\ of\ \ R = \{a, b, c\}$$
$$Range\ of\ R = \{8, 9\}$$

Solution No 3:

$$A \times B = \{(5, x), (5, y), (9, x), (9, y)\}$$
$$B \times A = \{(x, 5), (x, 9), (y, 5), (y, 9)\}$$

$A \times B$ is not equal to $B \times A$. Since order matters in Cartesian product.

Solution No 4:



Solution No 5:

(a) R is not transitive because $(\mathrm{d},\mathrm{b}) \in R$ and $(\mathrm{b},\mathrm{c}) \in R$ but $(\mathrm{d},\mathrm{c}) \notin R$

(b) R is not anti-symmetric because (b,c) *and* $(\mathrm{c},b) \in R$ *but* $b \neq c$

Lecture 13-15

Question No 1:

Let R and S be the following relations on A= {1, 2, 3}

R= {(1, 1), (1, 2), (2, 3), (3, 1), (3, 3), (1, 3), (2, 2)}

S= {(1, 2), (1, 3), (2, 1), (3, 3), (2, 2)}

Find the following

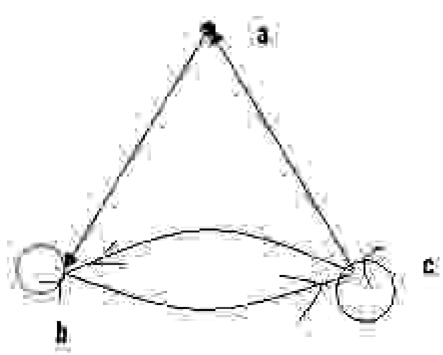
a) $R \cap S$
b) $R \cup S$
c) $R \circ S$
d) $S^2 = S \circ S$

Solution:

a) $R \cap S$ = {(1, 2),(3, 3),(1,3),(2,2)}
b) $R \cup S$ = {(1, 1),(1, 2),(1, 3),(2, 1),(2,2),(2, 3),(3, 1),(3, 3)}
c) $R \circ S$ = {(1,2),(1,1),(2,3),(3,2),(1, 3),(2,1),(2, 2),(3, 1),(3, 3)}
d) $S^2 = S \circ S$ = {(1, 1), (1, 2), (1, 3), (2, 2), (2, 1),(2,3),(3,3)}

Question No 2:

Determine whether the relation shown by the directed graph is reflexive, symmetric, anti-symmetric or transitive?



Solution:

Reflexive:

There is not a loop at every point of the directed graph so it is not reflexive.

Symmetric & Anti-symmetric:

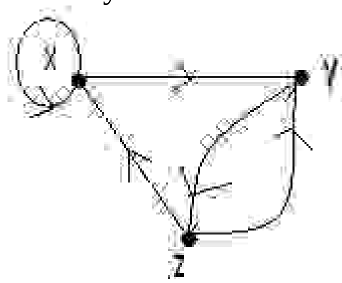
It is neither symmetric nor anti-symmetric, since there is an edge from a to b but not one from b to a. but there are edges in both directions b and c.

Transitive:

It is not transitive, since there is an edge from a to b and an edge from b to c, but no edge from a to c.

<u>Question No 3:</u>

In the following diagram
(a) Write the relation R as set of ordered pairs then write inverse of R.
(b) Whether it is symmetric or anti-symmetric relation?



<u>Solution:</u>

(a)
$$R = \{(x,x),(x,y),(y,z),(z,y),(z,x)\}$$
$$R^{-1} = \{(x,x),(y,x),(z,y),(y,z),(x,z)\}$$

(b) It is neither symmetric nor anti-symmetric, since there is an edge from z to x but not one from x to z. but there are edges in both directions z and y.

<u>Question No 4:</u>

Let $X = \{1,2,3,4\}$ be a set and $R = \{(x,y)|x \le y\}$ is a relation on X. Find all the ordered pairs of R and R^{-1}. Also draw the directed graph of R.

<u>Solution:</u>

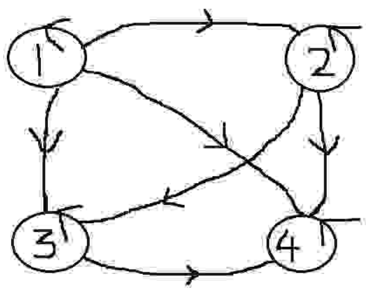
$X = \{1,2,3,4\}$

$$X \times X = \{1,2,3,4\} \times \{1,2,3,4\}$$

So
$$= \begin{Bmatrix} (1,1),(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),(2,4), \\ (3,1),(3,2),(3,3),(3,4),(4,1),(4,2),(4,3),(4,4) \end{Bmatrix}$$

$$R = \{(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4),(4,4)\}$$
$$R^{-1} = \{(1,1),(2,1),(3,1),(4,1),(2,2),(3,2),(4,2),(3,3),(4,3)(4,4)\}$$

The directed graph of R is



Question No 5:

Let R_1 and R_2 be the relations on a set $A = \{1,2,3,4\}$ given by

$R_1 = \{(1,1),(1,2),(3,2),(3,4),(4,2)\}$

$R_2 = \{(1,2),(2,1),(3,1),(4,4),(2,2)\}$

Find the ordered pairs of $R_1 \circ R_2$ and $R_2 \circ R_1$.

Solution:

$R_2 \circ R_1 = \{(1,2),(1,1),(4,1),(3,1),(3,2),(3,4),(4,2)\}$
$R_1 \circ R_2 = \{(2,1),(2,2),(3,1),(3,2),(4,2)\}$

Question No 6:

Find the matrix representing the relations $S \circ R$ where the matrices representing R and S are as follows

$$M_R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Solution:

$$M_{S\circ R} = M_R \odot M_S = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Question No 7:

If $f(x) = \dfrac{x}{2} - 3 \; and \; g(x) = \dfrac{3}{4}x + 2$ then find the value of $5f(-2) - 7g(-4)$.

Solution:

$f(-2) = -1 - 3 = -4$
$g(-4) = -3 + 2 = -1$
Then
$$\begin{aligned} 5f(-2) - 7g(-4) &= 5(-4) - 7(-1) \\ &= -20 + 7 = -13 \end{aligned}$$

Question No 8:

Determine whether $f : Z \rightarrow R$ such that $f(n) = 2\sqrt{n}$ is a function or not.

Solution:

f is not defined for $n < 0$, since f results in imaginary values (not real). So f is not a function.

Solution to Practice questions for Lecture No. 16 – 18

Solution 1:

Given function is

$$y = \frac{3x+2}{x+2}$$

We know that a function will be injective if for any two values $x_{1,}\ x_2$ from the domain of f,

$$f(x_1) = f(x_2)$$

Then

$$x_1 = x_2$$

Lets assume that for $x_{1,}\ x_2$, we have

$$\begin{aligned}
&\frac{3x_1+2}{x_1+2} = \frac{3x_2+2}{x_2+2} \\
&(3x_1+2)(x_2+2) = (3x_2+2)(x_1+2) \\
&3x_1x_2 + 6x_1 + 2x_2 + 4 = 3x_1x_2 + 6x_2 + 2x_1 + 4 \\
&6x_1 + 2x_2 = 6x_2 + 2x_1 \\
&4x_1 = 4x_2 \\
&x_1 = x_2
\end{aligned}$$

Thus, the given function is injective or one to one.

We know that, a function is surjective if for every value of $y \in Y$, there exist a $x \in X$ such that $f(x) = y$.

For that let's take

$$y = \frac{3x+2}{x+2}$$

And try to find x such that f(x) = y for all y belongs to Y.

$$
\begin{aligned}
&y = \frac{3x+2}{x+2} \\
&y(x+2) = 3x+2 \\
&xy + 2y = 3x+2 \\
&xy + 2y - 3x = 2 \\
&x(y-3) = 2-2y \\
&x = \frac{2-2y}{y-3}
\end{aligned}
$$

So, apparently, we see that for every y, there is an x as given above, but if you watch carefully, the value of x is undefined for y = 3. Saying so, in other words, we can very conveniently claim that for y = 3, there exist no x, Therefore, the given function is not surjective.

Solution 2:

Given that

$$f(x) = x^2 - 1 \text{ and } g(x) = 3x+5$$

Then

$$
\begin{aligned}
f \circ g(x) &= f(g(x)) \\
&= f(3x+5) \\
&= (3x+5)^2 - 1 \\
&= 9x^2 + 30x + 24
\end{aligned}
$$

$$
\begin{aligned}
g \circ f(x) &= g(f(x)) \\
&= g(x^2 - 1) \\
&= 3(x^2 - 1) + 5 \\
&= 3x^2 + 2
\end{aligned}
$$

Solution 3:

Given function is

$$
\begin{aligned}
&y = \frac{-2}{x-5} \\
&y(\mathrm{x}-5) = -2 \\
&xy - 5y = -2 \\
&xy = -2 + 5y \\
&x = \frac{5y-2}{y}
\end{aligned}
$$

Thus,

$$f^{-1}(\text{y}) = \frac{5y-2}{y}$$

Solution 4:

Given that

$$y = g(x) \ = (2x - 1)$$
$$\text{y}+1 = 2\,\text{x}$$
$$\text{x} = \frac{y+1}{2}$$

Which is the required inverse function.

Now, replacing x and y , we get

$$g^{-1}(\text{x}) = \frac{x+1}{2}$$
$$g^{-1}(5) = \frac{5+1}{2} = 3$$

Solution 5:

Given that

$f(x) = ax + b$ and $g(x) = cx + d$, where a, b, c and d are constants.

$$f \circ g(x) = f(g(x))$$
$$= f(cx+d)$$
$$= a(cx+d)+b$$
$$= acx + ad + b$$
$$g \circ f(x) = g(f(x))$$
$$= g(ax+b)$$
$$= c(ax+b)+d$$
$$= cax + cb + d$$
$$f \circ g(x) = g \circ f(x)$$
$$acx + ad + b = cax + cb + d$$

If

$$acx + ad + b = cax + cb + d$$
$$ad + b = cb + d$$
$$d(a-1) = b(c-1)$$
$$(a-1)/(c-1) = b/d$$

Solution File
Sequence, Series and Recursion
(Lecture # 19-22)

Question No 1:

Write the geometric sequence with positive terms whose third term is 3 and fourth term is 1.

Solution:

3^{rd} term = ar^2 =3…..1

4^{th} term = ar^3 =1……2

To find r and a, (2)/(1)

$ar^3/ ar^2 = 1/3$

r=1/3 and a = 27

So geometric sequence is, 27, 9, 3, 1,…

Question No 2:

Find the 15^{th} term of the Arithmetic sequence, if its 6^{th} term is 19 and 9^{th} term is 31.

Solution:

6^{th} term = a+5d = 19….(1)

9^{th} term = a+8d = 31…(2)

By subtracting (2) and (1), we get

3d =12, d =4 and a = -1

15^{th} term = a+14d = -1+14*4 = 55

Question No 3:

The sixth term of an arithmetic sequence is 22 and eighth term is 32 .Find the first four terms of this sequence.

Solution:

Do it by yourself.

Question No 4:

A man deposited Rs.20 in a bank in the first month; Rs.25 in the second month; Rs.30 in the third month and so on. Find how much he will have deposited in the bank by the 7th month.

Solution:

1^{st} term =20, 2^{nd} term =25, 3^{rd} term= 30, d=2^{nd} term - 1^{st} term = 5

In 7^{st} month = a+6d = 20+5*6 = 50.

Question No 5:

Find a common fraction for the recurring decimal $1.\overline{73}$.

Solution:

$1.\overline{73}$= 1.73737373….

= 1+ .73737373…. = 1+.73+ 0.0073+ 0.000073

Now using the geometric sequence on 0.73+ 0.0073+ 0.000073, here a = 0.73 and r = 0.0073/0.73 = 0.01

Formula = a / (1-r) = 0.73/ (1-0.01)

$1.\overline{73}$= 1+0.73/ (1-0.01)

Question No 6:

Sum the following series upto n terms.

$(1\times1)+(2\times3)+(3\times5)+...$

Solution:

It can be written as

$(1\times1)+(2\times3)+(3\times5)+... = (1\ ,2,3..)\ *(1,\ 3,\ 5..\ )$

Use Arithmetic Progression on (1,2,3..) and (1,3,5..), and then multiply both answers in order to get the sum of series upto n terms.

Question No 7:

Find the sum of first n terms of an arithmetic series.

Solution:

Do it by yourself

Question No 8:

How many terms are there in $-9-6-3+0+\cdots$ amount to 45?

Solution:

a =-9 and d = -6+9 = 3 and a_n = 45

a_n = 45 = a+(n-1)d = -9+(n-1)(3)

54 = (n-1) (3)

n-1 =18

n =19

Question No 9:

Let a and b be integers. Suppose a function Q is defined recursively as follows:

$$Q(a,b)=\begin{cases} 5 & \text{if } a\langle b \\ Q(a-b,b)-a & \text{if } b\le a \end{cases}$$

Find Q (25, 5)

Solution:

Now

Q (25, 5) = Q (25 – 5, 5) -25

= Q (20, 5) - 25

= [Q (20-5, 5) -20] - 25

= Q(15, 5) - 20 - 25

$= Q(15, 5) – 45$

$= [Q(15 - 5, 5) – 15] - 45$

$= Q(10 , 5) – 60$

$= [Q(10 - 5 , 5) - 10] - 60$

$= [Q(5 ,5) -70] = [Q(5 - 5, 5) – 5] – 70$

$= Q(0 ,5) – 75$

$= 5 - 75$

$= -70$

Question No 10:

Suppose that f is defined recursively by $f(0) = 2, f(n+1) = 4f(n) + 2$ then find $f(3)$.

Solution:

For n =0, f(0+1) = 4f(0)+2 = 4 (2) +2 =10

f(1) =10

For n =1, f(1+1) = 4f(1)+2 or f(2) = 4f(1)+2 = 4(10)+2 = 42

Similarly find the value for f(3)=170

Question No 12:

Show that the sequence $0,1,3,7,\ldots,2^n - 1,\ldots$, for $n \geq 0$ satisfies the recurrence relation $d_k = 3d_{k-1} - 2d_{k-2}$, for all integers $k \geq 2$.

Solution:

The sequence is given by the formula

$$d_n = 2^n – 1 \qquad \text{for } n \geq 0$$

Substituting k – 1 for n we get $d_{k-1} = 2^{k-1} – 1$

Substituting k – 2 for n we get $d_{k-2} = 2^{k-2} – 1$

We want to prove that

$$d_k = 3d_{k-1} - 2d_{k-2}$$

$$\begin{aligned}
\text{R.H.S.} &= 3(2^{k-1} - 1) - 2(2^{k-2} - 1) \\
&= 3 \cdot 2^{k-1} - 3 - 2 \cdot 2^{k-2} + 2 \\
&= 3 \cdot 2^{k-1} - 2^{k-1} - 1 \\
&= (3 - 1) \cdot 2^{k-1} - 1 \\
&= 2 \cdot 2^{k-1} - 1 = 2^{k} - 1 = d_k \quad = \text{L.H.S.}
\end{aligned}$$

Question No 11:

Define a sequence $a_0, a_1, a_2, \ldots$ by the formula $a_n = 2n + 1$, for all integers $n \geq 0$. Show that this sequence satisfies the recurrence relation $a_k = a_{k-1} + 2$, for all integers $k \geq 1$.

<u>Solution:</u>

Do it by yourself

Solution File Lecture No 23-25

Question No.1:

Use mathematical induction to prove that $4^n > 3^n + 4$ is true for integral values of $n \ge 2$

Solution:

Let P(n) = $4^n > 3^n + 4$

Let $P(n) = 4^n > 3^n + 4$

BASIS STEP:

$n = 2$

$4^2 > 3^2 + 4 = 16 > 13$

So $P(2)$ is true

INDUCTIVE STEP:

Suppose $P(k)$ is true for all $k \ge 2$

$$4^k > 3^k + 4 \qquad (1)$$

we are to prove $P(k+1)$ is true i.e.

$$4^{k+1} > 3^{k+1} + 4 \qquad (2)$$

Now

LHS $= 4^{k+1} = 4^k.4 > 4.(3^k + 4) = 4.3^k + 16 \;\; by\,(1) \;\; for \;\; k \ge 2.$

$4^{k+1} > 3^k.(3+1) + 16$

$4^{k+1} > 3^k.3 + 3^k + 16$

$4^{k+1} > 3^{k+1} + 3^k + 4 + 12$

$4^{k+1} > 3^{k+1} + 4 + 12 + 3^k$

$4^{k+1} > 3^{k+1} + 4 \qquad ignoring\, 12 + 3^k \quad as \;\; if \;\; a > b + c \;\; then \;\; a > b$

Hence it is proved by mathematical induction that $4^n > 3^n + 4$ is true for integral values of $n \ge 2$.

Question No.2

Use mathematical induction to prove that $3+6+9+...+3n=\frac{3n(n+1)}{2}$ for all positive integers n.

Solution:

Let $P(n)$: $\quad 3+6+9+...+3n=\frac{3n(n+1)}{2}$

BASIS STEP:

$LHS=3(1)=3$

$RHS=\frac{3(1+1)}{2}$ =3

So $P(1)$ is true

INDUCTIVE STEP:

Suppose $P(k)$ is true for all $k\geq 1$

$$3+6+9+...+3k=\frac{3k(k+1)}{2} \qquad (1)$$

we are to prove $P(k+1)$ is true i.e.

$$P(k+1): \quad 3+6+9+...+3(k+1)=\frac{3(k+1)((k+1)+1)}{2} \qquad (2)$$

Now

$$\text{LHS}=3+6+9+...+3k+3(k+1)=(3+6+9+...+3k)+3(k+1)$$

$$=\frac{3k(k+1)}{2}+3(k+1)$$

$$=3(k+1)\left(\frac{k}{2}+1\right)$$

$$=3(k+1)\frac{k+2}{2}$$

$$=\frac{3(k+1)\left[(k+1)+1\right]}{2}=\text{RHS}$$

Question No.3

Suppose $n^3 - n$ is divisible by 6 is true for any positive integer. Justify the statement for

n = k+1.

Solution:

Inductive Step:

Suppose for n=k, $k^3 - k$ is dividible by 6

then $k^3 - k = 6.q$

we need to prove $(k+1)^3 - (k+1)$ is dividible by 6

$$(k+1)^3 - (k+1) = (k^3 + 3k^2 + 3k + 1) - (k+1)$$
$$= k^3 - k + 3k(k+1)$$
$$= 6.q + 3k(k+1)$$

As the product $k(k+1)$ of integers is even so we can take as $2r$

$$= 6.q + 3k(k+1)$$
$$= 6.q + 3.2r$$
$$= 6(q+r)$$

So $(k+1)^3 - (k+1)$ is dividible by 6

Hence by induction method, $\mathrm{n}^3 - n$ is divisible 6

Question No 4:

Suppose $5^n - 1$ is divisible by 4 is true for $n = k$. Justify the statement for $n = k+1$.

Solution:

Inductive Step:

Let for $n = k$, $5^k - 1$ is divisible by 4 then

$5^k - 1 = 4.q$

we need to prove that $5^{k+1} - 1$ is divisible by 4

$$
\begin{aligned}
5^{k+1} - 1 &= 5.5^k - 1 \\
&= 5^k(4+1) - 1 \\
&= 5^k.4 + 5^k - 1 \\
&= 5^k.4 + 4.q \\
&= 4(5^k + q)
\end{aligned}
$$

Hence $5^{k+1} - 1$ is divisible by 4

Question No 5

Prove that if x divides y and y divides z then x divides z.

Solution:

By our assumptions, and the definition of divisibility, there are natural numbers k_1 and k_2 such that

$y = xk_1$ *and* $z = yk_2$.

Consequently,

$z = yk_2 = xk_1k_2$.

Let $k = k_1k_2$ Now k is a natural number and $z = xk$, so by the definition of divisibility, x divides z.

Question No 6:

Show that for all integers a and b, if $a+b$ is even then $a-b$ is also even.

Solution:

Take a+b = 2k
a= 2k-b
Subtracting ‘b’ from both sides
a-b =2k-b-b
a-b = 2k -2b
a-b = 2(k-b)
Hence a-b is also even.

Question No 7:

Assume that m and n are particular integers. Justify your answer to each of the following:

1) Is $4m+6n$ even?

2) Is $8mn+5$ odd?

Solution:

1) Yes: 4m+6n =2.(2m+3n) [by definition of even]

   (2m+3n) is an integer because 2, 3, m, n are integers and products and sums of integers are integers.

2) Yes: 8mn+5 =2.(4mn+2)+1 [by definition of odd]

   (4mn+2) is an integer because 4, 2, m, n are integers and products and sums of integers are integers.

Solution to Practice questions Lecture 26-28

Solution 1:

We are to prove by contradiction that $4+3\sqrt{2}$ is an irrational number. So, let's assume that $4+3\sqrt{2}$ is a rational number. Then by definition of rational number, it can be written in the following form

$$4+3\sqrt{2}=\frac{a}{b} \quad \text{for some integers a and b with } b\neq 0$$

$$4+3\sqrt{2}=\frac{a}{b}$$

$$3\sqrt{2}=\frac{a}{b}-4$$

$$\sqrt{2}=\frac{a-4b}{3b}$$

Since a and b are integers, so are a-4b and 3b with $3b\neq 0$. This shows that $\sqrt{2}$ is a rational number. Which is a contradiction to the f act that $\sqrt{2}$ is an irrational number. This contradiction shows that our initial supposition was incorrect and thus, $4+3\sqrt{2}$ is not an irrational number that is $4+3\sqrt{2}$ is a rational number.

Solution 2:

We are to prove by contradiction that if $5n+1$ is odd then n is even. Suppose for some integer n , $5n+1$ is odd but n is not even. That is n is odd. So it can be written as

$$n=2m+1 \quad \text{for some integer } m.$$

$$\begin{aligned} 5n+1 &= 5(2m+1)+1 \\ &= 10m+6 \\ &= 2(5m+3) \end{aligned}$$

This shows that $5n+1$ is equal to some multiple of 2 and so it is even which is a contradiction to our initial supposition. So, the correct supposition will be that if $5n+1$ is odd then n is even.

Solution 3:

We are to prove by contraposition that if n^2 is odd then n is odd. That is we are to prove that if n is even then n^2 is even.

Lets assume n is even. Then it can be written as a multiple of 2. That is

$$
\begin{aligned}
n &= 2k \text{ for some integer } k \\
n^2 &= (2k)^2 \\
&= 4k^2 \\
&= 2(2k^2)
\end{aligned}
$$

Clearly, n^2 is a multiple of 2 so n^2 is also even.

This Proves that if n^2 is odd then n is odd.

Solution 4:

The following are pre and post conditions.

Pre-condition: The input variables a and b are positive integers.
Post-condition: The output variable q and r are positive integers such that

$a = b \cdot q + r$ and $0 \le r < b$.

Solution 5:

We are to find the GCD (61114, 94).

Divide 61114 by 94

$$61114 = 94 \times 650 + 14$$

Divide 94 by 14

$$94 = 14 \times 6 + 10$$

Divide 14 by 10

$$14 = 10 \times 1 + 4$$

Divide 10 by 4

$$10 = 4 \times 2 + 2$$

Divide 4 by 2

$$4 = 2 \times 2 + 0$$

Thus, the GCD of 61114 and 94 is 2.

Solution to practice questions for Lecture No. 29-31

Solution 1:

For a round trip, the person will travel from city A to B, B to C then C to B and then B to A i.e.,

$$A \to B \to C \to B \to A$$

The person can travel 3 ways from A to B and 5 ways from B to C and back that is

$$A \xrightarrow{3} B \xrightarrow{5} C \xrightarrow{5} B \xrightarrow{3} A$$

Thus, there are

$$3 \times 5 \times 5 \times 3 = 225$$

ways to have a round trip.

Solution 2:

Each bit (binary digit) is either 0 or 1.
Hence, there are 2 ways to choose each bit. Since we have to choose seven bits therefore, the product rule shows, there are a total of $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 = 128$

Solution 3:

Given that

p(n,2)=42

That is

$$\frac{n!}{(n-2)!} = 42$$

$$\frac{n(n-1)(n-2)!}{(n-2)!} = 42$$

$$n(n-1) = 42$$

$$n^2 - n - 42 = 0$$

$$n^2 - 7n + 6n - 42 = 0$$

$$n(n-7) + 6(n-7) = 0$$

$$(n+6)(n-7) = 0$$

$$n = -6, \quad n = 7$$

Ignoring the negative value, the desired value of n is 7.

Solution 4:

5 boys can be selected from 10 in following number of ways

$$^{10}C_5 = 252$$

5 girls can be selected from 15 in following number of ways

$$^{15}C_5 = 3003$$

Total number of ways a team of 10 students be selected = 252×3003=756756

Solution 5:

There are 13 Spade cards in a deck of cards so C(13,2) = 78
There are 13 Heart cards in a deck of cards so C(13,2) = 78

$$C(13,2) \times C(13,2) = 78 \times 78 = 6084$$

Practice questions Lectures 32-34

Question No 1:

How many distinguishable ways can the letter of the word HULLABALOO be arranged?

Question No 2:

How many signals can be given by 7 flags of different colors using 3 flags at a time?

Question No 3:

How many 4-digit numbers can be formed by using each of the digits1, 2, 3,4,5,6 only once?

Question No 4:

Suppose that there are 2000 students at your university. Of these 500 are taking a course in computer science, 670 are taking a course in mathematics and 300 are taking course in both computer science and mathematics. How many are taking a course either in computer science or in mathematics?

Question No 5:

There are 24 students in a class, only 13 are willing to go to a tour for Murree. In how many ways can these be chosen?

Question No 6:

In an office, there are 75 faculty members can speak Malay(M) and 25 can speak Chinese (C), while only 10 can speak both Malay and Chinese. Using Inclusion-exclusion Principle find how many faculty members can speak either Malay (M) or Chinese(C)?

Practice Questions for Lecture No. 35 to 37

Question 1:

A box contains seven discs numbered 1 to 7. Find for each integer k (k is from 3 to 11), the probability that the numbers on two discs drawn without replacement have a sum equal to k.

Question 2:

A die is weighted so that the outcomes produce the following probability distribution:

Outcome	1	2	3	4	5	6
Probability	0.1	0.3	0.2	0.1	0.1	0.2

Consider the event A = {odd number occur on die} then find $P(\mathrm{A})$ and $P(\mathrm{A}^c)$.

Question 3:

There are 20 girls and 40 boys in a class. Half of the boys and half of the girls have blue eyes. Find the probability that one student chosen as monitor is either a girl or a student having blue eyes.

Question 4:

A pair of fair dice is thrown. Find the probability P that the sum is 9 or greater if

(i) 5 appear on first die

(ii) 5 appear on at least one die

Prcatice Question Lecture No 39-41

Question No 1:

Find the maximum number of edges for a complete graph $k_n$by using handshaking theorem.

Solution

The total degrees in graphK_n = 2 (number of edges in K_n)

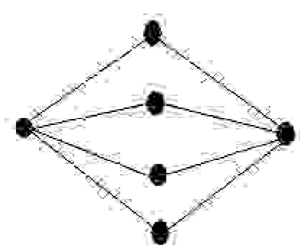
$\Rightarrow$ n(n-1) = 2(number of edges in K_n)

$\Rightarrow$

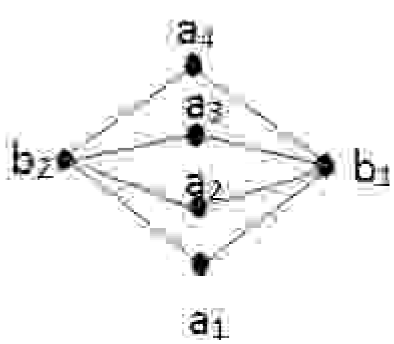
Number of edges in K_n $= \dfrac{n(n-1)}{2}$

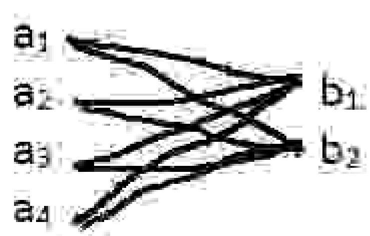
Question No 2:

Is the graph below is Complete Bipartite or not? Redraw the bipartite graph so that its bipartite nature is evident?



Solution:

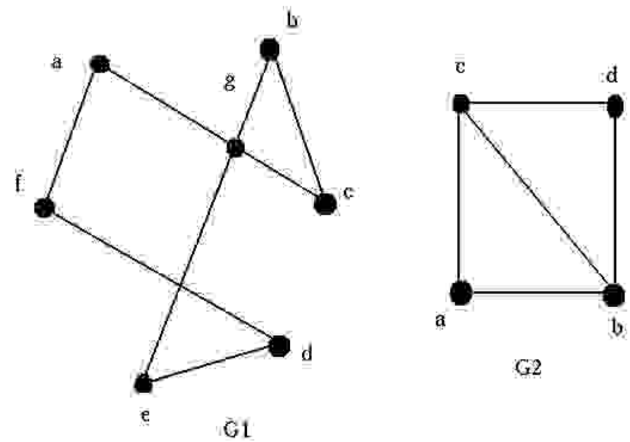




Yes it is a complete bipartite graph.

Question No 3:

Check whether the given graphs have an Euler circuit? If it does, find such a circuit, if it does not, give an argument to show why no such circuit exists.



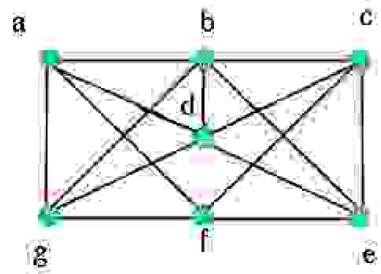
Solution:

In G1: each vertex has even degree so it an Euler circuit. In G2: vertices "c" and "b" has odd

degree so it is not an Euler circuit

Question No 4:

Find two Hamiltonian circuits of the following graph



Solution:

(i) abcdefga ii) abcefgda

Question No 5:

Find the product AB and BA of the matrices (if not possible then give reason).

$$A = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 4 \end{bmatrix}$$

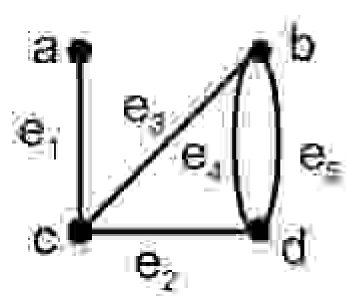
Solution:

$$AB = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 4 \end{bmatrix} = \begin{bmatrix} 2+0 & 4+0 & 6+0 \\ 1-3 & 2+2 & 3-4 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ -2 & 4 & -1 \end{bmatrix}$$

BA is not possible as B has 3 columns and A has two rows, so multiplication is not possible.

Question No 6:

Find the adjacency matrix of the graph shown below.



Solution:

$$\begin{array}{cc} & \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 \end{bmatrix} \end{array}$$

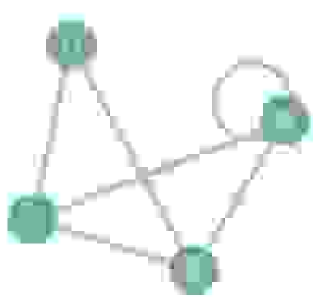
<u>Question No 7:</u>

Draw a graph with the adjacency matrix.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d.

Solution:



Lecture 41-43

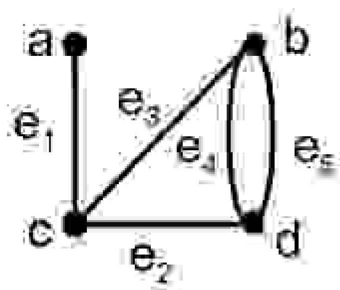
Question No 1:

Find the product AB and BA of the matrices (if not possible then give reason).

$$A = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} \quad \text{and} \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 4 \end{bmatrix}$$

Question No 2:

Find the adjacency matrix of the graph shown below.



Question No 3:

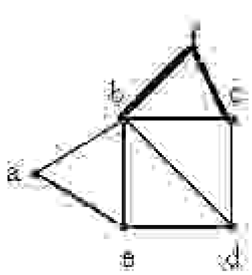
Draw a graph with the adjacency matrix.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d.

Question No 4 :

Find the degree sequence of the following graph.

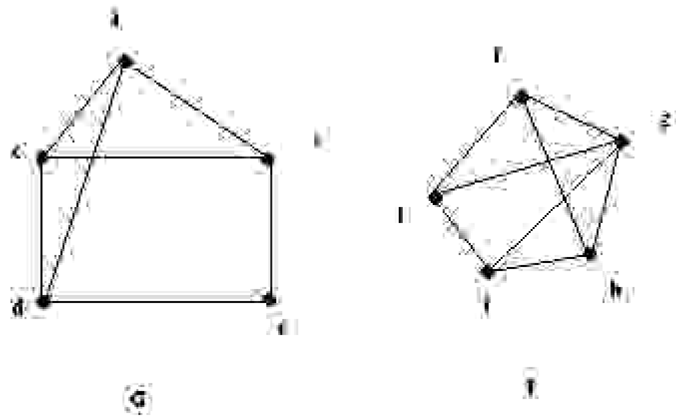


Question No 5:

Draw all possible simple graphs with 2 vertices, which are non-isomorphic to each other.

Question No 6:

Determine whether the given graphs are isomorphic? (justify your answer)

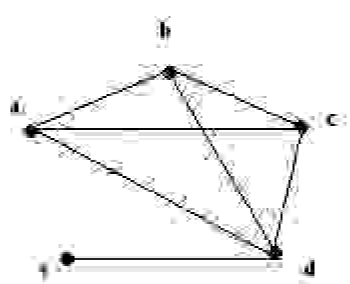


Question No 7:

Suppose that a connected planar simple graph has 22 edges. If you want to draw a graph having 11 faces, how many vertices does this graph have?

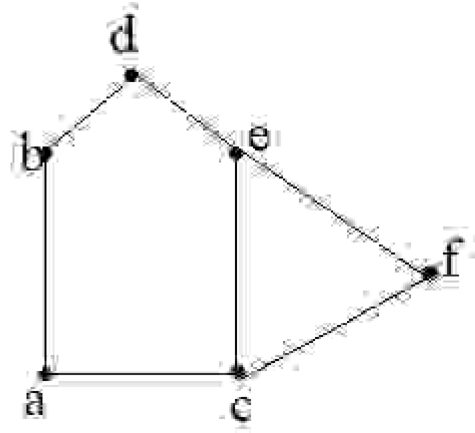
Question No 8:

Re-draw the given graph to prove it planar.



Question No 9:

Determine the chromatic number of the given graph.



Practice Questions for Lecture No. 44 and 45

Question 1:

Find the total number of internal and terminal vertices of a full binary tree with nine vertices

Question 2:

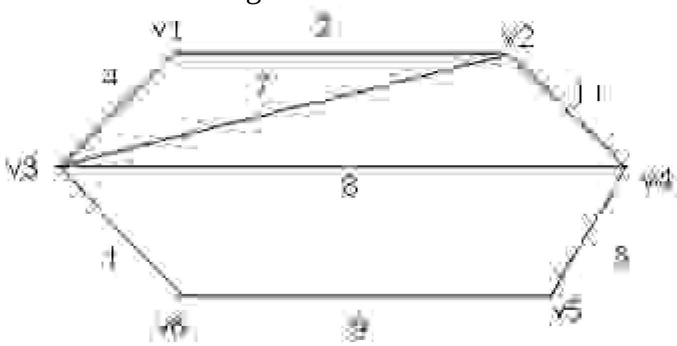
Draw a binary tree with height 3 and having seven terminal vertices.

Question 3:

Draw any two non-isomorphic trees of five vertices.

Question 4:

Use Kruskal's Algorithm to draw the minimal spanning tree for the graph given below. Indicate the order in which edges are added to form a tree.



Question 5:

Find a spanning tree for the graph $k_{1,5}$?

MTH202 FINAL TERM SOLVED MCQs

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Question No: 1 (Marks: 1) - Please choose one

When 5^k is even, then $5^k+5^k+5^k$ is odd.

► True

► False

Question No: 2 (Marks: 1) - Please choose one

An arrangement of objects without the consideration of order is called

► Combination

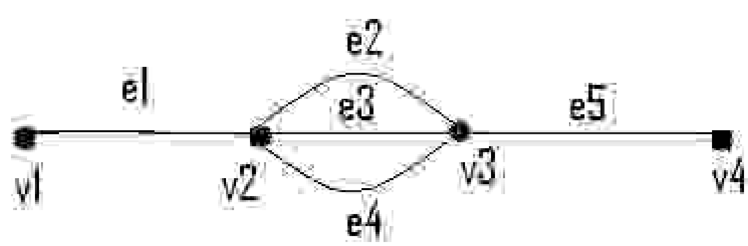
► Selection

► None of these

► Permutation

Question No: 3 (Marks: 1) - Please choose one

In the following graph



How many simple paths are there from v_1 to v_4

► 2
► 3
► 4

Question No: 4 (Marks: 1) - Please choose one

Changing rows of matrix into columns is called

► Symmetric Matrix
► Transpose of Matrix
► Adjoint of Matrix

Question No: 5 (Marks: 1) - Please choose one

The list of the degrees of the vertices of graph in non increasing order is called

► Isomorphic Invariant
► Degree Sequence
► Order of Graph

Question No: 6 (Marks: 1) - Please choose one

A vertex of degree greater than 1 in a tree is called a

► Branch vertex
► Terminal vertex
► Ancestor

Question No: 7 (Marks: 1) - Please choose one

The word "algorithm" refers to a step-by-step method for performing some action

► True

► False

► None of these

Question No: 8 (Marks: 1) - Please choose one

The sum of two irrational number must be an irrational number

► True
► False

Question No: 9 (Marks: 1) - Please choose one

An integer n is prime if, and only if, $n > 1$ and for all positive integers r and s, if $n = r \cdot s$, then

► $r = 1$ or $s = 1$.

► $r = 1$ or $s = 0$.

► $r = 2$ or $s = 3$.

► None of these

Question No: 10 (Marks: 1) - Please choose one

An integer n is even if, and only if, $n = 2k$ for some integer k.

► True

► False

► Depends on the value of k

Question No: 11 (Marks: 1) - Please choose one

For any two sets A and B, $A - (A - B) =$

► A Ç B

► A È B

► $A - B$

► None of these

Question No: 12 (Marks: 1) - Please choose one

A walk that starts and ends at the same vertex is called

► Simple walk

► Circuit

► Closed walk

Question No: 13 (Marks: 1) - Please choose one

Associative law of union for three sets is

► A È (B È C) = (A È B) È C

► A Ç (B Ç C) = (A Ç B) Ç C

► A È (B Ç C) = (A È B) Ç (A È B)

► None of these

Question No: 14 (Marks: 1) - Please choose one

Two distinct edges with the same set of end points are called

► Isolated

► Incident

► Parallel

Question No: 15 (Marks: 1) - Please choose one

The probability of getting 2 heads in two successive tosses of a balanced coin is

► $\frac{1}{4}$

► $\frac{1}{2}$

► $\frac{2}{3}$

Question No: 16 (Marks: 1) - Please choose one

What is the probability of getting a number greater than 4 when a die is thrown?

$\frac{1}{2}$

□□□□□□□□▶□

$\frac{3}{2}$

□□□□□□□□▶□

$\frac{1}{3}$

□□□□□□□□▶□

Question No: 17 (Marks: 1) - Please choose one

If two relations are reflexive then their composition is

≡ **Antisymmetric**

≡ **Reflexive**

≡ **Irreflexive**

≡ **Symmetric**

Question No: 18 (Marks: 1) - Please choose one

If p and q are statement variables, the biconditional of p and q is

denoted by

▶ p«q

▶ ~q ®~p

▶ ~p ®~q

▶ None of these

Question No: 19 (Marks: 1) - Please choose one

Select the correct one

▶ A proof by contradiction is based on the fact that a statement can be true and false at the same time.

► A proof by contraposition is based on the logical equivalence between a statement and its contradiction.

► The method of loop invariants is used to prove correctness of a loop without any conditions.

► None of the given choices

Question No: 20 (Marks: 1) - Please choose one

According to Demorgan's law

$\neg(p \vee q) \equiv ?$

► $= \bar{p} \vee \neg q$

► $= \bar{p} \wedge \neg q$

► $= \bar{p} \wedge \bar{q}$

► $= \bar{p} \vee \bar{q}$

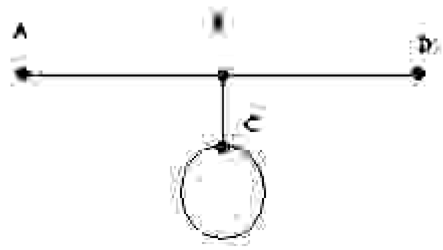
Question No: 21 (Marks: 2)

Find integers q and r so that a=bq+r , with 0≤r<b.
a=45 , b=6.

Ans:
If a =45 and b = 6 are two integers with $b \neq 0$ such that the q and r are non negitive integers.
a=bq+r
divides 45 by 6
this gives= 6.7 +3
divides 6 by 3
this gives = 3.2 +0
hence gcd of the (45,6) will be 3

Question No: 22 (Marks: 2)

Give the degree of each vertex in the figure (given below)



Ans: degree of A vertex = 1
Degree of B vertex = 3
Degree of C vertex = 3
Degree of D vertex = 1
Total degree of vertices = 8
Can be prove by formula
Degree of vertices = 2. no. of edges
= 2 . 4
= 8

Question No: 23 (Marks: 2)

What is the probability of getting a number greater than 2 when a dice is tossed?

As dice has 6 sides so possible event will be 36.
No. greater than 2 will be
3 ,4 ,5 ,6 = 18

P(E) =n(E)/(n(S))

= 18/36

= ½ will be the possibility to get no greater than 2

Question No: 24 (Marks: 3)

How many distinguishable ways can the letters of the word HULLABALOO be arranged if words are to begin with U and end with L

Question No: 25 (Marks: 3)

Write the Pre and Post Conditions of the

"Algorithm to compute a product of two negative integers".

Ans:
Pre condition: The input variables m and n are nonnegative integers.
Post condition: the out put variables p equals m.n.
Example:

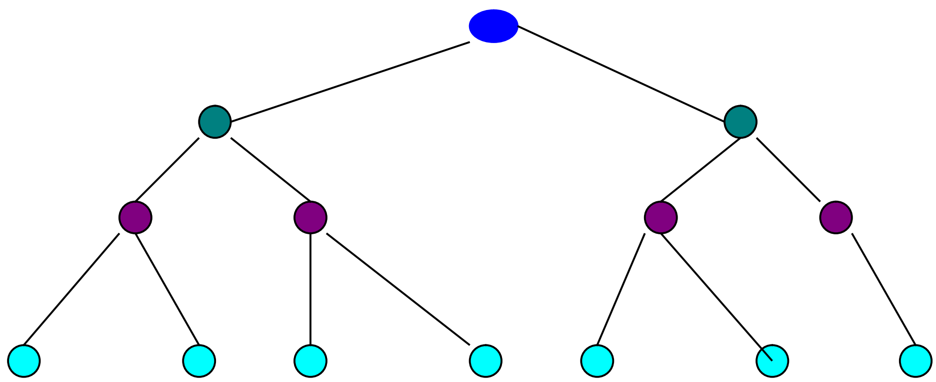
If $k \geq 0$, then $$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{1\cdot 2\cdots k} = (-1)^k\binom{-n+k-1}{k}$$
extends to all *n*.

extends to $k < 0$ via

$$\binom{n}{k} = \begin{cases} (-1)^{n-k}\binom{-k-1}{n-k} & \text{if } n \geq k, \\ (-1)^{n-k}\binom{-k-1}{-n-1} & \text{if } n \leq -1 \end{cases}$$

Question No: 26 (Marks: 3)

Draw a full binary tree with seven vertices.



Question No: 27 (Marks: 5)

Find n if

$P(n,2) = 72$

Given

$P(n,2) = 72$

$n.(n-1)=72$ by using the definition of permutation

$n^2-1=72$

$n^2-n-72=0$

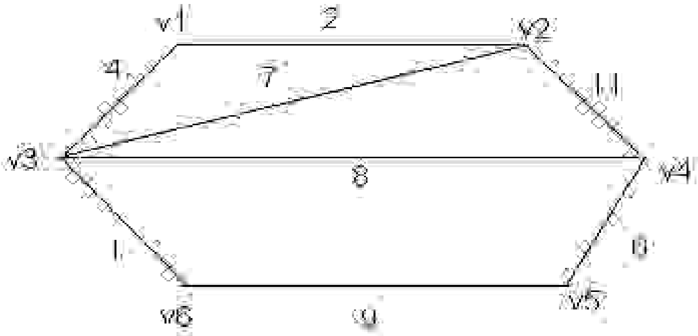
$n=9,-8$ since n must be positive so only the acceptable value for n is 9

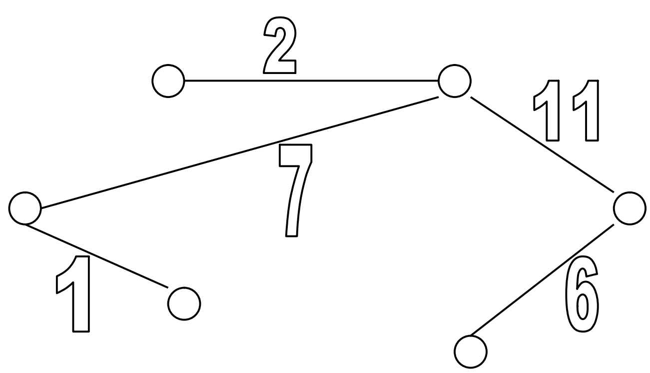
Question No: 28 (Marks: 5)

Five people are to be seated around a circular table. Two seating plans are considered as same if one is the rotation of other. How many different seating plans are possible?

Question No: 29 (Marks: 5)

Use Kruskal's Algorithm to draw the minimal spanning tree for the graph below. Indicate the order in which edges are added to form a tree.





Order of adding the edges:

{v3,v6},{v1,v2},{v4,v5},{v2,v3},{v2,v4},.

Question No: 30 (Marks: 10)

Show the sample space for tossing one penny and rolling one die.
(H = heads, T = tails) using tree diagram

Question No: 31 (Marks: 10)

,$10^{3n} + 13^{n+1}$ is divisible by 7 for all n>=1

Let $10^{3n} + 13^{n+1}$ is divisible by 7

Basis step:
P(1) is true.now
P(1):

(0.1)

$10^{3n}+13^{n+1}$ is divisible by 7

Since $10^{3.1}+13^{1+1} = 10^{(3)}+13^{2}$
Which is divisible by 7
Hence P(1) is true.now

Inductive step:
Suppose p(k is true)
$10^{3k}+13^{k+1} = 7.q$
To prove p(k+1) $10^{3n}+13^{n+1}$ is true is divdsible by 7

$10^{3k+1} + 13^{k+1+1} = 23^{4k+3}$
$= 23^{4k+3} +2 -2$
$=21^{4k+3}+2$
$=7.3^{4k+3} +2$
$= 7(3^{4k+3} +2)$

$=7. q$ where q is ant positive integer equal to 3^{4k+3} +2
So its proved that $10^{3n} + 13^{n+1}$ divisible by 7 for all n>=1

ASSALAM O ALAIKUM

All Dearz fellows

ALL IN ONE MTH202 Final term PAPERS & MCQz

FINALTERM EXAMINATION

Spring 2009

MTH202- Discrete Mathematics (Session - 2)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The negation of "Today is Friday" is

► Today is Saturday

► **Today is not Friday**

► Today is Thursday

Question No: 2 (Marks: 1) - Please choose one

An arrangement of rows and columns that specifies the truth value of a compound proposition for all possible truth values of its constituent propositions is called

► **Truth Table**

► Venn diagram

► False Table

► None of these

Question No: 3 (Marks: 1) - Please choose one

The converse of the conditional statement **p ® q** is

► **q ®p**

► ~q ®~p

► ~p ®~q

► None of these

Question No: 4 (Marks: 1) - Please choose one

Contrapositive of given statement "*If it is raining, I will take an umbrella*" is

- ► **I will not take an umbrella if it is not raining.**
- ► I will take an umbrella if it is raining.
- ► It is not raining or I will take an umbrella.
- ► None of these.

Question No: 5 (Marks: 1) - Please choose one

Let A= {1, 2, 3, 4} and R = {(1, 1), (2, 2), (3, 3),(4,4)} then

- ► R is symmetric.
- ► R is anti symmetric.
- ► R is transitive.
- ► **R is reflexive.**
- ► ***All given options are true***

Question No: 6 (Marks: 1) - Please choose one

A binary relation R is called Partial order relation if

- ► It is Reflexive and transitive
- ► It is symmetric and transitive
- ► It is reflexive, symmetric and transitive
- ► **It is reflexive, antisymmetric and transitive**

Question No: 7 (Marks: 1) - Please choose one

How many functions are there from a set with three elements to a set with two elements?

► 6
► **8**
► 12

Question No: 8 (Marks: 1) - Please choose one

$1, 10, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7, \ldots\ldots\ldots\ldots\ldots$ is

► **Arithmetic series**
► Geometric series
► Arithmetic sequence
► Geometric sequence

Question No: 9 (Marks: 1) - Please choose one

$\lceil x \rceil$ for x = -2.01 is

► -2.01
► -3
► **-2**
► -1.99

Question No: 10 (Marks: 1) - Please choose one

If A and B are two disjoint (mutually exclusive) events then

P(AUB) =

► P(A) + P(B) + P(AÇB)
► P(A) + P(B) + P(AUB)
► P(A) + P(B) - P(AÇB)
► P(A) + P(B) - P(AÇB)
► **P(A) + P(B)**

Question No: 11 (Marks: 1) - Please choose one

If a die is thrown then the probability that the dots on the top are prime numbers or odd numbers is

► 1
► $\frac{1}{3}$
► $\frac{2}{3}$

Question No: 12 (Marks: 1) - Please choose one

If $P(A \cap B) \neq P(A)P(B)$ then the events A and B are called

► Independent
► Dependent page 270
► Exhaustive

Question No: 13 (Marks: 1) - Please choose one

A rule that assigns a numerical value to each outcome in a sample space is called

► One to one function
► Conditional probability
► ***Random variable***

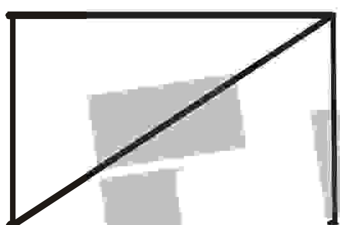
Question No: 14 (Marks: 1) - Please choose one

The expectation of x is equal to

► Sum of all terms
► Sum of all terms divided by number of terms
► $\sum xf(x)$

Question No: 15 (Marks: 1) - Please choose one

The degree sequence {a, b, c, d, e} of the given graph is



► 2, 2, 3, 1, 1
► **2, 3, 1, 0,1**
► 0, 1, 2, 2, 0
► **2, 3,1,2,0** page305

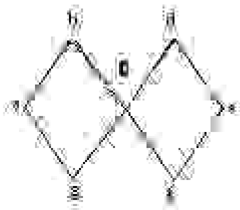
Question No: 16 (Marks: 1) - Please choose one

Which of the following graph is not possible?

► Graph with four vertices of degrees 1, 2, 3 and 4.
► **Graph with four vertices of degrees 1, 2, 3 and 5.**
► Graph with three vertices of degrees 1, 2 and 3.
► Graph with three vertices of degrees 1, 2 and 5.

Question No: 17 (Marks: 1) - Please choose one

The graph given below



► Has Euler circuit
► Has Hamiltonian circuit
► **Does not have Hamiltonian circuit**

Question No: 18 (Marks: 1) - Please choose one

Let n and d be integers and d ^{1} 0. Then n is divisible by d or d divides n
If and only if

► **n= k.d for some integer k**

► n=d

► n.d=1

► none of these

Question No: 19 (Marks: 1) - Please choose one

The contradiction proof of a statement pàq involves

► ***Considering p and then try to reach q***

► Considering ~q and then try to reach ~p

► Considering p and ~q and try to reach contradiction

► None of these

Question No: 20 (Marks: 1) - Please choose one

An integer n is prime if, and only if, n > 1 and for all positive integers r and s, if
n = r·s, then

► **r = 1 or s = 1.**

► r = 1 or s = 0.

► r = 2 or s = 3.

► None of these

Question No: 21 (Marks: 1) - Please choose one

The method of loop invariants is used to prove correctness of a loop with respect to certain pre and post-conditions.

► **True**

► False

► None of these

Question No: 22 (Marks: 1) - Please choose one

The greatest common divisor of 27 and 72 is

- ► 27
- ► **9**
- ► 1
- ► None of these

Question No: 23 (Marks: 1) - Please choose one

If a tree has 8 vertices then it has

- ► 6 edges
- ► **7 edges**
- ► 9 edges

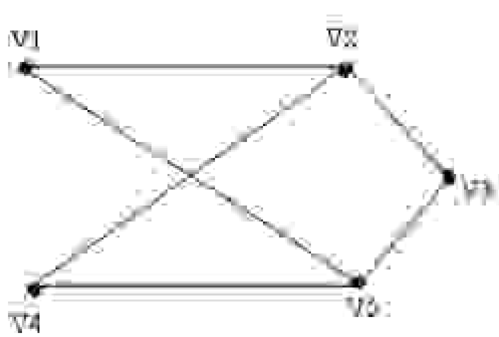
Question No: 24 (Marks: 1) - Please choose one

Complete graph is planar if

- ► **n = 4**
- ► n>4
- ► $n \leq 4$

Question No: 25 (Marks: 1) - Please choose one

The given graph is



- ► **Simple graph**
- ► Complete graph
- ► Bipartite graph
- ► Both (i) and (ii)
- ► Both (i) and (iii)

Question No: 26 (Marks: 1) - Please choose one

The value of 0! Is

- ► 0
- ► **1** pg160
- ► Cannot be determined

Question No: 27 (Marks: 1) - Please choose one

Two matrices are said to confirmable for multiplication if

- ► Both have same order
- ► **Number of columns of 1st matrix is equal to number of rows in 2nd matrix**
- ► Number of rows of 1st matrix is equal to number of columns in 2nd matrix

Question No: 28 (Marks: 1) - Please choose one

The value of (-2)! Is

- ► 0
- 1
- ► **Cannot be determined**

Question No: 29 (Marks: 1) - Please choose one

The value of $\frac{(n+1)!}{(n-1)!}$ is

- 0
- n(n-1)
- $n^2 + n$
- Cannot be determined

Question No: 30 (Marks: 1) - Please choose one

The number of *k*-combinations that can be chosen from a set of *n* elements can be written as

- $^{n}C_{k}$ pg223
- $^{k}C_{n}$
- $^{n}P_{k}$
- $^{k}P_{k}$

Question No: 31 (Marks: 1) - Please choose one

If the order does not matter and repetition is allowed then total number of ways for selecting k sample from n. is

- n^k
- **C(n+k-1,k)** page 228
- **P(n,k)**
- C(n,k)

Question No: 32 (Marks: 1) - Please choose one

If the order matters and repetition is not allowed then total number of ways for selecting k sample from n. is

► n^k

► C(n+k-1,k)

► **P(n,k)** page 228

► C(n,k)

Question No: 33 (Marks: 1) - Please choose one

To find the number of unordered partitions, we have to count the ordered partitions and then divide it by suitable number to erase the order in partitions

► **True** pg231

► False

► None of these

Question No: 34 (Marks: 1) - Please choose one

A tree diagram is a useful tool to list all the logical possibilities of a sequence of events where each event can occur in a finite number of ways.

► **True**

► False

Question No: 35 (Marks: 1) - Please choose one

If A and B are finite (overlapping) sets, then which of the following **must be** true

► n(AÈB) = n(A) + n(B)

► n(AÈB) = n(A) + n(B) - n(AÇB)

► n(AÈB)= ø

► None of these

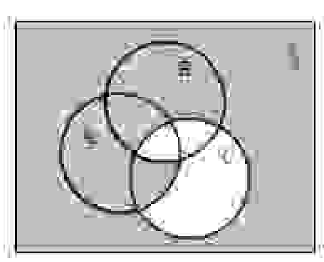
Question No: 36 (Marks: 1) - Please choose one

What is the output state of an OR gate if the inputs are 0 and 1?

► 0
► 1
► 2
► 3

Question No: 37 (Marks: 1) - Please choose one

In the given Venn diagram shaded area represents:



- ► (A Ç B) È C
- ► (A È B^c) È C
- ► **(AÇ B^c) È C^c** page 53
- ► (A Ç B) Ç C^c

Question No: 38 (Marks: 1) - Please choose one

Let A,B,C be the subsets of a universal set U.
Then $(A \cup B) \cup C$ is equal to:

- ► $A \cap (B \cup C)$
- ► $A \cup (B \cap C)$
- ► $\varnothing$
- ► $A \cup (B \cup C)$

Question No: 39 (Marks: 1) - Please choose one

n ! >2^n for all integers n ³ 4.

- ► True
- ► **False**

Question No: 40 (Marks: 1) - Please choose one

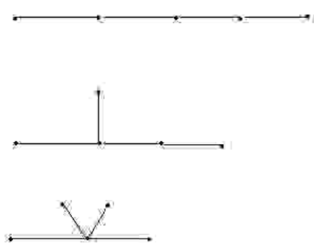
$+,-,\times,\div$ are

- ► Geometric expressions
- ► **Arithmetic expressions**
- ► Harmonic expressions

Question No: 41 (Marks: 2)

Find a non-isomorphic tree with five vertices.

There are three non-isomorphic trees with five vertices as shown (where every tree with five vertices has 5-1=4 edges).



Question No: 42 (Marks: 2)

Define a predicate.

A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables.
The domain of a predicate variable is the set of all values that may be substituted in place of the variable.

Let the declarative statement:
"x is greater than 3".
We denote this declarative statement by P(x) where

x is the variable,

P is the predicate "is greater than 3".
The declarative statement P(x) is said to be the value of the propositional function P at x.

Question No: 43 (Marks: 2)

Write the following in the factorial form:

(n +2)(n+1) n

$$\frac{(n+2)(n+1)n}{n!}$$

Question No: 44 (Marks: 3)

Determine the probability of the given event

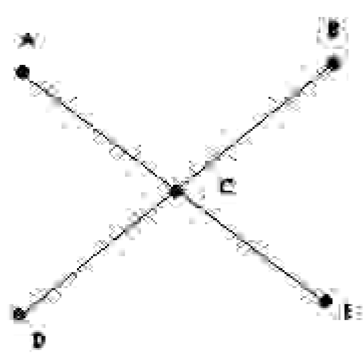
"An odd number appears in the toss of a fair die"

Sample space will be..S={1,2.3,4,5,6}...there are 3 odd numbers so,
For odd numbers, probability will be

$\frac{3}{6}$...Ans

Question No: 45 (Marks: 3)

Determine whether the following graph has Hamiltonian circuit.



This graph is not a Hamiltonian circuit, because it does not satisfy all conditions of it.

E.g. it has unequal number of vertices and edges. And its path cannot be formed without repeating vertices.

Question No: 46 (Marks: 3)

Prove that If the sum of any two integers is even, then so is their difference.

Theorem: ∀ integers m and n, if m + n is even, then so is m - n.

Proof:

Suppose m and n are integers so that m + n is even. By definition of even, m + n = 2k for some integer k. Subtracting n from both sides gives m = 2k - n. Thus,

m - n	= (2k - n) - n	by substitution
	= 2k - 2n	combining common terms
	= 2(k - n)	by factoring out a 2

But (k - n) is an integer because it is a difference of integers. Hence, (m - n) equals 2 times an integer, and so by definition of even number, (m - n) is even.

This completes the proof.

Question No: 47 (Marks: 5)

Show that if seven colors are used to paint 50 heavy bikes, at least 8 heavy bikes will be the same color.

N=50
K=7
C(7+50-1,7)

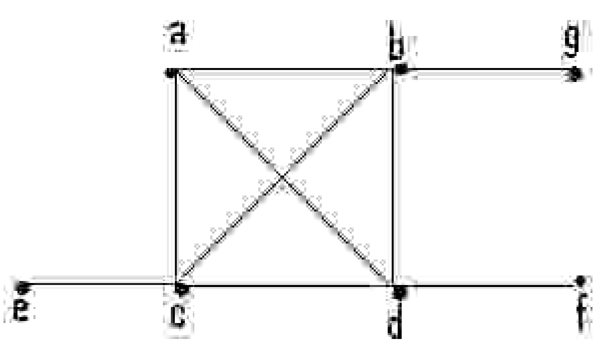
C(56,7)

56!/(56-7)!7!
56!/49!.7!

Question No: 48 (Marks: 5)

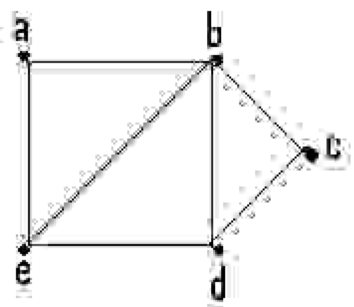
Determine whether the given graph has a Hamilton circuit? If it does, find such a circuit, if it does not , given an argument to show why no such circuit exists.

(a)



This graph does not have Hamiltonian circuit, because it does not satisfy the conditions. Circuit may not be completed without repeating edges. It has also unequal values of edges and vertices.

(b)



This graph is a Hamiltonian circuit ..Its path is a b c d e a

Question No: 49 (Marks: 5)

Find the GCD of **11425 , 450** using Division Algorithm.

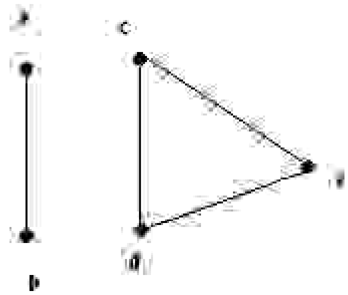
LCM = 205650

11425 = 450x25 + 175
450 = 175x2 + 100
175 = 100x1 + 75
100 = 75x1 + 25
75 = 25x3 + 0

Linear combination= 25 = 127x450 + -5x11425
GCD= 25...Ans

Question No: 50 (Marks: 10)

Write the adjacency matrix of the given graph also find transpose and product of adjacency matrix and its transpose (if not possible then give reason)



Adjacency matrix=
$$\begin{matrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{matrix}$$

Transpose =
$$\begin{matrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{matrix}$$

Its transpose is not possible…it's same! Because there is no loop.
It is not directed graph.

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Fall 2009

MTH202- Discrete Mathematics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

Let A = {a, b, c} and

R = {(a, c), (b, b), (c, a)} be a relation on A. Is R

► Transitive

► Reflexive

► **Symmetric**

► Transitive and Reflexive

Question No: 2 (Marks: 1) - Please choose one

Symmetric and antisymmetric are

► **Negative of each other**

► Both are same

► Not negative of each other

Question No: 3 (Marks: 1) - Please choose one

The statement $p \leftrightarrow q \equiv q \leftrightarrow p$

describes

► **Commutative Law: page 27**

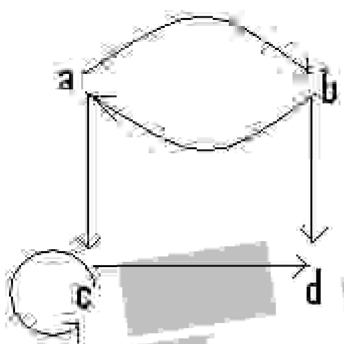
► Implication Laws:

► Exportation Law:

► Equivalence:

Question No: 4 (Marks: 1) - Please choose one

The relation as a set of ordered pairs as shown in figure is



► {(a,b),(b,a),(b,d),(c,d)}

► {(a,b),(b,a),(a,c),(b,a),(c,c),(c,d)}

► **{(a,b), (a,c), (b,a),(b,d), (c,c),(c,d)}**

► {(a,b), (a,c), (b,a),(b,d),(c,d)}

Question No: 5 (Marks: 1) - Please choose one

The statement $p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$

Describes

► Commutative Law:

► Implication Laws:

► Exportation Law:

► **Reductio ad absurdum** **page 27**

Question No: 6 (Marks: 1) - Please choose one

A circuit with one input and one output signal is called.

► **NOT-gate (or inverter)**

► OR- gate

► AND- gate

► None of these

Question No: 7 (Marks: 1) - Please choose one

If $f(x) = 2x+1$, $g(x)=x^2 -1$ then $fg(x) =$

► x^2 -1

► $2x^2$ -1

► $2x^3$ -1

Question No: 8 (Marks: 1) - Please choose one

Let g be the functions defined by
g(x)= 3x+2 then gog(x) =

► $9x^2+4$

► 6x+4

► 9x+8

Question No: 9 (Marks: 1) - Please choose one

How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5?

► 333

► 467

► 533

► 497

Question No: 10 (Marks: 1) - Please choose one

What is the smallest integer N such that $\left\lceil \frac{N}{6} \right\rceil = 9$

► 46

► 29

► 49

Question No: 11 (Marks: 1) - Please choose one

What is the probability of getting a number greater than 4 when a die is thrown?

► $\frac{1}{2}$

► $\frac{3}{2}$

► $\frac{1}{3}$

Question No: 12 (Marks: 1) - Please choose one

If A and B are two disjoint (mutually exclusive) events then

P(A□B) =

► P(A) + P(B) + P(A□B)

- ► P(A) + P(B) + P(AUB)
- ► P(A) + P(B) - P(A□B)
- ► P(A) + P(B) - P(A□B)
- ► **P(A) + P(B)**

Question No: 13 (Marks: 1) - Please choose one

If a die is thrown then the probability that the dots on the top are prime numbers or odd numbers is

- ► 1
- ► $\frac{1}{3}$
- ► $\frac{2}{3}$

Question No: 14 (Marks: 1) - Please choose one

The probability of getting 2 heads in two successive tosses of a balanced coin is

► $\frac{1}{4}$

► $\frac{1}{2}$

► $\frac{2}{3}$

Question No: 15 (Marks: 1) - Please choose one

The probability of getting a 5 when a die is thrown?

► $\frac{1}{6}$

► $\frac{5}{6}$

► $\frac{1}{3}$

Question No: 16 (Marks: 1) - Please choose one

If a coin is tossed then what is the probability that the number is 5

► $\frac{1}{2}$

► 0

► 1

Question No: 17 (Marks: 1) - Please choose one

If A and B are two sets then the set of all elements that belong to both A and B, is

► A U B

► A ▯ B

► A--B

► None of these

Question No: 18 (Marks: 1) - Please choose one

What is the expectation of the number of heads when three fair coins are tossed?

► 1

► 1.34

► 2

► **1.5** **page 275**

misuse

Question No: 19 (Marks: 1) - Please choose one

If A, B and C are any three events, then

$P(A \cup B \cup C)$ = is equal to

► $P(A) + P(B) + P(C)$

► **$P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$**

pg262

► $P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C)$

► $P(A) + P(B) + P(C) + P(A \cap B \cap C)$

Question No: 20 (Marks: 1) - Please choose one

A rule that assigns a numerical value to each outcome in a sample space is called

► One to one function

► Conditional probability

► **Random variable**

Question No: 21 (Marks: 1) - Please choose one

The power set of a set A is the set of all subsets of A, denoted P(A).

► False

► True

Question No: 22 (Marks: 1) - Please choose one

A walk that starts and ends at the same vertex is called

► Simple walk

► Circuit

► Closed walk

Question No: 23 (Marks: 1) - Please choose one

If a graph has any vertex of degree 3 then

► It must have Euler circuit

► It must have Hamiltonian circuit

► It does not have Euler circuit (becz 3 is odd)

Question No: 24 (Marks: 1) - Please choose one

The square root of every prime number is irrational

► True

► False

► Depends on the prime number given

Question No: 25 (Marks: 1) - Please choose one

A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables

► True pg200

► False

► None of these

Question No: 26 (Marks: 1) - Please choose one

If r is a positive integer then gcd(r,0)=

► r

► 0

► 1

► None of these

Question No: 27 (Marks: 1) - Please choose one

Combinatorics is the mathematics of counting and arranging objects

► True

► False

► Cannot be determined

Question No: 28 (Marks: 1) - Please choose one

A circuit that consist of a single vertex is called

► Trivial

► Tree

► Empty

Question No: 29 (Marks: 1) - Please choose one

In the planar graph, the graph crossing number is

► **0** **page 312**

► 1

► 2

► 3

Question No: 30 (Marks: 1) - Please choose one

How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?

► **C(10,5)**

► C(5,10)

► P(10,5)

► None of these

Solution: The answer is given by the number of 5-combinations of a set with ten elements. By Theorem 2, the number of such combinations is

$$C(10, 5) = \frac{10!}{5!\,5!} = 252.$$

Question No: 31 (Marks: 1) - Please choose one

The value of 0! Is

► 0

► 1

► Cannot be determined

Question No: 32 (Marks: 1) - Please choose one

If the transpose of any square matrix and that matrix are same then matrix is called

► Additive Inverse

► Hermition Matrix

► Symmetric Matrix

Question No: 33 (Marks: 1) - Please choose one

The value of $\frac{(n-1)!}{(n+1)!}$ is

► 0

► n(n-1)

► $\frac{1}{(n^2+n)}$

► Cannot be determined

Question No: 34 (Marks: 1) - Please choose one

If A and B are two disjoint sets then which of the following **must be** true

► **n(AB) = n(A) + n(B)**

► n(AB) = n(A) + n(B) - n(AB)

► n(AB)= ø

► None of these

Question No: 35 (Marks: 1) - Please choose one

Any two spanning trees for a graph

► Does not contain same number of edges

► Have the same degree of corresponding edges

► **contain same number of edges**

► May or may not contain same number of edges

Question No: 36 (Marks: 1) - Please choose one

When P(k) and P(k+1) are true for any positive integer k, then P(n) is not true for all +ve Integers.

► True

► **False by ali**

Question No: 37 (Marks: 1) - Please choose one

$n^2 > n+3$ for all integers $n \geq 3$.

► True

► False

Question No: 38 (Marks: 1) - Please choose one

Quotient –Remainder Theorem states that for any positive integer d, there exist unique integer q and r such that ________________ and 0≤r<d.

► **n=d.q+ r**

► n=d.r+ q

► n=q.r+ d

► None of these

Question No: 39 (Marks: 1) - Please choose one

Euler formula for graphs is

► f = e-v

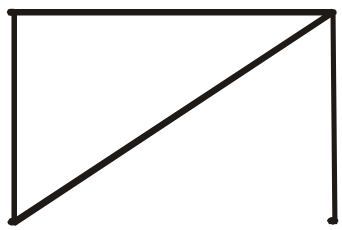
► f = e+v +2

► f = e-v-2

► **f = e-v+2**

Question No: 40 (Marks: 1) - Please choose one

The degrees of {a, b, c, d, e} in the given graph is



► 2, 2, 3, 1, 1

► 2, 3, 1, 0, 1

► 0, 1, 2, 2, 0

► 2, 3,1,2,0

Question No: 41 (Marks: 2)

Let $A = \begin{bmatrix} 1 & 3 & 7 \\ 5 & 2 & 9 \end{bmatrix}$ then find A^t

Question No: 42 (Marks: 2)

Write the contra positive of the following statements:

1. For all integers n, if n^2 is odd then n is odd.
2. If m and n are odd integers, then m+n is even integer.

Question No: 43 (Marks: 2)

How many distinguishable ways can the letter of the word HULLABALOO be arranged.

Question No: 44 (Marks: 3)

Find the variance σ^2 of the distribution given in the following table.

x_i	1	3	4	5
$f(x_i)$	0.4	0.1	0.2	0.3

Ans:
3

Question No: 45 (Marks: 3)

Prove that every integer is a rational number.

Question No: 46 (Marks: 3)

a. Evaluate P(5,2)
b. How many 5-permutations are there of a set of five objects?

Question No: 47 (Marks: 5)

Is it possible to have a simple graph with four vertices of degree 1, 1, 3, and 3.If no then give reason?(Justify your answer)

Question No: 48 (Marks: 5)

Find the GCD of **500008, 78** using Division Algorithm.

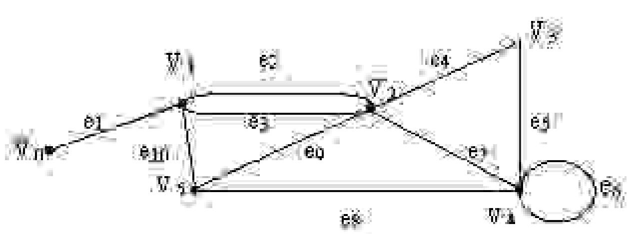
Question No: 49 (Marks: 5)

Find the M number of ways that ten chocolates can be divided among three children if the youngest child is to receive four chocolates and each of the others three chocolates.

Question No: 50 (Marks: 10)

In the graph below, determine whether the following walks are paths, simple paths, closed walks, circuits,

simple circuits, or are just walk?



i) $v_0\ e_1\ v_1\ e_{10}\ v_5\ e_9\ v_2\ e_2\ v_1$

ii) $v_4\ e_7\ v_2\ e_9\ v_5\ e_{10}\ v_1\ e_3\ v_2\ e_9\ v_5$

iii) v_2

iv) $v_5\ v_2\ v_3\ v_4\ v_4 v_5$

v) $v_2\ v_3\ v_4\ v_5\ v_2 v_4\ v_3\ v_2$

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ALL IN ONE MTH202 Final term PAPERS & MCQz

FINALTERM EXAMINATION

Spring 2010

MTH202- Discrete Mathematics (Session - 1)

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

Whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric, or transitive, where $(x, y) \in R$ if and only if $xy \geq 1$

- ► Antisymmetric
- ► Transitive
- ► **Symmetric**
- ► Both Symmetric and transitive

Question No: 2 (Marks: 1) - Please choose one

For a binary relation R defined on a set A , if for all $t \in A, (t, t) \notin R$ then R is

- ► **Antisymmetric**
- ► Symmetric
- ► Irreflexive

Question No: 3 (Marks: 1) - Please choose one

If ($A \cup B$) = A, then ($A \cap B$) = B

- ► True
- ► **False**
- ► Cannot be determined

Question No: 4 (Marks: 1) - Please choose one

Let

$$a_0 = 1, a_1 = -2 \text{ and } a_2 = 3$$

$$\text{then } \sum_{j=0}^{2} a_j =$$

► -6

► 2
► 8

Question No: 5 (Marks: 1) - Please choose one

The part of definition which can be expressed in terms of smaller versions of itself is called

► Base
► Restriction
► **Recursion**
► Conclusion

Question No: 6 (Marks: 1) - Please choose one

What is the smallest integer N such that $\left\lceil \frac{N}{6} \right\rceil = 9$

► 46
► 29
► **49**

Question No: 7 (Marks: 1) - Please choose one

In probability distribution random variable f satisfies the conditions

► $f(x_i) \leq 0 \quad and \quad \sum_{i=1}^{n} f(x_i) \neq 1$

► $f(x_i) \geq 0 \quad and \quad \sum_{i=1}^{n} f(x_i) = 1$

► $f(x_i) \geq 0 \quad and \quad \sum_{i=1}^{n} f(x_i) \neq 1$

► $f(x_i) \text{p} 0 \quad and \quad \sum_{i=1}^{n} f(x_i) = 1$

Question No: 8 (Marks: 1) - Please choose one

What is the probability that a hand of five cards contains four cards of one kind?

► 0.0018

► $\frac{1}{2}$

► **0.0024**

Question No: 9 (Marks: 1) - Please choose one

A rule that assigns a numerical value to each outcome in a sample space is called

► One to one function

► Conditional probability

► **Random variable**

Question No: 10 (Marks: 1) - Please choose one

A walk that starts and ends at the same vertex is called

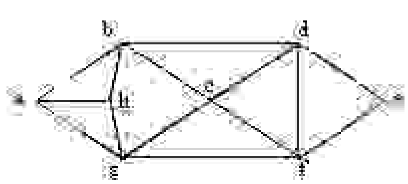
► Simple walk

► Circuit

► **Closed walk**

Question No: 11 (Marks: 1) - Please choose one

The Hamiltonian circuit for the following graph is



► abcdefgh
► abefgha
► abcdefgha

Question No: 12 (Marks: 1) - Please choose one

Distributive law of union over intersection for three sets

► A È (B È C) = (A È B) È C

► A Ç (B Ç C) = (A Ç B) Ç C

► A È (B Ç C) = (A È B) Ç (A È B)

► None of these

Question No: 13 (Marks: 1) - Please choose one

The indirect proof of a statement pàq involves

►Considering ~q and then try to reach ~p

►Considering p and ~q and try to reach contradiction

*►Both 2 and 3 above

► Considering p and then try to reach q

Question No: 14 (Marks: 1) - Please choose one

The square root of every prime number is irrational

► **True**

► False

► Depends on the prime number given

Question No: 15 (Marks: 1) - Please choose one

If a and b are any positive integers with b≠0 and q and r are non negative integers such that a= b.q+r then

► **gcd(a,b)=gcd(b,r)**

► gcd(a,r)=gcd(b,r)

► gcd(a,q)=gcd(q,r)

Question No: 16 (Marks: 1) - Please choose one

The greatest common divisor of 27 and 72 is

► 27

► **9**

► 1

► None of these

Question No: 17 (Marks: 1) - Please choose one

In how many ways can a set of five letters be selected from the English Alphabets?

► **C(26,5)**

► C(5,26)

► C(12,3)

► None of these

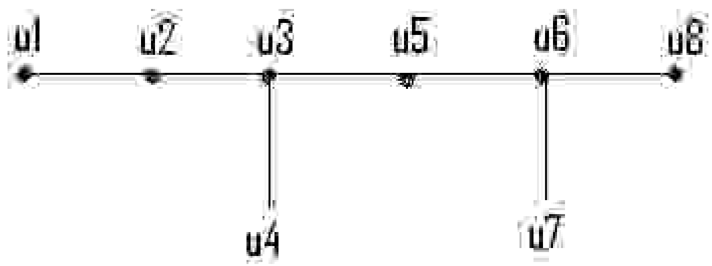
Question No: 18 (Marks: 1) - Please choose one

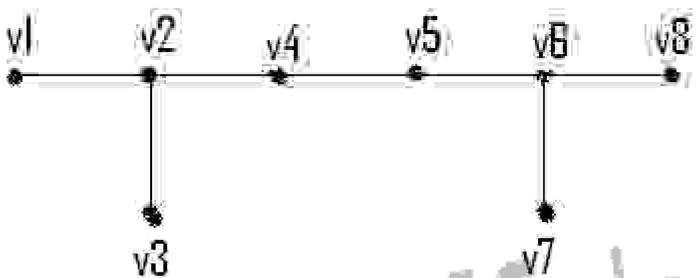
A vertex of degree greater than 1 in a tree is called a

► **Branch vertex**
► Terminal vertex
► Ancestor

Question No: 19 (Marks: 1) - Please choose one

For the given pair of graphs whether it is





► Isomorphic

► **Not isomorphic**

Question No: 20 (Marks: 1) - Please choose one

The value of (-2)! Is

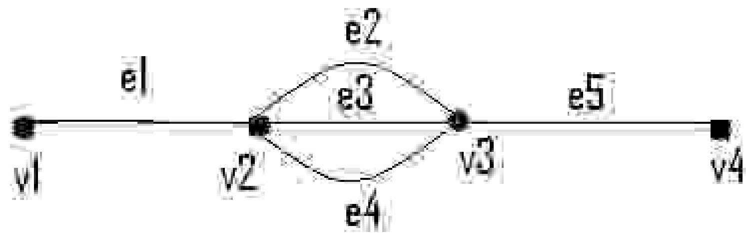
► 0

► 1

► **Cannot be determined**

Question No: 21 (Marks: 1) - Please choose one

In the following graph



How many simple paths are there from v_1 to v_4

► 2
► **3**
► 4

Question No: 22 (Marks: 1) - Please choose one

The value of $\frac{(n+1)!}{(n-1)!}$ is

► 0
► **n(n-1)**
► $n^2 + n$
► Cannot be determined

Question No: 23 (Marks: 1) - Please choose one

If A and B are finite (overlapping) sets, then which of the following **must be** true

► n(AÈB) = n(A) + n(B)

► **n(AÈB) = n(A) + n(B) - n(AÇB) page238**

► n(AÈB)= ø

► None of these

Question No: 24 (Marks: 1) - Please choose one

Any two spanning trees for a graph

► Does not contain same number of edges
► Have the same degree of corresponding edges
► contain same number of edges
► May or may not contain same number of edges

Question No: 25 (Marks: 1) - Please choose one

When 3^k is even, then $3^k+3^k+3^k$ is an odd.

► True
► **False**

Question No: 26 (Marks: 1) - Please choose one

Quotient -Remainder Theorem states that for any positive integer d, there exist unique integer q and r such that n=d.q+ r and __________________.

► **$0 \le r < d$**
► $0 < r < d$
► $0 \le d < r$
► None of these

Question No: 27 (Marks: 1) - Please choose one

The value of $\lceil x \rceil$ for x = -3.01 is

► **-3.01**
► -3
► -2

► -1.99

Question No: 28 (Marks: 1) - Please choose one

If p= A Pentium 4 computer,

q= attached with ups.

Then "no Pentium 4 computer is attached with ups" is denoted by

► ~ (pÙq)

► ~ pÚq

► ~ pÙq

► None of these

Question No: 29 (Marks: 1) - Please choose one

An integer n is prime if and only if n > 1 and for all positive integers r and s, if

n = r·s, then

► r = 1 or s = 2.

► r = 1 or s = 0.

► r = 2 or s = 3.

► None of these

Question No: 30 (Marks: 1) - Please choose one

If $P(A \cap B) \neq P(A)P(B)$ then the events A and B are called

► Independent
► **Dependent**
► Exhaustive

Question No: 31 (Marks: 2)

Let A and B be the events. Rewrite the following event using set notation

"Only A occurs"

Question No: 32 (Marks: 2)

Suppose that a connected planar simple graph has 15 edges. If a plane drawing of this graph has 7 faces, how many vertices does this graph have?

Answer:

Given,
Edges = e =15
Faces = f = 7
Vertices = v =?
According toEuler Formula, we know that,

f= e – v +2

Putting values, we get

7 = 15 – v + 2

7 = 17 – v

Simplifying

v =1 7-7 =10

Question No: 33 (Marks: 2)

How many ordered selections of two elements can be made from the set {0,1,2,3}?

Answer

The order selection of two elements from 4 is as

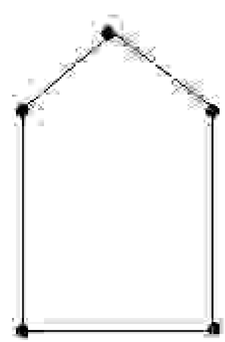
P(4,2) = 4!/(4-2)!
= (4.3.2.1)/2!
= 12

Question No: 34 (Marks: 3)

Consider the following events for a family with children:
A={children of both sexes}, B={at most one boy}.Show that A and B are dependent events if a family has only two children.

Question No: 35 (Marks: 3)

Determine the chromatic number of the given graph by inspection.



Question No: 36 (Marks: 3)

A cafeteria offers a choice of two soups, five sandwiches, three desserts and three drinks. How many different lunches, each consisting of a soup, a sandwiche, a dessert and a drink are possible?

Question No: 37 (Marks: 5)

A box contains 15 items, 4 of which are defective and 11 are good. Two items are selected. What is probability that the first is good and the second defective?

Answer

Question No: 38 (Marks: 5)

Draw a binary tree with height 3 and having seven terminal vertices.

Answer: page324

Given height=h=3

Any binary tree with height 3 has almost 2^3=8 terminal vertices. But here terminal vertices are 7 and Internal vertices=k=6 so binary trees exist:

Question No: 39 (Marks: 5)

Find n if

P(n,2) = 72

(a) P(n,2) = 72

SOLUTION:

(a) Given P(n,2) = 72

$\Rightarrow$ n · (n-1) = 72 (by using the definition of permutation)

$\Rightarrow$ n^2 -n = 72

$\Rightarrow$ n^2 - n - 72 = 0

$\Rightarrow$ n = 9, -8

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ALL IN ONE MTH202 Final term PAPERS & MCQz

FINALTERM EXAMINATION

Fall 2009

MTH202- Discrete Mathematics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The negation of "Today is Friday" is

► Today is Saturday

► Today is not Friday

► Today is Thursday

Question No: 2 (Marks: 1) - Please choose one

In method of proof by contradiction, we suppose the statement to be proved is false.

► True

► False

Question No: 3 (Marks: 1) - Please choose one

Whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric, or transitive, where $(x, y) \in R$ if and only if $xy \geq 1$

► Antisymmetric

► Transitive

► Symmetric

► Both Symmetric and transitive

Question No: 4 (Marks: 1) - Please choose one

The inverse of given relation R = {(1,1),(1,2),(1,4),(3,4),(4,1)} is

► {(1,1),(2,1),(4,1),(2,3)}

► {(1,1),(1,2),(4,1),(4,3),(1,4)}

► {(1,1),(2,1),(4,1),(4,3),(1,4)}

Question No: 5 (Marks: 1) - Please choose one

A circuit with one input and one output signal is called.

► NOT-gate (or inverter)

► OR- gate

► AND- gate

► None of these

Question No: 6 (Marks: 1) - Please choose one

A sequence in which common difference of two consecutive terms is same is called

► geometric mean

► harmonic sequence

► geometric sequence

► arithmetic progression page147

Question No: 7 (Marks: 1) - Please choose one

If the sequence $\{a_n\} = 2.(-3)^n + 5^n$ then the term a_1 is

► -1

► 0

► 1

► 2

Question No: 8 (Marks: 1) - Please choose one

How many integers from 1 through 100 must you pick in order to be sure of getting one that is divisible by 5?

► 21

► 41

► 81

► 56

Question No: 9 (Marks: 1) - Please choose one

What is the probability that a randomly chosen positive two-digit number is a multiple of 6?

► 0.5213

► 0.167 pg252

► 0.123

Question No: 10 (Marks: 1) - Please choose one

If a pair of dice is thrown then the probability of getting a total of 5 or 11 is

► $\frac{1}{18}$

► $\frac{1}{9}$

► $\frac{1}{6}$ pg256

Question No: 11 (Marks: 1) - Please choose one

If a die is rolled then what is the probability that the number is greater than 4

► $\frac{1}{3}$

► $\frac{3}{4}$

► $\frac{1}{2}$

Question No: 12 (Marks: 1) - Please choose one

If a coin is tossed then what is the probability that the number is 5

► $\frac{1}{2}$

► 0

► 1

Question No: 13 (Marks: 1) - Please choose one

If A and B are two sets then The set of all elements that belong to both A and B , is

► A ∩ B

► A U B

► A--B

► None of these

Question No: 14 (Marks: 1) - Please choose one

If A and B are two sets then The set of all elements that belong to A but not B , is

► A □ B

► A □ B

► None of these

► A-B

Question No: 15 (Marks: 1) - Please choose one

If A, B and C are any three events, then

P(A□B□C) is equal to

► P(A) + P(B) + P(C)

► P(A) + P(B) + P(C)- P(A□B) - P (A □C) - P(B □C) + P(A □B □C)

► P(A) + P(B) + P(C) - P(A□B) - P (A □C) - P(B □C)

► P(A) + P(B) + P(C) + P(A □B □C)

Question No: 16 (Marks: 1) - Please choose one

If a graph has any vertex of degree 3 then

► **It must have Euler circuit**

► **It must have Hamiltonian circuit**

► **It does not have Euler circuit**

Question No: 17 (Marks: 1) - Please choose one

The contradiction proof of a statement p q involves

► **Considering p and then try to reach q**

► Considering ~q and then try to reach ~p

► Considering p and ~q and try to reach contradiction

► None of these

Question No: 18 (Marks: 1) - Please choose one

How many ways are there to select a first prize winner a second prize winner, and a third prize winner from 100 different people who have entered in a contest.

► None of these

► **P(100,3)**

► P(100,97)

► P(97,3)

Question No: 19 (Marks: 1) - Please choose one

A vertex of degree 1 in a tree is called a

► **Terminal vertex**

► **Internal vertex**

Question No: 20 (Marks: 1) - Please choose one

Suppose that a connected planar simple graph has 30 edges. If a plane drawing of this graph has 20 faces, how many vertices does the graph have?

► **12**

► **13**

► **14**

Question No: 21 (Marks: 1) - Please choose one

How many different ways can three of the letters of the word BYTES be chosen if the first letter must be B ?

► P(4,2)

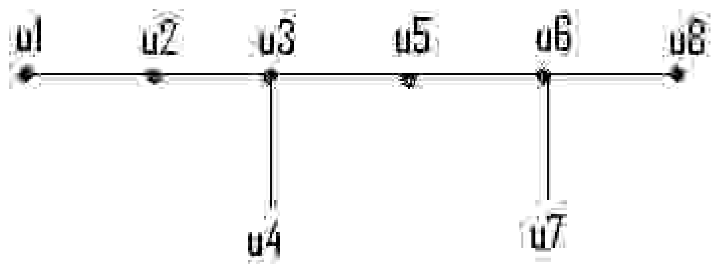
► P(2,4)

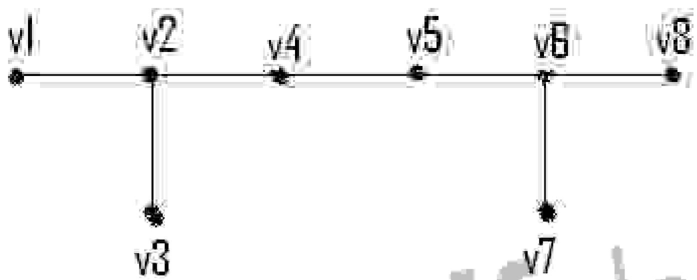
► C(4,2)

► None of these

Question No: 22 (Marks: 1) - Please choose one

For the given pair of graphs whether it is





► Isomorphic

► Not isomorphic

Question No: 23 (Marks: 1) - Please choose one

On the set of graphs the graph isomorphism is

► Isomorphic Invariant

► Equivalence relation pg304

► Reflexive relation

Question No: 24 (Marks: 1) - Please choose one

A matrix in which number of rows and columns are equal is called

► Rectangular Matrix

► Square Matrix

► Scalar Matrix

Question No: 25 (Marks: 1) - Please choose one

If the transpose of any square matrix and that matrix are same then matrix is called

► Additive Inverse

► Hermition Matrix

► Symmetric Matrix

Question No: 26 (Marks: 1) - Please choose one

The number of *k*-combinations that can be chosen from a set of *n* elements can be written as

► $^{n}C_{k}$ pg223

► $^{k}C_{n}$

► $^{n}P_{k}$

► $^{k}P_{k}$

Question No: 27 (Marks: 1) - Please choose one

The value of $C(n, 0)$ =

► 1

► 0

► n

► None of these

Question No: 28 (Marks: 1) - Please choose one

If the order does not matter and repetition is not allowed then total number of ways for selecting k sample from n. is

► P(n,k)

► C(n,k) pg228

► n^k

► C(n+k-1,k)

Question No: 29 (Marks: 1) - Please choose one

If A and B are two disjoint sets then which of the following must be true

► $n(A \cup B) = n(A) + n(B)$

► $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

► n(A□B)= ø

► None of these

Question No: 30 (Marks: 1) - Please choose one

Among 200 people, 150 either swim or jog or both. If 85 swim and 60 swim and jog, how many jog?

► 125 pg239

► 225

► 85

► 25

Question No: 31 (Marks: 1) - Please choose one

If two sets are disjoint, then P□Q is

- ▯
- P
- Q
- P▯Q

Question No: 32 (Marks: 1) - Please choose one

Every connected tree

- does not have spanning tree
- may or may not have spanning tree
- has a spanning tree

Question No: 33 (Marks: 1) - Please choose one

When P(k) and P(k+1) are true for any positive integer k, then P(n) is not true for all +ve Integers.

- True
- False

Question No: 34 (Marks: 1) - Please choose one

When 3^k is even, then $3^k+3^k+3^k$ is an odd.

► True

► False

Question No: 35 (Marks: 1) - Please choose one

$5^n - 1$ is divisible by 4 for all positive integer values of n.

► True

► False

Question No: 36 (Marks: 1) - Please choose one

Quotient –Remainder Theorem states that for any positive integer d, there exist unique integer q and r such that $n=d.q+r$ and ________________.

► $0 \le r < d$

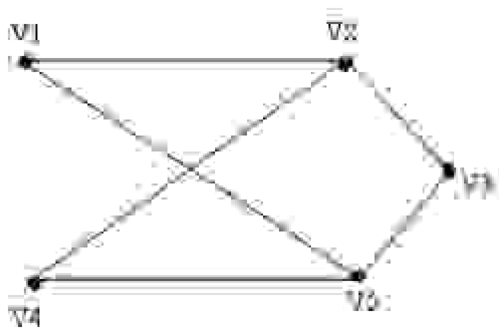
► $0 < r < d$

► $0 \le d < r$

► None of these

Question No: 37 (Marks: 1) - Please choose one

The given graph is



► Simple graph

► Complete graph

► Bipartite graph

► Both (i) and (ii)

► Both (i) and (iii)

Question No: 38 (Marks: 1) - Please choose one

An integer n is even if and only if n = 2k for some integer k.

► True

► False

► Depends on the value of k

Question No: 39 (Marks: 1) - Please choose one

The word "algorithm" refers to a step-by-step method for performing some action.

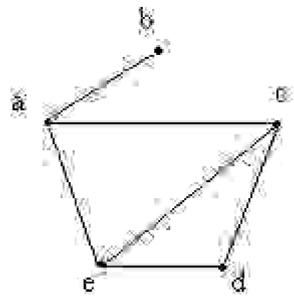
► True

► False

► None of these

Question No: 40 (Marks: 1) - Please choose one

The adjacency matrix for the given graph is



► $\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$

► $\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$

► $\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$

► None of these

Question No: 41 (Marks: 2)

Let A and B be events with

$P(A)=\frac{1}{2}, P(B)=\frac{1}{3}$ and $P(A \cap B)=\frac{1}{4}$

Find

$P(B \mid A)$

Ans =

P(A ∩ B) = P(A) P(B|A)

P(B|A)= P(A ∩ B)/ P(A)

= 0.5

Question No: 42 (Marks: 2)

Suppose that a connected planar simple graph has 15 edges. If a plane drawing of this graph has 7 faces, how many vertices does this graph have?

Vertices-edges+faces=2
V - 15 +7=2
V -8 = 2
V =10

Question No: 43 (Marks: 2)

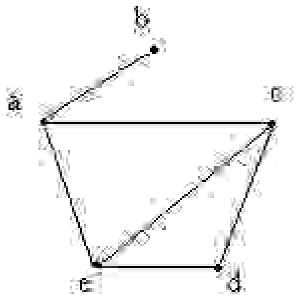
Find integers q and r so that a=bq+r , with 0≤r<b.
a=45 , b=6.

Question No: 44 (Marks: 3)

Draw a graph with six vertices, five edges that is not a tree.

Asn:

Here is the graph with six vertices, five edges that is not a tree



Question No: 45 (Marks: 3)

Prove that every integer is a rational number.

Ans:
Every integer is a rational number, since each integer n can be written in the form n/1. For example 5 = 5/1 and thus 5 is a rational number. However, numbers like 1/2, 45454737/2424242, and -3/7 are also rational; since they are fractions whose numerator and denominator are integers.

Question No: 46 (Marks: 3)

b. **Evaluate P(5,2)**
c. **How many 4-permutations are there of a set of seven objects?**

Question No: 47 (Marks: 5)

Find the GCD of 500008, 78 using Division Algorithm.

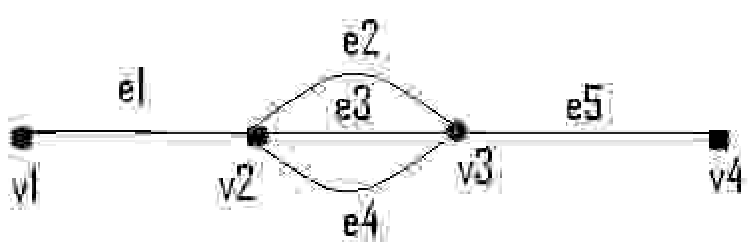
Question No: 48 (Marks: 5)

There are 25 people who work in an office together. Four of these people are selected to attend four different conferences. The first person selected will go to a conference in New York, the second will go to Chicago, the third to San Franciso, and the fourth to Miami. How many such selections are possible?

Ans= 12650

Question No: 49 (Marks: 5)

Consider the following graph



(a) How many simple paths are there from v_1 to v_4 ====1

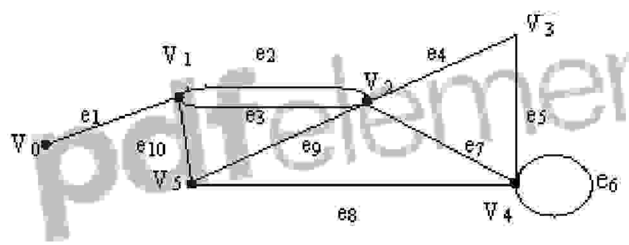
(b) How many paths are there from v_1 to v_4 ?
===========3

(c) How many walks are there from v_1 to v_4 ?==========3

Question No: 50 (Marks: 10)

In the graph below, determine whether the following walks are paths, simple paths, closed walks, circuits,

simple circuits, or are just walk?



vi) v_0 e_1 v_1 e_{10} v_5 e_9 v_2 e_2 v_1== paths

vii) v_4 e_7 v_2 e_9 v_5 e_{10} v_1 e_3 v_2 e_9 v_5========, circuits

viii) v_2========== closed walks

ix) v_5 v_2 v_3 v_4 v_4v_5=========== closed walks

x) v_2 v_3 v_4 v_5 v_2v_4 v_3 v_2========= paths

Subjective types short answer:
&
Important definitions:

Question: What does it mean by the preservation of edge end point function in the definition of isomorphism of graphs?

Answer: Since you know that we are looking for two functions (Suppose one function is "f" and other function is "g") which preserve the edge end point function and this preservation means that if we have vi as an end point of the edge ej then f(vi) must be an end point of the edge g(ej) and also the converse that is if f(vi) be an end point of the edge g(ej) then we must have vi as an end point of the edge ej. Note that vi and ej are the vertex and edge of one graph respectively where as f (vi) and g (ej) are the vertex and edge in the other graph respectively.

Question: Is there any method of identifying that the given graphs are isomorphic or not?(With out finding out two functions).

Answer: Unfortunately there is no such method which will identify whether the given graphs are isomorphic or not. In order to find out whether the two given graphs are isomorphic first we have to find out all the bijective mappings from the set vertices of one graph to the set of vertices of the other graph then find out all the bijective functions from the set of edges of one graph to the set of edges of the other graph. Then see which mappings preserve the edge end point function as defined in the definition of Isomorphism of graphs. But it is easy to identify that the two graphs are not isomorphic. First of all note that if there is any Isomorphic Invariant not satisfied by both the graphs, then we will say that the graphs are not Isomorphic. Note that if all the isomorphic Invariants are satisfied by two graphs then we can't conclude that the graphs are isomorphic. In order to prove that the graphs are isomorphic we have to find out two functions which satisfied the condition as defined in the definition of Isomorphism of graphs.

Question: What are Complementary Graphs?

Answer: Complementary Graph of a simple graph(G) is denoted by the (G bar) and has as many vertices as G but two vertices are adjacent in complementary Graph by an edge if and only if these two vertices are not adjacent in G .

Question: What is the application of isomorphism in real word?

Answer: There are many applications of the graph theory in computer Science as well as in the Practical life; some of them are given below. (1) Now you also go through the puzzles like that we have to go through these points without lifting the pencil and without repeating our path. These puzzles can be solved by the Euler and

Hamiltonian circuits. (2) Graph theory as well as Trees has applications in "DATA STRUCTURE" in which you will use trees, especially binary trees in manipulating the data in your programs. Also there is a common application of the trees is "FAMILY TREE". In which we represent a family using the trees. (3) Another example of the directed Graph is "The World Wide Web ". The files are the vertices. A link from one file to another is a directed edge (or arc). These are the few examples.

Question: Are Isomorphic graphs are reflexive, symmetric and transitive?

Answer: We always talk about " RELXIVITY"" SYMMETRIC" and TRANSIVITY of a relation. We never say that a graph is reflexive, symmetric or transitive. But also remember that we draw the graph of a relation which is reflexive and symmetric and the property of reflexivity and symmetric is evident from the graphs, we can't draw the graph of a relation such that transitive property of the relation is evident. Now consider the set of all graphs say it G, this being a set ,so we can define a relation from the set G to itself. So we define the relation of Isomorphism on the set G x G.(By the definition of isomorphism) Our claim is that this relation is an " Equivalence Relation" which means that the relation of Isomorphism's of two graphs is "REFLEXIVE" "SYMMETRIC" and "TRANSITIVE". Now if you want to draw the graph of this relation, then the vertices of this graph are the graphs from the set G.

Question: Why we can't use the same color in connected portions of planar graph?

Answer: We define the coloring of graph in such a manner that we can't assign the same color to the adjacent vertices because if we give the same colors to the adjacent vertices then they are indistinguishable. Also note that we can give the same color to the adjacent vertices but such a coloring is called improper coloring and the way which we define the coloring is known as the proper coloring. We are interested in proper coloring that's why all the books consider the proper coloring

Question: What is meant by isomorphic invariant?

Answer: A property "P" of a graph is known as Isomorphic invariant. if the same property is found in all the graphs which are isomorphic to it. And all these properties are called isomorphic invariant (Also it clear from the words Isomorphic Invariant that the properties which remain invariant if the two graphs are isomorphic to each other).

Question: What is an infinite Face?

Answer: When you draw a Planar Graph on a plane it divides the plane into different regions, these regions are known as the faces and the face which is not bounded by the edges of the graph is known as the Infinite face. In other words the region which is unbounded is known as Infinite Face.

Question: What is "Bipartite Graph"?

Answer: A graph is said to be Bipartite if it's set of vertices can be divided into two disjoint sets such that no two vertices of the same set are adjacent by some edge of the graph. It means that the edges of one set will be adjacent with the vertices of the other set.

Question: What is chromatic number?

Answer: While coloring a graph you can color a vertex which is not adjacent with the vertices you already colored by choosing a new color for it or by the same color which you have used for the vertices which are not adjacent with this vertex. It means that while coloring a graph you may have different number of colors used for this purpose. But the least number of colors which are being used during the coloring of Graphs is known as the Chromatic number.

Question: What is the role of Discrete mathematics in our prectical life. what advantages will we get by learning it.

Answer: In many areas people have to faces many mathematical problems which can,t be solved in computer so discrete mathematics provide the facility to overcome these problems. Discrete math also covers the wide range of topics, starting with the foundations of Logic, Sets and Functions. It moves onto integer mathematics and matrices, number theory, mathematical reasoning, probability graphs, tree data structures and Boolean algebra.So that is why we need discrete math.

Question: What is the De Morgan's law .

Answer: De Morgan law states " Negation of the conjunction of two statements is logiacally equivalent to the disjunction of their negation and Negation of the disjunction of two statements is logically equivalent to the conjucnction of their negation". i.e. ~(p^q) = ~p v ~q and ~(p v q)= ~p ^ ~q For example: " The bus was late and jim is waiting "(this is an example of conjuction of two statements) Now apply neaggation on this statement you will get through De Morgan's law " The bus was not late or jim is not waiting" (this is the disjunction of negation of two statements). Now see both statement are logically equivalent.Thats what De Morgan want to say

Question: What is Tauology?

Answer: A tautology is a statement form that is always true regardless of the truth values of the statement variables. i.e. If you want to prove that (p v q) is tautology ,you have to show that all values of statement (p v q) are true regardless of the values of p and q.If all the values of the satement (p v q) is not true then this statement is not tautology.

Question: What is binary relations and reflexive,symmetric and transitive.

Answer: Dear student! First of all ,I will tell you about the basic meaning of relation i.e It is a logical or natural association between two or more things; relevance of one to another; the relation between smoking and heart disease. The connection of people by blood or marriage. A person connected to another by blood or marriage; a relative. Or the way in which one person or thing is connected with another: the relation of parent to child. Now we turn to its mathematically definition, let A and B be any two sets. Then their cartesian product (or the product set) means a new set "A x B " which contains all the ordered pairs of the form (a,b) where a is in set A and b is in set B. Then if we take any subset say 'R' of "A x B" ,then 'R' is called the binary relation. Note All the subsets of the Cartesian product of two sets A and B are called the binary relations or simply a relation,and denoted by R. And note it that one raltion is also be the same as "A x B". Example: Let A={1,2,3} B={a,b} be any two sets. Then their Cartesian product means "A x B"={ (1,a),(1,b), (2,a),(2,b),(3,a),(3,b) } Then take any set which contains in "A x B" and denote it by 'R'. Let we take R={(2,b),(3,a),(3,b)} form "A x B". Clearly R is a subset of "A x B" so 'R' is called the binary relation. A reflexive relation defined on a set say 'A' means "all the ordered pairs in which 1st element is mapped or related to itself." For example take a relation say R1= {(1,1), (1,2), (1,3), (2,2) ,(2,1), (3,1) (3,3)} from "A x B" defined on the set A={1,2,3}. Clearly R1 is reflexive because 1,2 and 3 are related to itself. A relation say R on a set A is symmetric if whenever aRb then bRa,that is ,if whenever (a,b) belongs to R then (b,a) belongs to R for all a,b belongs to A. For example given a relation which is R1={(1,1), (1,2), (1,3), (2,2) ,(2,1), (3,1) (3,3)} as defined on a set A={1,2,3} And a relation say R1 is symmetric if for every (a, b) belongs to R ,(b, a) also belongs to R. Here as (a, b)=(1,1) belongs to R then (b, a)=(1,1)also belongs to R. as (a,b)=(1,2) belongs to R then (b,a)=(2,1)also belongs to R. as (a,b)=(1,3) belongs to R then (b,a)=(3,1)also belongs to R.etc So clearly the above relation R is symmetric. And read the definition of transitive relation from the handouts and the book. You can easily understand it.

Question: What is the matrix relation .

Answer: Suppose that A and B are finite sets.Then we take a relation say R from A to B. From a rectangular array whose rows are labeled by the elements of A and whose columns are labeled by the elements of B. Put a 1 or 0 in each position of the array according as a belongs to A is or is not related to b belongs to B. This array is called the matrix of the relation. There are matrix relations of reflexive and symmetric relations.

In reflexive relation, all the diagonal elements of relation should be equal to 1. For example if R = {(1,1), (1,3), (2,2), (3,2), (3,3)} defined on A = {1,2,3}. Then clearly R is reflexive. Simply in making matrix relation In the above example,as the defined set is A={1,2,3} so there are total three elements. Now we take 1, 2 and 3 horizontally and vertically.i.e we make a matrix from the relation R ,in the matrix you have now 3 columns and 3 rows. Now start to make the matrix ,as you have first order pair (1, 1) it means that 1 maps on itself and you write 1 in 1st row and in first column. 2nd order pair is (1, 3) it means that arrow goes from 1 to 3.Then you have to write 1 in 1st row and in 3rd column. (2, 2) means that arrow goes from 2 and ends itself. Here you have to write 1 in 2nd row and in 2nd column. (3,2) means arrow goes from 3 and ends at 2. Here you have to write 1 in 3rd row and in 2nd column. (3, 3) means that 3 maps on itself and you write 1 in 3rd row and in 3rd column. And where there is space empty or unfilled ,you have to write 0 there.

Question: what is binary relation.

Answer: Let A and B be any two sets. Then their cartesian product(or the product set) means a new set "A x B " which contains all the ordered pairs of the form (a,b) where a is in set A and b is in set B. Let we take any subset say 'R' of "A x B" ,then 'R' is called the binary relation. Note it that 'R' also be the same as "A x B". For example: Let A={1,2,3} B={a,b} be any two sets. Then their cartesian product means "A x B"={ (1,a), (1,b),(2,a),(2,b),(3,a),(3,b) } Then take any set which contains in "A x B" and denote it by 'R'. Let R={(2,b), (3,a),(3,b)} Clearly R is a subset of "A x B" so 'R' is called the binary relation.

Question: Role of ''Discrete Mathematics'' in our prectical life. what advantages will we get by learning it.

Answer: Discrete mathematics concerns processes that consist of a sequence of individual steps. This distinguishes it from calculus, which studies continuously changing processes. While the ideas of calculus were fundamental to the science and technology of the industrial revolution, the ideas of discrete mathematics underline the science and technology specific to the computer age. Logic and proof: An important goal of discrete mathematics is to develop students' ability to think abstractly. This requires that students learn to use logically valid forms of argument, to avoid common logical errors, to understand what it means to reason from definition, and to know how to use both direct and indirect argument to derive new results from those already known to be true. Induction and Recursion: An exciting development of recent years has been increased appreciation for the power and beauty of "recursive thinking": using the assumption that a given problem has been solved for smaller cases, to solve it for a given case. Such thinking often leads to recurrence relations, which can be "solved" by various techniques, and to verifications of solutions by mathematical induction. Combinatorics: Combinatorics is the mathematics of counting and arranging objects. Skill in using combinatorial techniques is needed in almost every discipline where mathematics is applied, from economics to biology, to computer science, to chemistry, to business management. Algorithms and their analysis: The word algorithm was largely unknown three decades ago. Yet now it is one of the first words encountered in the study of computer science. To solve a problem on a computer, it is necessary to find an algorithm or step-by-step sequence of instructions for the computer to follow. Designing an algorithm requires an understanding of the mathematics underlying the problem to be solved. Determining whether or not an algorithm is correct requires a sophisticated use of mathematical induction. Calculating the amount of time or memory space the algorithm will need requires knowledge of combinatorics, recurrence relations functions, and O-notation. Discrete Structures: Discrete mathematical structures are made of finite or count ably infinite collections of objects that satisfy certain properties. Those are sets, bolean of algebras, functions, finite start automata, relations, graphs and trees. The concept of isomorphism is used to describe the state of affairs when two distinct structures are the same intheir essentials and diffr only in the labeling of the underlying objects. Applications and modeling: Mathematics topic are best understood when they are seen ina variety of contexts and used to solve problems in a broad range of applied situations. One of the profound lessons of mathematics is that the same mathematical model can be used to solve problems in situations that appear superficially to be totally dissimilar. So in the end i want to say that discrete mathematics has many uses not only in computer science but also in the other fields too.

Question: what is the basic difference b/w sequences and series

Answer: A sequence is just a list of elements .In sequnce we write the terms of sequence as a list (seperated by comma's). e.g 2,3,4,5,6,7,8,9,... (in this we have terms 2,3,4,5,6,7,8,9 and so on).we write these in form of list seperated by comma's. And the sum of the terms of a sequence forms a series. e.g we have sequence 1,2,3,4,5,6,7 Now the series is sum of terms of sequence as 1+2+3+4+5+6+7.

Question: what is the purpose of permutations?

Answer: Permutation is an arrangement of objects in a order where repitition is not allowed. We need arrangments of objects in real life and also in mathematical problems.We need to know in how many ways we can arrange certain objects. There are four types of arrangments we have in which one is permutation.

Question: what is inclusion-exclusion principle

Answer: Inclusion-Exclusion principle contain two rules which are If A and B are disjoint finite sets, then n(AÈB) = n(A) + n(B) And if A and B are finite sets, then n(AÈB) = n(A) + n(B) - n(AÇB) For example If there are 15 girls students and 25 boys students in a class then how many students are in total. Now see if we take A ={ 15 girl students} and B={ 25 boys students} Here A and B are two disjoints sets then we can apply first rule n(AÈB) = n(A) + n(B) =15 + 25 =40 So in total there are 40 students in class. Take another Example for second rule. How many integers from 1 through 1000 are multiples of 3 or multiples of 5. Let A and B denotes the set of integers from 1 through 1000 that are multiples of 3 and 5 respectivly. n(A)= 333 n(B)=200 But these two sets are not disjoint because in A and B we have those elements which are multiple of both 3 and 5. so n(AÇB) =66 n(AÈB) = n(A) + n(B) - n(AÇB) =333 + 200 - 66 = 467

Question: How to use conditional probobility

Answer: Dear student In Conditional probability we put some condition on an event to be occur. e.g. A pair of dice is tossed. Find the probability that one of the dice is 2 if the sum is 6. If we have to find the probability that one of the dice is 2, then it is the case of simple probability. Here we put a condition that sum is six. Now A = { 2 appears in atleast one die} E = {sum is 6 } Here E = { (1,5), (2, 4), (3, 3), (4, 2), (5, 1) } Here two order pairs (2, 4) and (4, 2) satisfies the A. (i.e. belongs to A) Now A (intersection) B= { (2,4), (4,2)} Now by formula P(A/E) = P(A (intersection) E)/ P(E) = 2/5

Question: In which condition we use combination and in which condition permutation.

Answer: This depends on the statement of question. If in the statement of question you finds out that repetition of objects are not allowed and order matters then we use Permutation. e.g. Find the number of ways that a party of seven persons can arrange themselves in a row of seven chairs. See in this question repetition is no allowed because whenever a person is chosen for a particular seat r then he cannot be chosen again and also order matters in the arrangements of chairs so we use permutation here. If in the question repetition of samples are not allowed and order does not matters then we use combination. A student is to answer eight out of ten questions on an exam. Find the number m of ways that the student can choose the eight questions See in this question repetition is not allowed that is when you choose one question then you cannot choose it again and also order does not matters(i.e either he solved Q1 first or Q2 first) so you use combination in this question.

Question: What is the differnce between edge and vertex

Answer: Vertices are nodes or points and edges are lines/arcs which are used to connect the vertices. e.g If you are making the graph to find the shortest path or for nay purpose of cites and roads between them which contain Lahore, Islamabad, Faisalabad , Karachi, and Multan. Then cities Lahore, Islamabad, Faisalabad , Karachi, and Multan are vertices and roads between them are edges.

Question: What is the differnce between yes and allowed in graphs.

Answer: Allowed mean that specific property can be occurs in that case but yes mean that specific property always occurs in that case. e.g. In Walk you may start and end at same point and may not be (allowed). But in Closed Walk you have to start and end at same point (yes).

Question: what is the meanging of induction? and also Mathematical Induction?

Answer: Basic meaning of induction is: a)The act or an instance of inducting. b) A ceremony or formal act by which a person is inducted, as into office or military service. In Mathematics. A two-part method of proving a theorem involving a positive integral variable. First the theorem is verified for the smallest admissible value of the integer. Then it is proven that if the theorem is true for any value of the integer, it is true for the next greater value. The final proof contains the two parts. As you have studied. It also means that presentation of material, such as facts or evidence, in support of an argument or a proposition. Whether in Physics Induction means the creation of a voltage or current in a material by means of electric or magnetic fields, as in the secondary winding of a transformer when exposed to the changing magnetic field caused by an alternating current in the primary winding. In Biochemistry,it means that the process of initiating or increasing the production of an enzyme or other protein at the level of genetic transcription. In embryology,it means that the change in form or shape caused by the action of one tissue of an embryo on adjacent tissues or parts, as by the diffusion of hormones or chemicals.

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Question: What is "Hypothetical Syllogism".

Answer: Hypothetical syllogism is a law that if the argument is of the form p --> q q---> r Therefore p---> r Then it'll always be a tautology. i.e. if the p implies q and q implies r is true then its conclusion p implies r is always true.

Question: A set is define a well define collection of distinct objects so why an empty set is called a set although it has no element?

Answer: Some time we have collection of zero objects and we call them empty sets. e.g. Set of natural numbers greater than 5 and less than 5. A = { x belongs to N / 5< x < 5 } Now see this is a set which have collection of elements which are greater than 5 and less than 5 (from natural number).

Question: What is improper subset.

Answer: Let A and B be sets. A is proper subset of B, if, and only if, every element of A is in B but there is at least on element if B that is not in A. Now A is improper subset of B, if and only if, every element of A is in B and there is no element in B which is not in A. e.g. A= { 1, 2 , 3, 4} B= { 2, 1, 4, 3} Now A is improper subset of B. Because every element of A is in B and there is no element in B which is not in A

Question: FAQ's in document Form

Answer: .

Question: How to check validity and unvalidity of argument through diagram.

Answer: To check an argument is valid or not you can also use Venn diagram. We identify some sets from the premises . Then represent those sets in the form of diagram. If diagram satisfies the conclusion then it is a valid argument otherwise invalid. e.g. If we have three premises S1: all my friends are musicians S2: John is my friend. S3: None of my neighbor are musicians. conclusion John is not my neighbor. Now we have three sets Friends, Musicians, neighbors. Now you see from premises 1 and 2 that friends are subset of musicians .From premises 3 see that neighbor is an individual set that is disjoint from set musicians. Now represent then in form of Venn diagram. Musicians neighbour Friends Now see that john lies in set friends which is disjoint from set neighbors. So their intersection is empty.Which shows that john is not his neighbor. In that way you can check the validity of arguments

Question: why we used venn digram?

Answer: Venn diagram is a pictorial representation of sets. Venn diagram can sometime be used to determine whether or not an argument is valid. Real life problems can easily be illustrate through Venn diagram if you first convert them into set form and then in Venn diagram form. Venn diagram enables students to organize similarities and differences visually or graphically. A Venn diagram is an illustration of the relationships between and among sets, groups of objects that share something in common.

Question: what is composite relation .

Answer: Let A, B, and C be sets, and let R be relation from A to B and let S be a relation from B to C. Now by combining these two relations we can form a relation from A to C. Now let a belongs to A, b belongs to B, and c belongs to C. We can write relations R as a R b and S as b S c. Now by combining R and S we write a (R O S) c . This is called composition of Relations holding the condition that we must have a b belongs to B which can be write as a R b and b S c (as stated above) . e.g. Let A= {1,2,3,4}, B={a,b,c,d} , C ={x,y,z} and let R={ {1,a), (2, d), (3, a), (3, b), (3, d) } and S={ (b, x), (b, z), (c, y), (d, z)} Now apply that condition which is stated above (that in the composition R O S only those order pairs comes which have earlier an element is common in them e.g. from R we have (3, b) and from S we have (b, x) .Now one relation relate 3 to b and other relates b to x and our composite relation omits that common and relates directly 3 to x.) I do not understand your second question send it again. Now R O S ={(2,z), (3,x), (3,z)}

Question: What are the conditions to confirm functions .

Answer: The first condition for a relation from set X to a set Y to be a function is 1.For every element x in X, there is an element y in Y such that (x, y) belongs to F. Which means that every element in X should relate with distinct element of Y. e.g if X={ 1,2,3} and Y={x, y} Now if R={(1,x),(2,y),(1,y),(2,x)} Then R will not be a function because 3 belongs to X but is does not relates with any element of Y. so R={(1,x),(2,y),(3,y)} can be called a function because every element of X is relates with elements of Y. Second condition is : For all elements x in X and y and z in Y, if (x, y) belongs to F and (x, z) belongs to F, then y = z Which means that every element in X only relates with distinct element of Y. i.e. R={(1,x),(2,y),(2,x), (3,y)} cannot be called as function because 2 relates with x and y also.

Question: When a function is onto.

Answer: First you have to know about the concept of function. Function:It is a rule or a machine from a set X to a set Y in which each element of set X maps into the unique element of set Y. Onto Function: Means a function in which every element of set Y is the image of at least one element in set X. Or there should be no element left in set Y which is the image of no element in set X. If such case does not exist then the function is not called onto. For example:Let we define a function f :

R----R such that f(x)=x^2 (where ^ shows the symbol of power i.e. x raise to power 2). Clearly every element in the second set is the image of atleast one element in the first set. As for x=1 then f(x)=1^2=1 (1 is the image of 1 under the rule f) for x=2 then f(x)=2^2=4 (4 is the image of 2 under the rule f) for x=0 then f(x)=0^2=0 (0 is the image of 0 under the rule f) for x=-1 then f(x)=(-1)^2=1 (1 is the image of -1 under the rule f) So it is onto function.

Question: Is Pie an irrational number?

Answer: Pi π is an irrational number as its exact value has an infinite decimal expansion: Its decimal expansion never ends and does not repeat.

The numerical value of π truncated to 50 decimal places is:

3.14159 26535 89793 23846 26433 83279 50288 41971 69399 37510

Question: Difference between sentence and statement.

Answer: A sentence is a statement if it have a truth value otherwise this sentence is not a statement.By truth value i mean if i write a sentence "Lahore is capital of Punjab" Its truth value is "true".Because yes Lahore is a capital of Punjab. So the above sentence is a statement. Now if i write a sentence "How are you" Then you cannot answer in yes or no.So this sentence is not a statement. Every statement is a sentence but converse is not true.

Question: What is the truth table?

Answer: Truth table is a table which describe the truth values of a proposition. or we can say that Truth table display the complete behaviour of a proposition. There fore the purpose of truth table is to identify its truth values. A statement or a proposition in Discrete math can easily identify its truth value by the truth table. Truth tables are especially valuable in the determination of the truth values of propositions constructed from simpler propositions. The main steps while making a truth table are "first judge about the statement that how much symbols(or variables) it contain. If it has n symbols then total number of combinations=2 raise to power n. These all the combinations give the truth value of the statement from where we can judge that either the truthness of a statement or proposiotion is true or false. In all the combinations you have to put values either "F" or "T" against the variales.But note it that no row can be repeated. For example "Ali is happy and healthy" we denote "ali is happy" by p and "ali is healthy" by q so the above statement contain two variables or symbols. The total no of combinations are =2 raise to power 2(as n=2) =4 which tell us the truthness of a statement.

Question: how empty set become a subset of every set.

Answer: If A & B are two sets, A is called a subset of B, if, and only if, every element of A is also an element of B. Now we prove that empty set is subset of any other set by a contra positive statement(of above statement) i.e. If there is any element in the the set A that is not in the set B then A is not a subset of B. Now if A={} and B={1,3,4,5} Then you cannot find an element which is in A but not in B. So A is subset of B.

Question: What is rational and irrational numbers.

Answer: A number that can be expressed as a fraction p/q where p and q are integers and q\not=0, is called a rational number with numerator p and denominator q. The numbers which cannot be expressed as rational are called irrational number. Irrational numbers have decimal expansions that neither terminate nor become periodic where in rational numbers the decimal expansion either terminate or become periodic after some numbers.

Question: what is the difference between graphs and spanning tree?

Answer: First of all, a graph is a "diagram that exhibits a relationship, often functional, between two sets of numbers as a set of points having coordinates determined by the relationship. Also called plot". Or A pictorial device, such as a pie chart or bar graph, used to illustrate quantitative relationships. Also called chart. And a tree is a connected graph that does not contain any nontrivial circuit. (i.e., it is circuit-free) Basically, a graph is a nonempty set of points called vertices and a set of line segments joining pairs of vertices called edges. Formally, a graph G consists of two finite sets: (i) A set V=V(G) of vertices (or points or nodes) (ii) A set E=E(G) of edges; where each edge corresponds to a pair of vertices. Whereas, a spanning tree for a graph G is a subgraph of G that contains every vertex of G and is a tree. It is not neccesary for a graph to always be a spanning tree. Graph becomes a spanning tree if it satisfies all the properties of a spanning tree.

Question: What is the probability ?

Answer: The definition of probability is : Let S be a finite sample space such that all the outcomes are equally likely to occur. The probability of an event E, which is a subset of S, is P(E) = (the number of outcomes in E)/ (the number of total outcomes in S) P(E) = n (E) / n (S) This definition is due to 'Laplace.' Thus probability is a concept which measures numerically the degree of certainty or uncertainty of the occurrence of an event. Explaination The basic steps of probability that u have to remember are as under 1. First list out all possible out comes. That is called the sample space S For example when we roll a die the all possible outcomes are the set S i.e. S = {1,2,3,4,5,6} 2. Secondly we have to find out all that possible outcomes, in which the probability is required . For example we are asked to find the probability of even numbers. First we decide any name of that event i.e E Now we check all the even numbers in S which are E = {2,4,6} Remember Event is always a sub-set of Sample space S. 3. Now we apply the definition of probability P(E) = (the number of outcomes in E)/ (the number of total outcomes in S) P(E) = n (E) / n (S) So from above two steps we have n (E) = 3 and n (S) = 6 then P(E) = 3 / 6 = 1/2 which is probability of an even number.

Question: what is permutation?

Answer: Permutation comes from the word permute which means " to change the order of." Basically permutation means a "complete change." Or the act of altering a given set of objects in a group. In Mathematics point of view it means that a ordered arrangement of the elements of a set (here the order of elements matters but repetition of the elements is not allowed).

Question: What is a function.

Answer: A function say 'f' is a rule or machine from a set A to the set B if for every element say a of A, there exist a unique element say b of set such that b=f(a) Where b is the image of a under f,and a is the pre-image. Note it that set A is called the domain of f and Y is called the codomain of f. As we know that function is a rule or machine in which we put an input,and we get an output.Like that a juicer machine.We take some apples(here apples are input) and we apply a rule or a function of juicer machine on it,then we get the output in the form of juice.

Question: What is p implies q.

Answer: p--- >q means to "go from hypothesis to a conclusion" where p is a hypothesis and q is a conclusion. And note it that this statement is conditioned because the "truth ness of statement p is conditioned on the truth ness of statement q". Now the truth value of p--->q is false only when p is true and q is false otherwise it will always true. E.g. consider an implication "if you do your work on Sunday ,I will give you ten rupees." Here p=you do your work on Sunday (is the hypothesis) , q=I will give you ten rupees (the conclusion or promise). Now the truth value of p---->q will false only when the promise is braked. i.e. You do your work on Sunday but you do not get ten rupees. In all other conditions the promise is not braked.

Question: What is valid and invalid arguments.

Answer: As "an argument is a list of statements called premises (or assumptions or hypotheses) which is followed by a statement called the conclusion. " A valid argument is one in which the premises entail(or imply) the conclusion. 1)It cannot have true premises and a false conclusion. 2)If its premises are true, its conclusion must be true. 3)If its conclusion is false, it must have at least one false premise. 4)All of the information in the conclusion is also in the premises. And an invalid agrument is one in which the premises do not entail(or imply) the conclusion. It can have true premises and a false conclusion. Even if its premises are true, it may have a false conclusion. Even if its conclusion is false, it may have true premises. There is information in the conclusion that is not in the premises. To know them better,try to solve more and more examples and exercises.

Question: What is domain and co -domain.

Answer: Domain means "the set of all x-coordinates in a relation". It is very simple,Let we take a function say f from the set X to set Y. Then domain means a set which contain all the elements of the set X. And co domain means a set which contain all the elements of the set Y. For example: Let we define a function "f" from the set X={a,b,c,d} to Y={1,2,3,4}. such that f(a)=1, f(b)=2, f(c)=3, f(d)=1 Here the domain set is {a,b,c,d} And the co-domain set is {1,2,3,4} Where as the image set is {1,2,3}.Because f(a)=1 as 1 is the image of a under the rule 'f'. f(b)=2 as 2 is the image of b under the rule 'f'. f(c)=3 as 3 is the image of c under the rule 'f'. f(d)=1as 1 is the image of d under the rule 'f'. because "image set contains only those elements which are the images of elements found in set X". Note it that here f is one -one but not onto,because there is one element '4' left which is the image of nothing element under the rule 'f'.

Question: What is the difference between k-sample,k-selection, k-permutation and k-combination?

Answer: Actually, these all terms are related to the basic concept of choosing some elements from the given collection.

For it, two things are important:

1) Order of elements .i.e. which one is first, which one is second and so on.
2) Repetition of elements

So we can get 4 kinds of selections:

1) The elements have both order and repetition. (It is called k-sample)
2) The elements have only order, but no repetition. (It is called k-permutation)
3) The elements have only repetition, but no order. (It is called k-selection)
4) The elements have no repetition and no order. (It is called k-combination)

Question: What is a combination?

Answer: A combination is an un-ordered collection of unique elements. Given S, the set of all possible unique elements, a combination is a subset of the elements of S. The order of the elements in a combination is not important (two lists with the same elements in different orders are considered to be the same combination). Also, the elements cannot be repeated in a combination (every element appears uniquely once

Question: why is 0! equal to 1?

Answer: Since $n! = n(n-1)!$

Put n =1 in it.

1! = 1x(1 – 1)!

1! =1x0!

1! = 0!

Since 1! = 1

So 1 = 0!

0! = 1.

Question: What is the basic idea if Mathematical Induction?

Answer: Mathematical Induction

Question: Define symmetric and anti-symmetric.

Answer: Click here.

Question: What is the main deffernce between Calculus and Discrete Maths?

Answer: Discrete mathematics is the study of mathematics which concerns to the study of discrete objects. Discrete math build students approach to think abstractly and how to handle mathematical models problems in computer While Calculus is a mathematical tool used to analyze changes in physical quantities. Or "Calculus is sometimes described as the mathematics of change." Also calculus played an important role in industrial area as well discrete math in computer.

Discrete mathematics concerns processes that consist of a sequence of individual steps. This distinguishes it from calculus, which studies continuously changing processes. the ideas of discrete mathematics underline the science and technology specific to the computer age. An important goal of discrete mathematics is to develop students' ability to think abstractly.

Question: Explain Valid Arguments.

Answer: When some statement is said on the basis of a set of other statements, meaning that this statement is derived from that set of statements, this is called an argument. The formal definition is "an argument is a list of statements called **"premises"** (or assumptions or hypotheses) which is followed by a statement called the "**conclusion.**"

A **valid argument** is one in which the premises imply the conclusion.

1) It cannot have true premises and a false conclusion.

2) If its premises are true, its conclusion must be true.

3) If its conclusion is false, it must have at least one false premise.

4) All of the information in the conclusion is also in the premises.

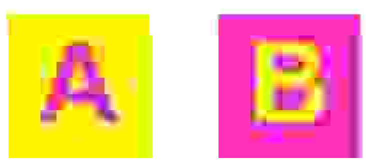
Question: What is the Difference between combinations and permutations?

Answer: When we talk of permutations and combinations in everyday talk we often use the two terms interchangeably. In mathematics, however, the two each have very specific meanings, and this distinction often causes problems

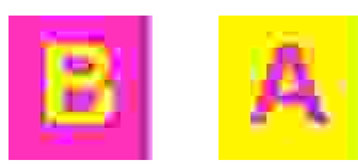
In brief, the permutation of a number of objects is the number of different ways they can be ordered; i.e. which one is first, which one is second or third etc. For example, you see, if we have two digits 1 and 2, then 12 and 21 are different in meaning. So their order has its own importance in permutation.

On the other hand, in combination, the order is not necessary. you can put any object at first place or second etc. For example, Suppose you have to put some pictures on the wall, and suppose you only have two pictures: A and B.

You could hang them



or



We could summarise permutations and combinations (very simplistically) as

Permutations - position important (although choice may also be important)
Combinations - chosen important,
which may help you to remember

Question: What is the use of kruskal's algorithn in our daily life?

Answer: The Kruskal's algorithm is usually used to find minimum spanning tree i.e. the possible smallest tree that contains all the vertices. The standard application is to a problem like phone network design. Suppose, you have a business with several offices; you want to lease phone lines to connect them up with each other; and the phone company charges different amounts of money to connect different pairs of cities. You want a set of lines that connects all your offices with a minimum total cost. It should be a spanning tree, since if a network isn't a tree you can always remove some edges and save money. A less obvious application is that the minimum spanning tree can be used to approximately solve the traveling salesman problem. A convenient formal way of defining this problem is to find the shortest path that visits each point at least once.

Question: What is irrational number?

Answer: Irrational number An irrational number can not be expressed as a fraction. In decimal form, irrational numbers do not repeat in a pattern or terminate. They "go on forever" (infinity). Examples of irrational numbers are: pi= 3.141592654...

Question: Define membership table and truth table.

Answer: Membership table: A table displaying the membership of elements in sets. Set identities can also be proved using membership tables. An element is in a set, a 1 is used and an element is not in a set, a 0 is used. Truth table: A table displaying the truth values of propositions.

Question: Define function and example for finding domain and range of a function.

Answer: Click here.

Question: Why do we use konigsberg bridges problem?

Answer: Click on it.

Question: Explain the intersection of two sets?

Answer: Click on it.

Question: What is absurdity With example?

Answer: Click here.

Question: What is sequence and series?

Answer: Sequence A sequence of numbers is a function defined on the set of positive integer. The numbers in the sequence are called terms. Another way, the sequence is a set of quantities u1, u2, u3... stated in a definite order and each term formed according to a fixed pattern. U r =f(r) In example: 1,3,5,7,... 2,4,6,8,... 1 2 ,– 2 2 ,3 2 ,– 4 2 ,... Infinite sequence:- This kind of sequence is unending sequence like all natural numbers: 1, 2, 3, ... Finite sequence:- This kind of sequence contains only a finite number of terms. One of good examples are the page numbers. Series:- The sum of a finite or infinite sequence of expressions. 1+3+5+7+...

Question: Differntiate contigency and contradiction.

Answer: Click here.

Question: What is conditional statement, converse, inverse and contra-positive?

Answer: Click here.

Question: What is Euclidean algorithm?

Answer: In number theory, the **Euclidean algorithm** (also called **Euclid's algorithm**) is an algorithm to determine the greatest common divisor (GCD) of two integers.

Its major significance is that it does not require factoring the two integers, and it is also significant in that it is one of the oldest algorithms known, dating back to the ancient Greeks.

Question: what is the circle definition?

Answer: A circle is the locus of all points in a plane which are equidistant from a fixed point. The fixed point is called centre of that circle and the distance is called radius of that circle

Question: What is bi-conditional statement?

Answer: Click here.

Question: Explain the difference between k-sample, k-selection, k-combination and k-permutation.

Answer: Click here.

Question: What is meant by Discrete?

Answer:

A type of data is discrete if there are only a finite number of values possible. Discrete data usually occurs in a case where there are only a certain number of values, or when we are counting something (using whole numbers). For example, 5 students, 10 trees etc.

Question: Explain D'Morgan Law.

Answer: Click here.

Question: What are digital circuits?

Answer: Digital circuits are electric circuits based on a number of discrete voltage levels.

In most cases there are two voltage levels: one near to zero volts and one at a higher level depending on the supply voltage in use. These two levels are often represented as L and H.

Question: What is absurdity or contradiction?

Answer: A statement which is always false is called an absurdity.

Question: What is contingency?

Answer: A statement which can be true or false depending upon the truth values of the variables is called a contingency.

Question: Is there any particular rule to solve Inductive Step in the mathematical Induction?

Answer: In the Inductive Step, we suppose that the result is also true for other integral values k. If the result is true for n = k, then it must be true for other integer value k +1 otherwise the statement cannot be true.

In proving the result for n = k +1, the procedure changes, as it depends on the shape of the given statement.

Following steps are main:

1) You should simply replace n by k+1 in the left side of the statement.

2) Use the supposition of n = k in it.

3) Then you have to simplify it to get right side of the statement. This is the step,

where students usually feel difficulty.

Here sometimes, you have to open the brackets, or add or subtract some terms

or take some term common etc. This step of simplification to get right side of the given statement for n = n + 1 changes from question to question.

Now check this step in the examples of the Lessons 23 and 24.

Question: What is Inclusion Exclusion Principle?

Answer: Click on Inclusion Exclusion Principle.

Question: What is recusion?

Answer: Click here.

Question: Different notations of conditional implication.

Answer: If p than q. P implies q. If p , q. P only if q. P is sufficient for q.

Question: What is cartesion product?

Answer: Cartesian product of sets:- Let A and B be sets. The Cartesian product of A and B, denoted A x B (read "A cross B") is the set of all ordered pairs (a, b), where a is in A and b is in B. For example: A = {1, 2, 3, 4, 5, 6} B = {a} A x B = {(1,a), (2,a), (3,a), (4,a), (5,a)}

Question: Define fraction and decimal expansion.

Answer: Fraction:- A number expressed in the form a/b where a is called the numerator and b is called the denominator. Decimal expansion:- The decimal expansion of a number is its representation in base 10 The number 3.22 3 is its integer part and 22 is its decimal part The number on the left of decimal point is integer part of the number and the number on the right of the decimal point is decimal part of the number.

Question: Explane venn diagram.

Answer: Venn diagram is a pictorial representation of sets. Venn diagram can sometime be used to determine whether or not an argument is valid. Real life problems can easily be illustrate through Venn diagram if you first convert them into set form and then in Venn diagram form. Venn diagram enables students to organize similarities and differences visually or graphically. A Venn diagram is an illustration of the relationships between and among sets, groups of objects that share something in common

Question: Write the types of functions.

Answer: Types of function:- Following are the types of function 1. One to one function 2. Onto function 3. Into function 4. Bijective function (one to one and onto function) One to one function:- A function f : A to B is said to be one to one if there is no repetition in the second element of any two ordered pairs. Onto function:- A function f : A to B is said to be onto if Range of f is equal to set B (co-domain). Into function:- A function f : A to B is said to be into function of Range of f is the subset of set B (co domain) Bijective function: Bijective function:- A function is said to be Bijective if it is both one to one and onto.

Question: **Explain the pigeonhole principle.**

Answer: Click here.

Question: **What is conditional probability with example?.**

Answer: Click here.

Question: **Explain combinatorics.**

Answer: Branch of mathematics concerned with the selection, arrangement, and combination of objects chosen from a finite set.

The number of possible bridge hands is a simple example; more complex problems include scheduling classes in classrooms at a large university and designing a routing system for telephone signals. No standard algebraic procedures apply to all combinatorial problems; a separate logical analysis may be required for each problem.

Question: **How the tree diagram use in our real computer life?**

Answer: Tree diagrams are used in data structure, compiler construction, in making algorithms, operating system etc.

Question: **Write detail of cards.**

Answer: Diamond Club Heart Spade A A A A 1 1 1 1 2 2 2 2 3 3 3 3 4 4 4 4 5 5 5 5 6 6 6 6 7 7 7 7 8 8 8 8 9 9 9 9 10 10 10 10 J J J J Q Q Q Q K K K K Where 26 cards are black & 26 are red. Also 'A' stands for 'ace' 'J' stands for 'jack' 'Q' stands for 'queen' 'K' stands for 'king'

Question: **what is the purpose of permutations?**

Answer: Definition:- Possible arrangements of a set of objects in which the order of the arrangement makes a difference. For example, determining all the different ways five books can be arranged in order on a shelf. In mathematics, especially in abstract algebra and related areas, a permutation is a bijection, from a finite set X onto itself. Purpose of permutation is to

FINALTERM EXAMINATION
MTH202- Discrete Mathematics
Time: 120 min
Marks: 80

1) If A and B are two disjoint (mutually exclusive) events then

P(AUB) =

► P(A) + P(B) + P(AÇB)
► P(A) + P(B) + P(AUB)
► P(A) + P(B) - P(AÇB)
► P(A) + P(B) - P(AÇB)
► P(A) + P(B)

2) If p=It is red,

q=It is hot

Then, It is not red but hot is denoted by ~ *p*Ù ~ *q*

► True

► **False**

3) If (*A*U*B*) = A, then (*A*n*B*) = B

► True

► **False**

► Cannot be determined

How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5?

► 333

► 467

► **533**

► 497

The value of $\lfloor x \rfloor$ for -2.01 is

► -3

► 1

► ***-2***

What is the expectation of the number of heads when three fair coins are tossed?

► 1

► 1.34

► 2

► **1.5**

Every relation is

- **function**
- may or may not function
- bijective mapping
- Cartesian product set

The statement p « q ° (p ®q)Ù(q ®p) describes

- Commutative Law
- Implication Laws
- Exportation Law
- **Equivalence**

The square root of every prime number is irrational

- **True**
- False
- Depends on the prime number given

A predicate is a sentence that contains a finite number of variables and becomes a statement when specific values are substituted for the variables

- True
- False
- None of these

If r is a positive integer then gcd(r,0)=

- r
- 0
- 1
- None of these

Associative law of union for three sets is

- A È (B È C) = (A È B) È C
- A Ç (B Ç C) = (A Ç B) Ç C
- A È (B Ç C) = (A È B) Ç (A È B)

► None of these

Values of X and Y, if the following order pairs are equal. (4X-1, 4Y+5)= (3,5) will be

► (x,y) = (3,5)
► (x,y) = (1.5,2.5)
► (x,y) = (1,0)
► None of these

The expectation of x is equal to

► Sum of all terms
► Sum of all terms divided by number of terms
► **å*xf* (*x*)**

A line segment joining pair of vertices is called

► Loop
► Edge
► Node

The indirect proof of a statement p□q involves

► Considering ~q and then try to reach ~p
► Considering p and ~q and try to reach contradiction
► Considering p and then try to reach q
► **Both 2 and 3 above**

The greatest common divisor of 5 and 10 is

► 5
► 0
► 1
► None of these

Suppose that there are eight runners in a race first will get gold medal the second will get siver and third will get bronze. How many different ways are there to award these medals if all possible outcomes of race can occur and there is no tie.

► **P(8,3)**
► P(100,97)
► P(97,3)
► None of these

The value of 0! Is

► 0
► **1**
► Cannot be determined

A sub graph of a graph G that contains every vertex of G and is a tree is called

► Trivial tree
► empty tree
► **Spanning tree**

In the planar graph, the graph crossing number is

► **0**
► 1
► 2
► 3

A matrix in which number of rows and columns are equal is called

► Rectangular Matrix
► **Square Matrix pg296**
► Scalar Matrix

Changing rows of matrix into columns is called

► Symmetric Matrix

► Transpose of Matrix
► Adjoint of Matrix

If A and B are finite (overlapping) sets, then which of the following must be true
► n(AUB) = n(A) + n(B)
► n(AÈB) = n(A) + n(B) - n(AÇB)
► n(AÈB)= ø
► None of these

When 3k is even, then 3k+3k+3k is an odd.
► True
► False

When 5k is even, then 5k+5k+5k is odd.
► True
► False

The product of the positive integers from 1 to n is called
► Multiplication
► n factorial
► Geometric sequence
The expectation m for the following table is
xi 1 3
f(xi) 0.4 0.1
► 0.5
► 3.4
► 0.3
► 0.7
If p= A Pentium 4 computer,
q= attached with ups.

The given graph is

► Simple graph
► Complete graph
► Bipartite graph
► Both (i) and (ii)
► Both (i) and (iii)

P(*n*) is called proposition or statement.

► True
► False

An integer n is odd if and only if n = 2k + 1 for some integer k.

► True
► False
► Depends on the value of k

MTH202 FINAL TERM SOLVED OBJECTIVE FILE

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Question No: 1 (Marks: 1) - Please choose one

If p = It is raining

q = She will go to college

"It is raining and she will not go to college" will be denoted by

► **$p \wedge \sim q$**

► $p \wedge q$

► $\sim (p \wedge q)$

► $\sim p \wedge q$

Question No: 2 (Marks: 1) - Please choose one

In a directed graph of a Irreflexive relation, there should be

► Loop on a one point

► **No loop at any point**

► No point connected

Question No: 3 (Marks: 1) - Please choose one

The statement p ®q º ~p Ú q º ~(p Ù ~q) describes

► **Commutative Law**

► Implication Laws

► Exportation Law

► Equivalence

Question No: 4 (Marks: 1) - Please choose one

How many functions are there from a set with three elements to a set with two elements?

► 6

► 8

2*2*2 =8

► 12

Question No: 5 (Marks: 1) - Please choose one

If a set contains exactly *m* distinct elements where *m* denotes some non negative integer then the set is .

► Finite

► Infinite

► None of these

Question No: 6 (Marks: 1) - Please choose one

Let f and g be the functions defined by
f(x)= 2x+3 & g(x)= 3x+2 then composition of f and g is

► 6x+6

► 5x+5

► 6x+7 **F(X)=2X(3X+2)+3 => 6X+7**

Question No: 7 (Marks: 1) - Please choose one

Let f is defined recursively by

F(0)=3

F(n+1)=2f(n)+2

Then f(2)=

► 8

► 10

► 18 **F(2) = 2F(1)+2 => 2(8)+2 => 18**

► 21

Question No: 8 (Marks: 1) - Please choose one

If A and B are two disjoint (mutually exclusive) events then

P(AÈB) =

► P(A) + P(B) + P(AÇB)
► P(A) + P(B) + P(AUB)
► P(A) + P(B) - P(AÇB)
► P(A) + P(B) - P(AÇB)
► P(A) + P(B)

Question No: 9 (Marks: 1) - Please choose one

If a pair of dice is thrown then the probability of getting a total of 5 or 11 is

► $\frac{1}{18}$

► $\frac{1}{9}$

► $\frac{1}{6}$ **6*6 =36**

4/36+2/36=6/36=1/6

Question No: 10 (Marks: 1) - Please choose one

If a die is rolled then what is the probability that the number is greater than 4

► $\frac{1}{3}$

► $\frac{3}{4}$

► $\frac{1}{2}$ **(5,6)**

2/6 = 1/2

Question No: 11 (Marks: 1) - Please choose one

What is the expectation of the number of heads when three fair coins are tossed?

► 1

► 1.34

► 2

► 1.5

Question No: 12   (Marks: 1)   - Please choose one

If A, B and C are any three events, then

P(AÈBÈC) is equal to

► P(A) + P(B) + P(C)

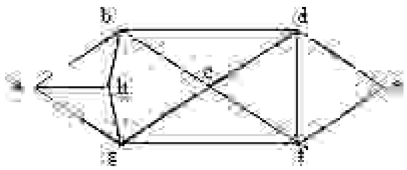
► P(A) + P(B) + P(C)- P(AÇB) - P (A ÇC) - P(B ÇC) + P(A ÇB ÇC)

► P(A) + P(B) + P(C) - P(AÇB) - P (A ÇC) - P(B ÇC)

► P(A) + P(B) + P(C) + P(A ÇB ÇC)

Question No: 13   (Marks: 1)   - Please choose one

The Hamiltonian circuit for the following graph is



► abcdefgh

► abefgha

► abcdefgha

Question No: 14 (Marks: 1) - Please choose one

Let n and d be integers and d ¹ 0. Then n is divisible by d or d divides n
If and only if

► n= k.d for some integer k

► n=d

► n.d=1

► none of these

Question No: 15 (Marks: 1) - Please choose one

The contra positive proof of a statement pàq involves

► Considering p and then try to reach q

► Considering ~q and then try to reach ~p

► Considering p and ~q and try to reach contradiction

► None of these

Question No: 16 (Marks: 1) - Please choose one

The sum of two irrational number must be an irrational number

► False

► True

Question No: 17 (Marks: 1) - Please choose one

The square root of every prime number is irrational

► True

► False

► Depends on the prime number given

Question No: 18 (Marks: 1) - Please choose one

The greatest common divisor of 27 and 72 is

► 27

► 9

72 = 27 · 2 + 18 => 27 = 18 · 1 + 9 => 18 = 9 · 2 + 0 =>(72, 27) = 9.

► 1

► None of these

Question No: 19 (Marks: 1) - Please choose one

If T is a full binary tree and has 5 internal vertices then the total vertices of T are

► 11

2(K+1)=2(5+1)=11

► 12

► 13

► None of the these

Question No: 20 (Marks: 1) - Please choose one

Suppose that a connected planar simple graph has 30 edges. If a plane drawing of this graph has 20 faces, how many vertices does the graph have?

► **12**

F=e-v+2

20=30-v+2

v=10+2=12

► 13

► 14

Question No: 21 (Marks: 1) - Please choose one

How many different ways can three of the letters of the word BYTES be chosen if the first letter must be B ?

► **P(4,2)**

► P(2,4)

► C(4,2)

► None of these

Question No: 22 (Marks: 1) - Please choose one

The value of 0! Is

► 0

► **1**

► Cannot be determined

Question No: 23 (Marks: 1) - Please choose one

An arrangement of objects with the consideration of order is called

► **Permutation**

► Combination

► Selection

► None of these

Question No: 24   (Marks: 1)   - Please choose one

If A and B are two disjoint sets then which of the following must be true

► **n(AÈB) = n(A) + n(B)** **because p(AUB)=P(A)+P(B)**

► n(AÈB) = n(A) + n(B) - n(AÇB)

► n(AÈB)= ø

► None of these

Question No: 25   (Marks: 1)   - Please choose one

Among 200 people, 150 either swim or jog or both. If 85 swim and 60 swim and jog, how many jog?

► **125**

► 225

► 85

► 25

Question No: 26   (Marks: 1)   - Please choose one

If a graph is a tree then

► it has 2 spanning trees

► it has only 1 spanning tree

► it has 4 spanning trees

► it has 5 spanning trees

Question No: 27 (Marks: 1) - Please choose one

Euler formula for graphs is

► f = e-v

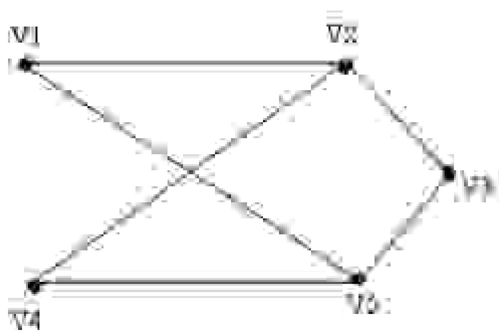
► f = e+v +2

► f = e-v-2

► f = e-v+2

Question No: 28 (Marks: 1) - Please choose one

The given graph is



► Simple graph

► Complete graph

► Bipartite graph

► Both (i) and (ii)

► Both (i) and (iii)

Question No: 29 (Marks: 1) - Please choose one

An integer n is odd if and only if n = 2k + 1 for some integer k.

► True

► False

► Depends on the value of k

Question No: 30 (Marks: 1) - Please choose one

If $P(A \cap B) \neq P(A)P(B)$ then the events A and B are called

► Independent

► Dependent

► Exhaustive

Question No: 31 (Marks: 2)

Let A and B be the events. Rewrite the following event using set notation

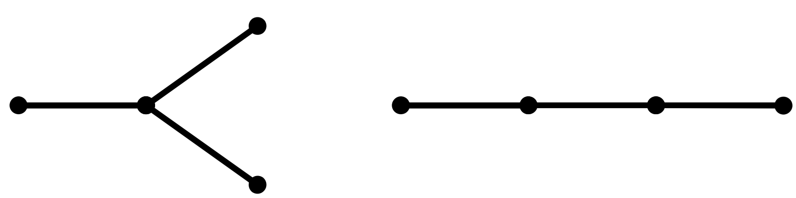
"A or not B occurs"

Question No: 32 (Marks: 2)

Find a non-isomorphic tree with four vertices.

Any tree with four vertices has (4-1=3) three edges. Thus, the total degree of a tree with 4 vertices must be 6 [by using total degree=2(total number of edges)].

Also, every tree with more than one vertex has at least two vertices of degree 1, so the only possible combinations of degrees for the vertices of the trees are 1, 1, 1, 3 and 1, 1, 2, 2.
The corresponding trees (clearly non-isomorphic, by definition) are



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Question No: 33 (Marks: 2)

Write the following in the factorial form:

n(n -1)(n-2)…(n-r+1)

$$n(n-1)(n-2)\ldots(n-r+1)=\frac{n(n-1)(n-2)\ldots(n-r+1)-(n-r)!}{(n-r)!}$$

$$=\frac{n!}{(n-r)!}$$

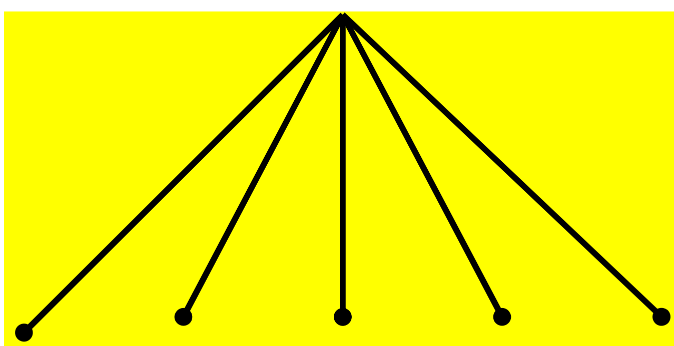
Question No: 34 (Marks: 3)

Compute ëxû and éxù for $x = 25/4$

Question No: 35 (Marks: 3)

Find a spanning tree for the graph $k_{1,5}$?

$k_{1,5}$ represents a complete bipartite graph on (1,5) vertices, drawn below:



Clearly the graph itself is a tree (six vertices and five edges). Hence the graph is itself a spanning tree.

Page number 325 lecture number 45…

Question No: 36 (Marks: 3)

The members of a club are 12 boys and 8 girls. In how many ways can a committee of 3 boys and 2 girls be formed?

Question No: 37 (Marks: 5)

Is it possible to have a simple graph with four vertices of degree 1, 1, 3, and 3.If no then give reason?(Justify your answer)

Question No: 38 (Marks: 5)

Draw a binary tree to represent the following expression

a/(b-c.d)

The internal vertices are arithmetic operators, the terminal vertices are variables and the operator at each vertex acts on its left and right sub trees in left-right order.

Question No: 39 (Marks: 5)

There are 25 people who work in an office together. Four of these people are selected to attend four different conferences. The first person selected will go to a conference in New York, the second will go to Chicago, the third to San Francisco, and the fourth to Miami. How many such selections are possible?

MTH202 FINAL TERM SOLVED LATEST PAPER

Question No: 1   (Marks: 1)   - Please choose one

The negation of "Today is Friday" is

► Today is Saturday

► **Today is not Friday**

► Today is Thursday

Question No: 2   (Marks: 1)   - Please choose one

An arrangement of rows and columns that specifies the truth value of a compound proposition for all possible truth values of its constituent propositions is called

► **Truth Table**

► Venn diagram

► False Table

► None of these

Question No: 3   (Marks: 1)   - Please choose one

The converse of the conditional statement **p ® q** is

► **q ®p**

► ~q ®~p

► ~p ®~q

► None of these

Question No: 4 (Marks: 1) - Please choose one

Contrapositive of given statement "***If it is raining, I will take an umbrella***" is

► **I will not take an umbrella if it is not raining.**

► I will take an umbrella if it is raining.

► It is not raining or I will take an umbrella.

► None of these.

Question No: 5 (Marks: 1) - Please choose one

Let A= {1, 2, 3, 4} and R = {(1, 1), (2, 2), (3, 3),(4,4)} then

► R is symmetric.
► R is anti symmetric.
► R is transitive.
► R is reflexive.
► **All given options are true**

Question No: 6 (Marks: 1) - Please choose one

A binary relation R is called Partial order relation if

► It is Reflexive and transitive
► It is symmetric and transitive
► It is reflexive, symmetric and transitive
► **It is reflexive, antisymmetric and transitive**

Question No: 7 (Marks: 1) - Please choose one

How many functions are there from a set with three elements to a set with two elements?

► 6
► **8**
► 12

Question No: 8 (Marks: 1) - Please choose one

$1, 10, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7, \ldots\ldots\ldots\ldots\ldots$ is

► **Arithmetic series**

Arithmetic series

► Geometric series

► Arithmetic sequence

► Geometric sequence

Question No: 9 (Marks: 1) - Please choose one

$\lceil x \rceil$ for x = -2.01 is

► -2.01
► -3
► **-2**
► -1.99

Question No: 10 (Marks: 1) - Please choose one

If A and B are two disjoint (mutually exclusive) events then

P(AÈB) =

► P(A) + P(B) + P(AÇB)
► P(A) + P(B) + P(AUB)
► P(A) + P(B) - P(AÇB)
► P(A) + P(B) - P(AÇB)

► **P(A) + P(B)**

Question No: 11 (Marks: 1) - Please choose one

If a die is thrown then the probability that the dots on the top are prime numbers or odd numbers is

► 1

► $\frac{1}{3}$

► $\frac{2}{3}$

Question No: 12 (Marks: 1) - Please choose one

If $P(A \cap B) \neq P(A)P(B)$ then the events A and B are called

► **Independent**

► Dependen

► Exhaustive

Question No: 13 (Marks: 1) - Please choose one

A rule that assigns a numerical value to each outcome in a sample space is called

► One to one function

► Conditional probability

► **Random variable**

Question No: 14 (Marks: 1) - Please choose one

The expectation of x is equal to

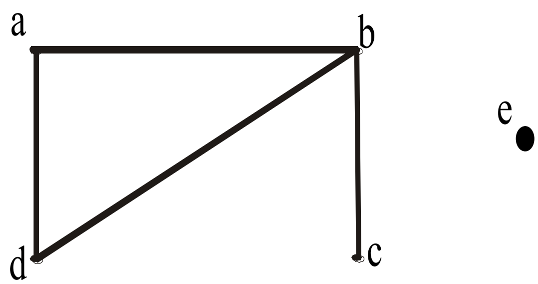
► Sum of all terms

► Sum of all terms divided by number of terms

► $\sum xf(x)$

Question No: 15 (Marks: 1) - Please choose one

The degree sequence {a, b, c, d, e} of the given graph is



- ► 2, 2, 3, 1, 1
- ► **2, 3, 1, 0, 1**
- ► 0, 1, 2, 2, 0
- ► 2,3,1,2,0

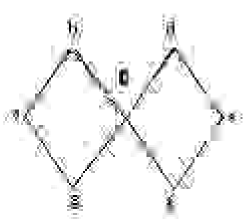
Question No: 16 (Marks: 1) - Please choose one

Which of the following graph is not possible?

- ► Graph with four vertices of degrees 1, 2, 3 and 4.
- ► **Graph with four vertices of degrees 1, 2, 3 and 5**.
- ► Graph with three vertices of degrees 1, 2 and 3.
- ► Graph with three vertices of degrees 1, 2 and 5.

Question No: 17 (Marks: 1) - Please choose one

The graph given below



- ► Has Euler circuit
- ► Has Hamiltonian circuit
- ► **Does not have Hamiltonian circuit**

Question No: 18 (Marks: 1) - Please choose one

Let n and d be integers and d ¹ 0. Then n is divisible by d or d divides n
If and only if

- ► **n= k.d for some integer k**

- ► n=d

- ► n.d=1

► none of these

Question No: 19 (Marks: 1) - Please choose one

The contradiction proof of a statement pàq involves

► **Considering p and then try to reach q**

► Considering ~q and then try to reach ~p

► Considering p and ~q and try to reach contradiction

► None of these

Question No: 20 (Marks: 1) - Please choose one

An integer n is prime if, and only if, n > 1 and for all positive integers r and s, if n = r·s, then

► **r = 1 or s = 1.**

► r = 1 or s = 0.

► r = 2 or s = 3.

► None of these

Question No: 21 (Marks: 1) - Please choose one

The method of loop invariants is used to prove correctness of a loop with respect to certain pre and post-conditions.

► **True**

► False

► None of these

Question No: 22 (Marks: 1) - Please choose one

The greatest common divisor of 27 and 72 is

► 27
► **9**
► 1
► None of these

Question No: 23 (Marks: 1) - Please choose one

If a tree has 8 vertices then it has

► 6 edges
► **7 edges**
► 9 edges

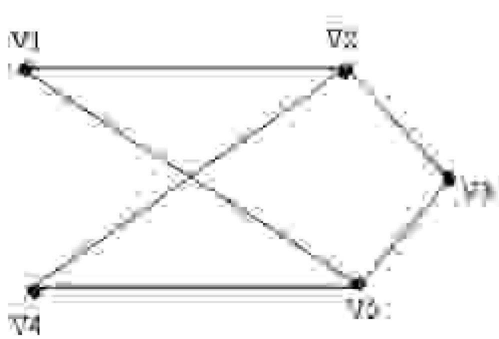
Question No: 24 (Marks: 1) - Please choose one

Complete graph is planar if

► **n = 4**
► n>4
► $n \leq 4$

Question No: 25 (Marks: 1) - Please choose one

The given graph is



- **Simple graph**
- Complete graph
- Bipartite graph
- Both (i) and (ii)
- Both (i) and (iii)

Question No: 26 (Marks: 1) - Please choose one

The value of 0! Is

- 0
- **1**
- Cannot be determined

Question No: 27 (Marks: 1) - Please choose one

Two matrices are said to confirmable for multiplication if

- Both have same order
- **Number of columns of 1^{st} matrix is equal to number of rows in 2^{nd} matrix**
- Number of rows of 1^{st} matrix is equal to number of columns in 2^{nd} matrix

Question No: 28 (Marks: 1) - Please choose one

The value of (-2)! Is

- 0
- 1
- **Cannot be determined**

Question No: 29 (Marks: 1) - Please choose one

The value of $\frac{(n+1)!}{(n-1)!}$ is

- 0
- n(n-1)
- n^2+n
- Cannot be determined

Question No: 30 (Marks: 1) - Please choose one

The number of *k*-combinations that can be chosen from a set of *n* elements can be written as

- **$^{n}C_{k}$**
- $^{k}C_{n}$

► $^{n}P_{k}$

► $^{k}P_{k}$

Question No: 31 (Marks: 1) - Please choose one

If the order does not matter and repetition is allowed then total number of ways for selecting k sample from n. is

► n^{k}

► **C(n+k-1,k)**

► P(n,k)

► C(n,k)

Question No: 32 (Marks: 1) - Please choose one

If the order matters and repetition is not allowed then total number of ways for selecting k sample from n. is

► n^{k}

► **C(n+k-1,k)**

► P(n,k)

► C(n,k)

Question No: 33 (Marks: 1) - Please choose one

To find the number of unordered partitions, we have to count the ordered partitions and then divide it by suitable number to erase the order in partitions

► True

► **False**

► None of these

Question No: 34 (Marks: 1) - Please choose one

A tree diagram is a useful tool to list all the logical possibilities of a sequence of events where each event can occur in a finite number of ways.

► **True**

► False

Question No: 35 (Marks: 1) - Please choose one

If A and B are finite (overlapping) sets, then which of the following **must be** true

► **n(AÈB) = n(A) + n(B)**

► n(AÈB) = n(A) + n(B) - n(AÇB)

► n(AÈB)= ø

► None of these

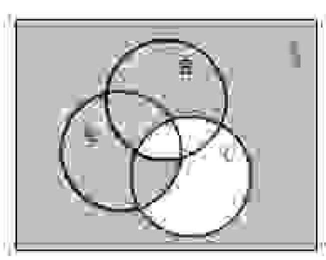
Question No: 36 (Marks: 1) - Please choose one

What is the output state of an OR gate if the inputs are 0 and 1?

► 0
► **1**
► 2
► 3

Question No: 37 (Marks: 1) - Please choose one

In the given Venn diagram shaded area represents:



► (A Ç B) È C
► (A È B^c) È C
► (AÇ B^c) È C^c
► **(A Ç B) Ç C^c**

Question No: 38 (Marks: 1) - Please choose one

Let A,B,C be the subsets of a universal set U.

Then $(A \cup B) \cup C$ is equal to:

► $A \cap (B \cup C)$

► $A \cup (B \cap C)$

► $\varnothing$

► $A \cup (B \cup C)$

Question No: 39   (Marks: 1)   - Please choose one

n ! >2ⁿ for all integers n ³4.

► True

► **False**

Question No: 40   (Marks: 1)   - Please choose one

$+, -, \times, \div$ are

► Geometric expressions

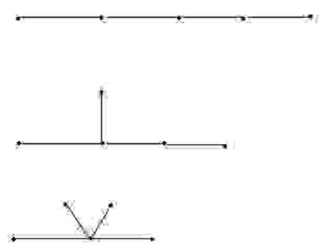
► **Arithmetic expressions**

► Harmonic expressions

Question No: 41   (Marks: 2)

Find a non-isomorphic tree with five vertices.

There are three non-isomorphic trees with five vertices as shown (where every tree with five vertices has 5-1=4 edges).



Question No: 42 (Marks: 2)

Define a predicate.

Let the declarative statement:
"x is greater than 3".
We denote this declarative statement by P(x) where

x is the variable,

P is the predicate "is greater than 3".
The declarative statement P(x) is said to be the value of the propositional function P at x.

Question No: 43 (Marks: 2)

Write the following in the factorial form:

(n +2)(n+1) n

Question No: 44 (Marks: 3)

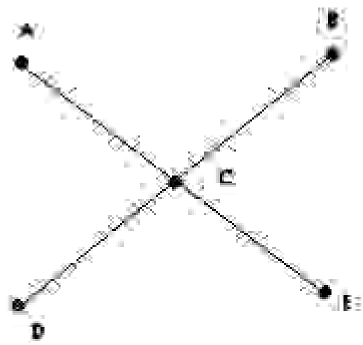
Determine the probability of the given event
"An odd number appears in the toss of a fair die"

Sample space will be..S={1,2.3,4,5,6}…there are 3 odd numbers so,
For odd numbers, probability will be

…Ans

Question No: 45 (Marks: 3)

Determine whether the following graph has Hamiltonian circuit.



This graph is not a Hamiltonian circuit, because it does not satisfy all conditions of it.
E.g. it has unequal number of vertices and edges. And its path cannot be formed without repeating vertices.

Question No: 46 (Marks: 3)

Prove that If the sum of any two integers is even, then so is their difference.

Theorem: ∀ integers m and n, if m + n is even, then so is m - n.

Proof:

Suppose m and n are integers so that m + n is even. By definition of even, m + n = 2k for some integer k. Subtracting n from both sides gives m = 2k - n. Thus,

m - n	= (2k - n) - n	by substitution
	= 2k - 2n	combining common terms
	= 2(k - n)	by factoring out a 2

But (k - n) is an integer because it is a difference of integers. Hence, (m - n) equals 2 times an integer, and so by definition of even number, (m - n) is even.

This completes the proof.

Question No: 47 (Marks: 5)

Show that if seven colors are used to paint 50 heavy bikes, at least 8 heavy bikes will be the same color.

N=50
K=7
C(7+50-1,7)

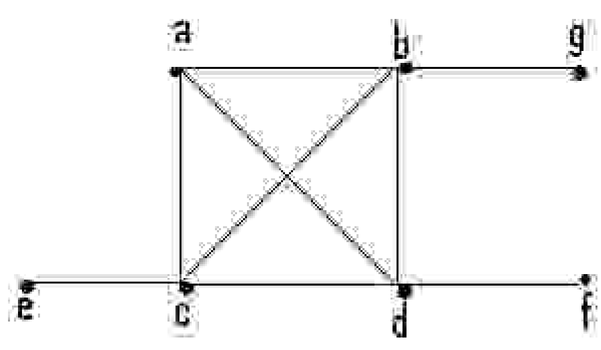
C(56,7)

56!/(56-7)!7!
56!/49!.7!

Question No: 48 (Marks: 5)

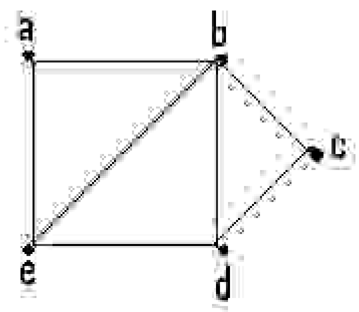
Determine whether the given graph has a Hamilton circuit? If it does, find such a circuit, if it does not , given an argument to show why no such circuit exists.

(a)



This graph does not have Hamiltonian circuit, because it does not satisfy the conditions. Circuit may not be completed without repeating edges. It has also unequal values of edges and vertices.

(b)



This graph is a Hamiltonian circuit ..Its path is a b c d e a

Question No: 49 (Marks: 5)

Find the GCD of **11425 , 450** using Division Algorithm.

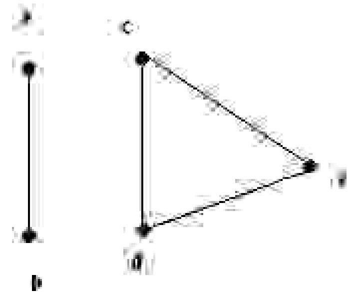
LCM = 205650

11425 = 450x25 + 175
450 = 175x2 + 100
175 = 100x1 + 75
100 = 75x1 + 25
75 = 25x3 + 0

Linear combination= 25 = 127x450 + -5x11425
GCD= 25…Ans

Question No: 50 (Marks: 10)

Write the adjacency matrix of the given graph also find transpose and product of adjacency matrix and its transpose (if not possible then give reason)



Adjacency matrix= 0 1 0 0 0
1 0 0 0 0
0 0 0 1 1
0 0 1 0 1
0 0 1 1 0

Transpose = 0 1 0 0 0
1 0 0 0 0
0 0 0 1 1
0 0 1 0 1
0 0 1 1 0

Its transpose is not possible…it’s same. Because there is no loop. It is not directed graph.

FINALTERM EXAMINATION
Spring 2011

Q1:Let $R \rightarrow R$ be defined by **5marks**

$$f(x)=\frac{2x+1}{2x+2}$$

Is f one-to-one?

Solution:

$$\frac{2x_1+1}{2x_1+2}=\frac{2x_2+1}{2x_2+2}$$
$$(2x_1+1)(2x_2+2)=(2x_2+1)(2x_1+2)$$
$$4x_1x_2+4x_1+2x_2+2=4x_1x_2+4x_2+2x_1+2$$
$$\cancel{4x_1x_2}+4x_1+2x_2+\cancel{2}-\cancel{4x_1x_2}-4x_2-2x_1-\cancel{2}=0$$
$$4x_1+2x_2-4x_2-2x_1=0$$
$$4x_1-2x_1=4x_2-2x_2$$
$$2x_1=2x_2$$
$$x_1=x_2$$

$x_1 = x_2$ Therefore this function is one to one function

Q2: Find the gcd of 1075, 45 using dividing algorithm. **5marks**

Solution:

1.Divide 1075 by 45:

This gives 1075 = 45 * 23 + 40

2.Divide 45 by 40:

This gives 45 = 40 · 1 + 5

3.Divide 40 by 5:

This gives 40 = 5 * 8 + 0

Hence gcd (1075, 45) = 5.

Q 3. Compute $\lfloor x \rfloor$ and $\lceil x \rceil$ for x=-2.01 **3 marks**

Solution: Page 249

$\lfloor -2.01 \rfloor = \lfloor -3 + 0.99 \rfloor = -3$
$\lceil -2.01 \rceil = \lceil -3 + 0.999 \rceil = -3 + 1 = -2$

Q 4: Find the greatest common division for the following pair of integer: 30,105 2marks

Solution:

1.Divide 105 by 30:

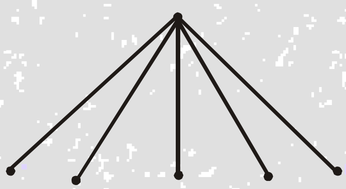
This gives 105 = 30 * 5 +0

Hence gcd (105,30) = 30

Q 5: Find the Spanning tree for the graph $K_{1,5}$? 2marks

Solution: Page 332

$k_{1,5}$ represents a complete bipartite graph on (1,5) vertices, drawn below:



Clearly the graph itself is a tree (six vertices and five edges). Hence the graph is itself a spanning tree.

Q 6 Determine which f is a function?

$$f(x) = \frac{1}{n^2 - 4}$$

Q7. What is the difference between? {a,b} and {{a,b}}?

Solution:

{a,b} is a set while {{a,b}} is a subset of some set.

Q8. How many 3-digits can be formed by using each one of the digits 2,3,5,7,9 only once?

Solution:

5* 4* 3 = 60

Q9 what is the smallest integer N such that [N/9] =6?

$N = 9 \times (6-1) + 1$

$= 9 \times 5 + 1 = 46$

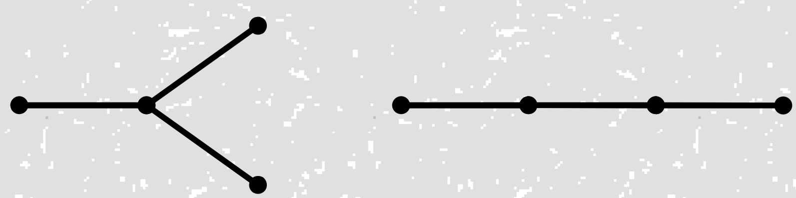
FINALTERM EXAMINATION
Spring 2010
MTH202- Discrete Mathematics (Session - 2)

Question No: 31 (Marks: 2)
Let A and B be the events. Rewrite the following event using set notation
"A or not B occurs"

Question No: 32 (Marks: 2)
Find a non-isomorphic tree with four vertices.
Solution: Page 323

Any tree with four vertices has (4-1=3) three edges. Thus, the total degree of a tree with 4 vertices must be 6 [by using total degree=2(total number of edges)].

Also, every tree with more than one vertex has at least two vertices of degree 1, so the only possible combinations of degrees for the vertices of the trees are 1, 1, 1, 3 and 1, 1, 2, 2.
The corresponding trees (clearly non-isomorphic, by definition) are



Question No: 33 (Marks: 2)
Write the following in the factorial form:
n(n -1)(n-2)…(n-r+1)
Solution Page 217:

$$n(n-1)(n-2)\ldots(n-r+1) = \frac{n(n-1)(n-2)\ldots(n-r+1)-(n-r)!}{(n-r)!}$$

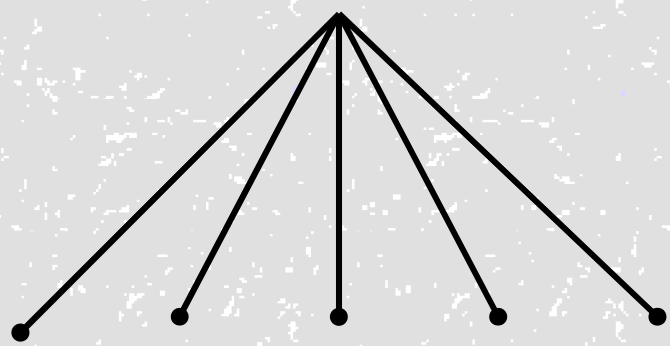
$$= \frac{n!}{(n-r)!}$$

Question No: 34 (Marks: 3)
Compute ëxû and éxù for $x = 25/4$

Question No: 35 (Marks: 3)
Find a spanning tree for the graph $k_{1,5}$?

$k_{1,5}$ represents a complete bipartite graph on (1,5) vertices, drawn below:

Solution (Page 332):



Clearly the graph itself is a tree (six vertices and five edges). Hence the graph is itself a spanning tree.

Question No: 36 (Marks: 3)
The members of a club are 12 boys and 8 girls. In how many ways can a committee of 3 boys and 2 girls be formed?

Solution:
$C(12,3)\times C(8,2)=220\times 28=6160$

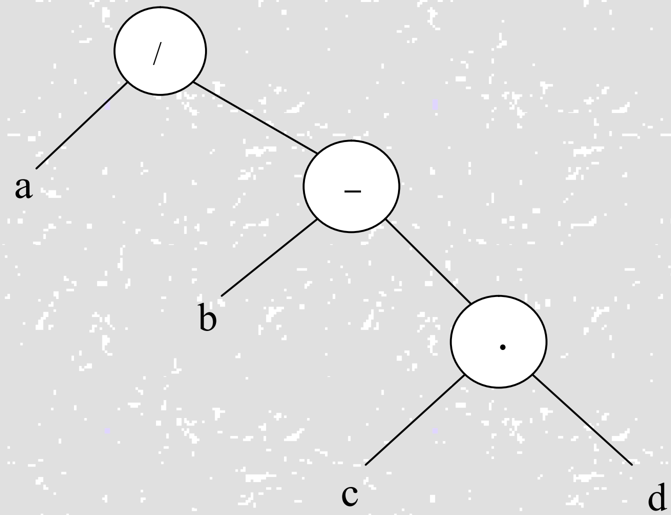
Question No: 37 (Marks: 5)
Is it possible to have a simple graph with four vertices of degree 1, 1, 3, and 3.If no then give reason?(Justify your answer)
Solution:

Yes, It is possible to make a graph with four vertices of degree 1, 1, 3, 3
Because 1+1+3+3=8
And According to handshaking theorem, the sum of the degrees of all the vertices of G equals twice the number of edges of G.
2*4=8
So it is possible.

Question No: 38 (Marks: 5)
Draw a binary tree to represent the following expression
a/(b-c.d)

The internal vertices are arithmetic operators, the terminal vertices are variables and the operator at each vertex acts on its left and right sub trees in left-right order.

Solution:



Question No: 39 (Marks: 5)
There are 25 people who work in an office together. Four of these people are selected to attend four different conferences. The first person selected will go to a conference in New York, the second will go to Chicago, the third to San Francisco, and the fourth to Miami. How many such selections are possible?

FINALTERM EXAMINATION
Spring 2010
MTH202- Discrete Mathematics (Session - 1)

Question No: 31 (Marks: 2)
Let A and B be the events. Rewrite the following event using set notation
"Only A occurs"
AB^c

Question No: 32 (Marks: 2)
Suppose that a connected planar simple graph has 15 edges. If a plane drawing of this graph has 7 faces, how many vertices does this graph have?

Answer:

Given,
Edges = v =15
Faces = f = 7

Vertices = v =?
According toEuler Formula, we know that,

$f = e - v + 2$

Putting values, we get

$7 = 15 - v + 2$

$7 = 17 - v$

Simplifying

$v = 17 - 7 = 10$

Question No: 33 (Marks: 2)

How many ordered selections of two elements can be made from the set {0,1,2,3}?

Answer:

The order selection of two elements from 4 is as

$P(4, 2) = 4! / (4-2)!$

$= (4.3.2.1)/2!$

$= 12$

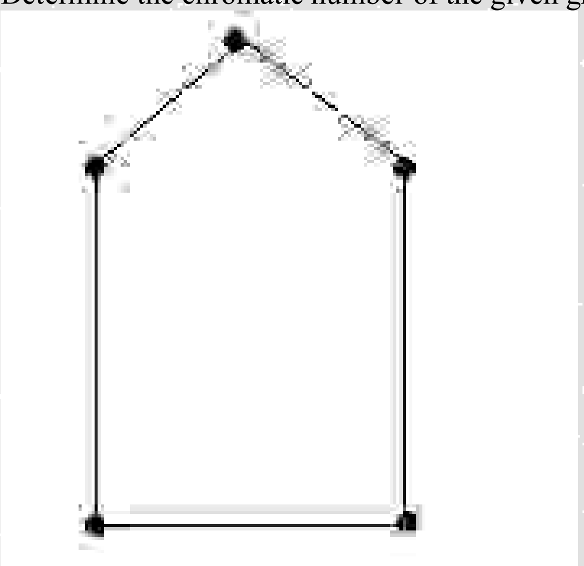
Question No: 34 (Marks: 3)

Consider the following events for a family with children:

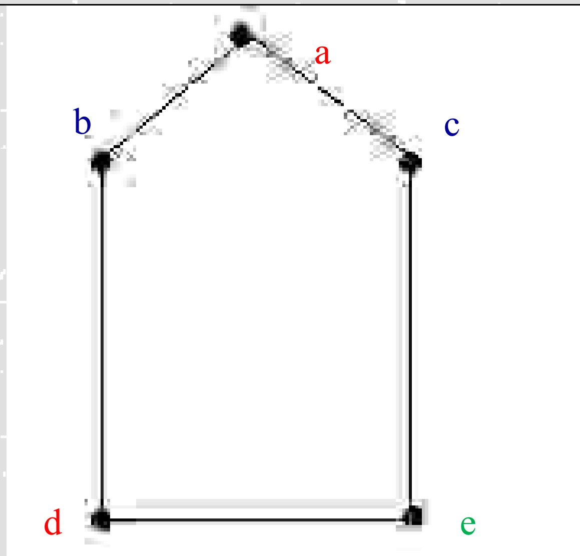
A={children of both sexes}, B={at most one boy}.Show that A and B are dependent events if a family has only two children.

Question No: 35 (Marks: 3)

Determine the chromatic number of the given graph by inspection.



Solution:



The **chromatic number** of a graph is the least (minimum) number of colors for coloring of this graph.
So chromatic number in this graph is 3

Question No: 36 (Marks: 3)
A cafeteria offers a choice of two soups, five sandwiches, three desserts and three drinks. How many different lunches, each consisting of a soup, a sandwich, a dessert and a drink are possible?
Solution:

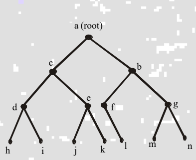
$$C(13,4) = \frac{13!}{4! \times (13-4)!}$$

$$= \frac{13 \times 12 \times 11 \times 10 \times \cancel{9!}}{4! \times \cancel{9!}} = \frac{17160}{24} = 715$$

Question No: 37 (Marks: 5)
A box contains 15 items, 4 of which are defective and 11 are good. Two items are selected. What is probability that the first is good and the second defective?

Question No: 38 (Marks: 5)
Draw a binary tree with height 3 and having seven terminal vertices.
Solution: On Page 327

Given height=h=3
Any binary tree with height 3 has atmost 2^3=8 terminal vertices.
But here terminal vertices are 7
and Internal vertices=k=6 so binary tree exists and is as follows:



Question No: 39 (Marks: 5)
Find n if
P(n,2) = 72
Solution:

$P(n,2) = 72$

$n(n-1) = 72$ by using the definition of permutation

$n^2 - n = 72$

$n^2 - n - 72 = 0$

$n^2 + 8n - 9n - 72 = 0$

$n(n+8) - 9(n+8) = 0$

$(n-9)(n+8) = 0$

$n - 9 = 0 \qquad n + 8 = 0$

$n = 9 \qquad n = -8$

$n = 9$ or -8

since n must be positive so only the acceptable value for n is 9

FINALTERM EXAMINATION
Fall 2009
MTH202- Discrete Mathematics

(Marks: 2)
Find the degree sequence of the following graph

(Marks: 2)
Let A and B be events with

$P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$

Find

$P(A \mid B)$

Solution:

$$P(A|B)=\frac{P(A\cap B)}{P(B)}$$

$$=\frac{\frac{1}{4}}{\frac{1}{3}}$$

$$=\frac{1}{4}\times\frac{3}{1}=\frac{3}{4}$$

(Marks: 3)

Find the greatest common divisor of the following pair of integer:
72,63

Solution:

1.Divide 72 by 63:

This gives 72 = 63 * 1 + 9

2.Divide 63 by 9:

This gives 63 = 9 * 7 +0

Hence gcd (72, 63) = 9.

(Marks: 2)

Find all non isomorphic simple connected graphs with three vertices.

(Marks: 3)

How many 3-digit numbers can be formed by using each one of the digits 2,3,5,7,9 only once?

Solution:

5*4*3=60

(Marks: 3)

How many permutations of the letter of the word PANAMA can be made, if P is to be the first letter in each arrangement?

Solution:

Total letter =6
Like letters = A =3
First letter is P already selected , remaining =5
Therefore,
P (5,3) = 6

(Marks: 5) incomplete Question

A die is weighted so that the outcomes produce the following probability distribution:

Outcome	1	2	3	4	5	6
Probability	0.1	0.3	0.2	0.1	0.1	0.2

Consider the event

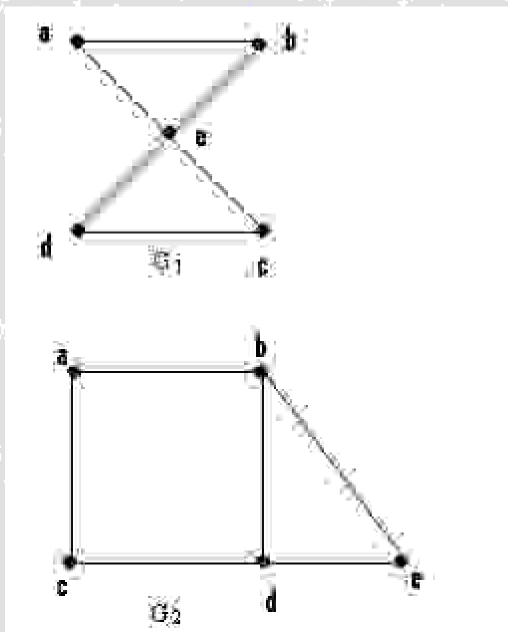
A= {even number} then find the following

a) P(A)

b) P (A^c)

(Marks: 5)

Determine whether the given graphs have an Euler circuit? If it does, find such a circuit, if it does not, give an argument to show why no such circuit exists.



(Marks: 5)

By using Mathematical induction prove that for all positive integers n

$1^2 + 2^2 + 3^2 + ... + n^2 = n(n+1)(2n+1) / 6$

(Marks: 10)

Prove by mathematical induction that $3^{(2n-1)} + 1$ is divisible by 4 for all $n \geq 1$.

Question No: 21 (Marks: 2)

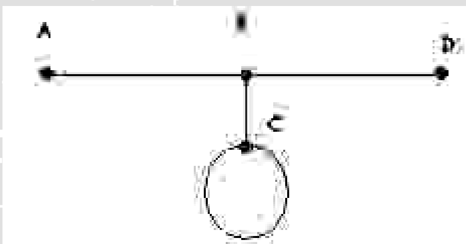
Find integers q and r so that a=bq+r , with 0≤r<b.

a=45 , b=6.

Solution:

If a =45 and b = 6 are two integers with b $\neq$ 0 such that the q and r are non negative integers.
a=bq+r
divides 45 by 6
this gives= 6*7 +3
divides 6 by 3
this gives = 3*2 +0
hence gcd of the (45,6) will be 3

Question No: 22 (Marks: 2)
Give the degree of each vertex in the figure (given below)



Solution:

degree of A vertex = 1
Degree of B vertex = 3
Degree of C vertex = 3
Degree of D vertex = 1
Total degree of vertices = 8
Can be prove by formula
Degree of vertices = 2. no. of edges
= 2 . 4
= 8

Question No: 23 (Marks: 2)
What is the probability of getting a number greater than 2 when a dice is tossed?

Solution:
As dice has 6 sides so possible event will be 36.
No. greater than 2 will be
3 ,4 ,5 ,6 = 4 outcomes are greater than 2

P(E) =n(E)/(n(S))

= 4/36

= 1/9 will be the possibility to get no greater than 2

Question No: 24 (Marks: 3)
How many distinguishable ways can the letters of the word HULLABALOO be arranged if words are to begin

with U and end with L

Solution:

If the words are to begin with U and end with L, then there are eight positions left to fill.
Where,
There are 3 L alike
2 O alike
2 A alike.
Therefore, the permutation becomes.

$$=\frac{8!}{3!*2!*2!}=\frac{40320}{24}=1680$$

Question No: 26 (Marks: 3)

Draw a full binary tree with seven vertices.
exact example hai Lecture 44 ki

Solution:

EXERCISE:

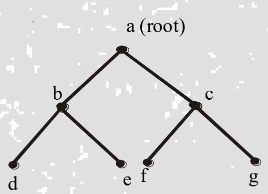
Draw a full binary tree with seven vertices.

SOLUTION:

Total vertices= 2k + 1 = 7 (by using the above theorem)

$\Rightarrow$ k = 3

Hence, total number of internal vertices (i.e. a vertex of degree greater than 1)=k=3
and total number of terminal vertices(i.e. a vertex of degree 1 in a tree) =k+1=3+1=4
Hence, a full binary tree with seven vertices is



Question No: 27 (Marks: 5)

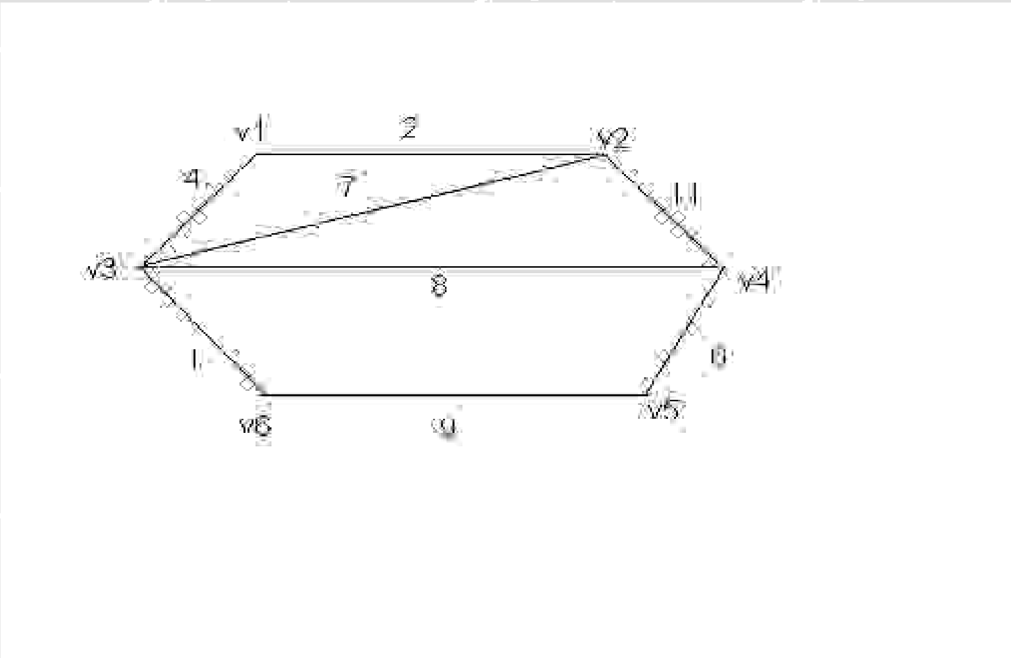
Find n if
P(n,2) = 72

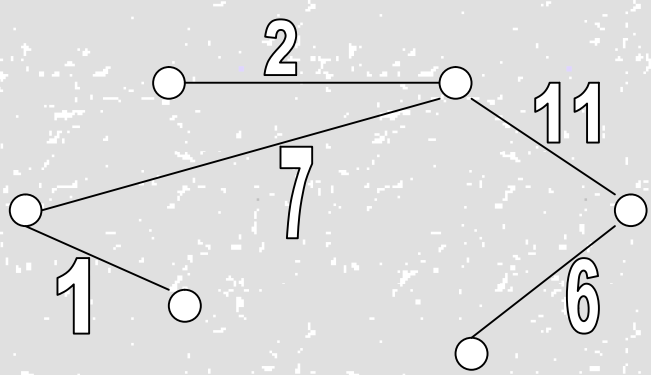
Solution:

Given
P(n,2) = 72
n.(n-1)=72 by using the definition of permutation
n^2-1=72
n^2-n-72=0
n=9,-8 since n must be positive so only the acceptable value for n is 9

Question No: 28 (Marks: 5)
Five people are to be seated around a circular table. Two seating plans are considered as same if one is the rotation of other. How many different seating plans are possible?

Question No: 29 (Marks: 5)
Use Kruskal's Algorithm to draw the minimal spanning tree for the graph below. Indicate the order in which edges are added to form a tree.





Order of adding the edges:

{v3,v6},{v1,v2},{v4,v5},{v2,v3},{v2,v4},.

Question No: 30 (Marks: 10)
Show the sample space for tossing one penny and rolling one die.
(H = heads, T = tails) using tree diagram

Question No: 31 (Marks: 10)
,$10^{3n} + 13^{n+1}$ is divisible by 7 for all n>=1

Solution:

Let $10^{3n} + 13^{n+1}$ is divisible by 7

Basis step:
P(1) is true.now
P(1):

(1.1)

$10_{3n} + 13_{n+1}$ is divisible by 7

Since $10^{3.1}+13^{1+1} = 10^3+13^2$
This is divisible by 7
Hence P(1) is true.now

Inductive step:
Suppose p(k is true)
$10_{3k} + 13\ k_{+1} = 7.q$
To prove p(k+1) $10_{3n} + 13_{n+1}$ is true is divisible by 7

$10_{3k+1} + 13_{k+1+1} = 23_{4k+3}$
$= 23_{4k+3} + 2 - 2$
$= 21_{4k+3} + 2$
$= 7.3_{4k+3} + 2$
$= 7(3_{4k+3} + 2)$

$= 7.q$ where q is ant positive integer equal to $3_{4k+3} + 2$
So its proved that $10^{3n} + 13^{n+1}$ divisible by 7 for all n>=1

MTH202 Discrete Mathematics
Lecture Wise Questions and Answers
For Final Term Exam Preparation by Virtualians Social Network

Lecture No. 23.

Q. What is inductive step?
Ans.
In basis step we check that the given statement or proposition is true for n=1 and in Inductive step we first suppose that the proposition is true for n=k and prove that it is also true for n=k+1.
In the inductive step we suppose that a function is true for an arbitrary integer k and then we prove that the given statement is also hold for k+1. It means that the statement is true for all positive integers n
Q. How can we define propositional function ?
Ans.
In simple words propositional function is an expression in form of proposition but having undefined symbols (variables) and become a proposition by assigning appropriate values to these symbols.
Q. In the Basis step of mathematical Induction, I've seen in some problems P(0) is used while in some P(1) is used.

In the basic definition we've to find P(1) is true. So what's the reason of using P(0)?
Ans.
Principle of Mathematical Induction is defined as follow:
Let P(n) be a property that is defined for integers n, and let ' a ' be a fixed integer.

Suppose the following two statements are true:

1. P(a) is true.

2. For all integers $k \geq a$, if P(k) is true then P(k + 1) is true.

Then the statement for all integers $n \geq a$, P(n) is true.

Here P(1) is not fixed, we take P(a) for some fixed ' a ' and than suppose for some $k \geq a$, and so on.
Q. Explain the last exercise of the lecture (DeMorgan's Law)
Ans.
That is the DeMorgan's Law for the n sets. For understanding it, you should see first DeMorgan's Law for two sets A and B which is following:

$(A \cup B)^c = A^c \cap B^c$

For n number of sets A1, A2, ..., An, DeMorgan's Law is the following:

$(A_1 \cup A_2 \cup ... \cup A_n)^c = A_1^c \cap A_2^c \cap .... \cap A_n^c$

This is to be proved using Mathematical Induction in the lecture.
Q.In first example basic step why it is written that series is ending at one while interger are increasing?
Ans.
In the Mathematical Induction, we take some initial value of n in the basic step. This initial value can be 0 or 1.

In the example, we take n = 1 in the series. It means we take only one term in the series as an initial value for our Mathematical Induction.
Lecture No.24.
Q.what is consecutive positive integer and inequality?
Consecutive integers are integers that follow each other in order.
For Example
1,2,3,... are called consecutive positive integers. You can see, we will get next integer by adding 1 in the current integer.
Inequality is a statement in which equal sign is not involved or which is not equal.
For example:
$2 < 5$ or $7 > 4$.
Lecture No. 25
Q. What is the rule of direct proof?
In direct proof, the conclusion is established by logically combining the axioms, definitions, and earlier theorems. For example, direct proof can be used to establish that the sum of two even integers is always even.
Q. What is rational number and integer?
The natural numbers including 0 (0, 1, 2, 3, ...) together with the negatives of the non-zero natural numbers (−1, −2, −3, ...). they are numbers that can be written without a fractional or decimal component, and fall within the set {... −2, −1, 0, 1, 2, ...}. For example, 65, 7, and −756 are integers; 1.6 and 1½ are not integers.

A rational number is a number that can be in the form p/q where p and q are integers and q is not equal to zero.
Any whole number is rational. Its denominator is 1. For instance, 8 equals 8/1, which is the quotient of two integers. A number like square root of 16 is rational, since it can be expressed as

the quotient of two integers in the form 4/1. The following are also examples of rational numbers: 5/6,-6/7,3/1 etc

Q. Define Prime number?

A prime number is a whole number greater than 1, whose only two whole-number factors are 1 and itself. The first few prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23, and 29. Diviser is the number that can divide your desired number. e.g. 6 has the following divisor 1,2,3,6.

Lecture No.26

Q. What is Direct Method and Indirect method?

There are two methods of proof Direct Method and Indirect Methods.

In Direct Method we try to prove the statement directly as it is .

For example "Prove that the sum of two odd integers is even." We will take two odd integers and try to prove that their sum is even.

While in Indirect Methods we try to restate the given statement and use this to prove our objective.

Q. Define Geometric Sequence or Geometric Progression (G.P.)?

A sequence in which every term after the first is obtained from the preceding term by multiplying it with a constant number is called a geometric sequence or geometric progression (G.P.)

The constant number, being the ratio of any two consecutive terms is called the common ratio of the G.P. commonly denoted by “r”.

Example: 6, -2, 2/3, - 2/9, … (common ratio = r = -1/3).

The sum of the terms of a geometric sequence forms a geometric series (G.S.). In general, if a is the first term and r the common ratio of a geometric series, then the series is given as: $a + ar + ar^2 + ar^3 + \dots$

For Example: 6 -2 + 2/3 - 2/9 +… (common ratio = r = -1/3)

is geometric series.

For finite series the formula is

$S_n = (a(1-r^n)/(1-r))$ since r ¹1.

For the given example: 6 -2 + 2/3 - 2/9 +… (common ratio = r = -1/3) is an infinite geometric series so the formula to find sum is

$S_n = (a/(1-r))$ since $|r| < 1$.

$S_n = (a/(r-1))$ if $|r| > 1$.

Lecture No. 27

Q. Define RECURSION?

It is the process of describing an action in terms of itself. OR

The process of defining an object in terms of smaller versions of itself is called recursion.

A recursive definition has two parts:

1.BASE:

An initial simple definition which cannot be expressed in terms of smaller versions of itself.

2. RECURSION:

The part of definition which can be expressed in terms of smaller versions of itself

Q.PRE-CONDITIONS AND POST-CONDITIONS?
Pre-condition is a condition or predicate that must always be true just prior to the theorems, definitions or before an operation in a formal specification.
Post-condition is a condition that must always be tested just after the Pre-condition.
Lecture No.28
Q. The loop invariant?
The loop invariant is a method which is used to prove the correctness of the loop with respect to certain per and post conditions. It is based on the principle of mathematical induction
Q. Division Algorithm?

For any integer n and a positive integer d, there exist unique integers q and r such that $n = d \cdot q + r$ and $0 \leq r < d$.

For example, for integers 5 and 3, there exist 1 and 2 such that $5 = 3x1 + 2$

Q. The Euclidean Algorithm?

The Euclidean algorithm takes integers a and b with $a > b \geq 0$ and calculate their greatest common divisor.

See the example at page #205

Q. Lemma?

Lemma is a proven result or proposition that helps for drawing the larger results.

Q. Euclidean algorithm?
In mathematics, the Euclidean algorithm (also called Euclid's algorithm) is an efficient method for computing the greatest common divisor (GCD) of two integers, also known as the greatest common factor (GCF) or highest common factor (HCF). It is named after the Greek mathematician Euclid.
The Euclidean algorithm is an algorithm for finding the greatest common divisor (G.C.D) of two numbers a and b.
Example: Suppose that a = 2322, b = 654.
$2322 = 654 \cdot 3 + 360$
$654 = 360 \cdot 1 + 294$
$360 = 294 \cdot 1 + 66$
$294 = 66 \cdot 4 + 30$
$66 = 30 \cdot 2 + 6$
$30 = 6 \cdot 5$
So gcd(2322,654) = 6

Lecture No.29

Q .What is combinatores?

Combinatory is the mathematics of counting and arranging objects.Counting of objects with certain properties (enumeration) is required to solve many different types of problem .For example,counting is used to:

(i) Determine number of ordered or unordered arrangement of objects.

(ii)Generate all the arrangements of a specified kind which is important in computer simulations.

(iii)Compute probabilities of events.

(iv)Analyze the chance of winning games, lotteries etc.

(v)Determine the complexity of algorithms

RULES FOR SUM

If one event can occur in n_1 ways, a second event can occur in n_2 (different) ways, then the total number of ways in which exactly one of the events (i.e., first or second) can occur is $n_1 + n_2$.

EXAMPLE:

Suppose there are 7 different optional courses in Computer Science and 3 different optional courses in Mathematics. Then there are $7 + 3 = 10$ choices for a student who wants to take one optional course

EXERCISE:

A student can choose a computer project from one of the three lists. The three lists contain 23, 15 and 19 possible projects, respectively. How many possible projects are there to choose from?

SOLUTION:

The student can choose a project from the first list in 23 ways, from the second list in 15 ways, and from the third list in 19 ways. Hence, there are

$23 + 15 + 19 = 57$ projects to choose from

PRDUCT RULE IN TRMS OF SET

If $A_1, A_2, \ldots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements in each set.

If $n(A_i)$ denotes the number of elements in set A_i, then

$n(A_1 ´ A_2´ \ldots ´A_m) = n (A_1) \cdot n(A_2) \cdot\ldots \cdot n (A_{m,})$

EXERCISE:

Find the number n of ways that an organization consisting of 15 members can elect a president, treasurer, and secretary. (assuming no person is elected to more than one position)

SOLUTION:

The president can be elected in 15 different ways; following this, the treasurer can be elected in 14 different ways; and following this, the secretary can be elected in 13 different ways. Thus, by product rule, there are

$n = 15 \times 14 \times 13 = 2730$

different ways in which the organization can elect the officers

Lecture No.30

Q.FACTORIAL OF A POSITIVE INTEGER?

For each positive integer n, its factorial is defined to be the product of all the integers from 1 to n and is denoted n!. Thus n! = n(n - 1) (n - 2) … 3×2 ×1

In addition, we define

0! = 1

REMARK:

n! can be recursively defined as

Base: 0! = 1

Recursion n! = n (n - 1)! for each positive integer N

Lecture No.30

FACTORIAL OF A POSITIVE INTEGER:

For each positive integer n, its factorial is defined to be the product of all the integers from 1 to n and is denoted n!. Thus n! = n(n - 1) (n - 2) … 3×2 ×1

In addition, we define

0! = 1

REMARK:

n! can be recursively defined as

Base: 0! = 1

Recursion n! = n (n - 1)! for each positive integer N

Lecture No.31

A k-combination of a set of n elements is a choice of k elements taken from the set of n elements such that the order of the elements does not matter and elements can't be repeated.

The symbol $C(n, k)$ denotes the number of k-combinations that can be chosen from a set of n elements.

EXAMPLE:

Let $X = \{a, b, c\}$. Then 2-combinations of the 3 elements of the set X are:

$\{a, b\}$, $\{a, c\}$, and $\{b, c\}$. Hence $C(3,2) = 3$.

EXERCISE:

Let $X = \{a, b, c, d, e\}$.

List all 3-combinations of the 5 elements of the set X, and hence find the value of $C(5,3)$.

SOLUTION:

Then 3-combinations of the 5 elements of the set X are:

$\{a, b, c\}$, $\{a, b, d\}$, $\{a, b, e\}$, $\{a, c, d\}$, $\{a, c, e\}$,

$\{a, d, e\}$, $\{b, c, d\}$, $\{b, c, e\}$, $\{b, d, e\}$, $\{c, d, e\}$

Lecture No.32

DEFINITION:

A k-selection of a set of n elements is a choice of k elements taken from a set of n elements such that the order of elements does not matter and elements can be repeated.

REMARK:

1. k-selections are also called k-combinations with repetition allowed or multisets of size k.
2. With k-selections of a set of n elements repetition of elements is allowed. Therefore k need not to be less than or equal to n.

ORDERED AND UNORDERED PARTITIONS:

An unordered partition of a finite set S is a collection $[A_1, A_2, \ldots, A_k]$ of disjoint (nonempty) subsets of S (called cells) whose union is S.

The partition is ordered if the order of the cells in the list counts.

EXAMPLE:

Let S = {1, 2, 3, …, 7}

The collections

$P_1 = [\{1,2\}, \{3,4,5\}, \{6,7\}]$

And $P_2 = [\{6,7\}, \{3,4,5\}, \{1,2\}]$

determine the same partition of S but are distinct ordered partitions

Lecture No.34

PIGEONHOLE PRINCIPLE:

A function from a set of $k + 1$ or more elements to a set of k elements must have at least two elements in the domain that have the same image in the co-domain.

If $k + 1$ or more pigeons fly into k pigeonholes then at least one pigeonhole must contain two or more pigeons.

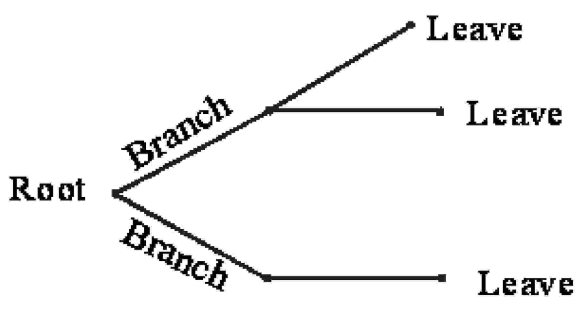
Lecture No.35

TREE DIAGRAM

A tree diagram is a useful tool to list all the logical possibilities of a sequence of events where each event can occur in a finite number of ways.

A tree consists of a root, a number of branches leaving the root, and possible additional branches leaving the end points of other branches. To use trees in counting problems, we use a branch to represent each possible choice. The possible outcomes are represented by the leaves (end points of branches).

A tree is normally constructed from left to right.



A TREE STRUCTURE

Q. Definition of tree diagram?

graphical tool which systematically breaks down, and then maps out in increasing detail, all components or elements of a condition phenomenon, process , or situation, at successive levels or stages . In case of a 'divergent tree,' it begins with a single entry that has one or more paths (branches) leading out from it, some or all of which subdivide into more branches. This process is repeated until all possibilities are exhausted. In case of a 'convergent tree,' this process works in reverse. family(genealogical) and organization charts are the common examples of a tree diagram. Also called chain of causes or dendrite diagram.

Lecture No.37

Q. Conditional probability?

CONDITIONAL PROBABILITY
MULTIPLICATION THEOREM
INDEPENDENT EVENTS

EXAMPLE:

a. What is the probability of getting a 2 when a dice is tossed?

b. An even number appears on tossing a die

(i) What is the probability that the number is 2?

(ii) What is the probability that the number is 3?

SOLUTION:

When a dice is tossed, the sample space is S={1,2,3,4,5,6} ? ? ? ? ? n(S)=6

a. Let "A" denote the event of getting a 2 i.e. A={ 2 } ? ? n(A)=1

$$P(\text{2 appears when the die is tossed}) = \frac{n(A)}{n(S)} = \frac{1}{6}$$

b (i) Let "S_1" denote the total number of even numbers from a sample space S when a dice is tossed (i.e $S_1 \subseteq S$)

S_1={ 2,4,6 } n(S_1)=3

Let "B" denote the event of getting a 2 from total number of even number i.e B={ 2 } ? n(B) = 1

$$P(\text{2 appears: given that the number is even}) = P(B) = \frac{n(B)}{n(S_1)} = \frac{1}{3}$$

(ii) Let "C" denote the event of getting a 3 in S_1 (among the even numbers) i.e C={ } n(C)=0

$$P(\text{3 appears: given that the number is even}) = P(C) = \frac{n(C)}{n(S_1)} = \frac{0}{3} = 0$$

EXAMPLE:

Suppose that an urn contains 3 red balls, 2 blue balls, and 4 white balls, and that a ball is selected at random.

Let E be the event that the ball selected is red.

Then P(E) = 3/9 (as there are 3 red balls out of total 9 balls)

Let F be the event that the ball selected is not white.

Then the probability of E if it is already known that the selected ball is not white would be

P(red ball selected, given that the selected ball is not white) = 3/5 (as we count no white ball so there are total 9 balls(i.e 2 blue and 3 red balls))

This is called the conditional probability of E given F and is denoted by P(E|F).

DEFINITION:

Let E and F be two events in the sample space of an experiment with $P(F) \neq 0$. The **conditional probability** of E given F, denoted by $P(E|F)$, is defined as

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

EXAMPLE:

Let A and B be events of an experiment such that

$P(B) = 1/4$ and $P(A \cap B) = 1/6$.

What is the conditional probability $P(A|B)$?

SOLUTION:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{1/4} = \frac{4}{6} = \frac{2}{3}$$

EXERCISE:

Let A and B be events with $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$

Find

(i) $P(A|B)$ (ii) $P(B|A)$

(iii) $P(A \cup B)$ (iv) $P(A'|B')$

SOLUTION:

Using the formula of the conditional Probability we can write

(i) $P(A|B)$ $= \frac{P(A \cap B)}{P(B)}$

$= \frac{1/4}{1/3} = \frac{3}{4}$

(ii) $P(B|A)$ $= \frac{P(B \cap A)}{P(A)}$

$= \frac{1/4}{1/2} = \frac{2}{4} = \frac{1}{2}$ (As $P(B \cap A) = P(A \cap B) = 1/4$)

(iii) $P(A \cup B)$ $= P(A) + P(B) - P(A \cap B)$

$= \frac{1}{2} + \frac{1}{3} - \frac{1}{4}$

$= \frac{7}{12}$

Lecture No.38

RANDOM VARIABLE

PROBABILITY DISTRIBUTION

EXPECTATION AND VARIANCE

INTRODUCTION:

Suppose S is the sample space of some experiment. The outcomes of the experiment, or the points in S, need not be numbers. For example in tossing a coin, the outcomes are H (heads) or T(tails), and in tossing a pair of dice the outcomes are pairs of integers. However, we frequently wish to assign a specific number to each outcome of the experiment. For example, in coin tossing, it may be convenient to assign 1 to H and 0 to T; or in the tossing of a pair of dice, we may want to assign the sum of the two integers to the outcome. Such an assignment of numerical values is called a random variable.

RANDOM VARIABLE:

A random variable X is a rule that assigns a numerical value to each outcome in a sample Space S.

OR

It is a function which maps each outcome of the sample space into the set of real numbers.

We shall let X(S) denote the set of numbers assigned by a random variable X, and refer to X(S) as the range space.

In formal terminology, X is a function from S(sample space) to the set of real numbers R, and X(S) is the range of X.

REMARK:

1. A random variable is also called a chance variable, or a stochastic variable(not called simply a variable, because it is a function).

2. Random variables are usually denoted by capital letters such as X, Y, Z; and the values taken by them are represented by the corresponding small letters.

EXAMPLE:

A pair of fair dice is tossed. The sample space S consists of the 36 ordered pairs i.e

$S = \{(1,1), (1,2), (1,3), \ldots, (6,6)\}$

Let X assign to each point in S the sum of the numbers; then X is a random variable with range space i.e

$X(S) = \{2,3,4,5,6,7,8,9,10,11,12\}$

Let Y assign to each point in S the maximum of the two numbers in the outcomes; then Y is a random variable with range space.

$Y(S) = \{1,2,3,4,5,6\}$

PROBABILITY DISTRIBUTION OF A RANDOM VARIABLE:

Let $X(S) = \{x_1, x_2, \ldots, x_n\}$ be the range space of a random variable X defined on a finite sample space S.

Define a function f on X(S) as follows:

$f(x_i) = P(X = x_i)$

= sum of probabilities of points in S whose image is x_i.
This function f is called the probability distribution or the probability function of X.
The probability distribution f of X is usually given in the form of a table.

x_1	x_2	…	x_n
$f(x_1)$	$f(x_2)$	…	$f(x_n)$

The distribution f satisfies the conditions.

(i) $f(x_i) \geq 0$ and (ii) $\sum_{i=1}^{n} f(x_i) = 1$

Q. Definition of random variable?
A random variable is a variable whose value is subject to variations due to chance (i.e. randomness, in a mathematical sense). As opposed to other mathematical variables, a random variable conceptually does not have a single, fixed value (even if unknown); rather, it can take on a set of possible different values, each with an associated probability.
A random variable's possible values might represent the possible outcomes of a yet-to-be-performed experiment or an event that has not happened yet, or the potential values of a

past experiment or event whose already-existing value is uncertain (e.g. as a result of incomplete information or imprecise measurements).

They may also conceptually represent either the results of an "objectively" random process (e.g. rolling a die), or the "subjective" randomness that results from incomplete knowledge of a quantity. The meaning of the probabilities assigned to the potential values of a random variable is not part of probability theory itself, but instead related to philosophical arguments over the interpretation of probability. The mathematics works the same regardless of the particular interpretation in use.

Lecture No.39

Q. Graph Theory?

Graph theory plays an important role in several areas of computer science such as:

- switching theory and logical design
- Artificial intelligence
- Formal languages
- Computer graphics
- Operating systems
- Compiler writing
- Information organization and retrieval.

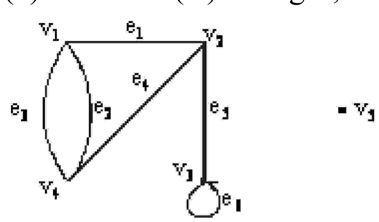
GRAPH

A graph is a non-empty set of points called vertices and a set of line segments joining pairs of vertices called edges.

Formally, a graph G consists of two finite sets:

(i) A set V=V(G) of vertices (or points or nodes)

(ii)A set E=E(G) of edges; where each edge corresponds to a pair of vertices.



The graph G with

$V(G) = \{v_1, v_2, v_3, v_4, v_5\}$ and

$E(G) = \{e_1, e_2, e_3, e_4, e_5, e_6\}$

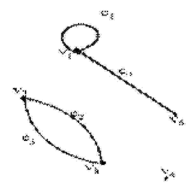
DRAWING PICTURE FOR A GRAPH

Draw picture of Graph H having vertex set $\{v_1, v_2, v_3, v_4, v_5\}$ and edge set $\{e_1, e_2, e_3, e_4\}$ with edge endpoint function

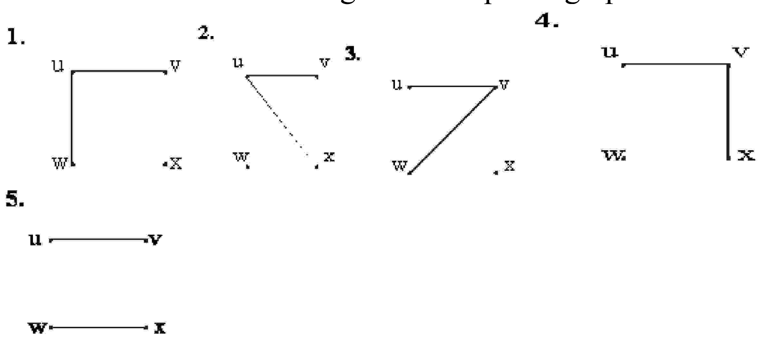
Edge	Endpoint
e_1	$\{v_1\}$
e_2	$\{v_2,v_3\}$
e_3	$\{v_2,v_3\}$

e_4	$\{v_1,v_5\}$

SOLUTION:
Given $V(H) = \{v_1, v_2, v_3, v_4, v_5\}$
and $E(H) = \{e_1, e_2, e_3, e_4\}$
with edge endpoint function



SOLUTION:
There are $C(4,2) = 6$ ways of choosing two vertices from 4 vertices. These edges may be listed as:
$\{u,v\},\{u,w\},\{u,x\},\{v,w\}, \{v,x\},\{w,x\}$
One edge of the graph is specified to be $\{u,v\}$, so any of the remaining five from this list may be chosen to be the second edge. This required graphs are:



Lecture No.40
PATHS AND CIRCUITS
KONIGSBERG BRIDGES PROBLEM
DEFINITIONS:
Let G be a graph and let v and w be vertices in graph G.
1. WALK
A walk from v to w is a finite alternating sequence of adjacent vertices and edges of G.
Thus a walk has the form
$v_0e_1v_1e_2 \dots v_{n-1}e_nv_n$
where the v's represent vertices, the e's represent edges $v_0=v$, $v_n=w$, and for all
$i = 1, 2 \dots n$, v_{i-1} and v_i are endpoints of e_i.
The trivial walk from v to v consists of the single vertex v.
2. CLOSED WALK
A closed walk is a walk that starts and ends at the same vertex.
3. CIRCUIT
A circuit is a closed walk that does not contain a repeated edge. Thus a circuit is a walk of the form

$v_0e_1v_1e_2 \ldots v_{n-1}\, e_n\, v_n$

where $v_0 = v_n$ and all the e_i,s are distinct.

Q.SIMPLE CIRCUIT?

A simple circuit is a circuit that does not have any other repeated vertex except the first and last. Thus a simple circuit is a walk of the form

$v_0e_1v_1e_2 \ldots v_{n-1}e_nv_n$

where all the e_i,s are distinct and all the v_j,s are distinct except that $v_0 = v_n$

5. PATH

A path from v to w is a walk from v to w that does not contain a repeated edge. Thus a path from v to w is a walk of the form

$v = v_0e_1v_1e_2 \ldots v_{n-1}e_nv_n = w$

where all the e_i,s are distinct (that is $e_i\ ^1\ e_k$ for any $i\ ^1k$).

6. SIMPLE PATH

A simple path from v to w is a path that does not contain a repeated vertex. Thus a simple path is a walk of the form

$v = v_0e_1v_1e_2 \ldots v_{n-1}e_nv_n= w$

where all the e_i,s are distinct and all the v_j,s are also distinct (that is, $v_j\ ^1\ v_m$ for any $j\ ^1\ m$).

Q. EULER PATH?

<u>DEFINITION:</u>

Let G be a graph and let v and w be two vertices of G. An Euler path from v to w is a sequence of adjacent edges and vertices that starts at v, end and w, passes through every vertex of G at least once, and traverses every edge of G exactly once.

Q. COROLLARY?

Let G be a graph and let v and w be two vertices of G. There is an Euler path from v to w if, and only if, G is connected, v and w have odd degree and all other vertices of G have even degree.

Q. HAMILTONIAN CIRCUITS?

<u>DEFINITION:</u>

Given a graph G, a Hamiltonian circuit for G is a simple circuit that includes every vertex of G. That is, a Hamiltonian circuit for G is a sequence of adjacent vertices and distinct edges in which every vertex of G appears exactly once.

Lecture No.41

Q. Matrix Representation of Graphs?

Definitions:

In this section, we introduce two kinds of matrix representations of a graph, that is, the adjacency matrix and incidence matrix of the graph.

A graph G with the vertex-set V (G) = {x1, x2, · · · , vv} can be described by means of matrices. The adjacency matrix of G is a v × v matrix

A(G) = (aij), where aij = μ(xi, xj) = |EG(xi, xj)|.

The adjacency matrix or the incidence matrix of a graph is another representation of the graph, and it is this form that a graph can be commonly stored in computers. The matrix representation of a graph is often convenient if one intends to use a computer to obtain some information or solve a problem concerning the graph. This kind of representation of a graph is conducive to study properties of the graph by means of algebraic methods.

It is not difficult to see that the adjacency matrices of two isomorphic graphs are per mutedly similar. In other words, assume that A and B are the adjacency matrices of two isomorphic graphs G and H, respectively, then there exists a $v \times v$ permutation matrix P such that $A = P-1BP$.

Similarly, the incidence matrices of two isomorphic graphs are per mutedly equivalent. In other words, assume that M and N are the incidence matrices of two isomorphic graphs G and H, respectively, then there exist a $v \times v$ permutation matrix P and an " × " permutation matrix Q such that $M = PNQ$.

Q. Adjacency matrix?

An adjacency matrix is a means of representing which vertices (or nodes) of a graph are adjacent to which other vertices. Another matrix representation for a graph is the incidence matrix. Specifically, the adjacency matrix of a finite graph G on n vertices is the $n \times n$ matrix where the non-diagonal entry aij is the number of edges from vertex i to vertex j, and the diagonal entry aii, depending on the convention, is either once or twice the number of edges (loops) from vertex i to itself. Undirected graphs often use the latter convention of counting loops twice, whereas directed graphs typically use the former convention. There exists a unique adjacency matrix for each isomorphism class of graphs (up to permuting rows and columns), and it is not the adjacency matrix of any other isomorphism class of graphs. In the special case of a finite simple graph, the adjacency matrix is a (0,1)-matrix with zeros on its diagonal. If the graph is undirected, the adjacency matrix is symmetric.

The relationship between a graph and the eigenvalues and eigen vectors of its adjacency matrix is studied in spectral graph theory

Lecture No.42

ISOMORPHISM OF GRAPHs

An isomorphism of graphs *G* and *H* is a bijection between the vertex sets of *G* and *H* such that any two vertices *u* and *v* of *G* are adjacent in *G* if and only if $f(u)$ and $f(v)$ are adjacent in *H*. This kind of bijection is commonly called "edge-preserving bijection", in accordance with the general notion of isomorphism being a structure-preserving bijection.

In the above definition, graphs are understood to be undirected non-labeled nonweighted graphs. However, the notion of isomorphism may be applied to all other variants of the notion of graph, by adding the requirements to preserve the corresponding additional elements of structure: arc directions, edge weights, etc., with the following exception. When spoken about graph labeling with *unique labels*, commonly taken from the integer range 1,...,*n*, where *n* is the number of the vertices of the graph, two labeled graphs are said to be isomorphic if the corresponding underlying unlabeled graphs are isomorphic.

If an isomorphism exists between two graphs, then the graphs are called isomorphic and we write . In the case when the bijection is a mapping of a graph onto itself, i.e., when *G* and *H* are one and the same graph, the bijection is called an automorphism of *G*.
The graph isomorphism is an equivalence relation on graphs and as such it partitions the class of all graphs into equivalence classes. A set of graphs isomorphic to each other is called an isomorphism class of graphs

ISOMORPHIC GRAPHS:

Let G and G' be graphs with vertex sets V(G) and V(G') and edge sets E(G) and E(G'), respectively.
G is isomorphic to G' if, and only if, there exist one-to-one correspondences g: V(G)®V(G') and h: E(G) ®E(G') that preserve the edge-endpoint functions of G
and G' in the sense that for all v ÎV(G) and eÎ E(G).
v is an endpoint of e Û g(v) is an endpoint of h(e).

EQUIVALENCE RELATION:

Graph isomorphism is an equivalence relation on the set of graphs.
1. Graphs isomorphism is Reflexive (It means that the graph should be isomorphic to itself).
2. Graphs isomorphism is Symmetric (It means that if G is isomorphic to G' then G' is also isomorphic to G).
3. Graphs isomorphism is Transitive (It means that if G is isomorphic to G' and G' is isomorphic to G'', then G is isomorphic to G'').

ISOMORPHIC INVARIANT:

A property P is called an isomorphic invariant if, and only if, given any graphs G and G', if G has property P and G' is isomorphic to G, then G' has property P.

THEOREM OF ISOMORPHIC INVARIANT:

Each of the following properties is an invariant for graph isomorphism, where n, m and k are all non-negative integers, if the graph:
1. has n vertices.
2. has m edges.
3. has a vertex of degree k.
4. has m vertices of degree k.
5. has a circuit of length k.
6. has a simple circuit of length k.
7. has m simple circuits of length k.
8. is connected.
9. has an Euler circuit.
10. has a Hamiltonian circuit.

DEGREE SEQUENCE:

The degree sequence of a graph is the list of the degrees of its vertices in non-increasing order.

GRAPH ISOMORPHISM FOR SIMPLE GRAPHS:

If G and G' are simple graphs (means the "graphs which have no loops or parallel edges") then G is isomorphic to G' if, and only if, there exists a one-to-one correspondence (1-1 and onto function) g from the vertex set V (G) of G to the vertex set
V (G') of G' that preserves the edge-endpoint functions of G and G' in the sense that for all vertices u and v of G,
{u, v} is an edge in G Û {g(u), g(v)} is an edge in G'.
OR
You can say that with the property of one-one correspondence, u and v are adjacent in graph G Û if g (u) and g (v) are adjacent in G'.
Note:
It should be noted that unfortunately, there is no efficient method for checking that whether two graphs are isomorphic(methods are there but take so much time in calculations).Despite that there is a simple condition. Two graphs are isomorphic if they have the same number of vertices(as there is a 1-1 correspondence between the vertices of both the graphs) and the same number of edges(also vertices should have the same degree.
Q.Degree sequence?
The degree sequence of an undirected graph is the non-increasing sequence of its vertex degrees. The degree sequence is a graph invariant so isomorphic graphs have the same degree sequence. However, the degree sequence does not, in general, uniquely identify a graph; in some cases, non-isomorphic graphs have the same degree sequence.
The degree sequence problem is the problem of finding some or all graphs with the degree sequence being a given non-increasing sequence of positive integers. (Trailing zeroes may be ignored since they are trivially realized by adding an appropriate number of isolated vertices to the graph.) A sequence which is the degree sequence of some graph, i.e. for which the degree sequence problem has a solution, is called a graphic or graphical sequence. As a consequence of the degree sum formula, any sequence with an odd sum, such as (3, 3, 1), cannot be realized as the degree sequence of a graph. The converse is also true: if a sequence has an even sum, it is the degree sequence of a multigraph.
Lecture No.43
PLANAR GRAPHS
GRAPH COLORING
Graph coloring is a special case of graph labeling; it is an assignment of labels traditionally called "colors" to elements of a graph subject to certain constraints. In its simplest form, it is a way of coloring the vertices of a graph such that no two adjacent vertices share the same color; this is called a vertex coloring. Similarly, an edge coloring assigns a color to each edge so that no two adjacent edges share the same color, and a face coloring of a planar graph assigns a color to each face or region so that no two faces that share a boundary have the same color.
Vertex coloring is the starting point of the subject, and other coloring problems can be transformed into a vertex version. For example, an edge coloring of a graph is just a vertex coloring of its line graph and a face coloring of a plane graph is just a vertex coloring of its dual However, non-vertex coloring problems are often stated and studied *as is*. That is partly for

perspective, and partly because some problems are best studied in non-vertex form, as for instance is edge coloring.
The convention of using colors originates from coloring the countries of a map, where each face is literally colored. This was generalized to coloring the faces of a graph embedded in the plane. By planar duality it became coloring the vertices, and in this form it generalizes to all graphs. In mathematical and computer representations, it is typical to use the first few positive or nonnegative integers as the "colors". In general, one can use any finite set as the "color set". The nature of the coloring problem depends on the number of colors but not on what they are.
In this graph, note it that each vertex is connected to every other vertex, but no edge is crossed.
Note: The graphs shown above are complete graphs with four vertices (denoted by K_4).

DEFINITION:

A graph is called planar if it can be drawn in the plane without any edge crossed (crossing means the intersection of lines). Such a drawing is called a plane drawing of the graph.

OR

You can say that a graph is called planar in which the graph crossing number is "0".

HOW TO DRAW A GRAPH FROM A MAP:

1. Each map in the plane can be represented by a graph.
2. Each region is represented by a vertex (in 1^{st} map as there are 7 regions, so 7 vertices are used in drawing a graph, similarly we can see 2^{nd} map).
3. If the regions connected by these vertices have the common border, then edge connect two vertices.
4. Two regions that touch at only one point are not adjacent.

DEFINITION:

1. A coloring of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.
2. The chromatic number of a graph is the least (minimum) number of colors for coloring of this graph

Lecture No.44

TREES

APPLICATION AREAS:

Trees are used to solve problems in a wide variety of disciplines. In computer science trees are employed to

1) construct efficient algorithms for locating items in a list.
2) construct networks with the least expensive set of telephone lines linking distributed computers.
3) construct efficient codes for storing and transmitting data.
4) model procedures that are carried out using a sequence of decisions, which are valuable in the study of sorting algorithms.

TREE:

A tree is a connected graph that does not contain any non-trivial circuit. (i.e. it is circuit-free).
A trivial circuit is one that consists of a single vertex.

SOME SPECIAL TREES

1. TRIVIAL TREE:

A graph that consists of a single vertex is called a trivial tree or degenerate tree.

2. EMPTY TREE

A tree that does not have any vertices or edges is called an empty tree.

3. FOREST

A graph is called a forest if, and only if, it is circuit-free.

OR "Any non-connected graph that contains no circuit is called a forest."

Hence, it clears that the connected components of a forest are trees.

PROPERTIES OF TREES:

1. A tree with n vertices has n - 1 edges (where n $\geq$0).
2. Any connected graph with n vertices and n - 1 edges is a tree.
3. A tree has no non-trivial circuit; but if one new edge (but no new vertex) is added to it, then the resulting graph has exactly one non-trivial circuit.
4. A tree is connected, but if any edge is deleted from it, then the resulting graph is not connected.
5. Any tree that has more than one vertex has at least two vertices of degree 1.
6. A graph is a tree iff there is a unique path between any two of its vertices

ROOTED TREE:

A rooted tree is a tree in which one vertex is distinguished from the others and is called the *root.*

The level of a vertex is the number of edges along the unique path between it and the root.

The heightof a rooted tree is the maximum level to any vertex of the tree.

The children of any internal vertex v are all those vertices that are adjacent to v and are one level farther away from the root than v.

If w is a child of v, then v is called the parent of w.

Two vertices that are both children of the same parent are called siblings.

Given vertices v and w, if v lies on the unique path between w and the root, then v is an ancestor of w and w is a descendant of v.

BINARY TREE

A *binary tree* is a rooted tree in which every internal vertex has at most two children.

Every child in a binary tree is designated either a left child or a right child (but not both).

A *full binary tree* is a binary tree in which each internal vertex has exactly two children.

Lecture No.45

SPANNING TREES:

SPANNING TREE:

A spanning tree for a graph G is a subgraph of G that contains every vertex of G and is a tree.

REMARK:

1. Every connected graph has a spanning tree.
2. A graph may have more than one spanning trees.
3. Any two spanning trees for a graph have the same number of edges.

4. If a graph is a tree, then its only spanning tree is itself.

MINIMAL SPANNING TREE:

A minimal spanning tree for a weighted graph is a spanning tree that has the least possible total weight compared to all other spanning trees of the graph.

If G is a weighted graph and e is an edge of G then w(e) denotes the weight of e and w(G) denotes the total weight of G.

KRUSKAL'S ALGORITHM:

Input: G [a weighted graph with n vertices]

Algorithm:

1. Initialize T(the minimal spanning tree of G) to have all the vertices of G and no edges.
2. Let E be the set of all edges of G and let m: = 0.
3. While (m < n - 1)

3a. Find an edge e in E of least weight.

3b. Delete e from E.

3c. If addition of e to the edge set of T does not produce a circuit then add e to the edge set of T and set m: = m + 1

end while

Output T

end Algorithm